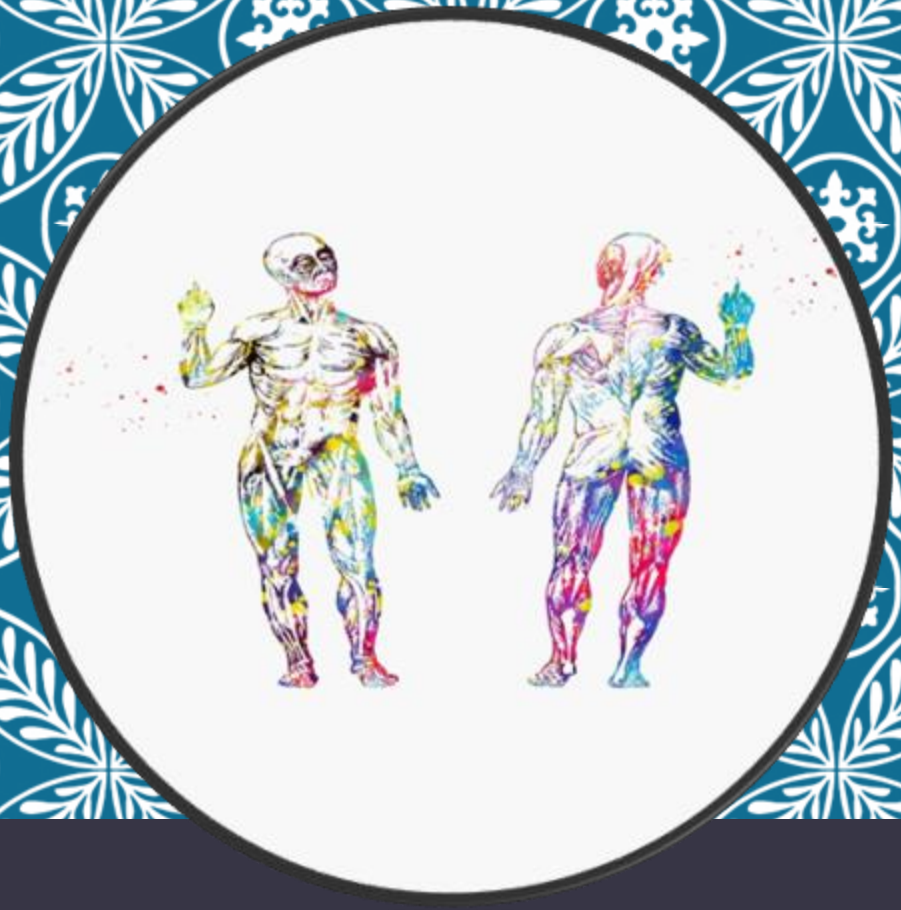


MUSCULAR SYSTEM

ANATOMY AND PHYSIOLOGY (MC1)
COLLEGE OF NURSING
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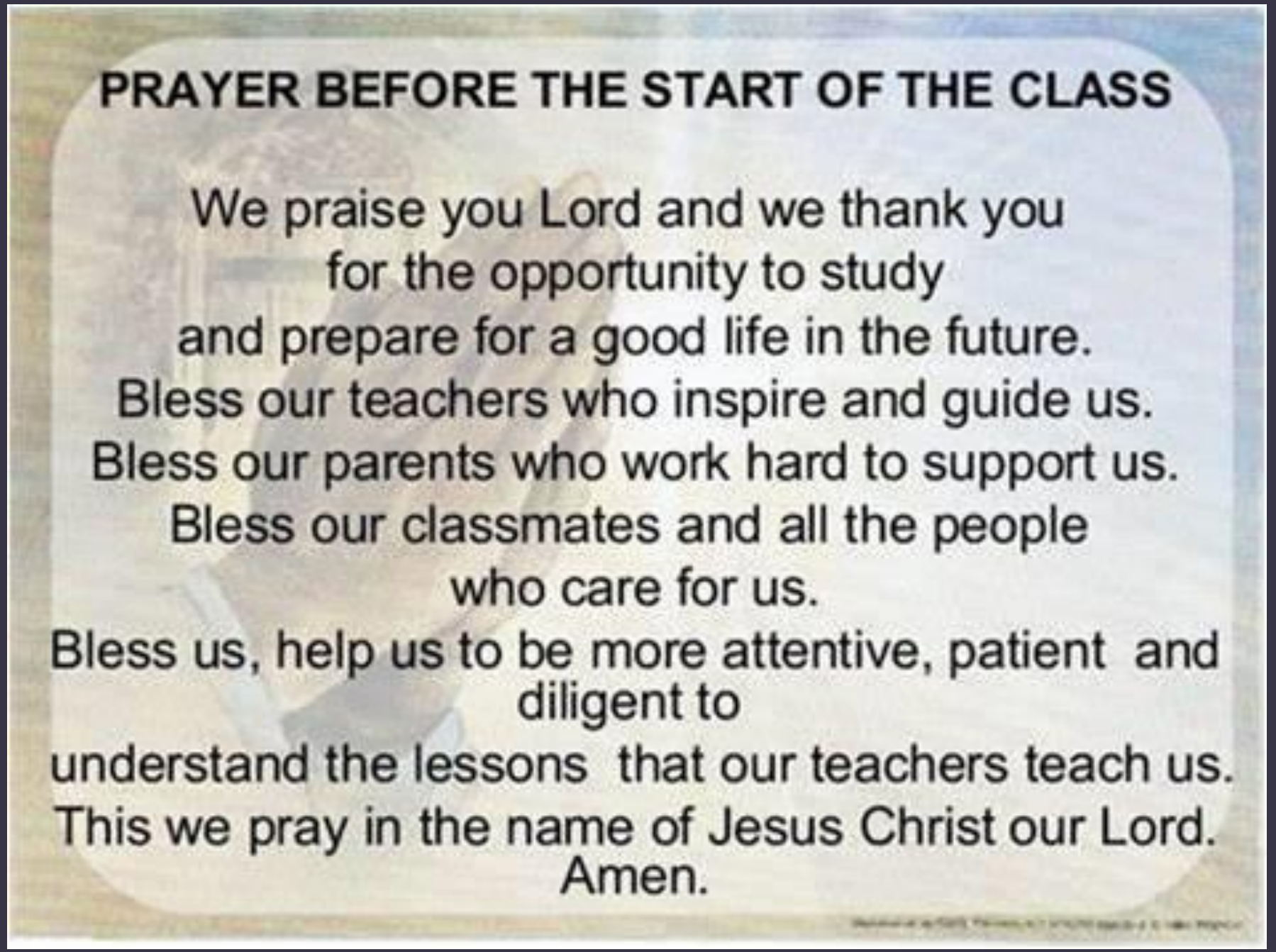
MARIA NOVIEL DOMINGUEZ, RN, MD

COLLEGE OF NURSING, PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY

ANATOMY AND PHYSIOLOGY

2020

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PRAYER BEFORE THE START OF THE CLASS
We praise you Lord and we thank you
for the opportunity to study
and prepare for a good life in the future.
Bless our teachers who inspire and guide us.
Bless our parents who work hard to support us.
Bless our classmates and all the people
who care for us.
Bless us, help us to be more attentive, patient and diligent to
understand the lessons that our teachers teach us.
This we pray in the name of Jesus Christ our Lord.
Amen.

OBJECTIVES

1. Differentiate the three types of muscle tissue: **Skeletal, Smooth, and Cardiac Muscles**
2. Enumerate and describe the **microscopic parts of skeletal muscle**
3. Identify and enumerate the **major skeletal muscles of the human body**

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THE MUSCULAR SYSTEM

Muscles and Body Movement

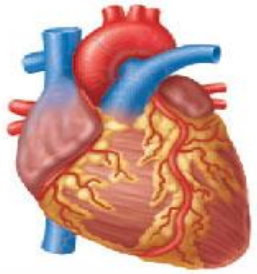
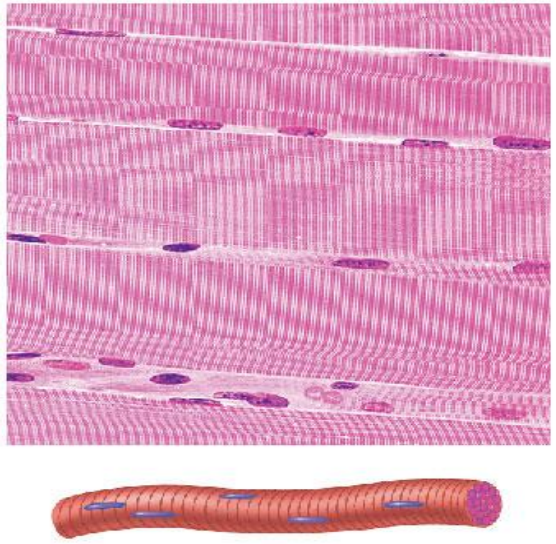
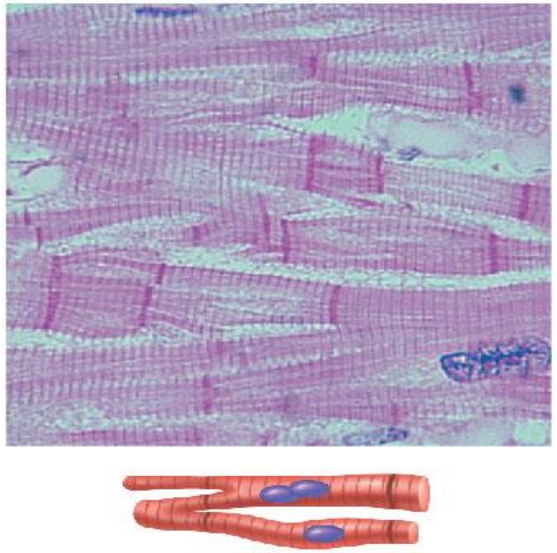
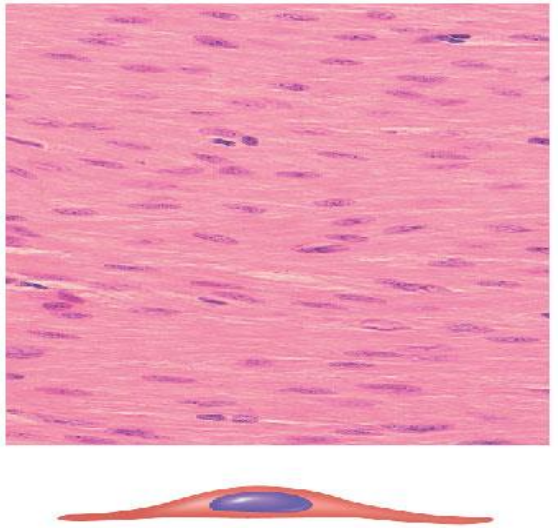
- Muscles are responsible for **all types of body movement**
- **Three basic types of muscle tissue:**
  - **Skeletal Muscle** – voluntary, striated, attached to bones
  - **Cardiac Muscle** – involuntary, striated, found only in the heart
  - **Smooth Muscle** – involuntary, non-striated, found in walls of hollow organs

Shared Terminology

- **Myo- / Mys-** → "muscle"
- **Sarco-** → "flesh," commonly used for muscle structures (e.g., sarcoplasm, sarcolemma)

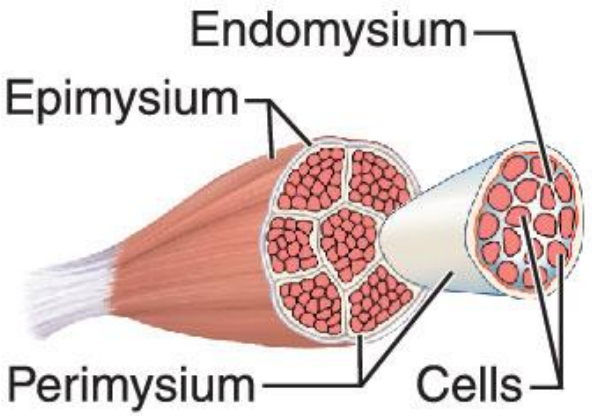
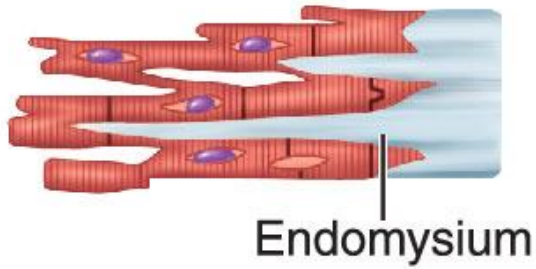
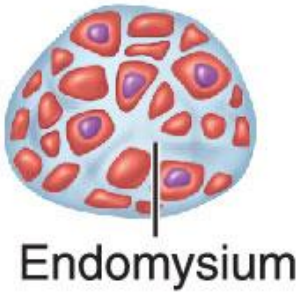
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Table 6.1 Comparison of Skeletal, Cardiac, and Smooth Muscles

Characteristic	Skeletal	Cardiac	Smooth
Body location	Attached to bones or, for some facial muscles, to skin 	Walls of the heart 	Mostly in walls of hollow visceral organs (other than the heart) 
Cell shape and appearance	Single, very long, cylindrical, multinucleate cells with very obvious striations 	Branching chains of cells; uninucleate, striations; intercalated discs 	Single, fusiform, uninucleate; no striations 

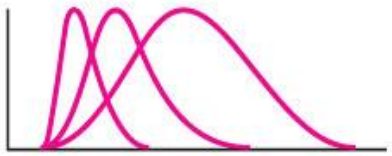
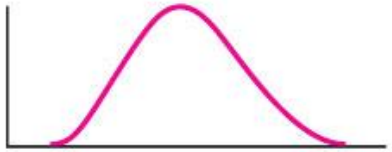
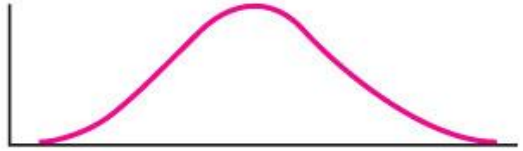
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Table 6.1 Comparison of Skeletal, Cardiac, and Smooth Muscles (*continued*)

Characteristic	Skeletal	Cardiac	Smooth
Connective tissue components	Epimysium, perimysium, and endomysium	Endomysium attached to the fibrous skeleton of the heart	Endomysium
			

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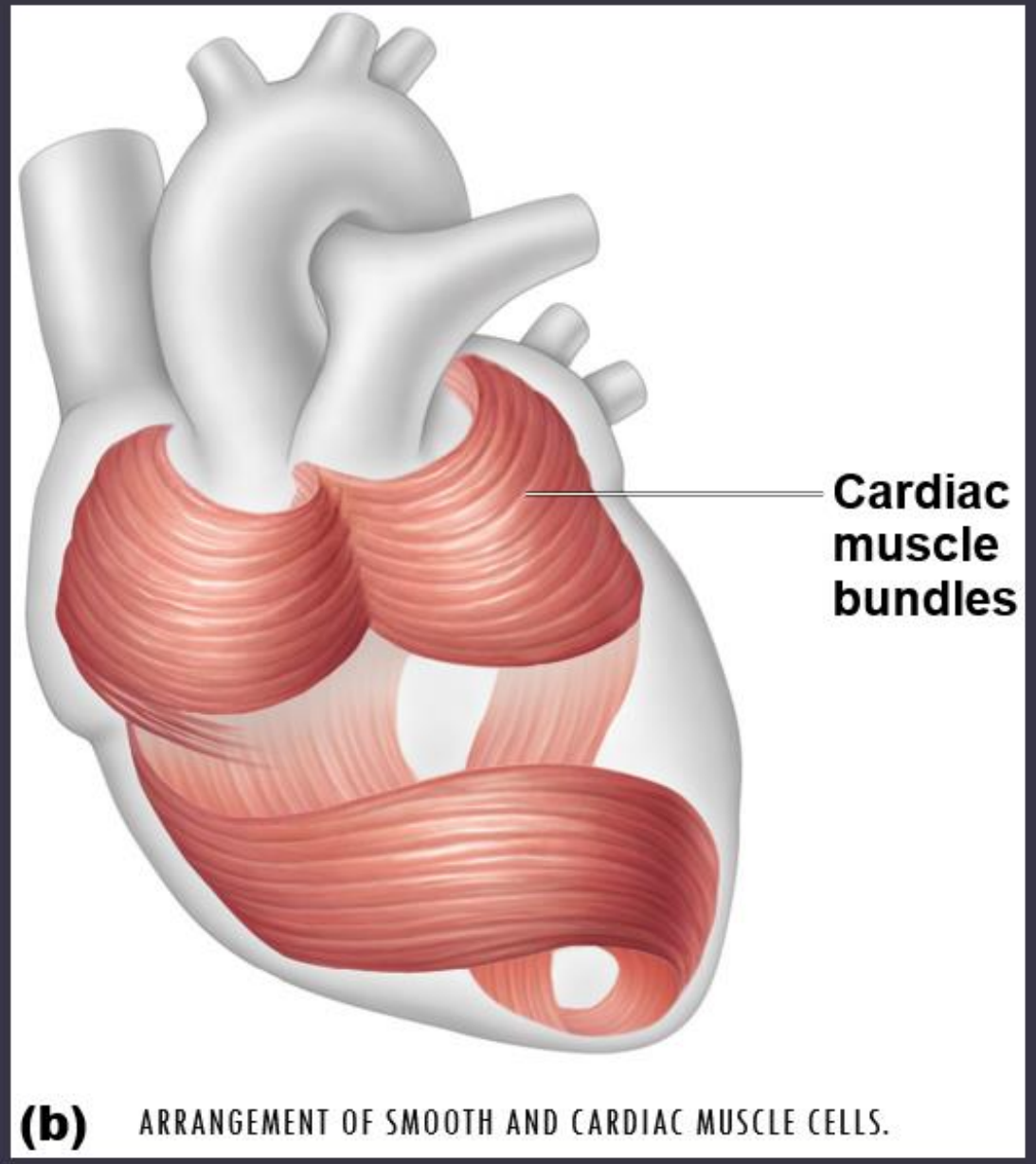
Table 6.1 Comparison of Skeletal, Cardiac, and Smooth Muscles (*continued*)

Characteristic	Skeletal	Cardiac	Smooth
Regulation of contraction	Voluntary; via nervous system controls	Involuntary; the heart has a pacemaker; also nervous system controls; hormones	Involuntary; nervous system controls; hormones, chemicals, stretch
Speed of contraction	Slow to fast 	Slow 	Very slow 
Rhythmic contraction	No	Yes	Yes, in some

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MUSCLE TYPES
CARDIAC MUSCLE

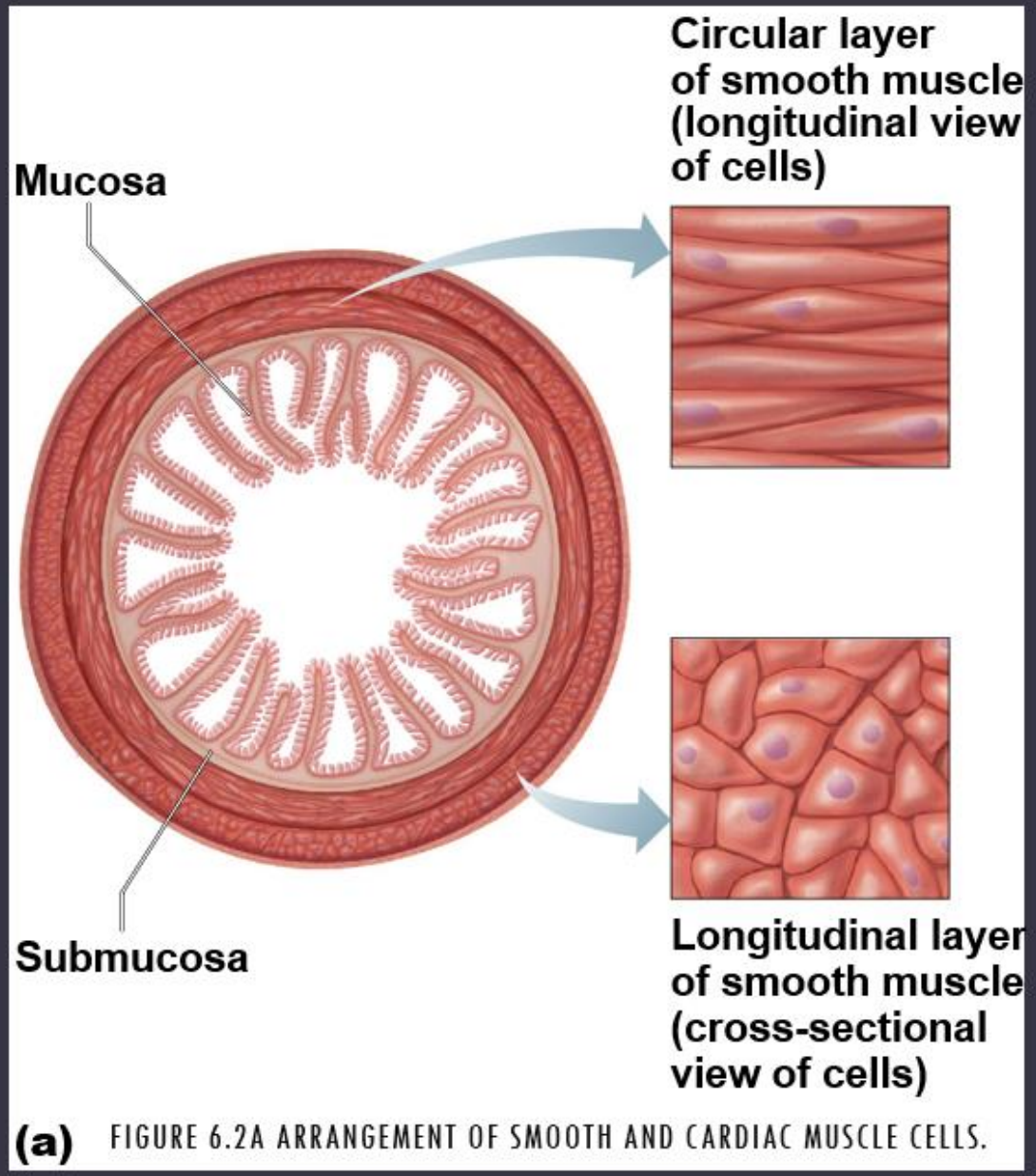
- **Striated** appearance under the microscope
- **Involuntary** control
- Found **only in the walls of the heart**
- Cells are **uninucleate** (single nucleus per cell)
- **Branching cells** joined by **intercalated discs** (gap junctions & desmosomes)
- Contracts at a **steady rate** set by the **pacemaker (sinoatrial node)**, but can be influenced by the nervous and endocrine systems



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MUSCLE TYPES
SMOOTH MUSCLE

- **No striations** (non-striated appearance)
- **Involuntary** control
- Found mainly in the **walls of hollow visceral organs**
- e.g., stomach, intestines, urinary bladder, blood vessels, respiratory passages
- **Spindle-shaped fibers** with a single nucleus (**uninucleate**)
- **Contractions** are **slow, sustained, and fatigue-resistant**



(a) FIGURE 6.2A ARRANGEMENT OF SMOOTH AND CARDIAC MUSCLE CELLS.

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MUSCLE TYPES
SKELETAL MUSCLE

- Most skeletal muscles are **attached to bones by tendons**
- **Cells**: large, elongated ("cigar-shaped"), and **multinucleate**
- Also called **striated muscle** due to visible stripes (sarcomere organization)
- Also called **voluntary muscle** because it is under **conscious control**

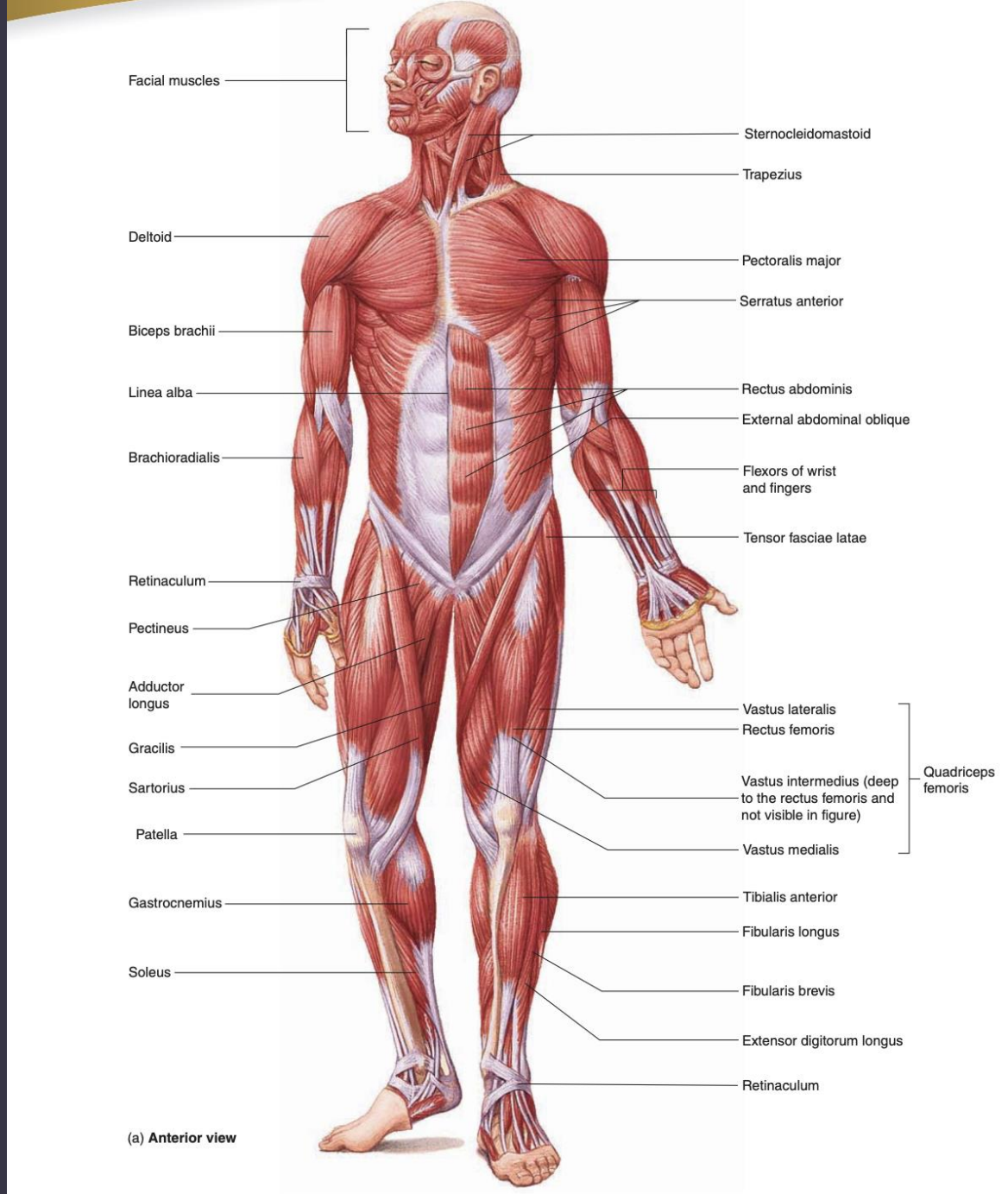


FIGURE 10.3 Overview of the Superficial Body Musculature

MUSCLE TYPES

SKELETAL MUSCLE

Arrangement of Skeletal Muscle Structures (Smallest → Largest)

1. **Muscle Fiber** – individual muscle cell
2. **Endomysium** – connective tissue that surrounds each muscle fiber
3. **Fascicle** – bundle of muscle fibers
4. **Perimysium** – connective tissue wrapping around each fascicle
5. **Epimysium** – connective tissue covering the entire skeletal muscle
6. **Fascia** – connective tissue layer on the outside of the epimysium, separating muscles from surrounding tissues

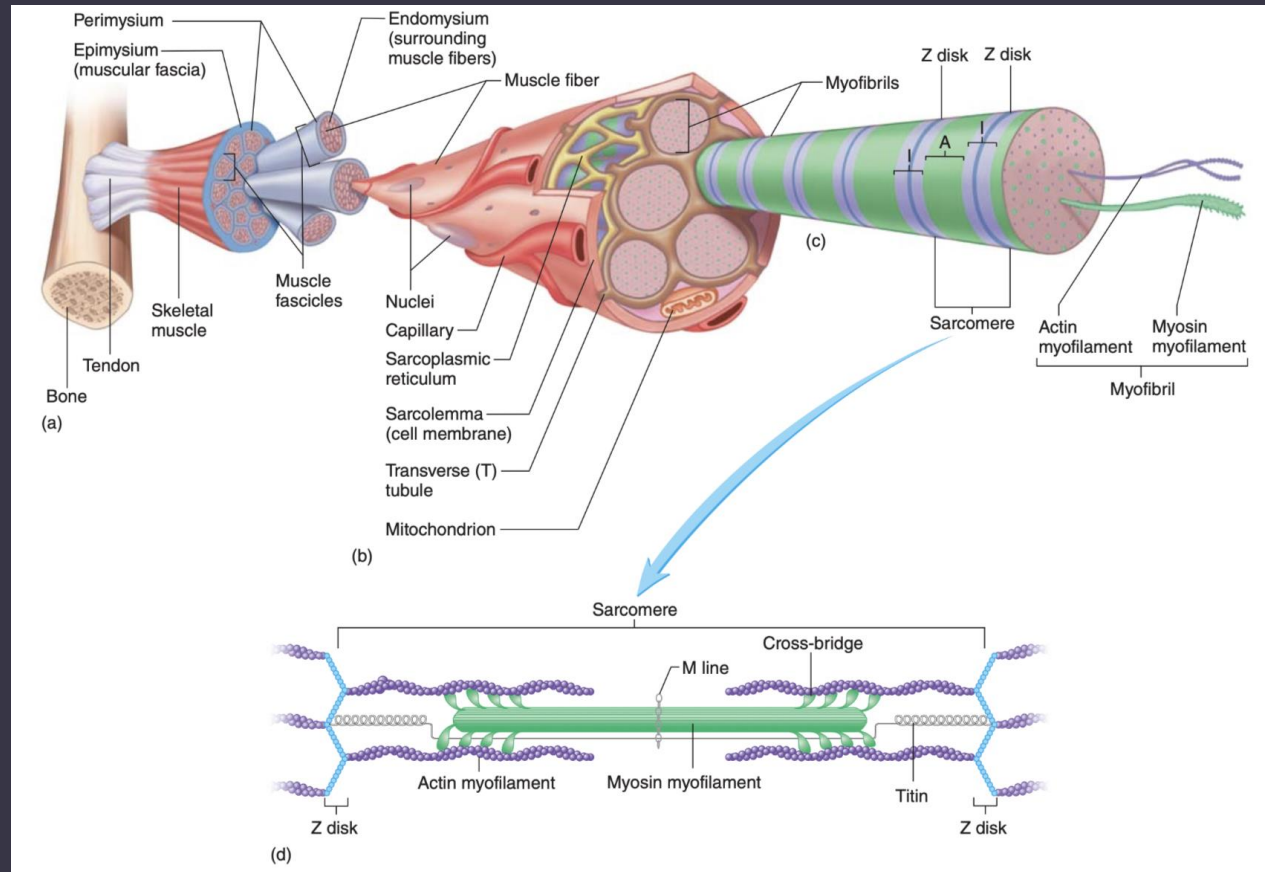


FIGURE 9.3 Parts of a Muscle

(*a*) Part of a muscle attached by a tendon to a bone. (*b*) Enlargement of one muscle fiber. The muscle fiber contains several myofibrils. (*c*) A myofibril extended out the end of the muscle fiber. The banding patterns of the sarcomeres are shown in the myofibril. (*d*) A single sarcomere of a myofibril is composed of actin myofilaments and myosin myofilaments. The Z disk anchors the actin myofilaments, and the myosin myofilaments are held in place by titin molecules and the M line.

MUSCLE TYPES
SKELETAL MUSCLE

Connective Tissue Coverings of Skeletal Muscle

- **Endomysium** – surrounds each individual muscle fiber
- **Perimysium** – wraps around a fascicle (bundle of fibers)
- **Epimysium** – covers the entire skeletal muscle
- **Fascia** – connective tissue on the outside of the epimysium, separating muscles from surrounding tissues

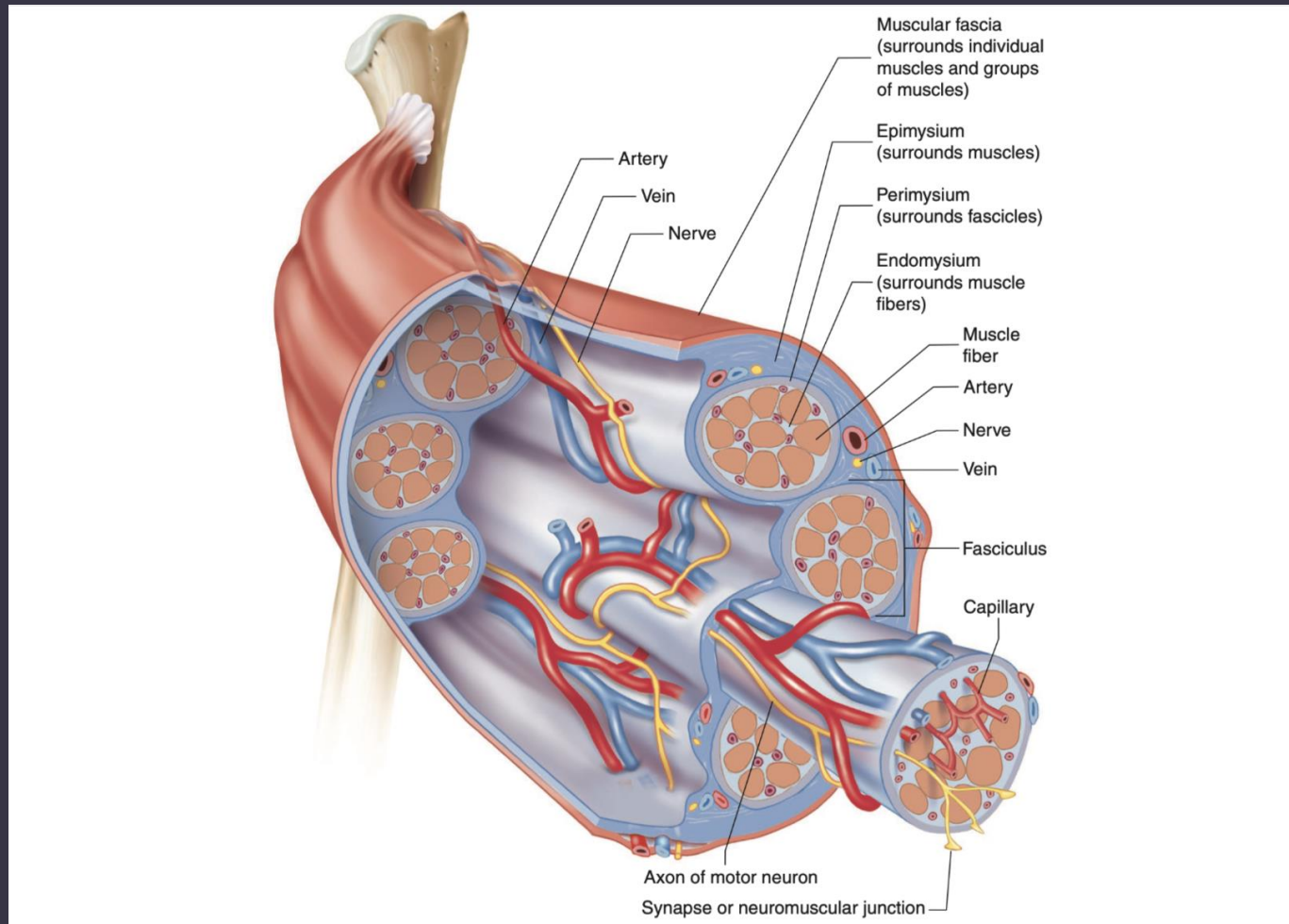
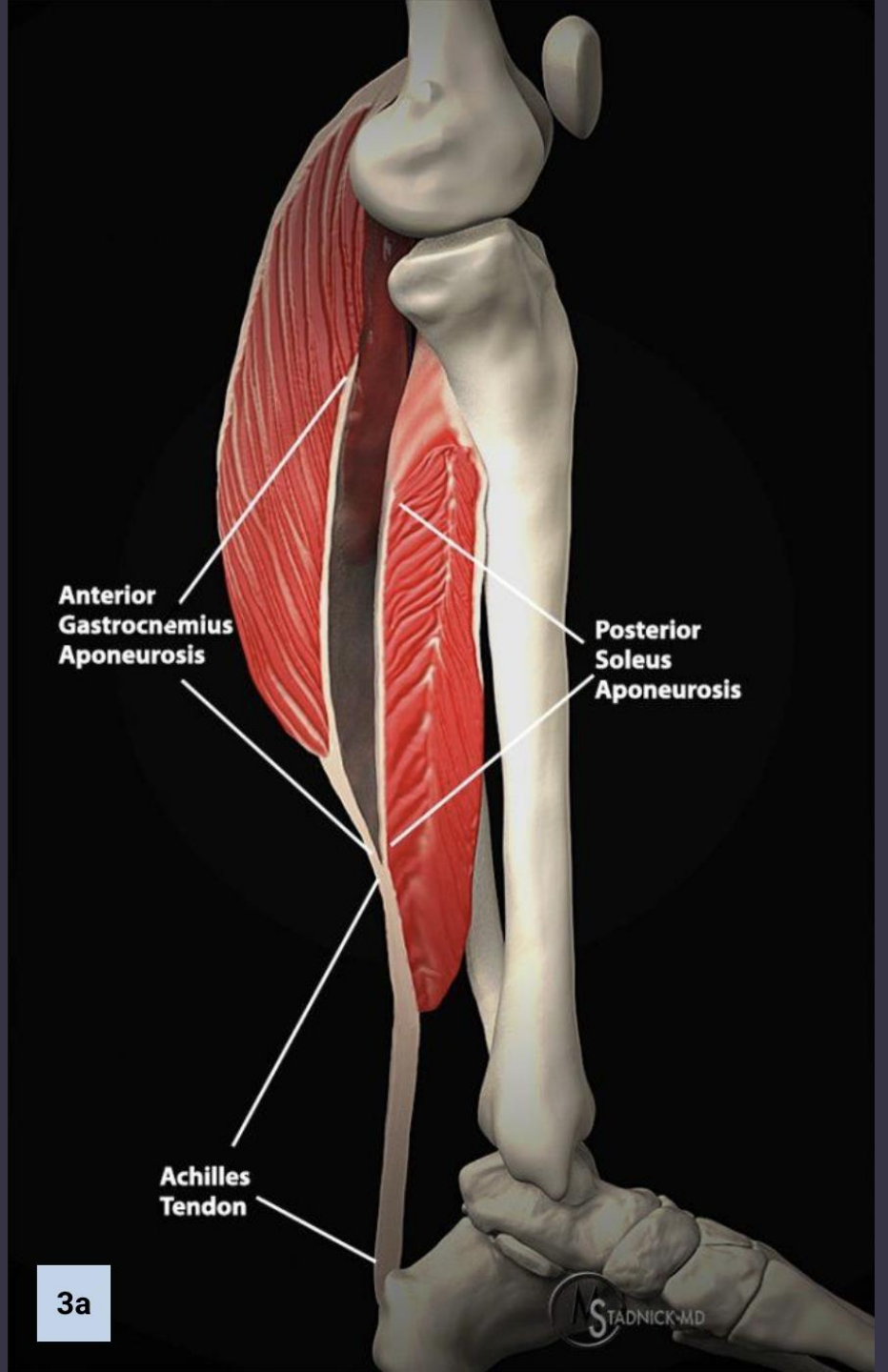


FIGURE 9.1 Whole Skeletal Muscle Structure: Connective Tissue, Innervation, and Blood Supply
A muscle is composed of muscle fascicles, each surrounded by perimysium. The fascicles are composed of bundles of individual muscle fibers (muscle cells), each surrounded by endomysium. This figure shows the relationship among muscle fibers, fascicles, and associated connective tissue layers: the epimysium, perimysium, and endomysium. Arteries, veins, and nerves course together through the connective tissue of muscles. They branch frequently as they approach individual muscle fibers. At the level of the perimysium, axons of neurons branch, and each branch extends to a muscle fiber. AP|R

MUSCLE TYPES
SKELETAL MUSCLE

Connective Tissue Attachments of Skeletal Muscle

- **Epimysium** blends into connective tissue attachments:
  - **Tendons**
    - Cordlike structures
    - Composed mostly of **collagen fibers**
    - Often cross joints due to toughness and small size
  - **Aponeuroses**
    - Broad, sheetlike structures
    - Attach muscles **indirectly** to bones, cartilage, or connective tissue coverings of other muscles



MUSCLE FUNCTIONS
SKELETAL MUSCLE

Functions of Skeletal Muscle

- All muscle types produce movement, but skeletal muscle also:
  - **Maintains posture and body position**
  - **Stabilizes joints**
  - **Generates heat** (thermogenesis)

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OBJECTIVES

1. Differentiate the three types of muscle tissue: **Skeletal, Smooth, and Cardiac Muscles**
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MICROSCOPIC ANATOMY

SKELETAL MUSCLE

- **Sarcolemma** – specialized plasma membrane of a muscle fiber
- **Myofibrils** – long, cylindrical organelles within the muscle fiber
- **Striations** – alternating **light (I) bands** and **dark (A) bands** give skeletal muscle its striped appearance

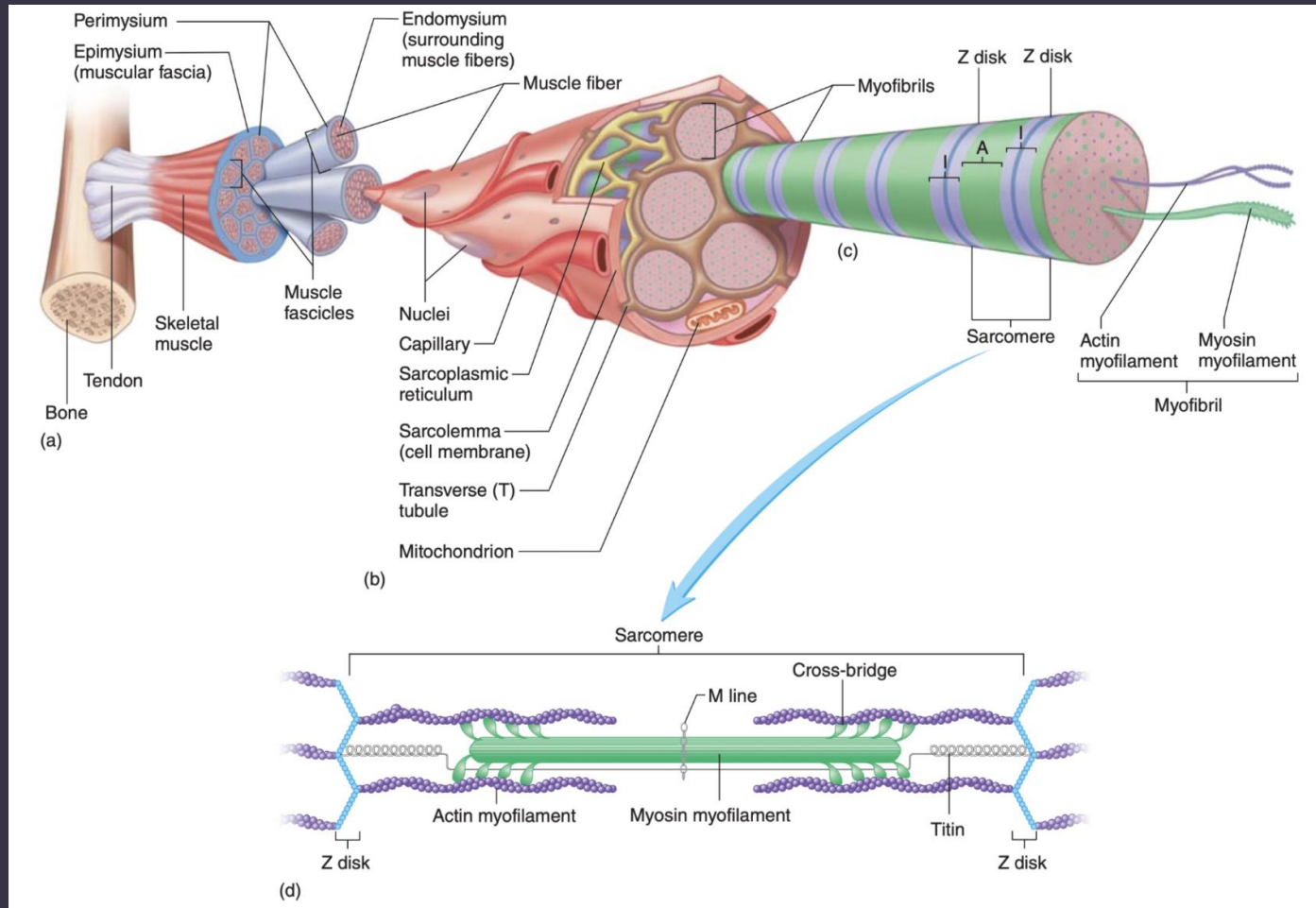


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MICROSCOPIC ANATOMY
SKELETAL MUSCLE

Sarcomere – Contractile Unit of Muscle Fiber

- Structural and functional unit of skeletal muscle
- **Organization:**
  - Myofilaments arranged in repeating units
  - **Thick filaments = myosin**
  - **Thin filaments = actin**
- **Banding Pattern:** Myofilaments produce striations (striped appearance)

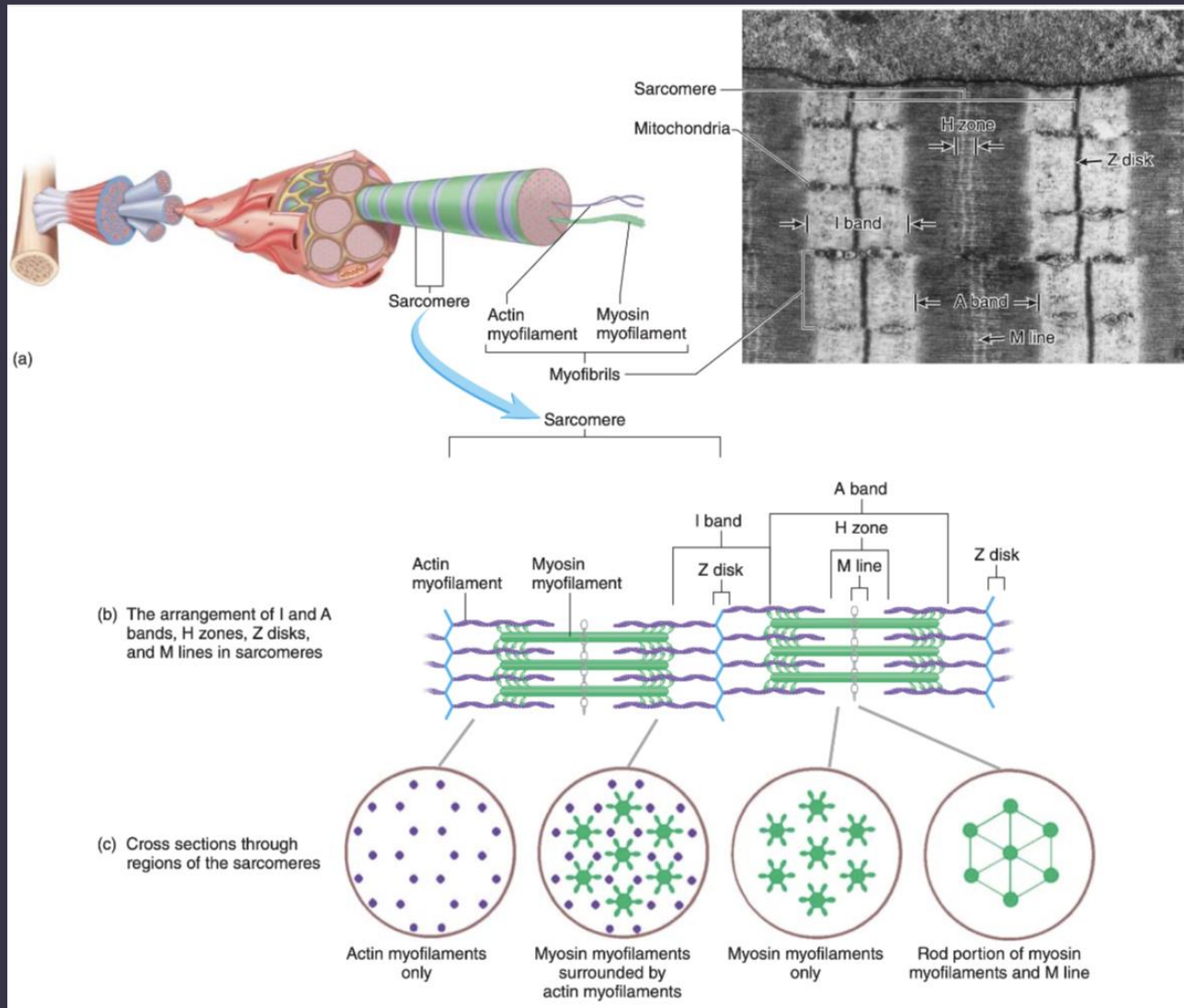


FIGURE 9.4 Organization of Sarcomeres
(*a*) Electron micrograph of a skeletal muscle. Several sarcomeres are shown in the myofibrils of a muscle fiber. (*b*) Diagram of two adjacent sarcomeres, depicting structures responsible for the banding pattern. The I band is between the ends of myosin myofilaments on each side of a Z disk. The A band is formed by the myosin myofilaments within a sarcomere. The H zone is between the ends of the actin myofilaments within a sarcomere. Myosin myofilaments are attached to the M line. (*c*) Cross sections through regions of the sarcomeres show the arrangement of proteins in three dimensions.

MICROSCOPIC ANATOMY

SKELETAL MUSCLE

(BANDING PATTERN OF MYOFIBRILS)

- **I band = light band**
  - Contains only thin filaments (ACTIN)
  - Z disc is a midline interruption

- **A band = dark band**
  - Contains the entire length of the thick filaments (MYOSIN)
  - **H zone:** lighter central region of the A band (no thin filaments)
  - **M line:** center of the H zone, where supporting proteins hold thick filaments together

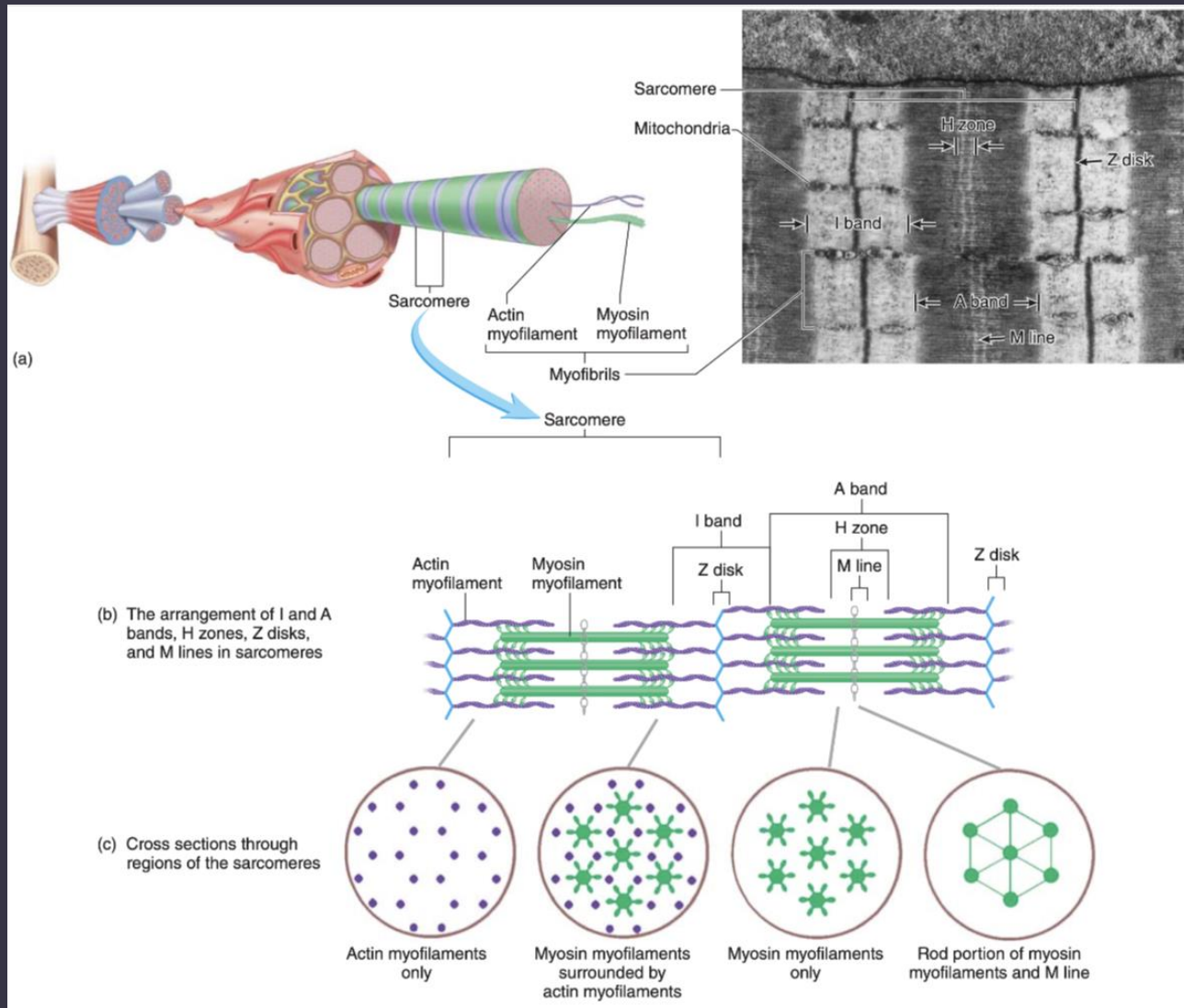


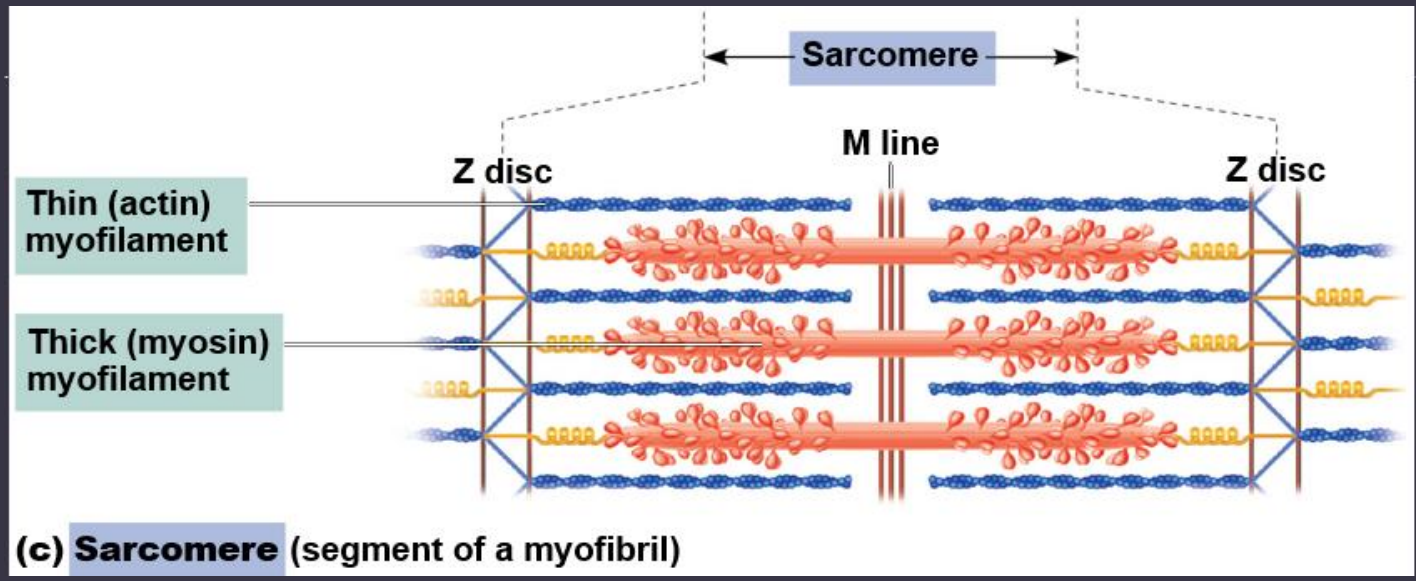
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MICROSCOPIC ANATOMY
SKELETAL MUSCLE

Thick Filaments – Myosin Filaments

- Composed of **myosin protein**
- Contain **ATPase enzymes** → split ATP for energy
- Possess projections called **myosin heads**
- **Cross-bridges:** myosin heads link thick and thin filaments during contraction

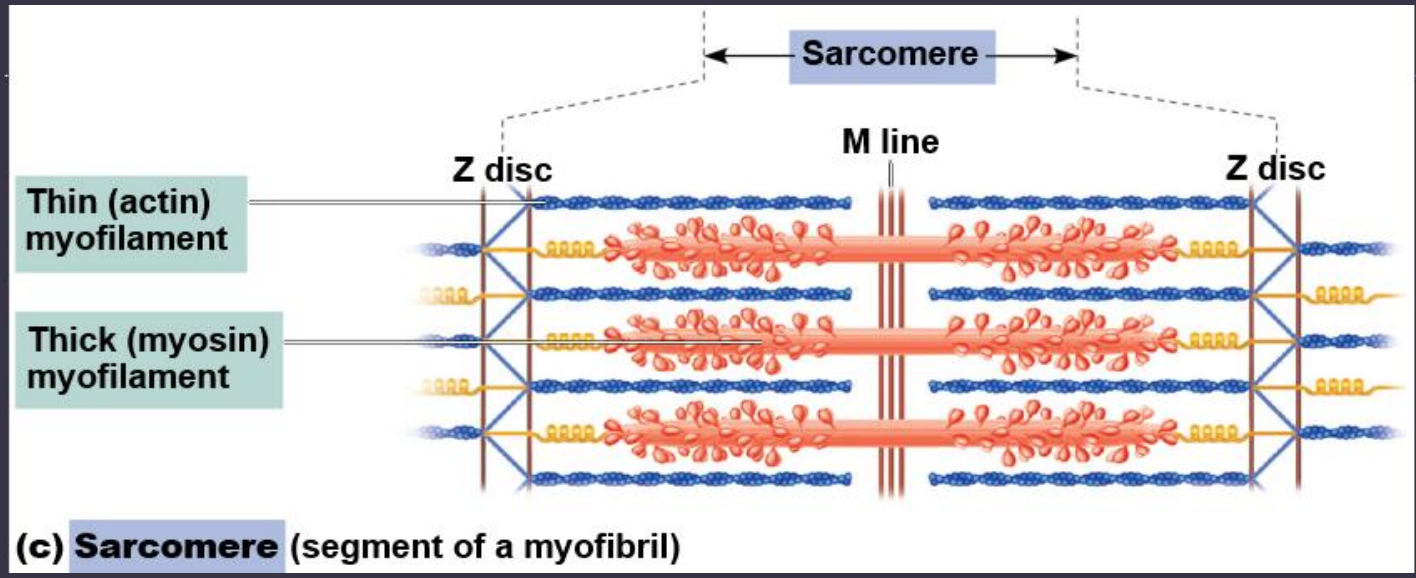


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MICROSCOPIC ANATOMY
SKELETAL MUSCLE

Thin Filaments – Actin Filaments

- Composed of **contractile protein actin**
- **Anchored to Z disc** at sarcomere boundaries
- **H zone:** central region of A band lacking actin (at rest)
- **Contraction:** H zones disappear as actin and myosin filaments overlap

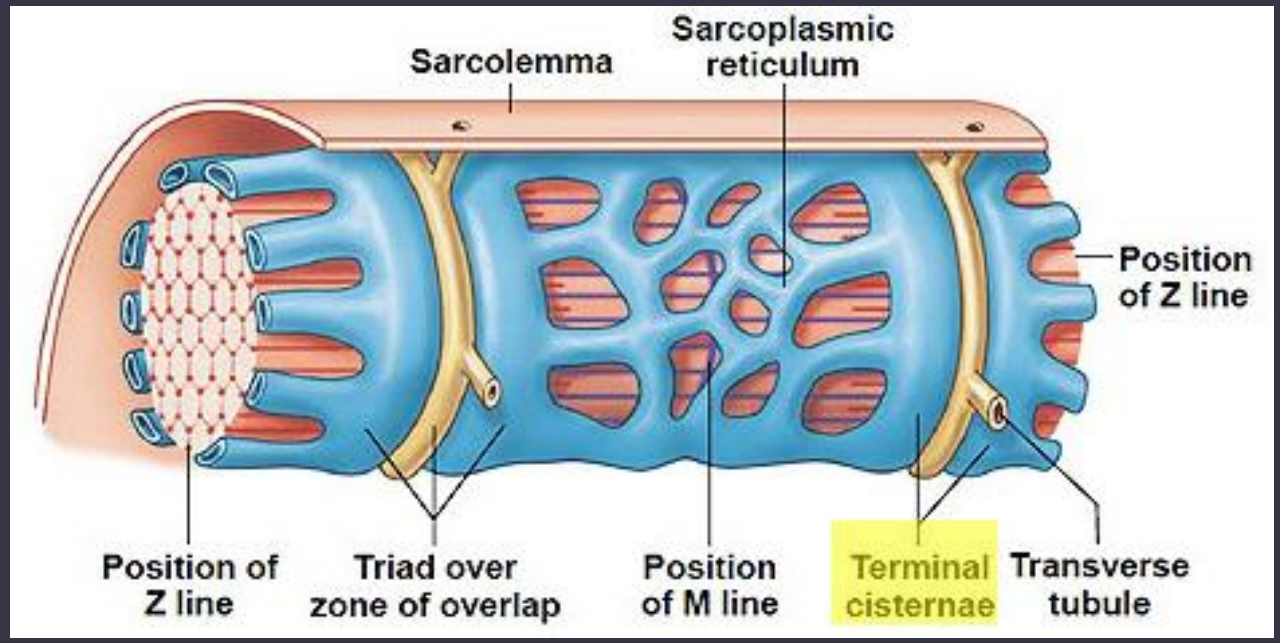


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MICROSCOPIC ANATOMY
SKELETAL MUSCLE

Sarcoplasmic reticulum (SR)

- Specialized form of **smooth endoplasmic reticulum**
- **Surrounds myofibrils** in skeletal muscle fibers
- **Stores calcium (Ca^{2+})** at rest
- **Releases calcium** to trigger muscle contraction



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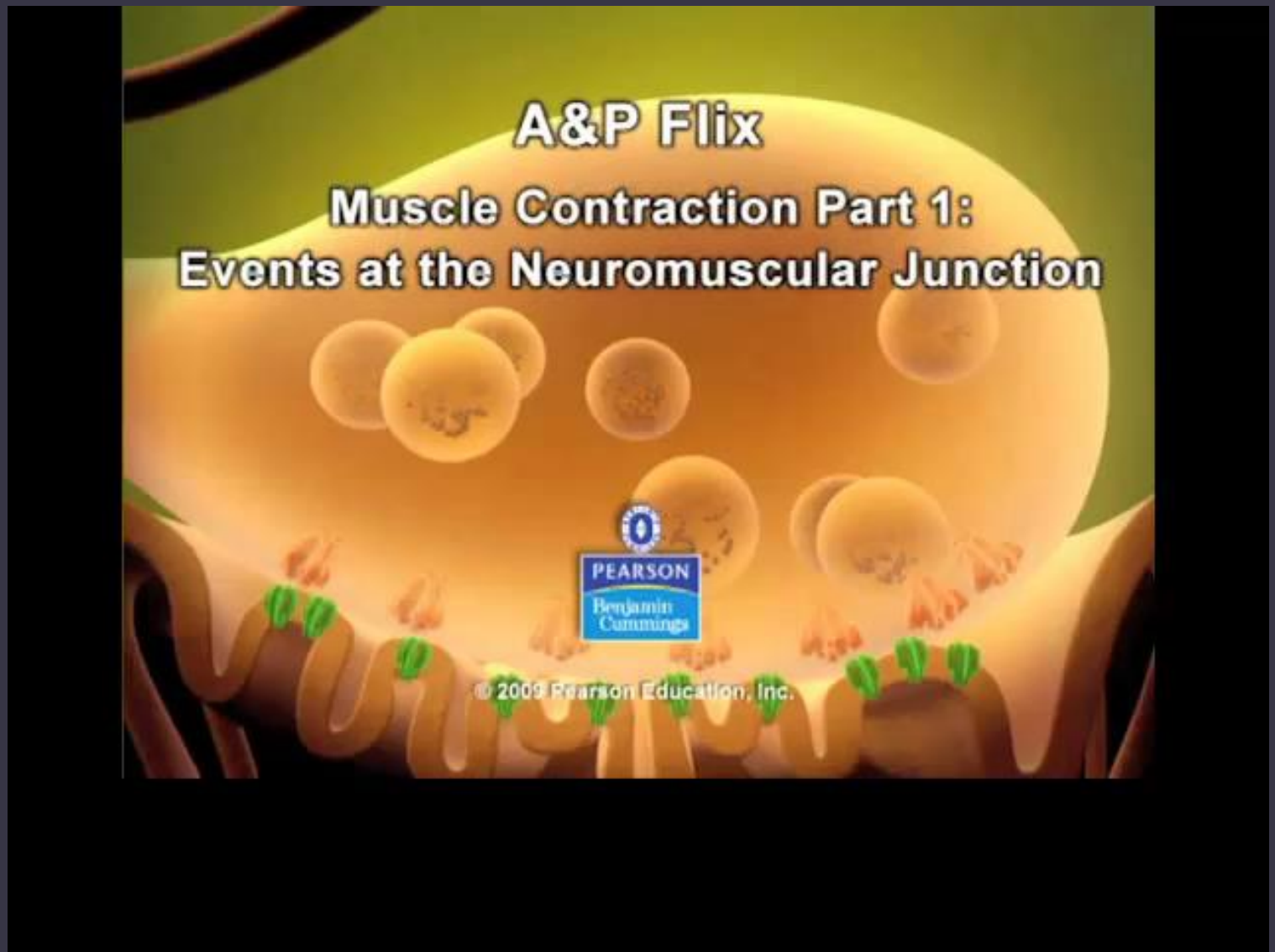
STIMULATION AND CONTRACTION OF SINGLE SKELETAL MUSCLE CELLS

Special Functional Properties of Skeletal Muscle

- **Irritability (Responsiveness):** ability to receive & respond to a stimulus
- **Contractility:** ability to shorten forcibly when adequately stimulated
- **Extensibility:** ability to be stretched without damage
- **Elasticity:** ability to recoil & return to resting length after stretching

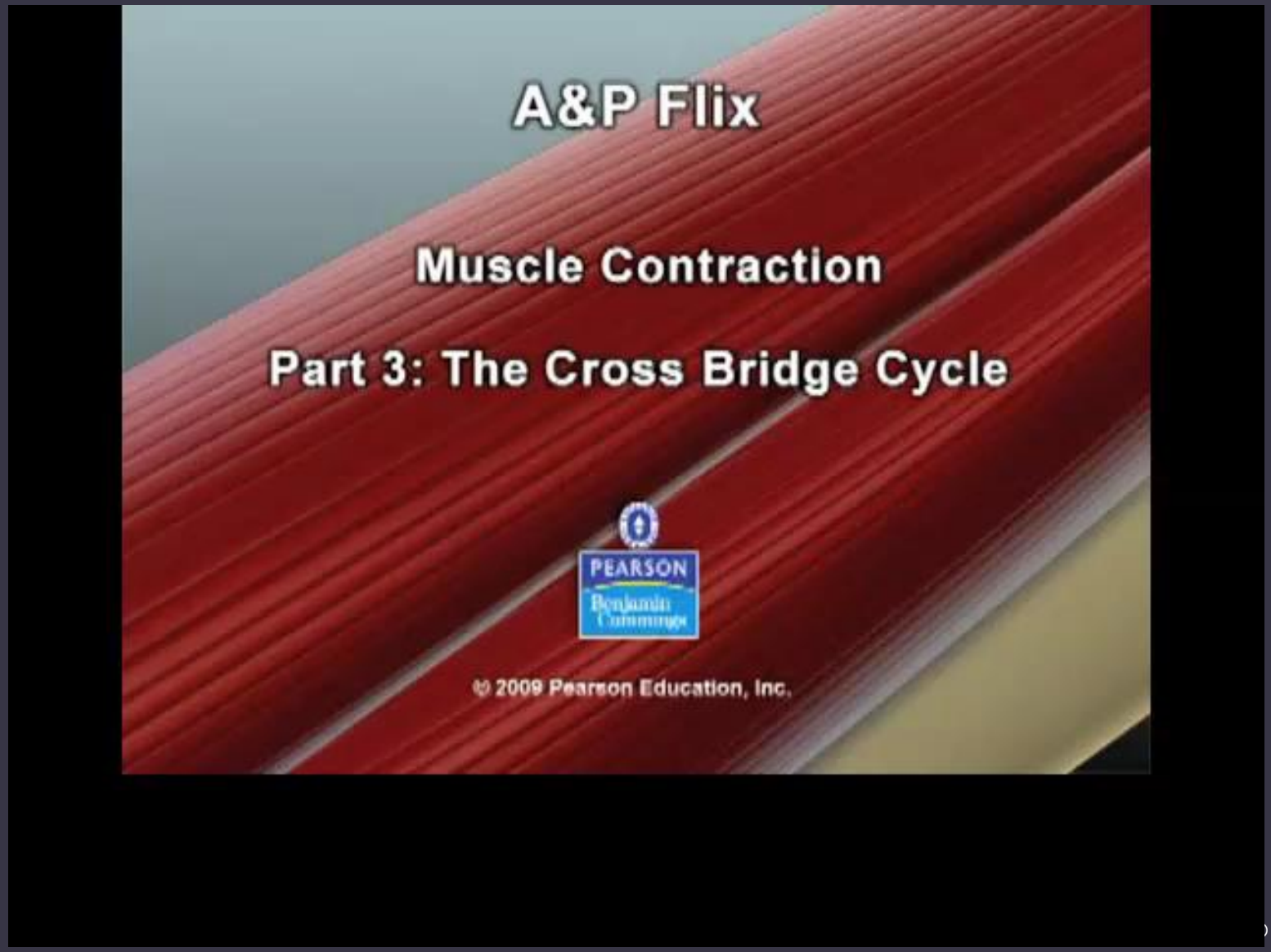
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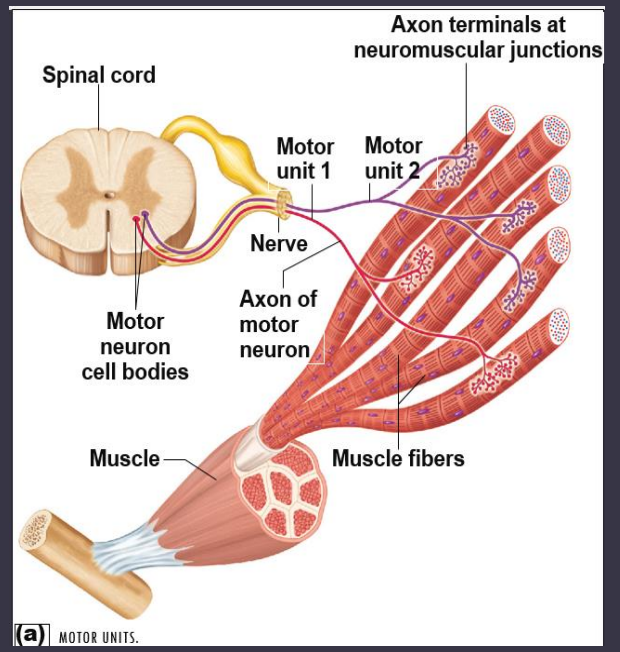
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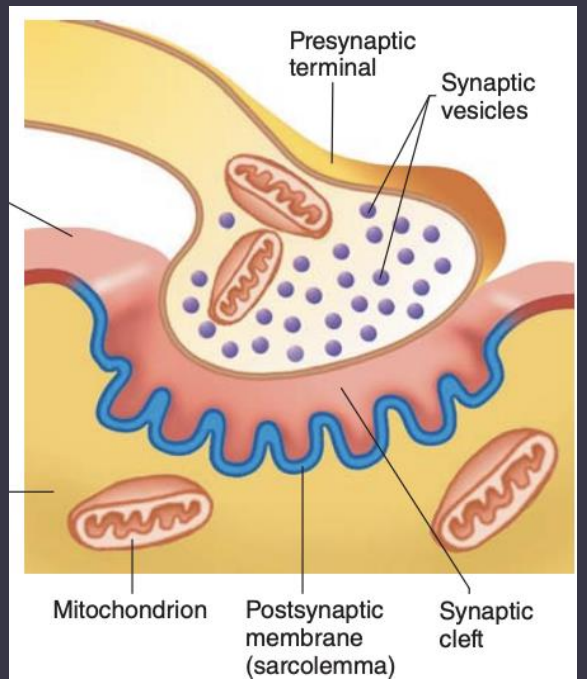
THE NERVE STIMULUS AND ACTION POTENTIAL

SKELETAL SYSTEM

Neuromuscular Junction (NMJ)

- Site where a motor neuron communicates with a skeletal muscle fiber
- **Components:**
  - **Axon terminal** of motor neuron
  - **Synaptic cleft** (space between neuron and muscle)
  - **Motor end plate** (specialized region of sarcolemma)
- **Neurotransmitter: Acetylcholine (ACh)** released into synaptic cleft → binds receptors on sarcolemma → triggers muscle action potential



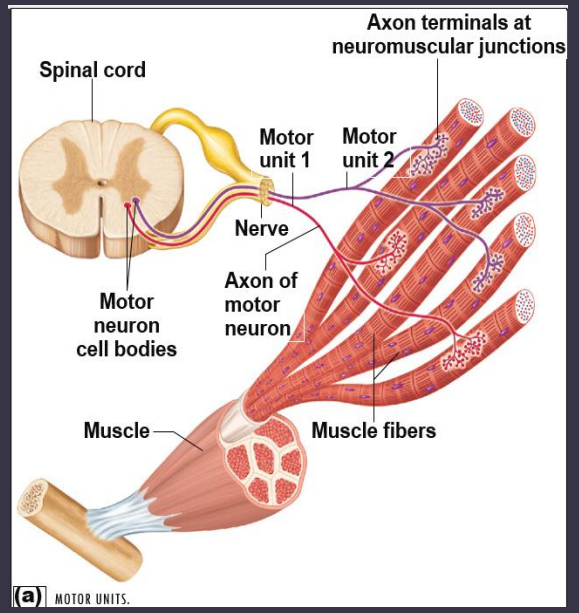


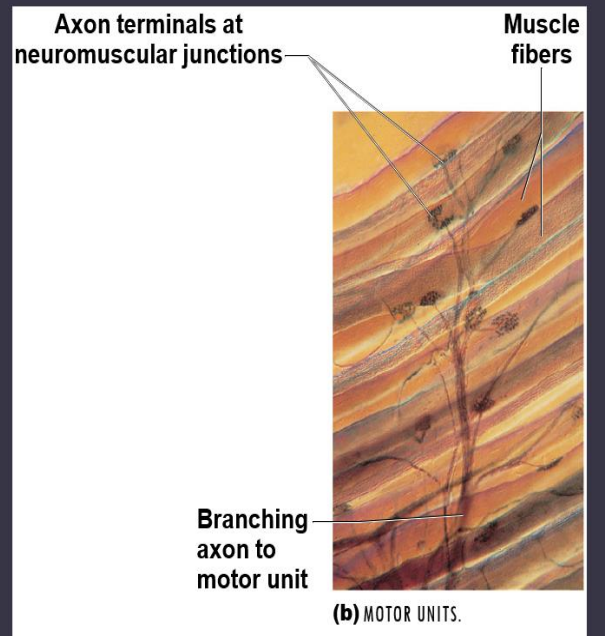
THE NERVE STIMULUS AND ACTION POTENTIAL

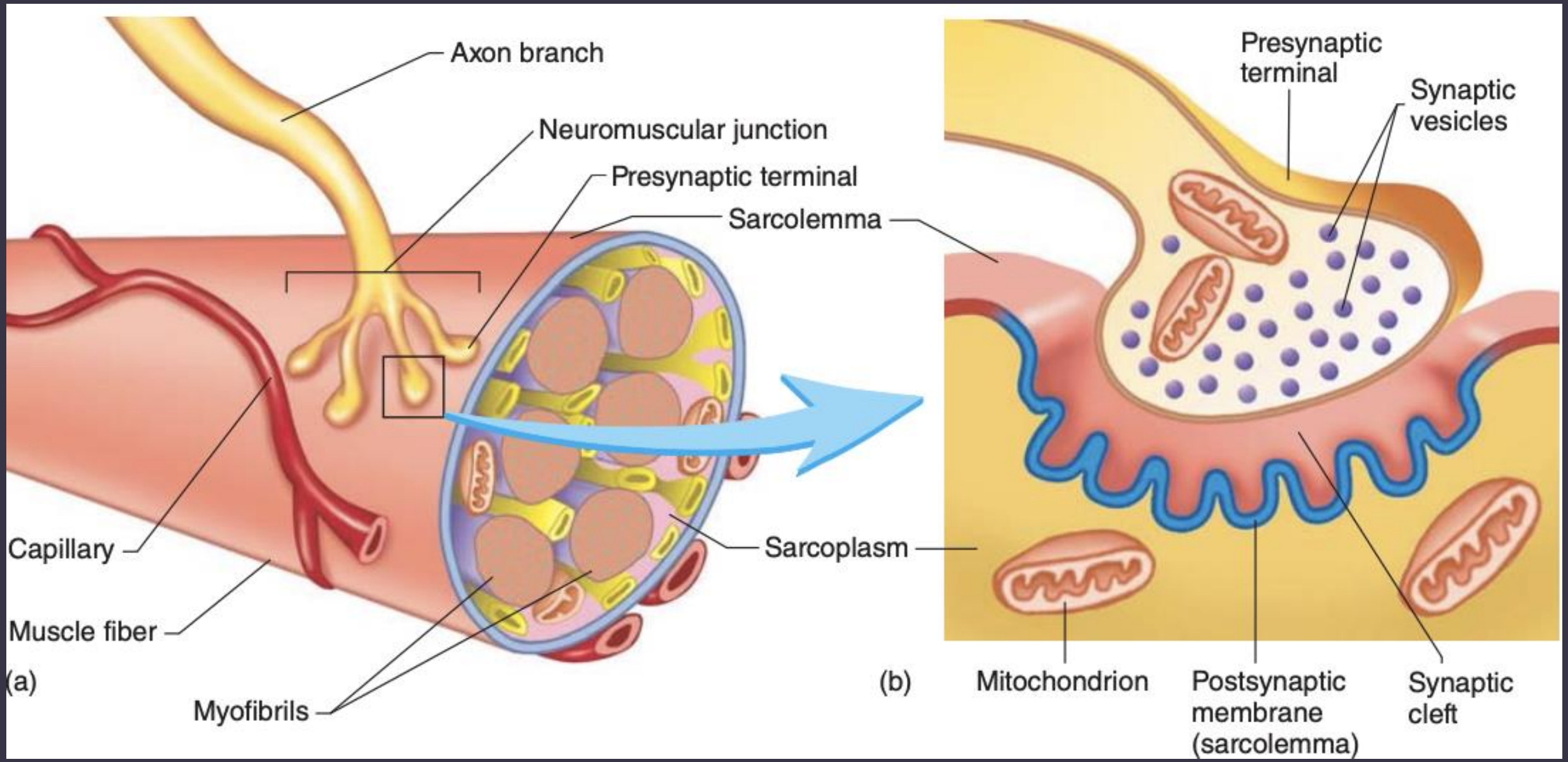
SKELETAL SYSTEM

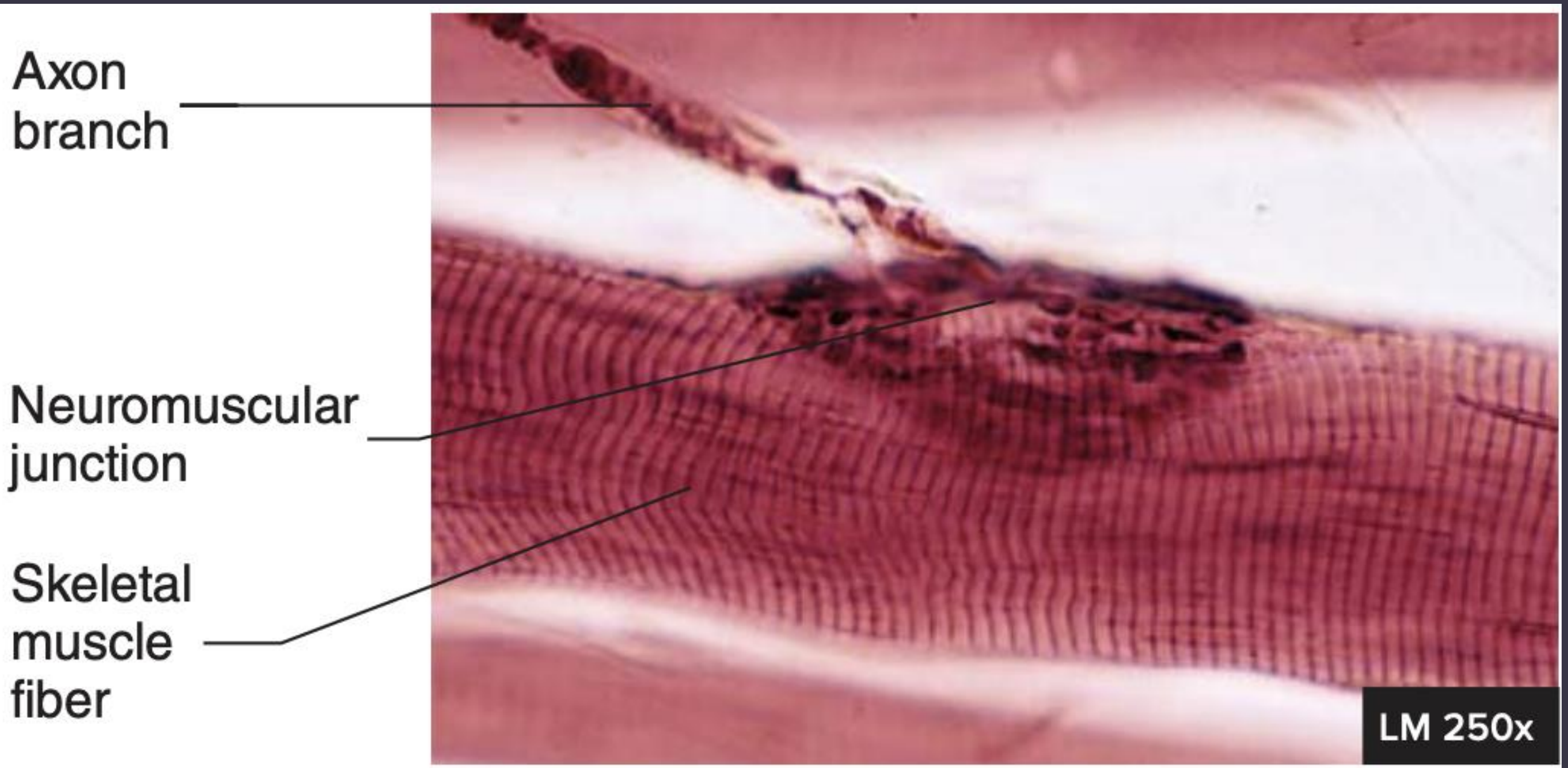
Motor Unit

- One motor neuron and all the skeletal muscle fibers it stimulates
- **Characteristics:**
  - Size varies depending on function
  - **Small motor units** → fine, precise control (e.g., eye muscles)
  - **Large motor units** → powerful contractions (e.g., thigh muscles)
- Motor units work together to coordinate graded muscle contractions






Axon branch
Neuromuscular junction
Presynaptic terminal
Sarcolemma
Capillary
Muscle fiber
Myofibrils
Sarcoplasm
(a)
Presynaptic terminal
Synaptic vesicles
(b)
Mitochondrion
Postsynaptic membrane (sarcolemma)
Synaptic cleft


Axon branch
Neuromuscular junction
Skeletal muscle fiber
LM 250x

THE NERVE STIMULUS AND ACTION POTENTIAL

SKELETAL MUSCLE

Events at the Neuromuscular Junction (NMJ): Nerve Impulse Transmission

Step 1: Arrival of Nerve Impulse → Calcium (Ca^{2+}) channels open in the axon terminal

Step 2: Neurotransmitter Release → Ca^{2+} entry triggers synaptic vesicles to release **acetylcholine (ACh)**

Step 3: ACh Binding → ACh diffuses across the synaptic cleft and binds to **ACh receptors** on the sarcolemma

Step 4: Depolarization of Sarcolemma

1. Sarcolemma becomes more permeable to **Na^+ influx** and **K^+ efflux**
2. More Na^+ enters than K^+ exits → **depolarization** occurs
3. Depolarization opens more Na^+ channels, propagating the action potential

THE NERVE STIMULUS AND ACTION POTENTIAL
SKELETAL MUSCLE

Events at the Neuromuscular Junction (NMJ): Continuation

Step 5: Action Potential Propagation

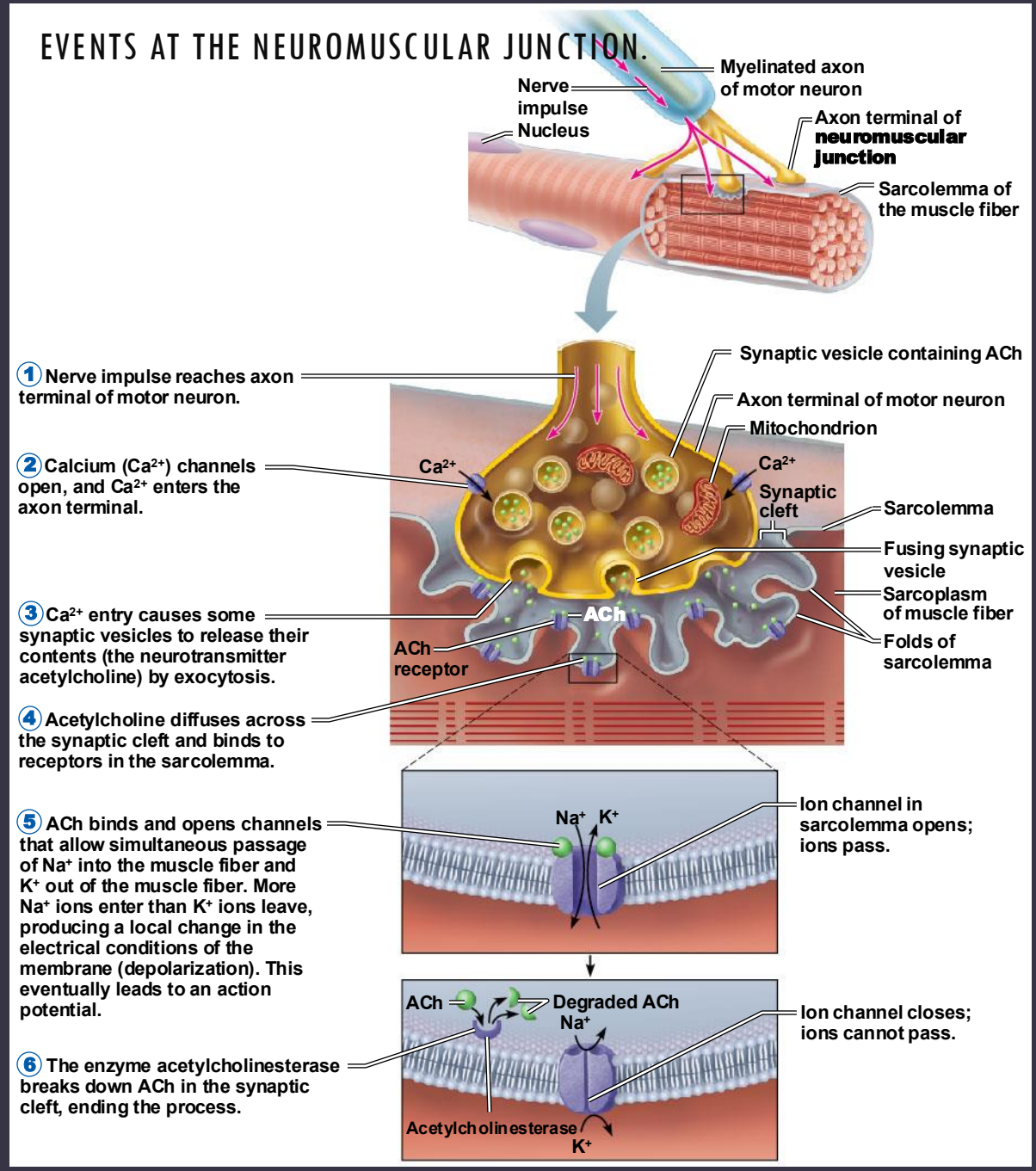
1. Depolarization opens more **voltage-gated Na^+ channels**
2. **Massive Na^+ influx** → full **action potential** generated
3. Action potential spreads along the **sarcolemma** and down **T-tubules**
4. Once initiated, the action potential is **all-or-none** (unstoppable)

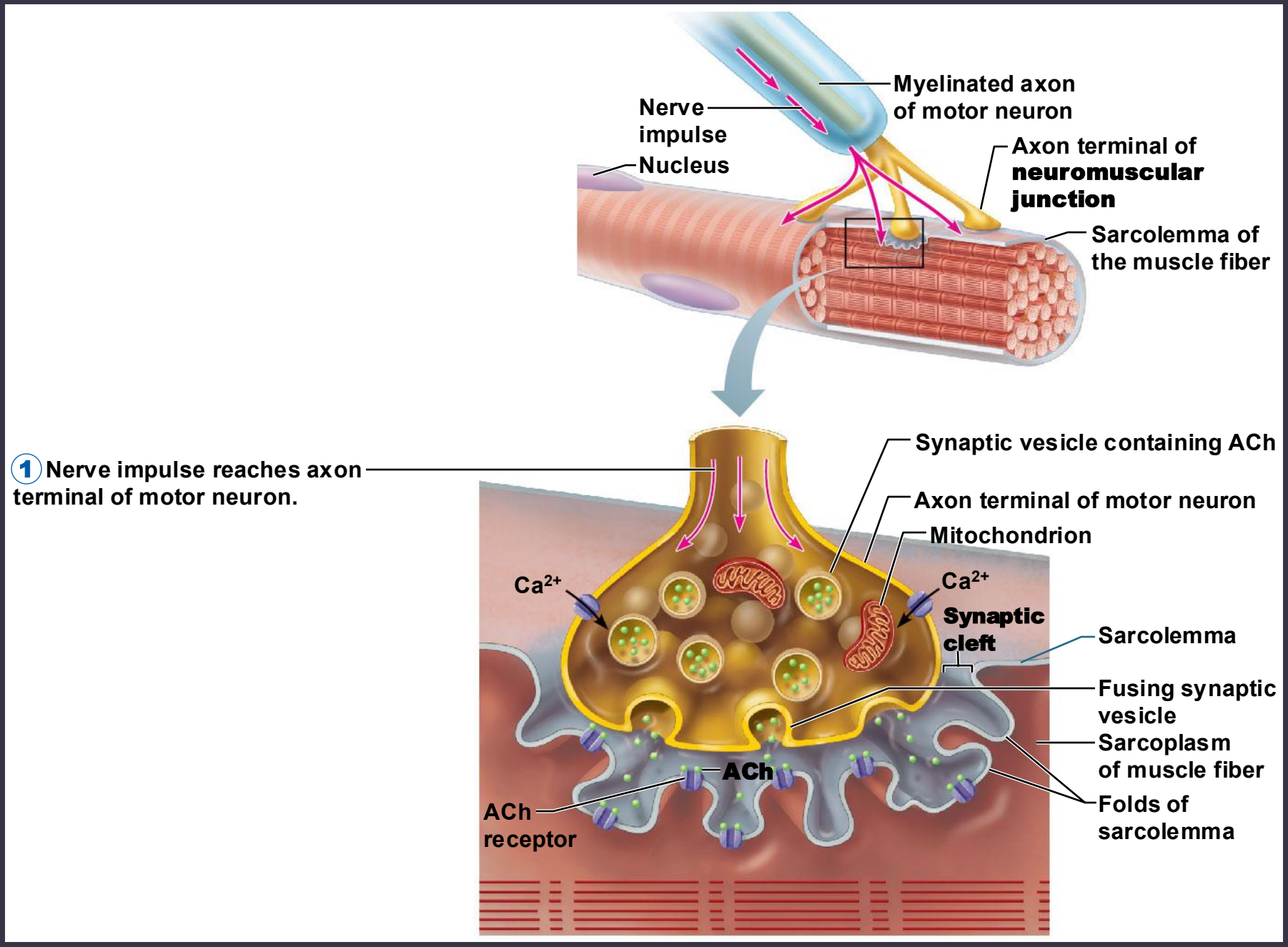
Step 6: Termination of Signal

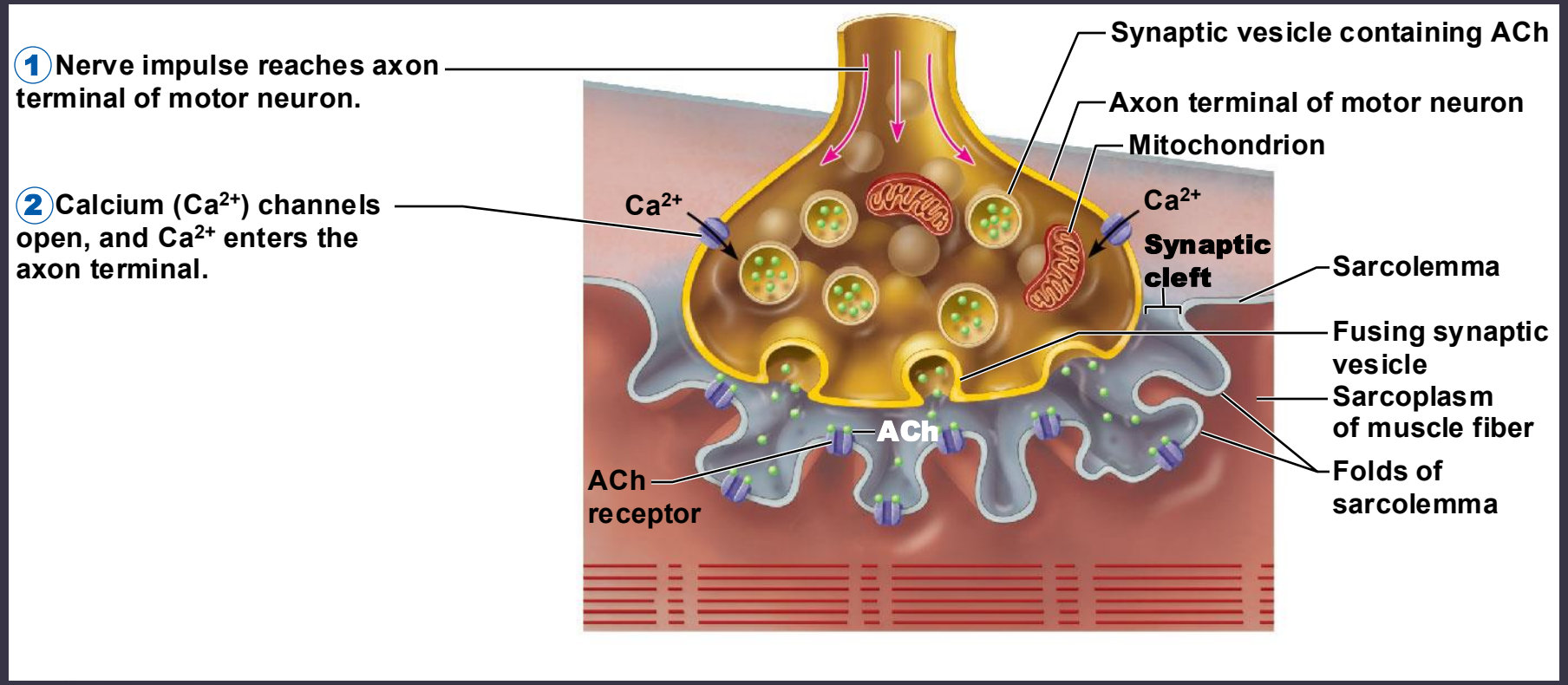
1. **Acetylcholinesterase (AChE)** in synaptic cleft breaks down ACh → **acetic acid + choline**
2. Breakdown prevents continuous stimulation
3. A single nerve impulse = **one muscle contraction (twitch)**

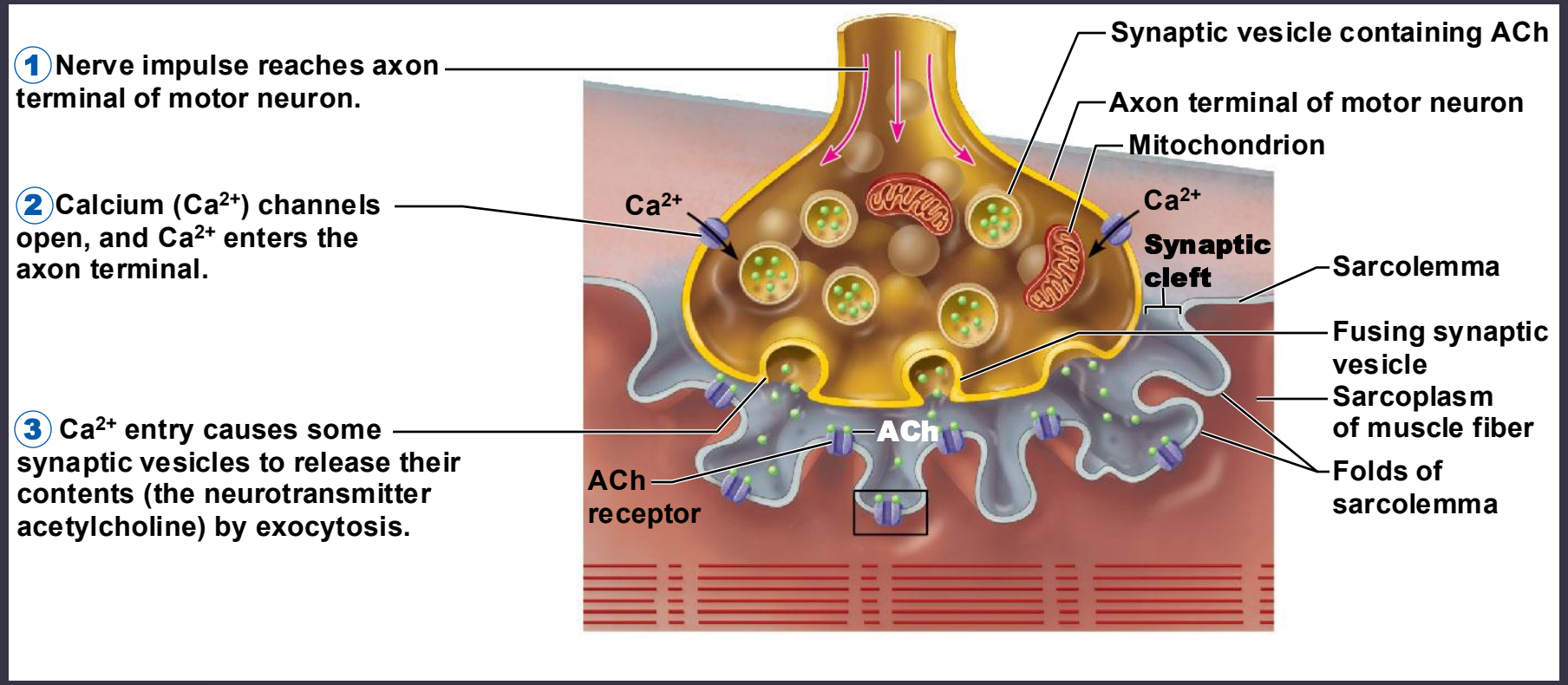
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EVENTS AT THE NEUROMUSCULAR JUNCTION.

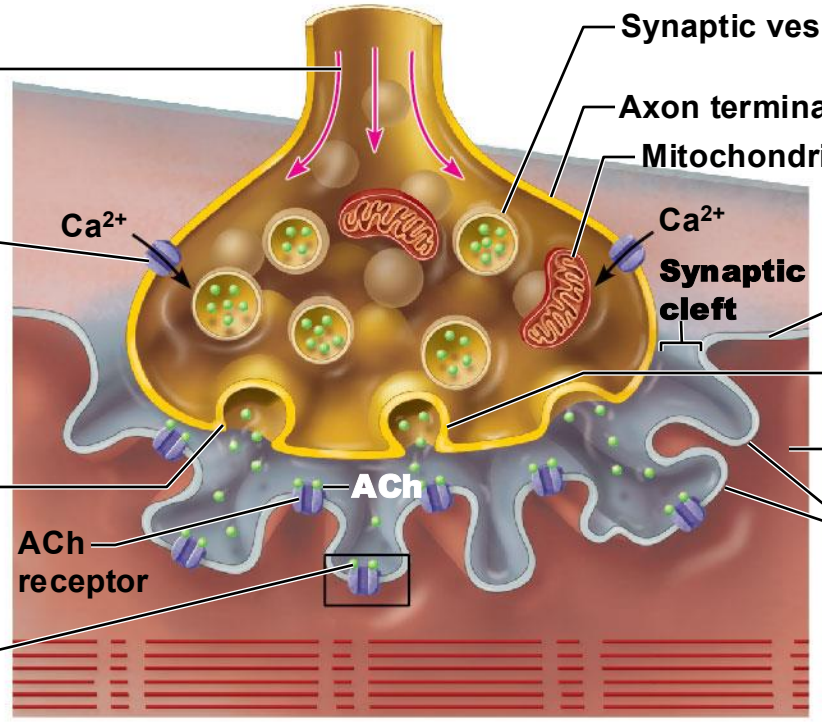



Myelinated axon of motor neuron
Nerve impulse
Nucleus
Axon terminal of neuromuscular junction
Sarcolemma of the muscle fiber
1 Nerve impulse reaches axon terminal of motor neuron.
Synaptic vesicle containing ACh
Axon terminal of motor neuron
Mitochondrion
Ca2+
Ca2+
Synaptic cleft
Sarcolemma
Fusing synaptic vesicle
Sarcoplasm of muscle fiber
ACh
ACh receptor
Folds of sarcolemma

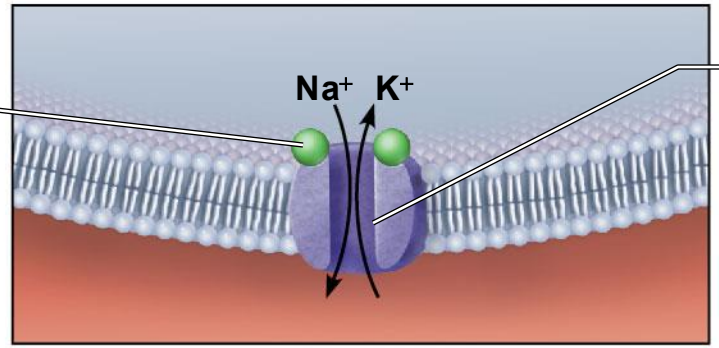

1 Nerve impulse reaches axon terminal of motor neuron.
2 Calcium (Ca^{2+}) channels open, and Ca^{2+} enters the axon terminal.
Synaptic vesicle containing ACh
Axon terminal of motor neuron
Mitochondrion
Ca^{2+}
Ca^{2+}
Synaptic cleft
Sarcolemma
Fusing synaptic vesicle
Sarcoplasm of muscle fiber
ACh
ACh receptor
Folds of sarcolemma


1 Nerve impulse reaches axon terminal of motor neuron.
2 Calcium (Ca^{2+}) channels open, and Ca^{2+} enters the axon terminal.
3 Ca^{2+} entry causes some synaptic vesicles to release their contents (the neurotransmitter acetylcholine) by exocytosis.
Synaptic vesicle containing ACh
Axon terminal of motor neuron
Mitochondrion
Ca^{2+}
Ca^{2+}
Synaptic cleft
Sarcolemma
Fusing synaptic vesicle
Sarcoplasm of muscle fiber
Folds of sarcolemma
ACh
ACh receptor

1. Nerve impulse reaches axon terminal of motor neuron.
2. Calcium (Ca^{2+}) channels open, and Ca^{2+} enters the axon terminal.
3. Ca^{2+} entry causes some synaptic vesicles to release their contents (the neurotransmitter acetylcholine) by exocytosis.
4. Acetylcholine diffuses across the synaptic cleft and binds to receptors in the sarcolemma.

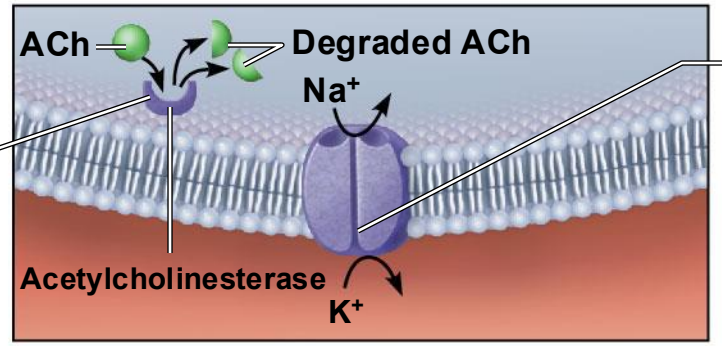


⑤ ACh binds and opens channels that allow simultaneous passage of Na^+ into the muscle fiber and K^+ out of the muscle fiber. More Na^+ ions enter than K^+ ions leave, producing a local change in the electrical conditions of the membrane (depolarization). This eventually leads to an action potential.

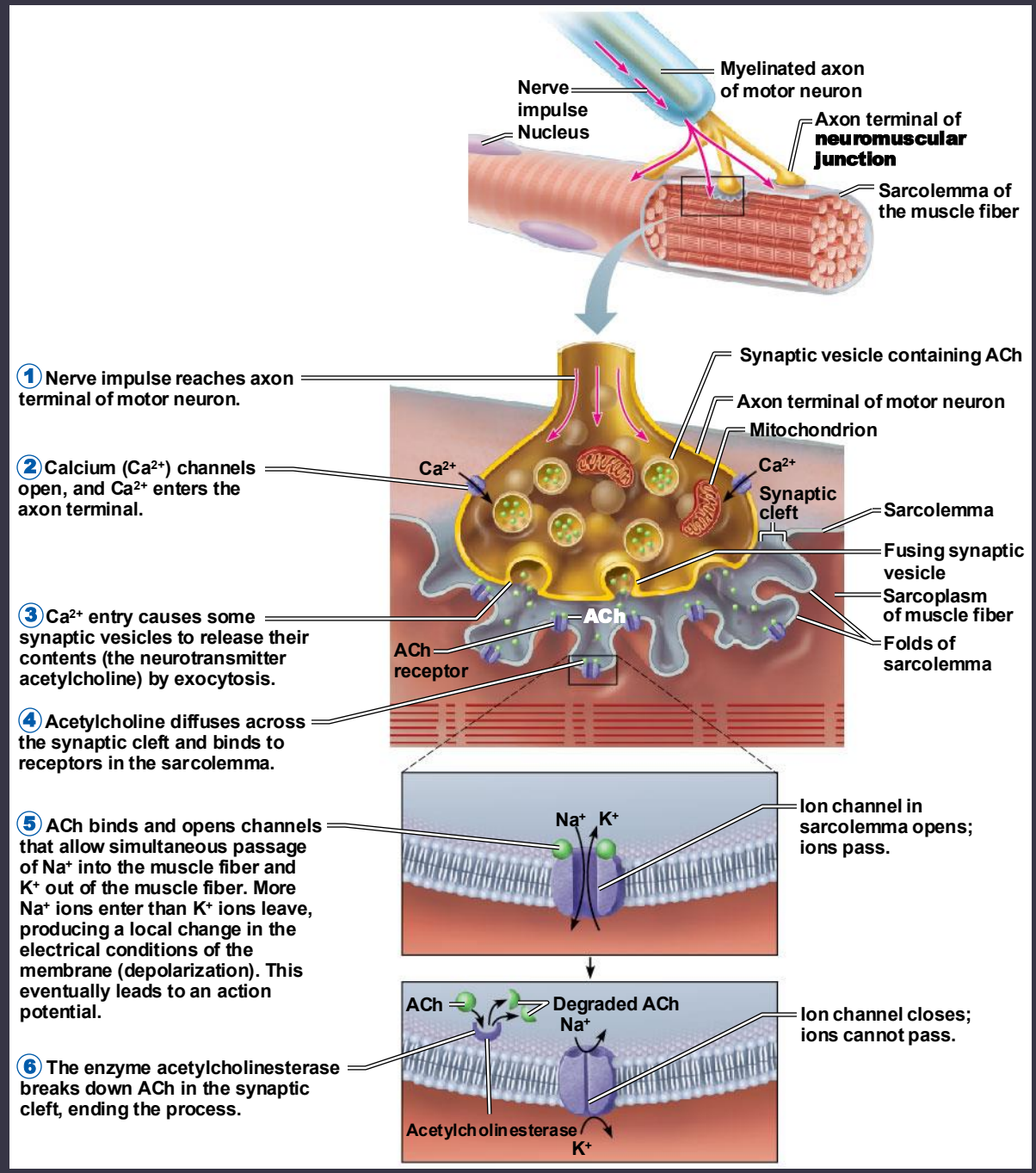


Ion channel in sarcolemma opens; ions pass.

⑥ The enzyme acetylcholinesterase breaks down ACh in the synaptic cleft, ending the process.



Ion channel closes; ions cannot pass.

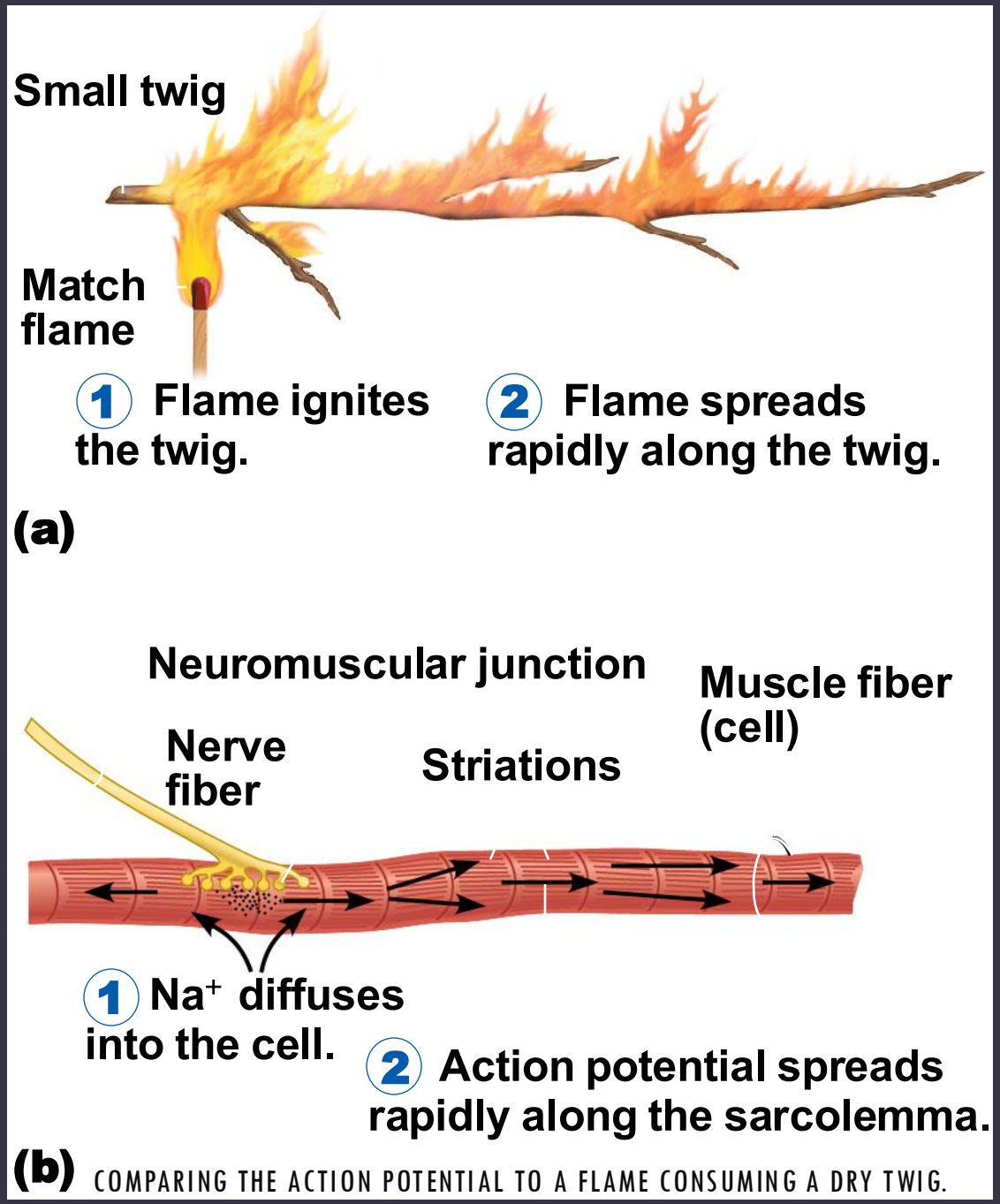

Myelinated axon of motor neuron
Nerve impulse
Nucleus
Axon terminal of neuromuscular junction
Sarcolemma of the muscle fiber
1 Nerve impulse reaches axon terminal of motor neuron.
Synaptic vesicle containing ACh
Axon terminal of motor neuron
Mitochondrion
2 Calcium (Ca^{2+}) channels open, and Ca^{2+} enters the axon terminal.
Ca^{2+}
Ca^{2+}
Synaptic cleft
Sarcolemma
Fusing synaptic vesicle
3 Ca^{2+} entry causes some synaptic vesicles to release their contents (the neurotransmitter acetylcholine) by exocytosis.
ACh
Sarcoplasm of muscle fiber
ACh receptor
Folds of sarcolemma
4 Acetylcholine diffuses across the synaptic cleft and binds to receptors in the sarcolemma.
5 ACh binds and opens channels that allow simultaneous passage of Na^+ into the muscle fiber and K^+ out of the muscle fiber. More Na^+ ions enter than K^+ ions leave, producing a local change in the electrical conditions of the membrane (depolarization). This eventually leads to an action potential.
Na^+ K^+
Ion channel in sarcolemma opens; ions pass.
ACh
Degraded ACh
Na^+
Ion channel closes; ions cannot pass.
6 The enzyme acetylcholinesterase breaks down ACh in the synaptic cleft, ending the process.
Acetylcholinesterase
K^+

THE NERVE STIMULUS AND ACTION POTENTIAL
SKELETAL MUSCLE

Return to Resting State (Repolarization):

- **Potassium ions (K^+) diffuse out** of the muscle cell → restores negative charge inside
- **Sodium–Potassium Pump (Na^+/K^+ ATPase):**
  - Actively transports **3 Na^+ out** and **2 K^+ in**
  - Restores original ion distribution
- Muscle fiber is ready for the **next action potential**

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COMPARING THE ACTION POTENTIAL TO A FLAME CONSUMING A DRY TWIG.

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MECHANISM OF MUSCLE CONTRACTION

THE SLIDING FILAMENT THEORY

Sliding Filament Mechanism of Muscle Contraction

Step 1: **Calcium ions (Ca^{2+})** → bind to regulatory proteins **troponin** on thin filaments → expose **myosin-binding sites**

Step 2: **Cross-bridge formation** → **Myosin heads (cross-bridges)** attach to actin filaments

Step 3: **Power stroke** → Each cross-bridge pivots, pulling thin filaments toward the **center of sarcomere**

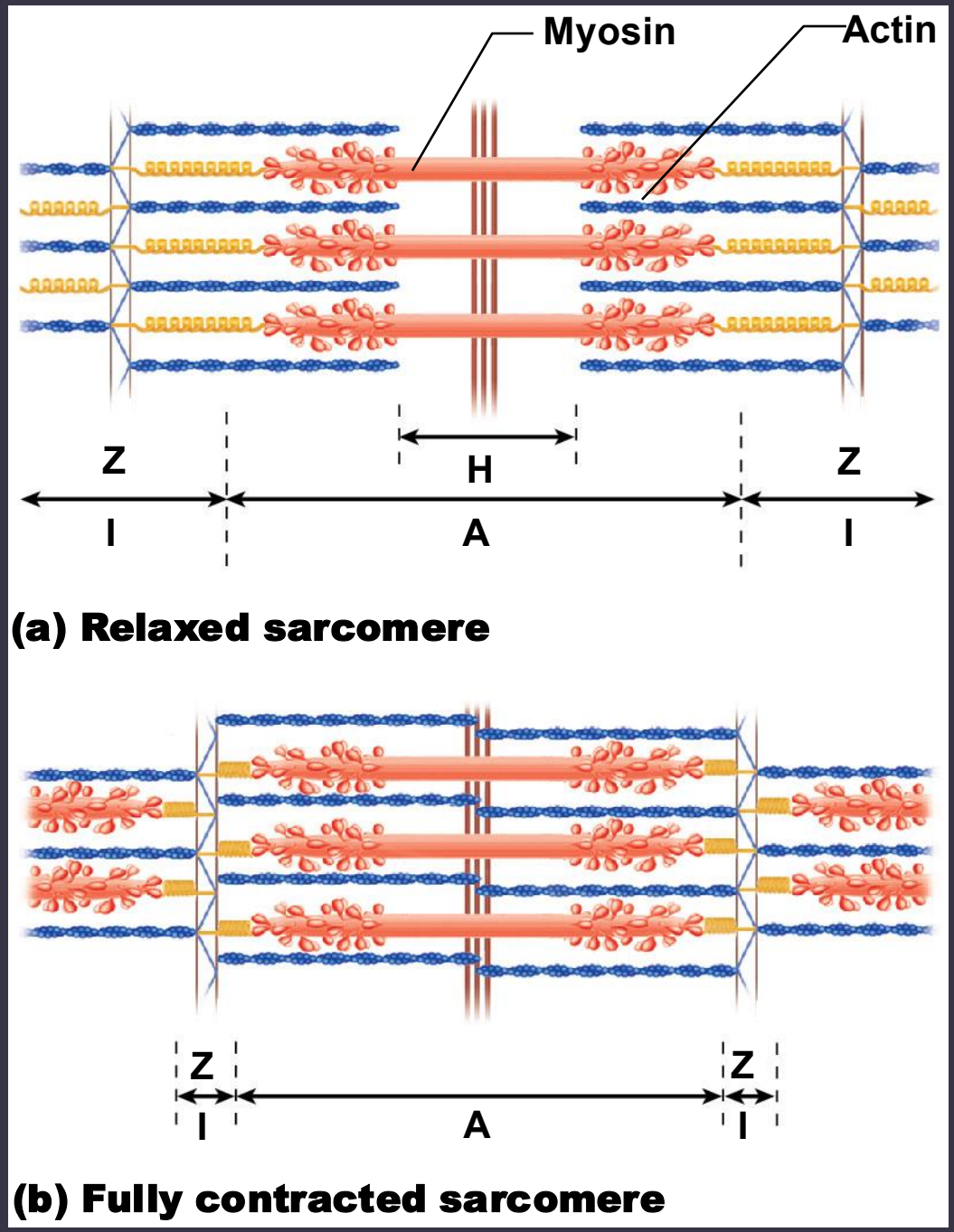
- Sarcomere shortens → **muscle contraction**

Step 4: **Detachment** → Cross-bridges repeatedly attach and detach during contraction

Step 5: **Cycling** → **ATP** powers cross-bridge cycling

- Process continues as long as **Ca^{2+} and ATP** are available

DIAGRAMMATIC VIEWS OF A SARCOMERE.



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CONTRACTION OF A SKELETAL MUSCLE AS A WHOLE

Graded Responses in Skeletal Muscle

- **Muscle fiber level:**
  - Contraction follows the **all-or-none principle** → contracts fully when threshold is reached
- **Whole muscle level:**
  - Not all fibers contract at the same time
  - Different numbers of fibers can be recruited
- **Graded responses:**
  - Varying strength of contraction by:
    - **Changing the frequency** of stimulation
    - **Changing the number** of muscle fibers stimulated
- Result → **different degrees of muscle shortening (force output)**

CONTRACTION OF A SKELETAL MUSCLE AS A WHOLE

Muscle Response to Stronger Stimuli

- **Muscle force** depends on the **number of fibers (motor units) activated**
- **Recruitment:** Increasing stimulus strength → more motor units activated
- **Greater fiber contraction = greater muscle tension**
- When **all motor units** are recruited → maximum contraction (muscle is at full strength)
- This mechanism allows precise control of movement and force

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SUMMATION VS. TETANUS IN MUSCLE CONTRACTIONS

- **Summation**
  - Occurs when a second stimulus arrives before the muscle has completely relaxed
  - Results in **increased contraction strength** (additive effect)
- **Incomplete Tetanus (Unfused Tetanus)**
  - Rapid, repeated stimuli with brief relaxation between contractions
  - Produces a **sustained but wavering contraction**
- **Complete Tetanus (Fused Tetanus)**
  - Stimuli delivered so rapidly that no relaxation occurs
  - Produces a **smooth, sustained contraction**
  - Rarely achieved under normal physiological conditions

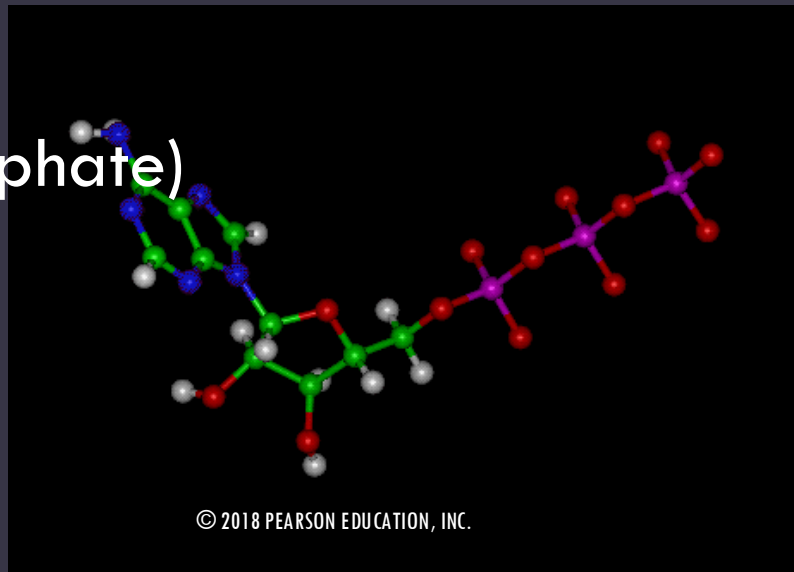
PROVIDING ENERGY FOR MUSCLE CONTRACTION: ATP

ATP (Adenosine Triphosphate)

- Only immediate energy source directly used for muscle contraction
- Stored in small amounts; depleted in ~4–6 seconds

Three Pathways to Regenerate ATP

1. Direct Phosphorylation of ADP (by Creatine Phosphate)
2. Aerobic Pathway (Cellular Respiration)
3. Anaerobic Glycolysis and Lactic Acid Formation

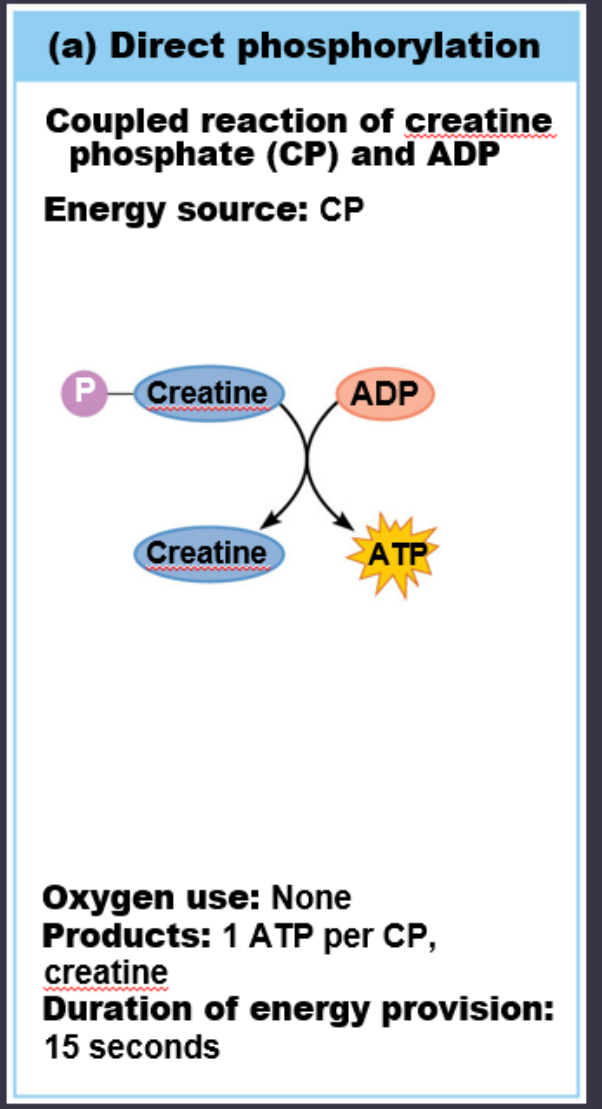


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PROVIDING ENERGY FOR MUSCLE CONTRACTION

Direct Phosphorylation of ADP by Creatine Phosphate (CP)

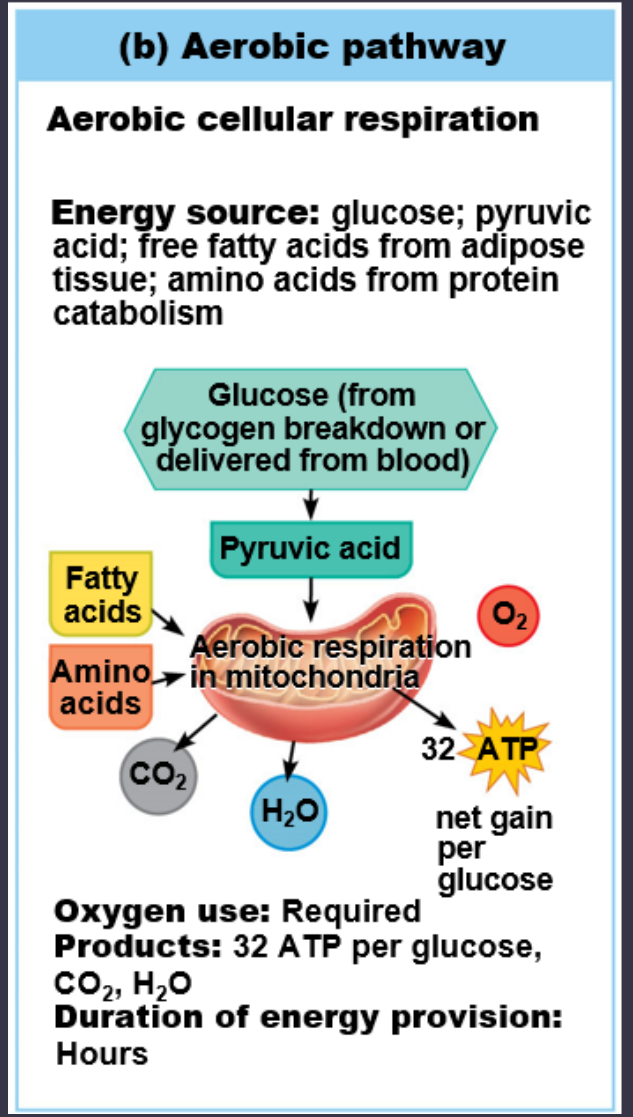
- **Fastest ATP regeneration pathway**
- Muscle fibers store **creatine phosphate (CP)**, a high-energy molecule
- After ATP is depleted, **ADP remains**
- **CP donates a phosphate group to ADP → ATP**
- Supplies exhausted in **<15 seconds**
- Yield: **1 ATP per CP molecule**



PROVIDING ENERGY FOR MUSCLE CONTRACTION

Aerobic Respiration

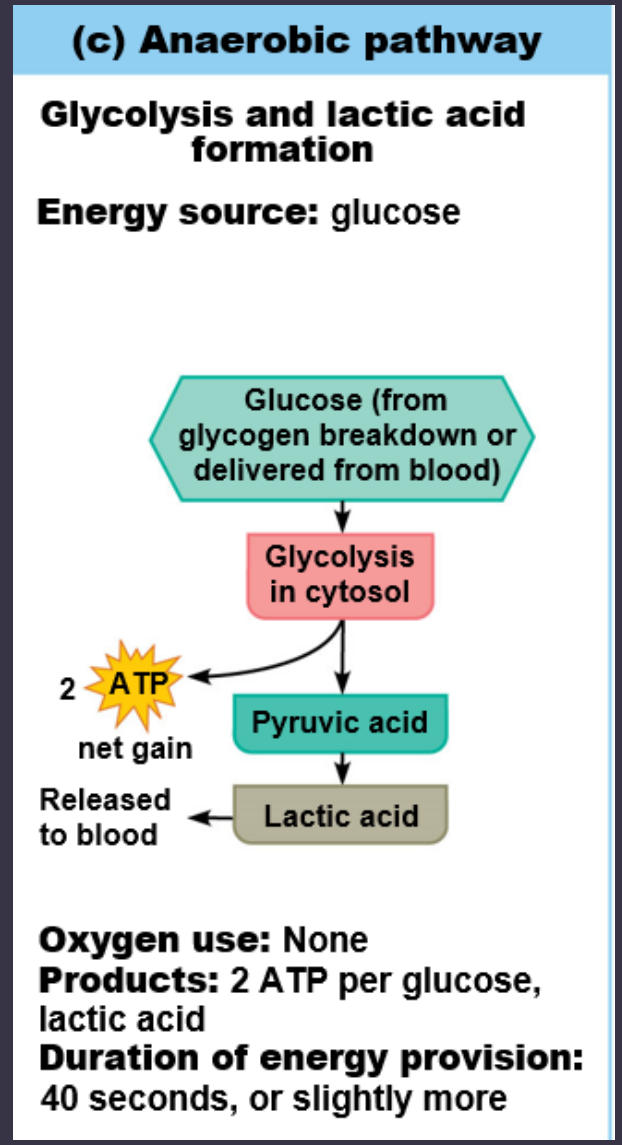
- **Primary ATP source** at rest and during light to moderate exercise
- Occurs in the **mitochondria** via **oxidative phosphorylation**
- Requires **oxygen and nutrients (glucose, fatty acids, amino acids)**
- Glucose $\rightarrow CO_2 + H_2O + \sim 32$ ATP
- **Slow process**, but efficient and long-lasting
- Depends on continuous **oxygen delivery**



PROVIDING ENERGY FOR MUSCLE CONTRACTION

Anaerobic Glycolysis and Lactic Acid Formation

- **ATP production without oxygen** (anaerobic)
- Glucose → **pyruvic acid** → **2 ATP per glucose molecule**
- Pyruvic acid → **lactic acid**
- Fast ATP generation, but **inefficient**
- Produces **muscle soreness** due to lactic acid accumulation
- Requires **large amounts of glucose** for sustained activity



MUSCLE FATIGUE AND OXYGEN DEFICIT

Muscle Fatigue:

- Inability of a muscle to maintain force despite continued stimulation
- Causes include:
  - ATP depletion
  - Accumulation of lactic acid
  - Ionic imbalances (Na^+, K^+)
  - Reduced calcium availability

Oxygen Debt (Excess Post-Exercise Oxygen Consumption, EPOC):

- Extra oxygen required after strenuous exercise to restore:
  - ATP and creatine phosphate levels
  - Oxygen bound to myoglobin
  - Lactic acid conversion back to pyruvate or glucose

Ensures **muscle recovery** and readiness for future contractions

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TYPES OF MUSCLE CONTRACTIONS

1. Isotonic Contractions

- Myofilaments **slide past each other** during contraction
- Muscle **shortens**, producing **movement**
- Examples:
  - Bending the knee
  - Lifting weights
  - Smiling

2. Isometric Contractions

- Muscle filaments attempt to slide, but muscle **cannot shorten**
- Tension **increases**, but **no movement** occurs
- Examples:
  - Pushing palms together
  - Holding a heavy object stationary

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MUSCLE TONE, STRENGTH, AND HYPERTROPHY/ATROPHY

1. Muscle Tone

- Continuous, **partial contraction** of resting muscles
- Maintains **posture and joint stability**
- Allows muscles to respond **quickly to stimulation**

2. Muscle Strength

- Determined by:
  - Number of muscle fibers recruited
  - Size of muscle fibers
  - Frequency of stimulation

3. Hypertrophy

- **Increase in muscle size** due to repetitive stimulation or exercise
- Increased **myofibril number and diameter**

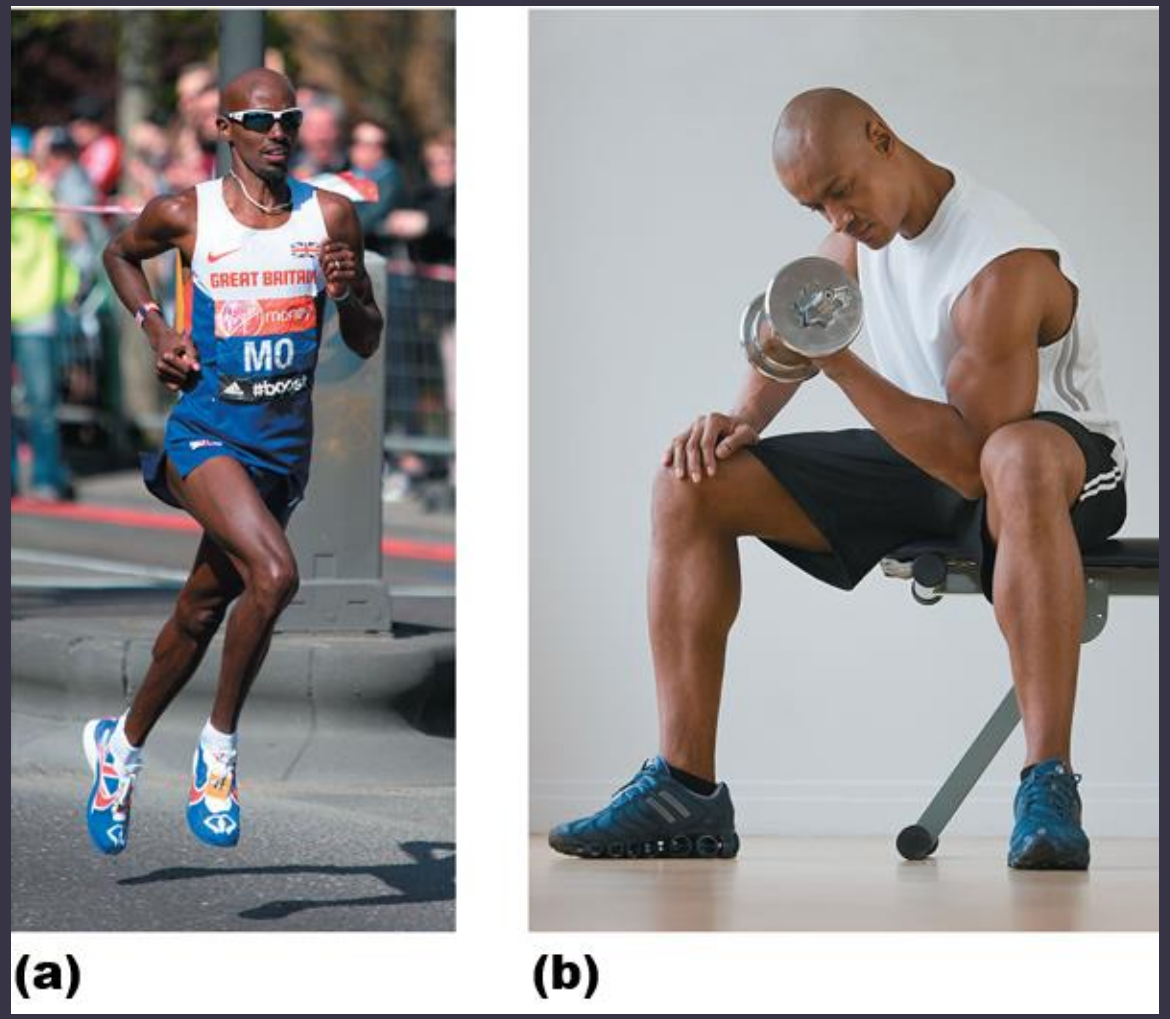
4. Atrophy

- **Decrease in muscle size and strength** due to disuse or denervation
- Muscle fibers shrink; can lead to **loss of function**

EFFECT OF EXERCISE ON MUSCLES

Exercise increases muscle:

- Size
- Strength
- Endurance



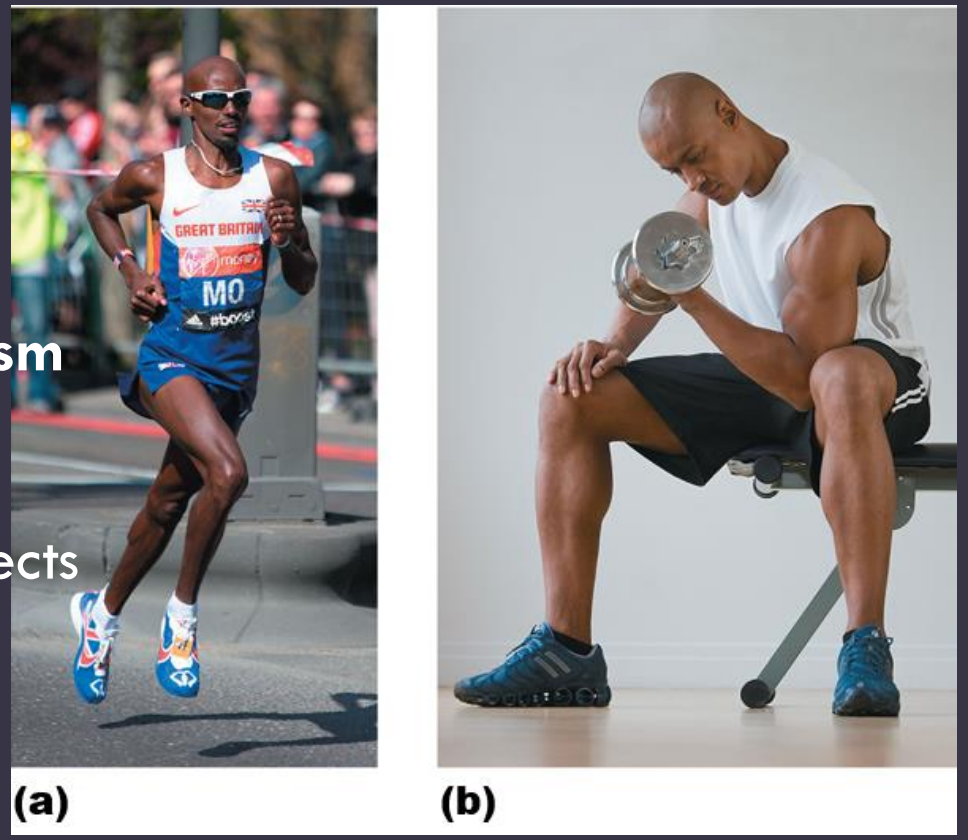
EFFECT OF EXERCISE ON MUSCLES

1. Aerobic (Endurance) Exercise

- Examples: biking, jogging, swimming
- Effects:
  - Stronger, more flexible muscles
  - Greater resistance to fatigue
  - Improved **cardiovascular efficiency and metabolism**
  - Enhances **coordination and digestion**

2. Resistance (Isometric) Exercise

- Examples: weight lifting, pushing against immovable objects
- Effects:
  - Increases **muscle size (hypertrophy)**
  - Enhances **strength**
  - **Individual muscle fibers enlarge**



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MUSCLE MOVEMENTS, ROLES, AND NAMES

Table 6.2 The Five Golden Rules of Skeletal Muscle Activity

1. With a few exceptions, all skeletal muscles cross at least one joint.
2. Typically, the bulk of a skeletal muscle lies proximal to the joint crossed.
3. All skeletal muscles have at least two attachments: the origin and the insertion.
4. Skeletal muscles can only pull; they never push.
5. During contraction, a skeletal muscle insertion moves toward the origin.

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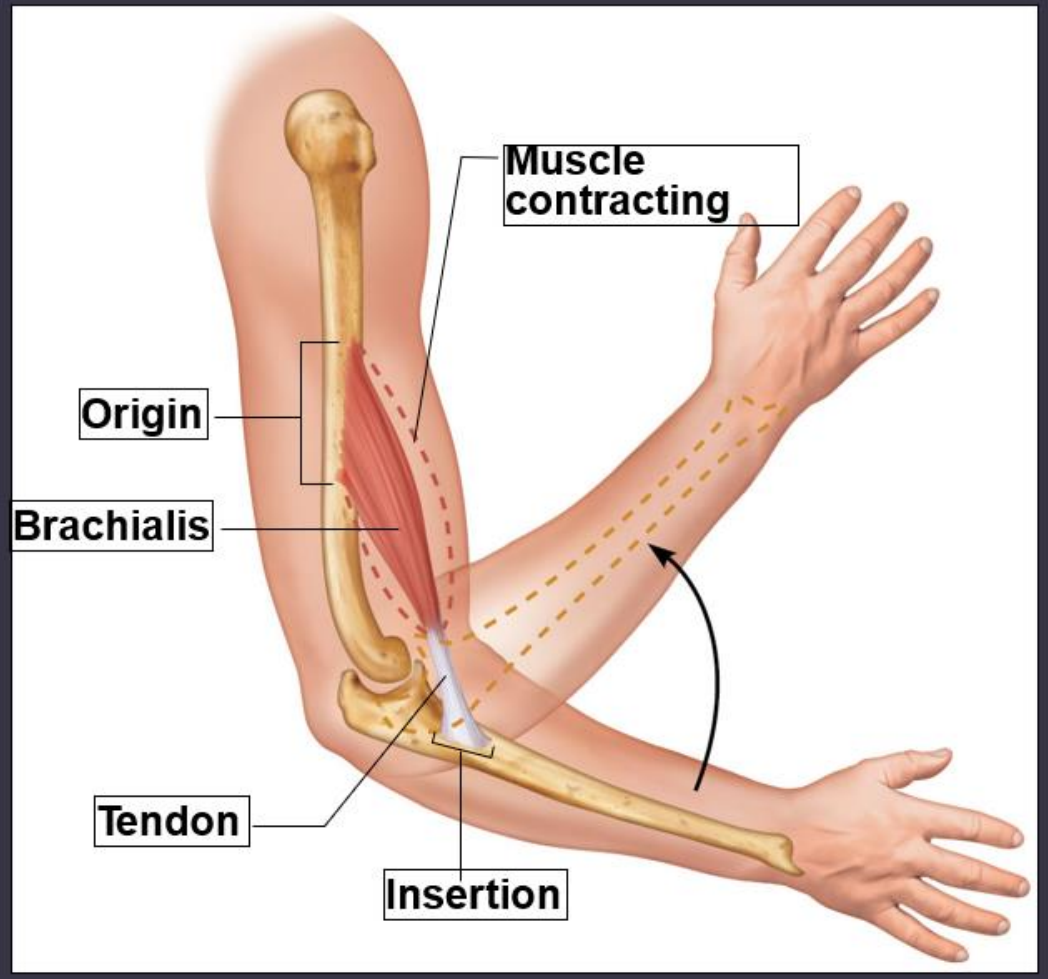
MUSCLE ATTACHMENTS AND MOVEMENTS

Muscles attach to at least two points:

- **Origin** → attachment to **immovable or less movable bone**
- **Insertion** → attachment to **movable bone**

Principle of movement:

- When a muscle contracts, the **insertion moves toward the origin**
- **Body movements occur** when muscles contract across joints



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INTERACTIONS OF SKELETAL MUSCLES IN THE BODY

Muscle Roles in Movement

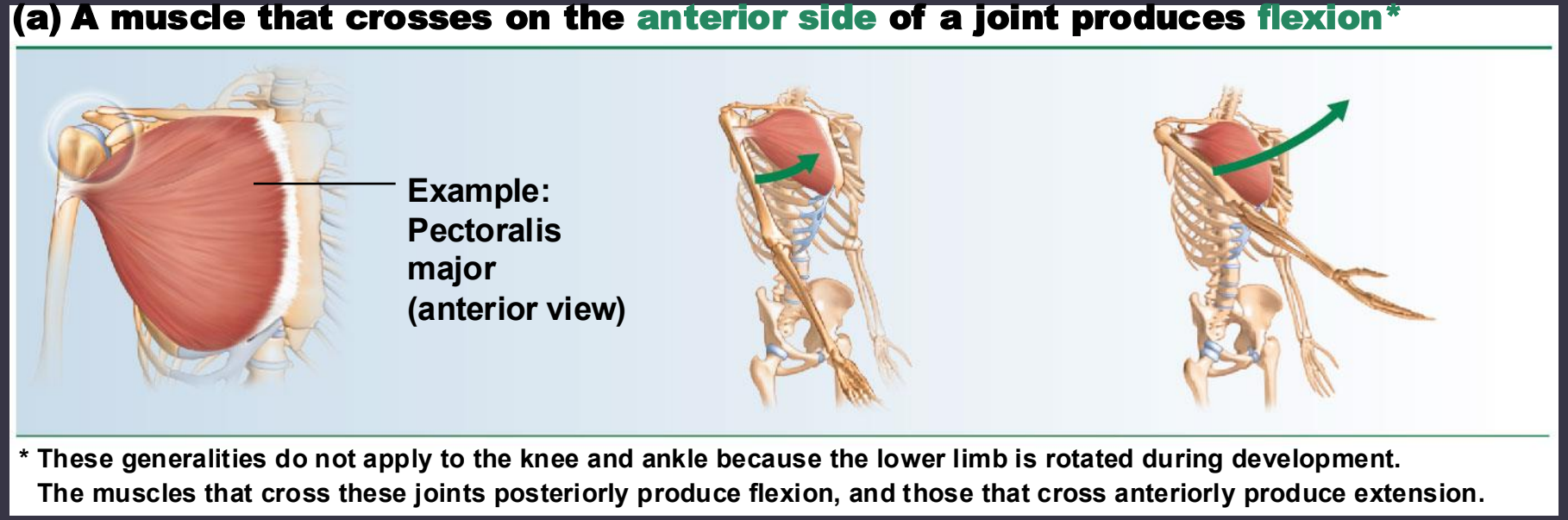
- **Muscles contract to pull, not push**
- Opposing muscle groups lie on **opposite sides of a joint**

Types of Muscle Roles:

- **Prime mover (agonist):** major responsibility for movement
- **Antagonist:** opposes or reverses a prime mover
- **Synergist:** assists prime mover; reduces undesirable movement
- **Fixator:** stabilizes origin of prime mover; holds a bone steady

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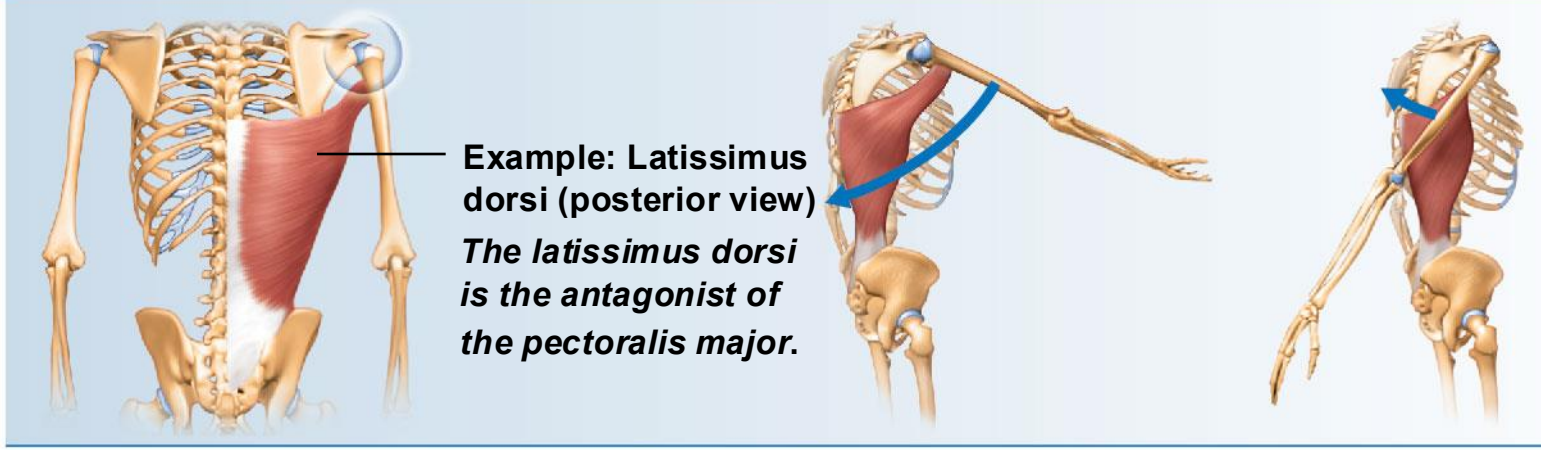
MUSCLE ACTION



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MUSCLE ACTION

(b) A muscle that crosses on the posterior side of a joint produces extension*



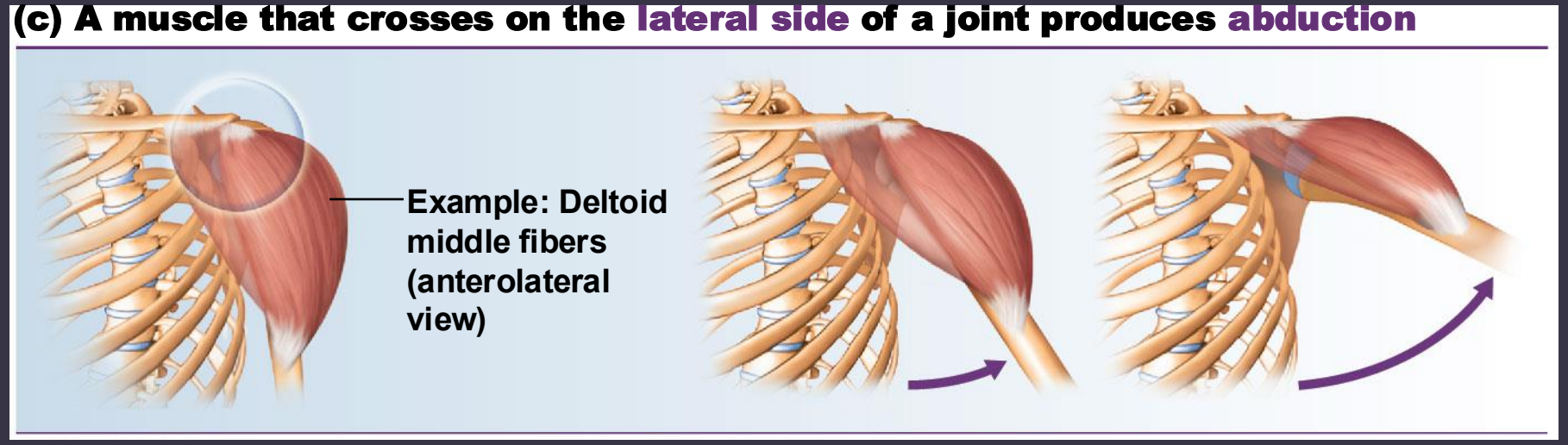
Example: Latissimus dorsi (posterior view)

The latissimus dorsi is the antagonist of the pectoralis major.

* These generalities do not apply to the knee and ankle because the lower limb is rotated during development. The muscles that cross these joints posteriorly produce flexion, and those that cross anteriorly produce extension.

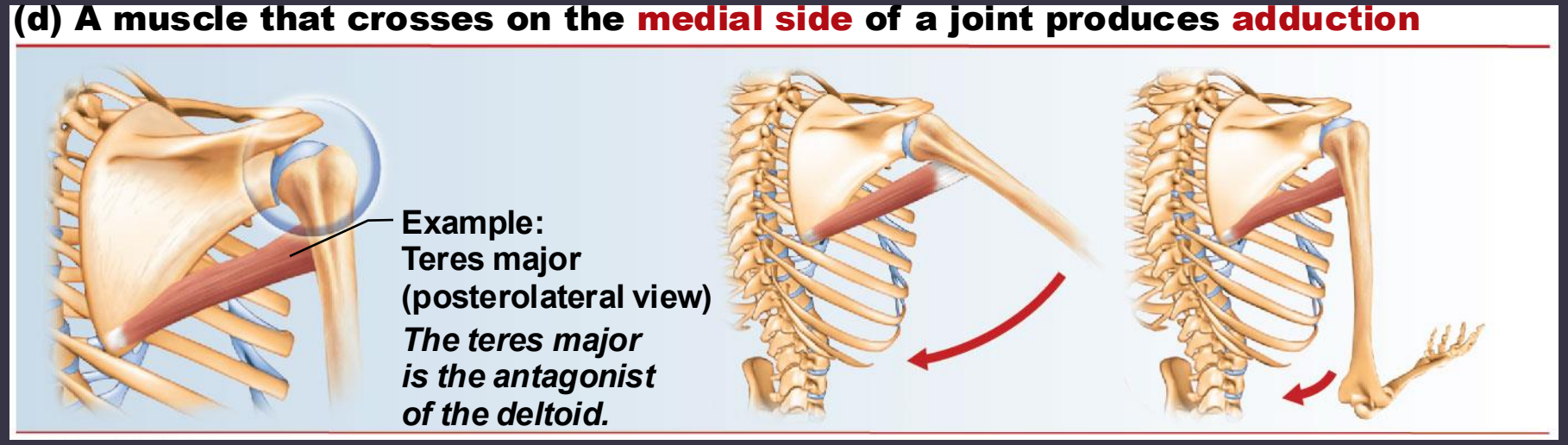
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MUSCLE ACTION



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MUSCLE ACTION



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OBJECTIVES

1. Differentiate the three types of muscle tissue: **Skeletal, Smooth, and Cardiac Muscles**
2. Enumerate and describe the **microscopic parts of skeletal muscle**
3. Identify and enumerate the **major skeletal muscles of the human body**

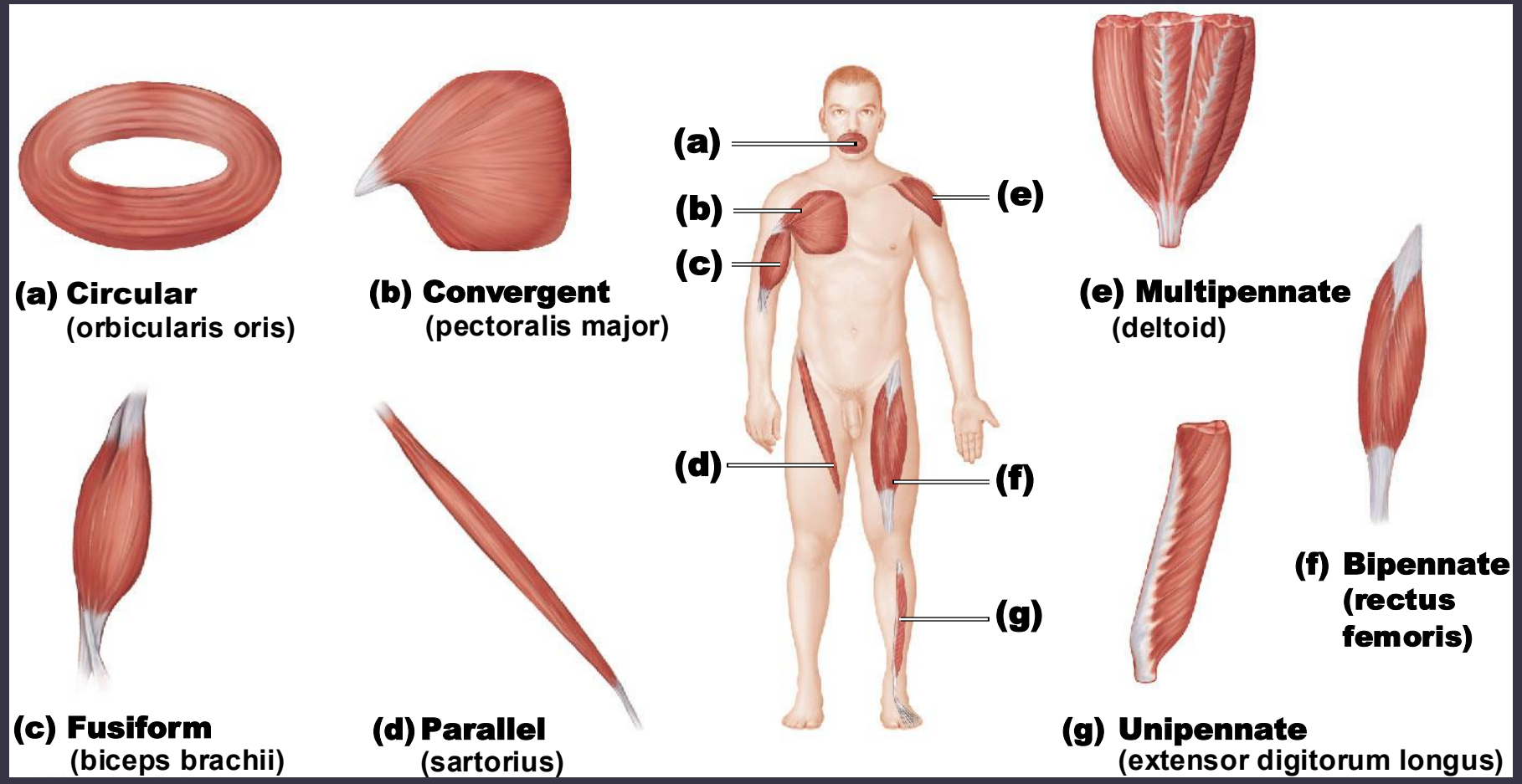
NAMING SKELETAL MUSCLES

Muscles are named on the basis of several criteria

- By direction of muscle fibers
  - Example: *rectus* (straight)
- By relative size of the muscle
  - Example: *maximus* (largest)
- By location of the muscle
  - Example: *temporalis* (temporal bone)
- By number of origins
  - Example: *triceps* (three heads)
- By location of the muscle's origin and insertion
  - Example: *sterno* (on the sternum)
- By shape of the muscle
  - Example: *deltoid* (triangular)
- By action of the muscle
  - Example: *flexor* and *extensor* (flexes or extends a bone)

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RELATIONSHIP OF FASCICLE ARRANGEMENT TO MUSCLE STRUCTURE.



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MAJOR SUPERFICIAL MUSCLES OF THE ANTERIOR SURFACE OF THE BODY.

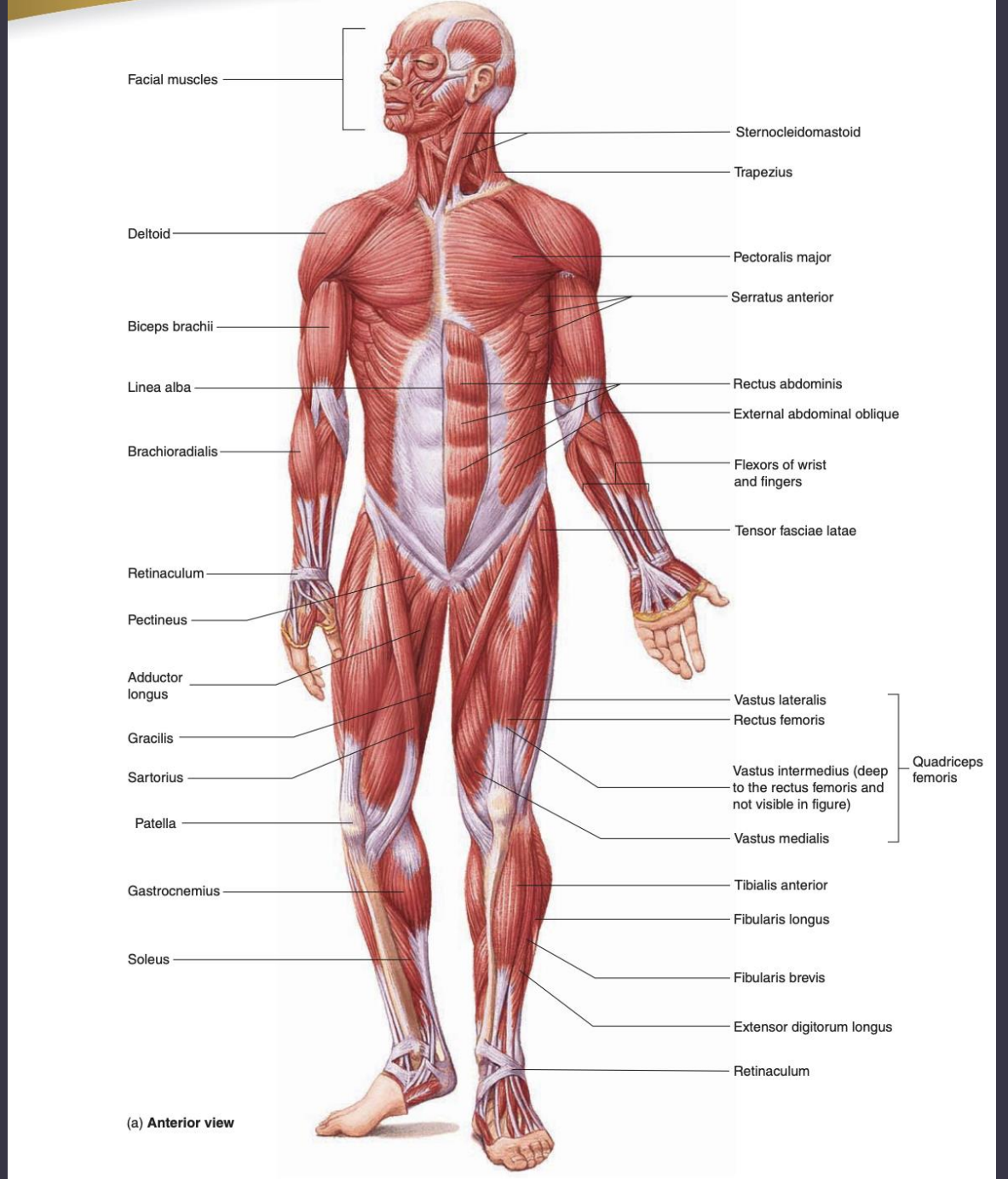
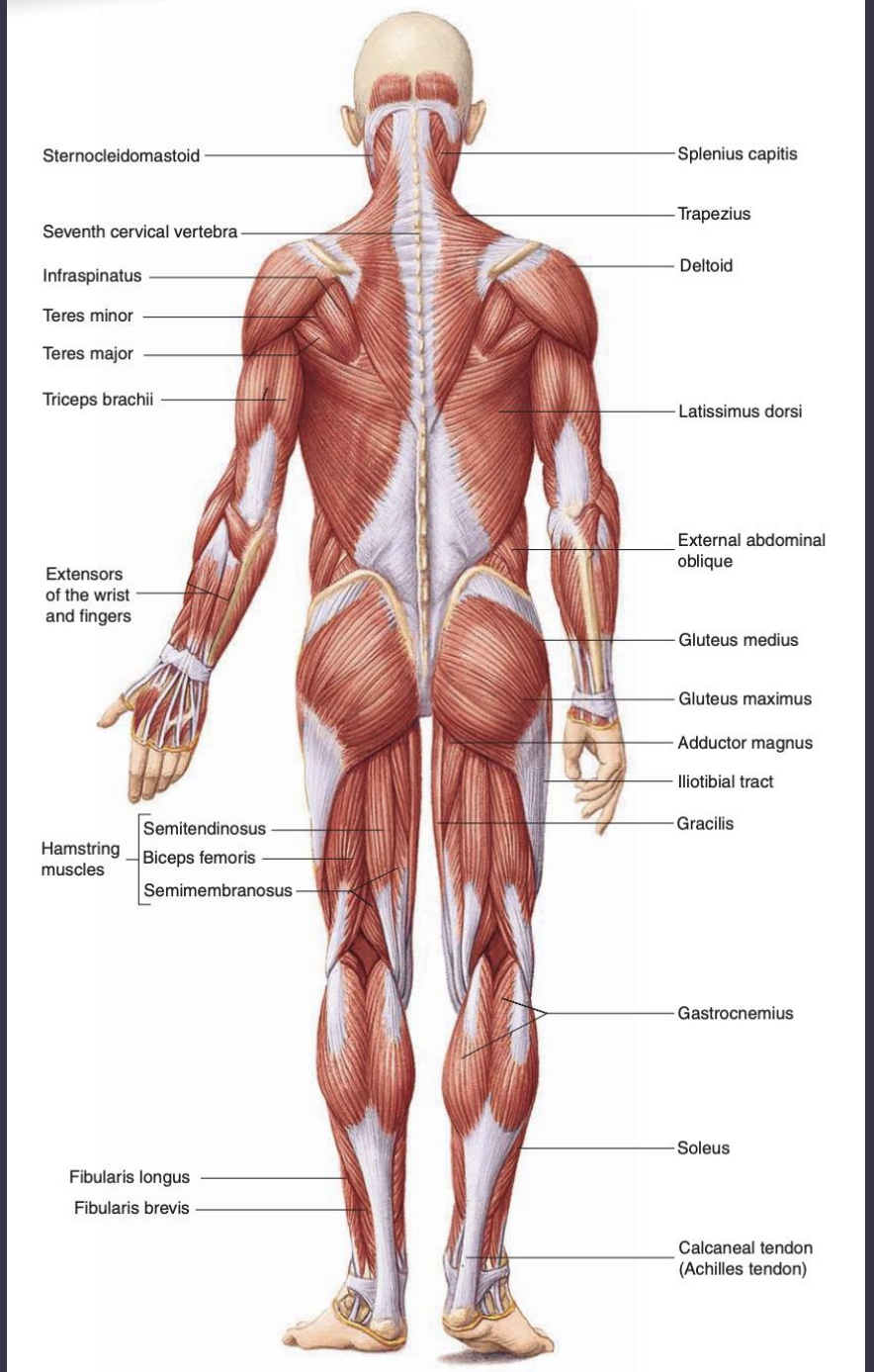


FIGURE 10.3 Overview of the Superficial Body Musculature

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MAJOR SUPERFICIAL MUSCLES OF THE POSTERIOR SURFACE OF THE BODY



(b) **Posterior view**

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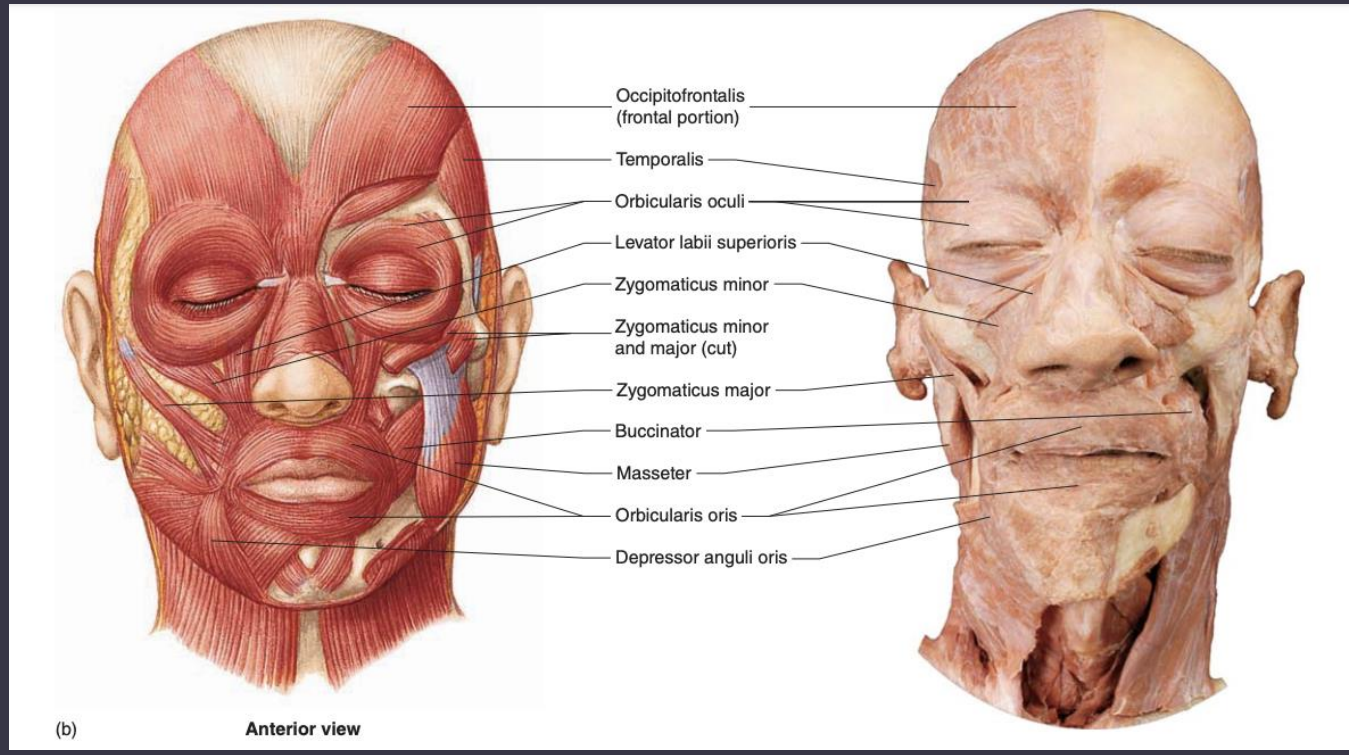
DETAILED ANATOMY AND PHYSIOLOGY OF MUSCLE GROUPS

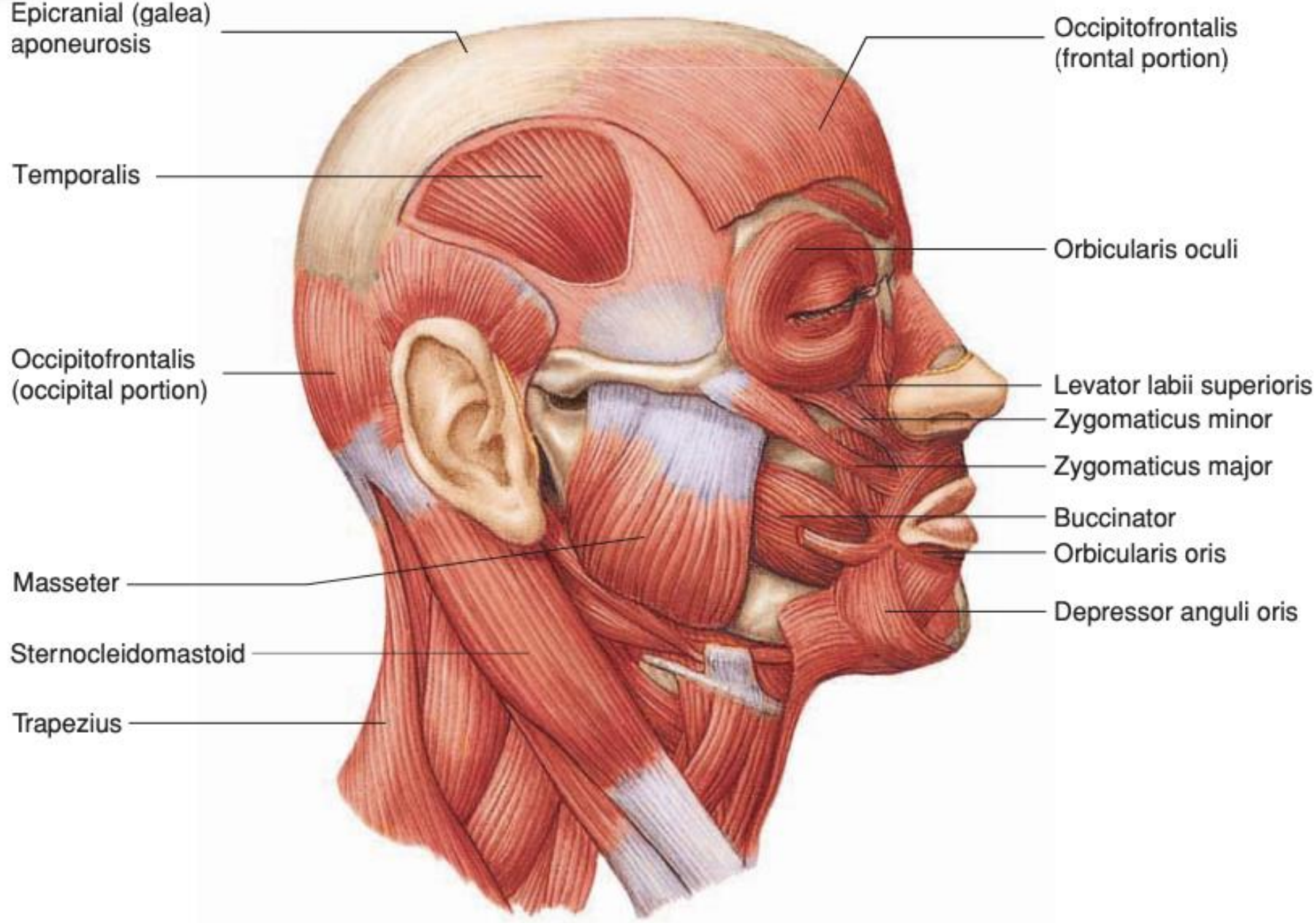
MUSCLES OF FACIAL EXPRESSION

- Insert into the skin → control facial movements & expressions
- Innervated by **cranial nerve VII (facial nerve)**

Key muscles:

- **Orbicularis oculi** – closes the eyelids ("blink, squint")
- **Orbicularis oris** – closes/puckers lips ("kissing muscle")
- **Zygomaticus (major & minor)** – elevates corners of mouth (smile)
- **Buccinator** – compresses cheek (whistling, blowing, assists in chewing)
- **Frontalis (occipitofrontalis)** – raises eyebrows, wrinkles forehead





(a) **Lateral view**

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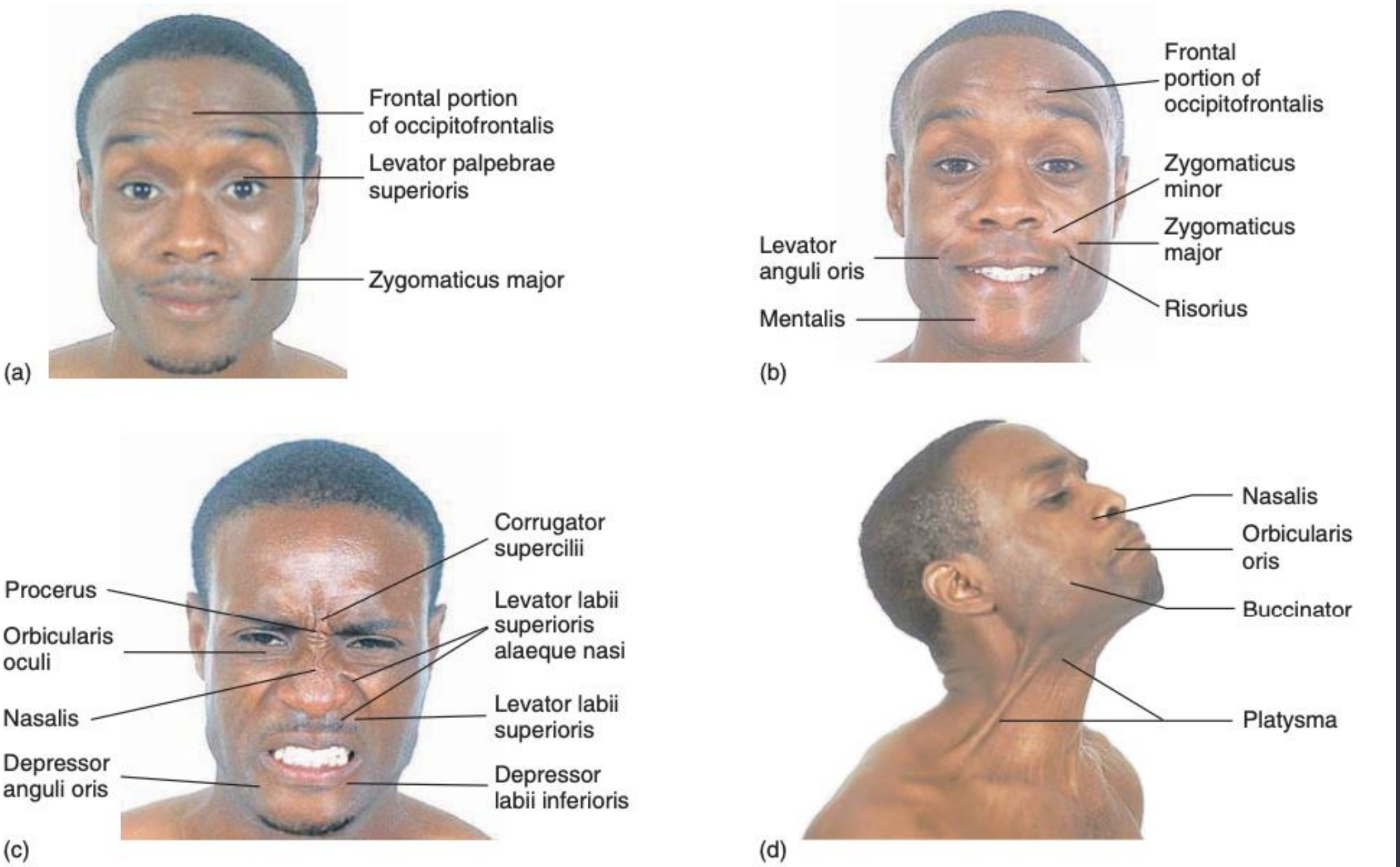


FIGURE 10.8 **Surface Anatomy, Muscles of Facial Expression**

TABLE 10.3 Muscles of Facial Expression (see figures 10.7 and 10.8)

Muscle	Origin	Insertion	Nerve	Action
Auricularis (aw-rik′ū-lăr′is)				
Anterior	Aponeurosis over head	Cartilage of auricle	Facial	Draws auricle superiorly and anteriorly
Posterior	Mastoid process	Posterior root of auricle	Facial	Draws auricle posteriorly
Superior	Aponeurosis over head	Cartilage of auricle	Facial	Draws auricle superiorly and posteriorly
Buccinator (buk′sĭ-nā′tōr)	Mandible and maxilla	Orbicularis oris at angle of mouth	Facial	Retracts angle of mouth; flattens cheek
Corrugator supercilii (kōr′ŭ′gā′ter soo′per-sil′ē-ī)	Nasal bridge and orbicularis oculi	Skin of eyebrow	Facial	Depresses medial portion of eyebrow; draws eyebrows together, as in frowning
Depressor anguli oris (dē-pres′ŏr ang′gū-lī ōr′is)	Lower border of mandible	Skin of lip near angle of mouth	Facial	Depresses angle of mouth
Depressor labii inferioris (dē-pres′ŏr lā′bē-ī in-fēr′ē-ōr-is)	Lower border of mandible	Skin of lower lip and orbicularis oris	Facial	Depresses lower lip
Levator anguli oris (lē-vā′tor, le-vā′ter ang′gū-lī ōr′is)	Maxilla	Skin at angle of mouth and orbicularis oris	Facial	Elevates angle of mouth
Levator labii superioris (lē-vā′tor, le-vā′ter lā′bē-ī sū-pēr′ē-ōr-is)	Maxilla	Skin and orbicularis oris of upper lip	Facial	Elevates upper lip
Levator labii superioris alaeque nasi (lē-vā′tor, le-vā′ter lā′bē-ī sū-pēr′ē-ōr-is ă-lak′ă nā′zī)	Maxilla	Ala at nose and upper lip	Facial	Elevates ala of nose and upper lip
Levator palpebrae superioris (lē-vā′tor, le-vā′ter pal-pē′brē sū-pēr′ē-ōr-is)	Lesser wing of sphenoid	Skin of eyelid	Oculomotor	Elevates upper eyelid

TABLE 10.3 Muscles of Facial Expression (see figures 10.7 and 10.8)

Muscle	Origin	Insertion	Nerve	Action
Mentalis (men-tā′lis)	Mandible	Skin of chin	Facial	Elevates and wrinkles skin over chin; protrudes lower lip
Nasalis (nā′zăˊ-lis)	Maxilla	Bridge and ala of nose	Facial	Dilates nostril
Occipitofrontalis (ok-sip′i-tō-frŭn′tā′lis)	Occipital bone	Skin of eyebrow and nose	Facial	Moves scalp; elevates eyebrows
Orbicularis oculi (ōr-bik′ū-lā′ris ok′ū-lī)	Maxilla and frontal bones	Circles orbit and inserts near origin	Facial	Closes eye
Orbicularis oris (ōr-bik′ū-lā′ris ōr′is)	Nasal septum, maxilla, and mandible	Fascia and other muscles of lips	Facial	Closes lips
Platysma (plă-tiz′mă)	Fascia of deltoid and pectoralis major	Skin over inferior border of mandible	Facial	Depresses lower lip; wrinkles skin of neck and upper chest
Procerus (prō-sē′rŭs)	Bridge of nose	Frontalis	Facial	Creates horizontal wrinkles between eyes, as in frowning
Risorius (ri-sōr′ē-ŭs)	Platysma and masseter fascia	Orbicularis oris and skin at corner of mouth	Facial	Abducts angle of mouth
Zygomaticus major (zī′gō-mat′i-kŭs)	Zygomatic bone	Angle of mouth	Facial	Elevates and abducts upper lip
Zygomaticus minor (zī′gō-mat′i-kŭs)	Zygomatic bone	Orbicularis oris of upper lip	Facial	Elevates and abducts upper lip

LIP MUSCLES

ELEVATE THE UPPER LIP

1. Levator anguli oris
2. Levator labii superioris
3. Levator labii superioris alaeque
4. Zygomaticus major
5. Zygomaticus minor

DEPRESSORS OF UPPER LIP

1. Depressor anguli oris
2. Depressor labii inferioris

PROTRUDES THE LOWER LIP

1. Mentalis

DEPRESSORS OF LOWER LIP

1. Depressor anguli oris
2. Depressor labii inferioris
3. Platysma

NECK MUSCLES

Anterior Neck Muscles

- Function: primarily in **lateral head movement**

Posterior Neck Muscles

- Function: **extension of the neck**
- Attach to **occipital bone** & **mastoid process**
- Assist in **rotation** & **lateral flexion**

Lateral Neck Muscles

- Function: **rotation & lateral flexion** of neck
- **Sternocleidomastoid** → **prime mover** of lateral neck movement

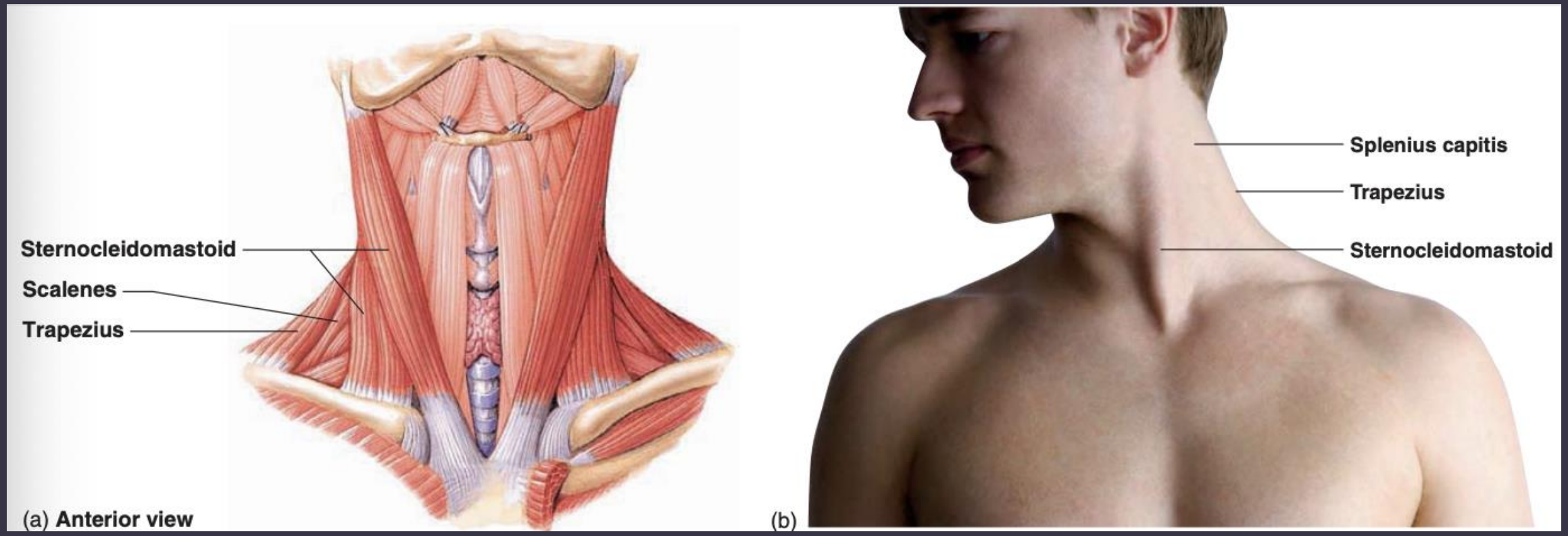
POSTERIOR and LATERAL NECK MUSCLES

Rotation and lateral flexion of the neck

TABLE 10.2* Muscles Moving the Head and Neck (see figures 10.4–10.6)

Muscle	Origin	Insertion	Nerve	Action
Anterior				
Longus capitis (lon′gŭs ka′pi-tis; not illustrated)	C3–C6	Occipital bone	C1–C3	Flexes neck
Rectus capitis anterior (rek′tŭs ka′pi-tis; not illustrated)	Atlas	Occipital bone	C1–C2	Flexes neck
Posterior				
Longissimus capitis (lon-gis′ĭ-mŭs ka′pi-tis)	Upper thoracic and lower cervical vertebrae	Mastoid process	Dorsal rami of cervical nerves	Extends, rotates, and laterally flexes neck
Oblique capitis superior (ka′pi-tis)	Atlas	Occipital bone (inferior nuchal line)	Dorsal ramus of C1	Extends and laterally flexes neck
Rectus capitis posterior (rek′tŭs ka′pi-tis)	Axis, atlas	Occipital bone	Dorsal ramus of C1	Extends and rotates neck
Semispinalis capitis	C4–T6	Occipital bone	Dorsal rami of cervical nerves	Extends and rotates neck
Splenius capitis	C4–T6	Superior nuchal line and mastoid process	Dorsal rami of cervical nerves	Extends, rotates, and laterally flexes neck
Trapezius	Occipital protuberance, nuchal ligament, spinous processes of C7–T12	Clavicle, acromion process, and scapular spine	Accessory (cranial nerve XI)	Extends and laterally flexes neck
Lateral				
Rectus capitis lateralis (not illustrated)	Atlas	Occipital bone	C1	Laterally flexes neck
Sternocleidomastoid (ster′nō-klī′dō-mas′toyd)	Manubrium and medial clavicle	Mastoid process and superior nuchal line	Accessory (cranial nerve XI)	One contracting alone: laterally flexes head and neck to same side and rotates head and neck to opposite side Both contracting together: flex neck
Scalene (skā′lēn) muscles	C2–C6	First and second ribs	Cervical and brachial plexuses	Flex, laterally flex, and rotate neck

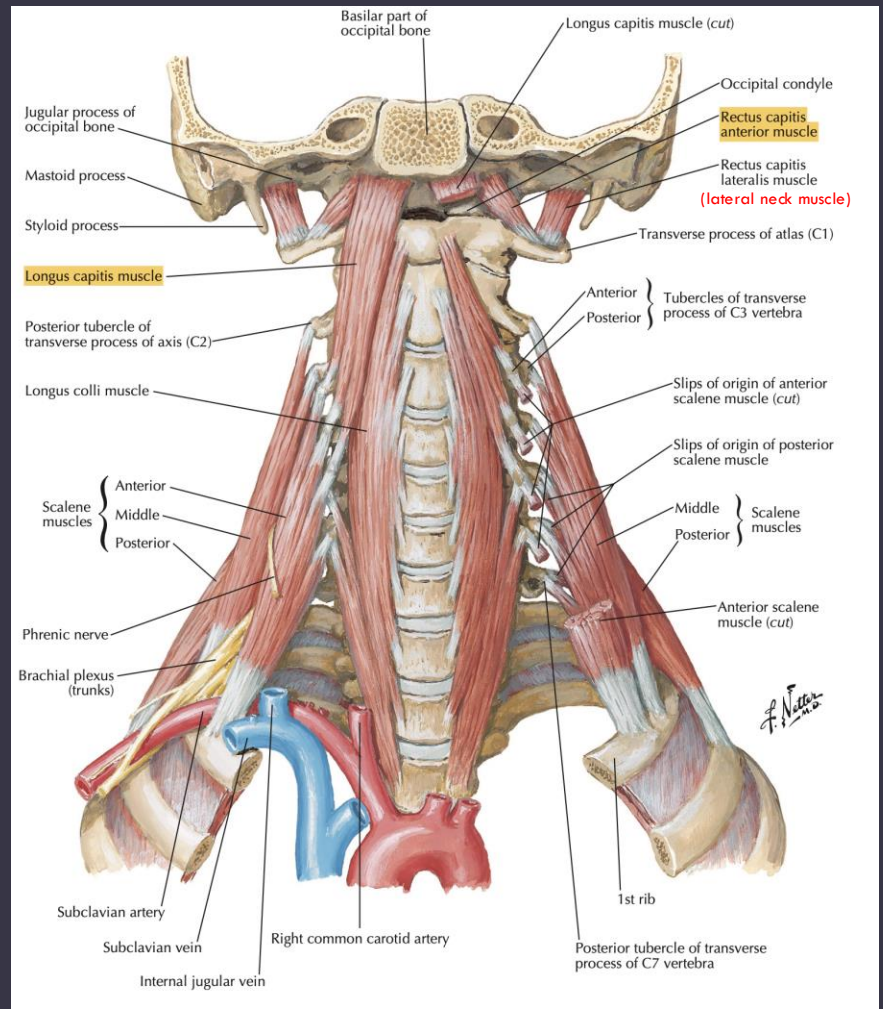
SUPERFACE ANATOMY OF ANTERIOR NECK MUSCLES



MUSCLES MOVING THE HEAD AND NECK

Anterior Neck Muscles (Deep Flexors)

- **Function:**
  - Flex the neck
  - Assist in **lateral head movement**
  - Located **deep** in the anterior cervical region
- **Longus capitis** – flexes head and neck
- **Rectus capitis anterior** – flexes head at the atlanto-occipital joint



MUSCLES MOVING THE HEAD AND NECK

Posterior Neck Muscles

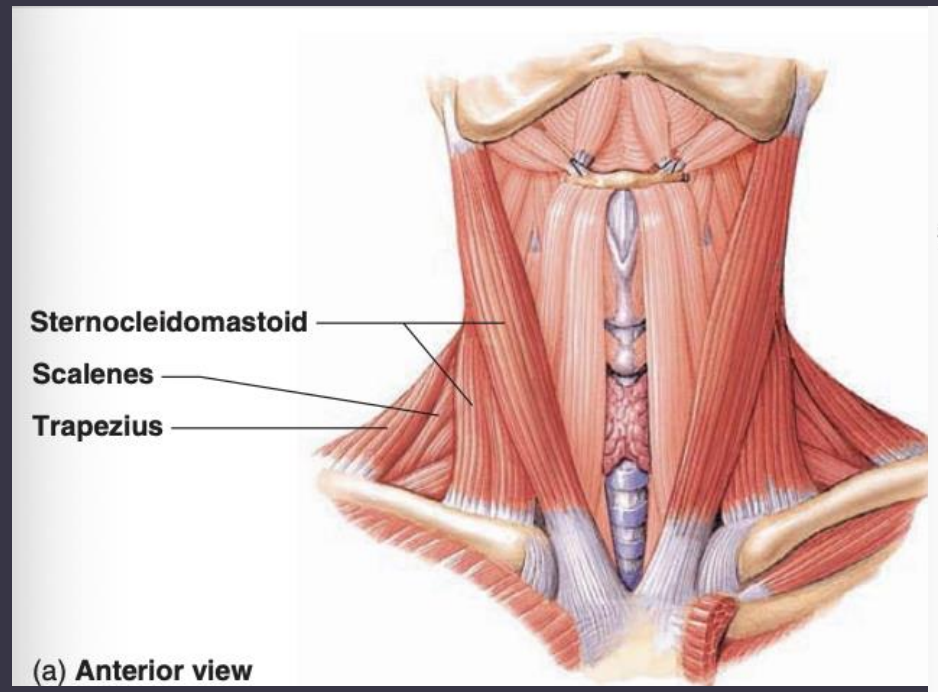
- **Function:**
  - **Extension** of the neck
  - **Rotation** & **lateral flexion**
  - Provide **postural support**

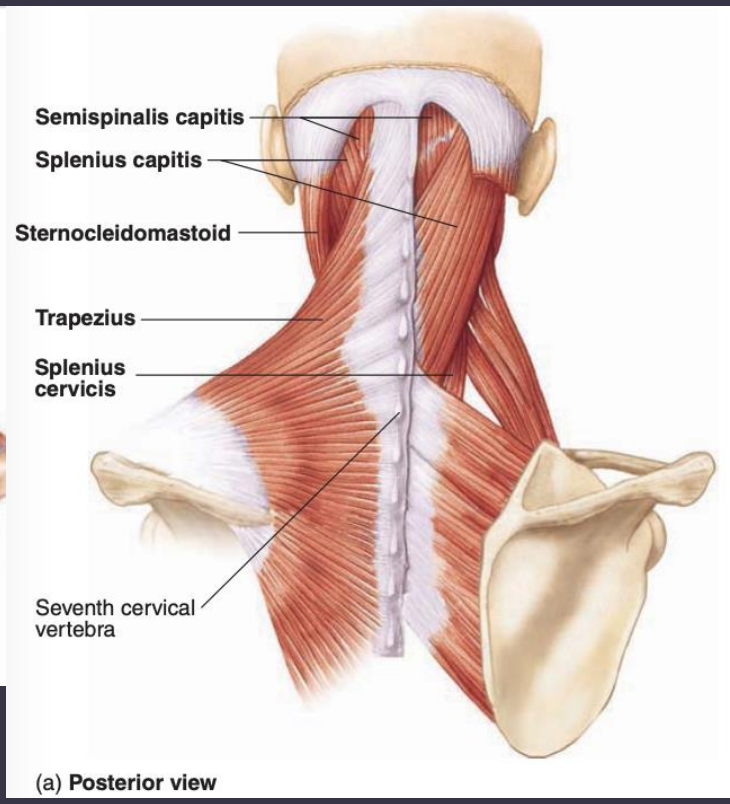
- **Splenius capitis** – extends, rotates, and laterally flexes head/neck
- **Semispinalis capitis** – powerful extensor of head/neck
- **Longissimus capitis** – extends and laterally flexes head, assists in rotation
- **Suboccipital muscles** (rectus capitis posterior, obliquus capitis) – fine control of head extension and rotation

MUSCLES MOVING THE HEAD AND NECK

Lateral Neck Muscles

- **Function:** Rotate and laterally flex the neck
  - **Rectus capitis lateralis** – stabilizes head, assists in lateral flexion
  - **Sternocleidomastoid (SCM)** – **prime mover** of lateral flexion & rotation
  - **Scalene muscles (anterior, middle, posterior)** – lateral flexion, elevate first and second ribs during inspiration





MUSCLES OF MASTICATION (CHEWING MUSCLES)

1. Temporalis
2. Masseter
3. Lateral Pterygoid
4. Medial Pterygoid

All innervated by: Trigeminal nerve (cranial nerve V, mandibular branch)

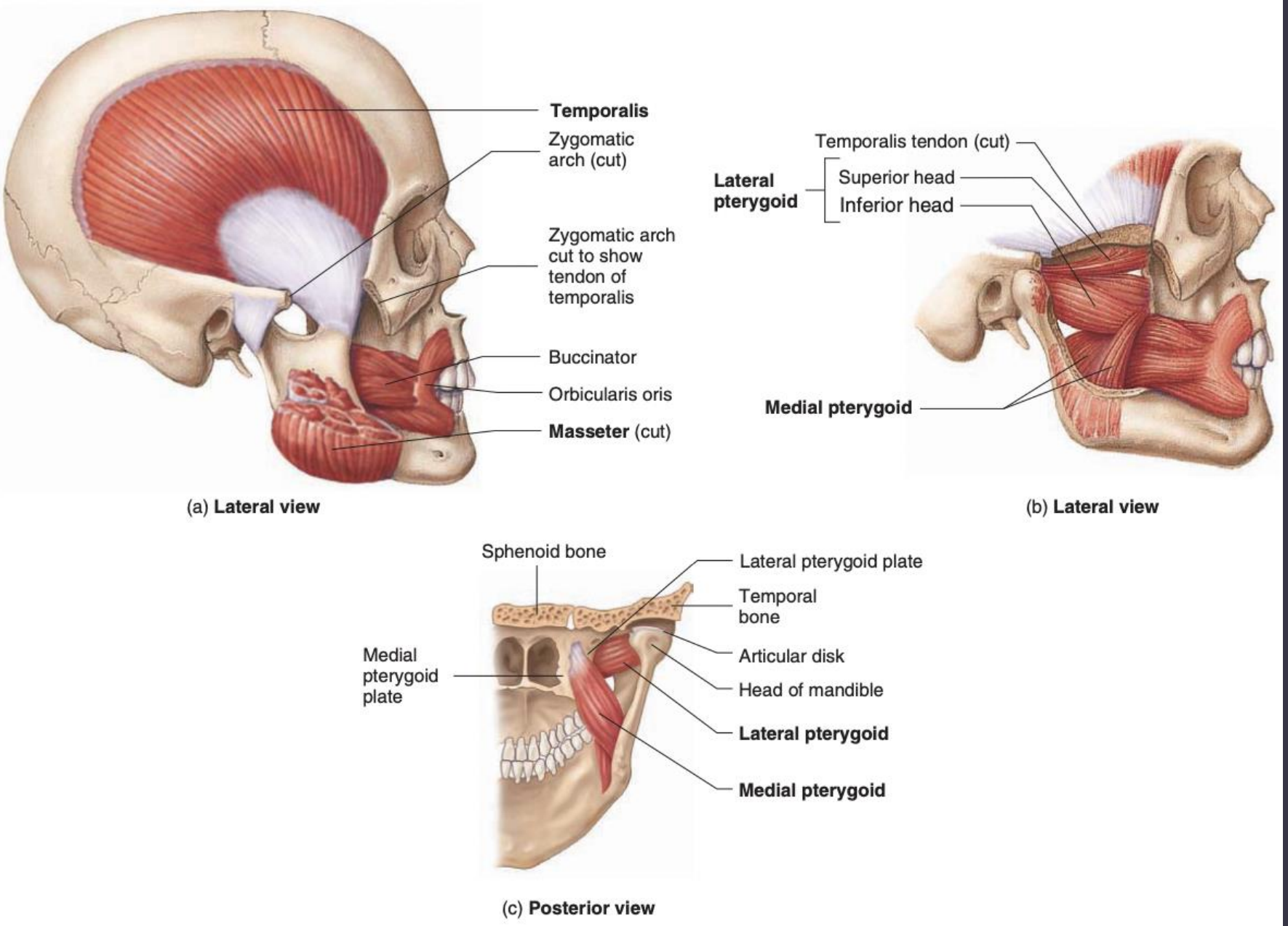
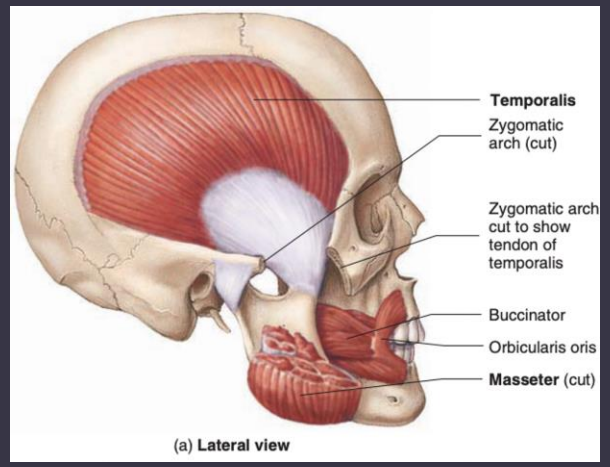


FIGURE 10.9 Muscles of Mastication

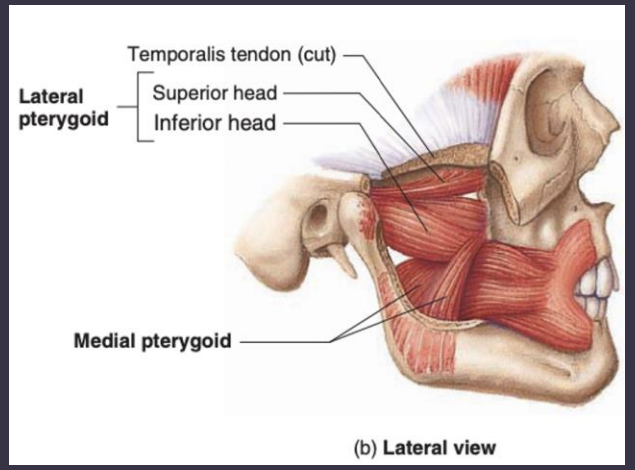
(*a*) The masseter and zygomatic arch are cut away to expose the temporalis. (*b*) The masseter and temporalis muscles are removed, and the zygomatic arch and part of the mandible are cut away to reveal the deeper muscles. (*c*) Frontal section of the skull, showing the pterygoid muscles. (Muscle names in bold are those involved in mastication.) AP|R

TABLE 10.4 Muscles of Mastication (see figures 10.7 and 10.9)

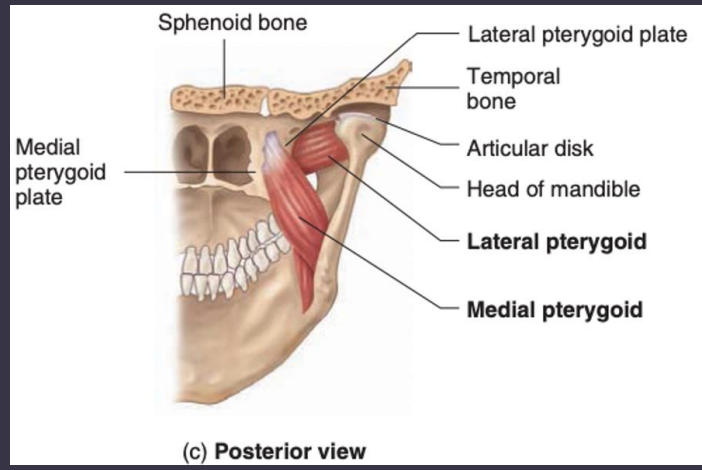
Muscle	Origin	Insertion	Nerve	Action
Temporalis (tem-pŏ-rā′lis)	Temporal fossa	Anterior portion of mandibular ramus and coronoid process	Mandibular division of trigeminal	Elevates and retracts mandible; involved in excursion
Masseter (ma′se-ter)	Zygomatic arch	Lateral side of mandibular ramus	Mandibular division of trigeminal	Elevates and protracts mandible; involved in excursion
Pterygoids (ter′i-goydz)				
Lateral	Lateral side of lateral pterygoid plate and greater wing of sphenoid	Condylar process of mandible and articular disk	Mandibular division of trigeminal	Protracts and depresses mandible; involved in excursion
Medial	Medial side of lateral pterygoid plate and tuberosity of maxilla	Medial surface of mandible	Mandibular division of trigeminal	Protracts and elevates mandible; involved in excursion



(a) Lateral view



(b) Lateral view

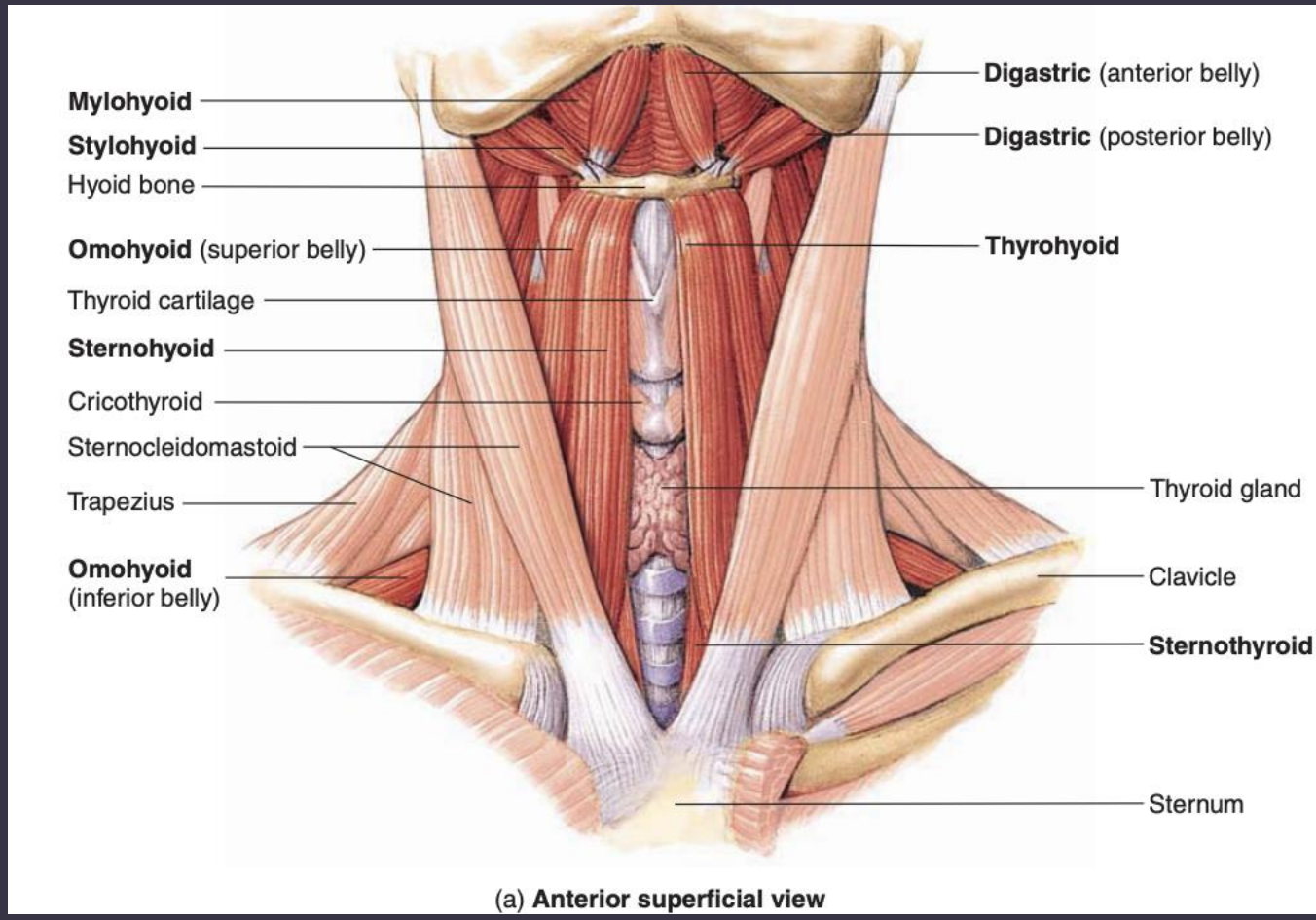


(c) Posterior view

SWALLOWING AND THE LARYNX

HYOID MUSCLES

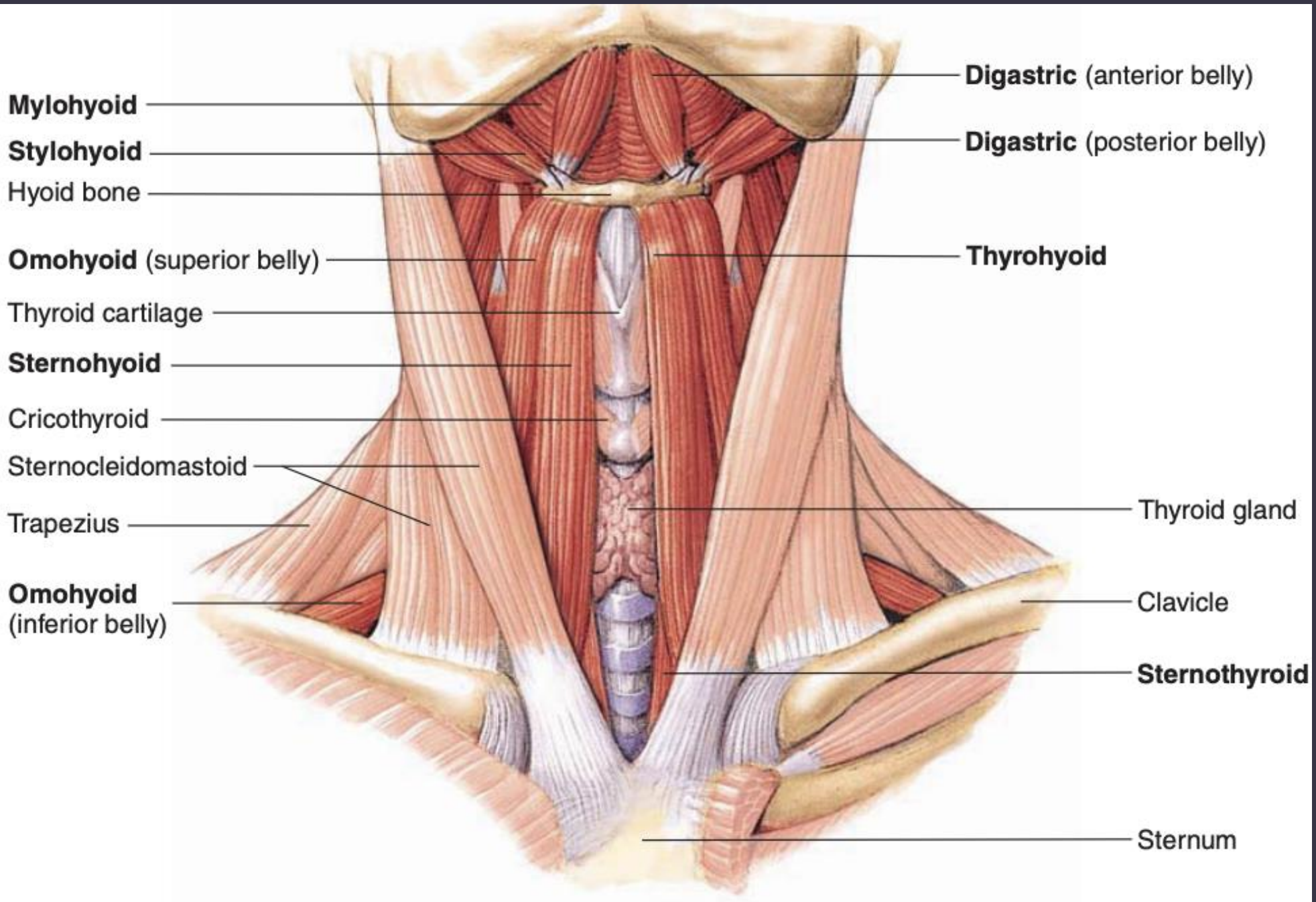
1. **Suprahyoid Group** (above hyoid bone)
   - Fix the hyoid from above → stabilize it
   - Help depress the mandible (assist in opening the mouth)
2. **Infrahyoid Group** (below hyoid bone)
   - Fix the hyoid from below → stabilize it during mandibular depression
   - **Thyrohyoid** (special role): elevates the larynx during swallowing and speech



(a) **Anterior superficial view**

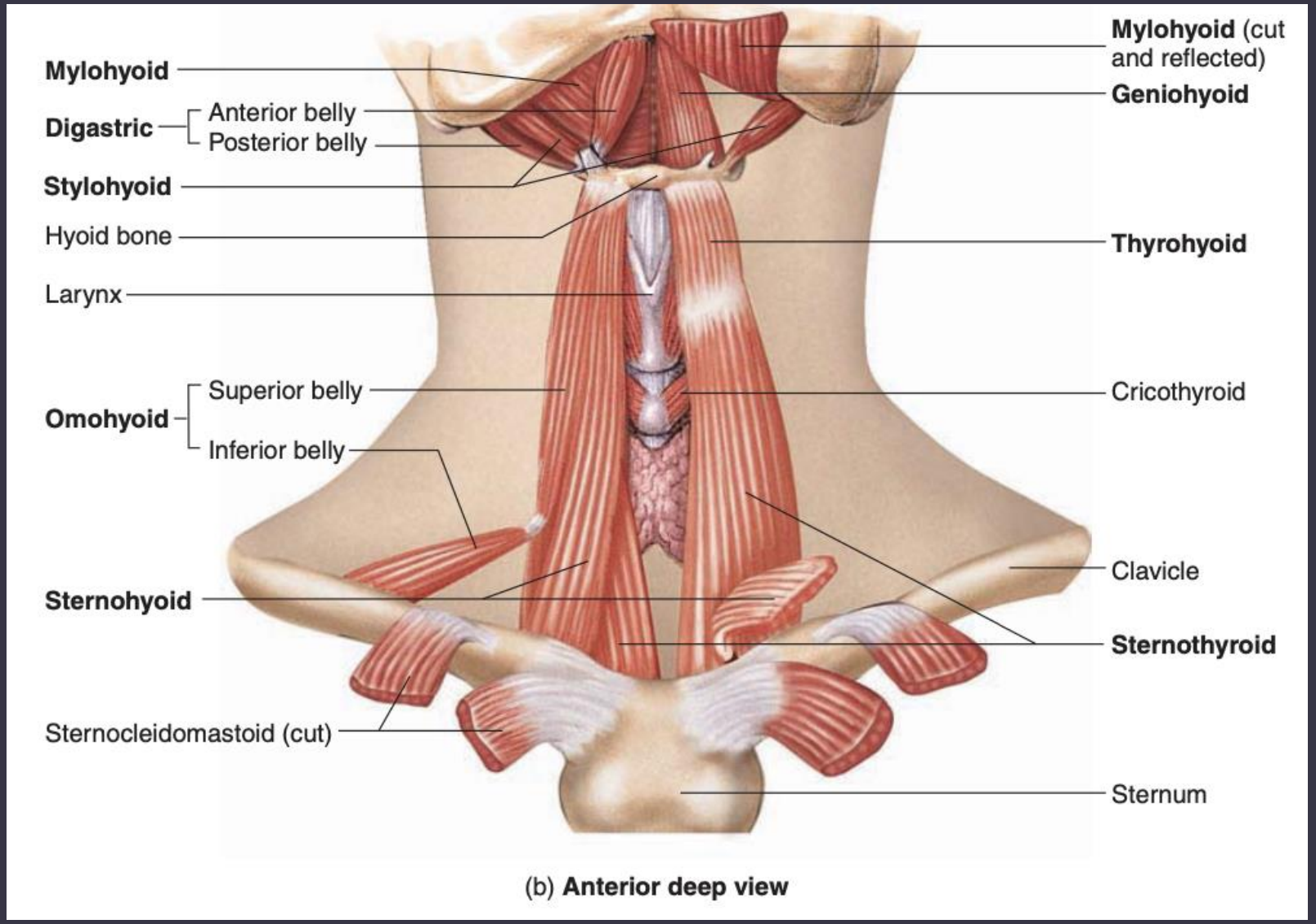
TABLE 10.5 Hyoid Muscles (see figure 10.10)

Muscle	Origin	Insertion	Nerve	Action
Suprahyoid Muscles				
Digastric (dī-gas′trik)	Mastoid process (posterior belly)	Mandible near midline (anterior belly)	Posterior belly—facial; anterior belly—mandibular division of trigeminal	Depresses and retracts mandible; elevates hyoid
Geniohyoid (jĕ-nī-ō-hī′oyd)	Mental protuberance of mandible	Body of hyoid	Fibers of C1 and C2 with hypoglossal	Protracts hyoid; depresses mandible
Mylohyoid (mī′lō-hī′oyd)	Body of mandible	Hyoid	Mandibular division of trigeminal	Elevates floor of mouth and tongue; depresses mandible when hyoid is fixed
Stylohyoid (stī-lō-hī′oyd)	Styloid process	Hyoid	Facial	Elevates hyoid
Infrahyoid Muscles				
Omohyoid (ō-mō-hī′oyd)	Superior border of scapula	Hyoid	Upper cervical through ansa cervicalis	Depresses hyoid; fixes hyoid in mandibular depression
Sternohyoid (ster′nō-hī′oyd)	Manubrium and first costal cartilage	Hyoid	Upper cervical through ansa cervicalis	Depresses hyoid; fixes hyoid in mandibular depression
Sternothyroid (ster′nō-thī′royd)	Manubrium and first or second costal cartilage	Thyroid cartilage	Upper cervical through ansa cervicalis	Depresses larynx; fixes hyoid in mandibular depression
Thyrohyoid (thī-rō-hī′oyd)	Thyroid cartilage	Hyoid	Upper cervical, passing with hypoglossal	Depresses hyoid and elevates thyroid cartilage of larynx; fixes hyoid in mandibular depression



(a) **Anterior superficial view**

HYOID MUSCLES



(b) **Anterior deep view**

SWALLOWING AND THE LARYNX

SOFT PALATE, PHARYNX (THROAT), AND LARYNX

contain several muscles involved in swallowing and speech

Soft Palate

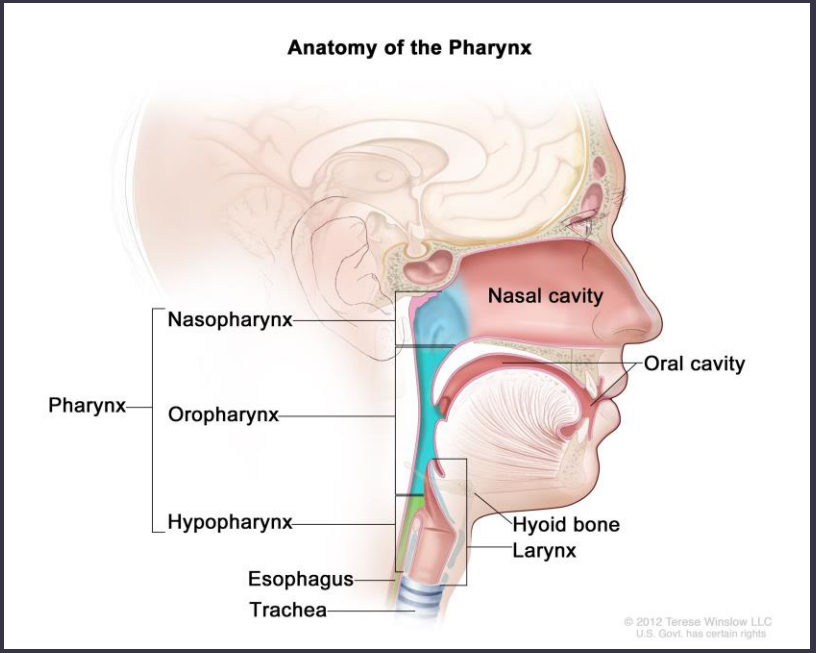
- Closes posterior opening to the nasal cavity during swallowing (prevents food/liquid from entering nasal passages)

Palatopharyngeus

- Elevates the pharynx
- Assists in swallowing
- Classified as an **extrinsic tongue muscle**

Salpingopharyngeus

- Constricts the pharynx (superior → inferior) to move bolus downward
- Opens auditory (Eustachian) tube → equalizes middle ear pressure with atmosphere
- Also an **extrinsic tongue muscle**



Soft Palate				
Levator veli palatini (lē-vā′tor, le-vā′ter vel′ī pal′ă-tē′nī)	Temporal bone and pharyngotympanic	Soft palate	Pharyngeal plexus	Elevates soft palate
Palatoglossus (pal-ă-tō-glos′ŭs)	Soft palate	Tongue	Pharyngeal plexus	Narrows fauces; elevates posterior tongue
Palatopharyngeus (pal′ă-tō-far-in-jē′ŭs)	Soft palate	Pharynx	Pharyngeal plexus	Narrows fauces; depresses palate; elevates pharynx
Tensor veli palatini (ten′sōr vel′ī pal′ă-tē′nī)	Sphenoid and auditory tube	Soft palate division of auditory tube	Mandibular, division of trigeminal	Tenses soft palate; opens auditory tube
Uvulae (ū′vū-lē)	Posterior nasal spine	Uvula	Pharyngeal plexus	Elevates uvula

- Muscular extension of the hard palate
- Separates oral cavity from nasal cavity
- Elevates during swallowing → closes off nasopharynx
- Contributes to speech (helps articulate sounds)
- Contains several muscles that aid swallowing and phonation

Pharynx				
Pharyngeal constrictors (fă-rin′jē-ăl)				
Inferior	Thyroid and cricoid cartilages	Pharyngeal raphe	Pharyngeal plexus and external laryngeal nerve	Narrows inferior portion of pharynx in swallowing
Middle	Stylohyoid ligament and hyoid	Pharyngeal raphe	Pharyngeal plexus	Narrows pharynx in swallowing
Superior	Medial pterygoid plate, mandible, floor of mouth, and side of tongue	Pharyngeal raphe	Pharyngeal plexus	Narrows superior portion of pharynx in swallowing
Salpingopharyngeus (sal-ping′gō-far-in-jē′ŭs)	Auditory tube	Pharynx	Pharyngeal plexus	Elevates pharynx; opens auditory tube in swallowing
Stylopharyngeus (stī′lō-far-in-jē′ŭs)	Styloid process	Pharynx	Glossopharyngeus	Elevates pharynx

- Muscular tube connecting **nasal and oral cavities** to the **larynx and esophagus**
- Functions:
- **Passageway for food and air**
- **Assists in swallowing (deglutition)**
- Contributes to **speech resonance**

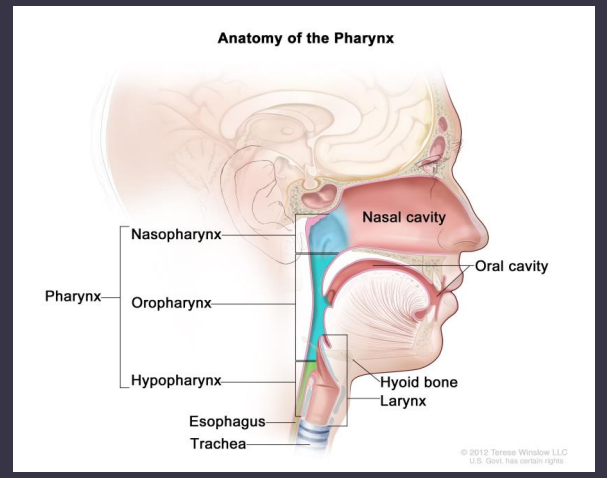


TABLE 10.7 Muscles of Swallowing and the Larynx (see figures 10.11 and 10.12)

Muscle	Origin	Insertion	Nerve	Action
Larynx				
Arytenoids (ar-i-tē′noydz)				
Oblique (not illustrated)	Arytenoid cartilage	Opposite arytenoid cartilage	Recurrent laryngeal	Narrows opening to larynx
Transverse (not illustrated)	Arytenoid cartilage	Opposite arytenoid cartilage	Recurrent laryngeal	Narrows opening to larynx
Cricoarytenoids (krī′kō-ar-i-tē′noydz)				
Lateral (not illustrated)	Lateral side of cricoid cartilage	Arytenoid cartilage	Recurrent laryngeal	Narrows opening to larynx
Posterior (not illustrated)	Posterior side of cricoid cartilage	Arytenoid cartilage	Recurrent laryngeal	Widens opening of larynx
Cricothyroid (krī-kō-thī′royd)	Anterior cricoid cartilage	Thyroid cartilage	Superior laryngeal	Lengthens (tenses) vocal cords
Thyroarytenoid (thī′rō-ar′i-tē′noyd; not illustrated)	Thyroid cartilage	Arytenoid cartilage	Recurrent laryngeal	Shortens (relaxes) vocal cords
Vocalis (vō-kal′ĭs; not illustrated)	Thyroid cartilage	Arytenoid cartilage	Recurrent laryngeal	Shortens (relaxes) vocal cords

LARYNX

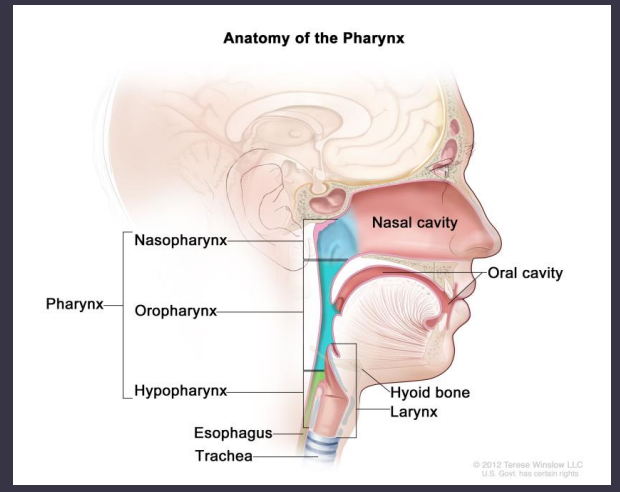
Muscular and cartilaginous structure located **anterior to the laryngopharynx**

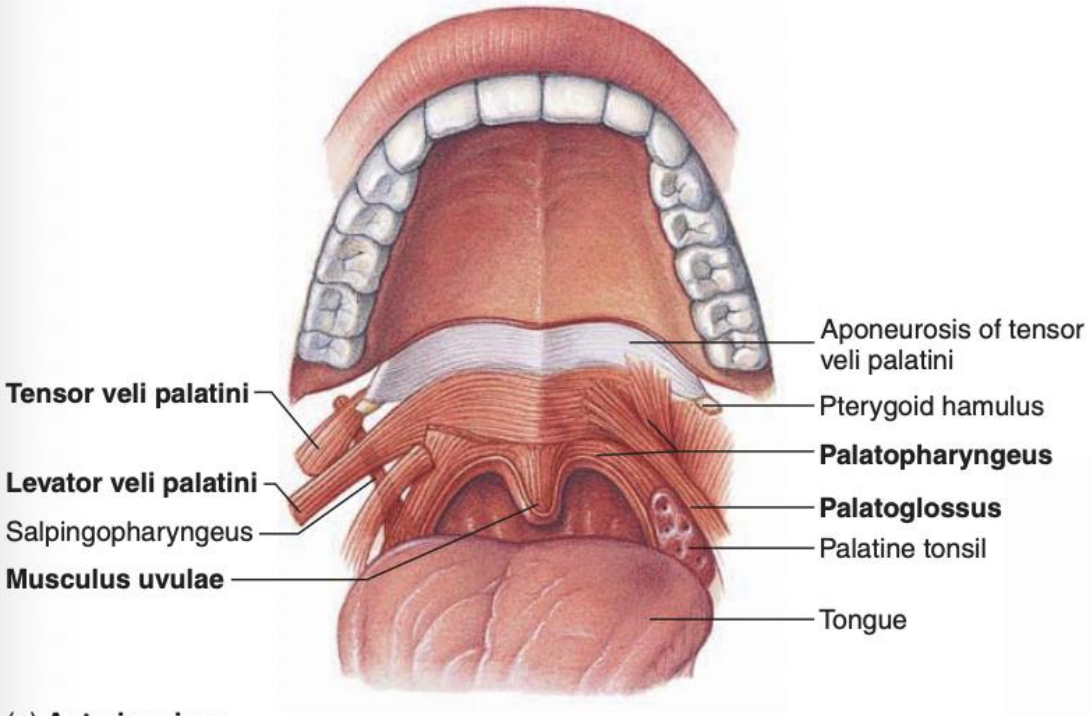
Functions:

- **Air passageway** between pharynx and trachea
- **Voice production** (houses vocal cords)
- **Protects airway** during swallowing

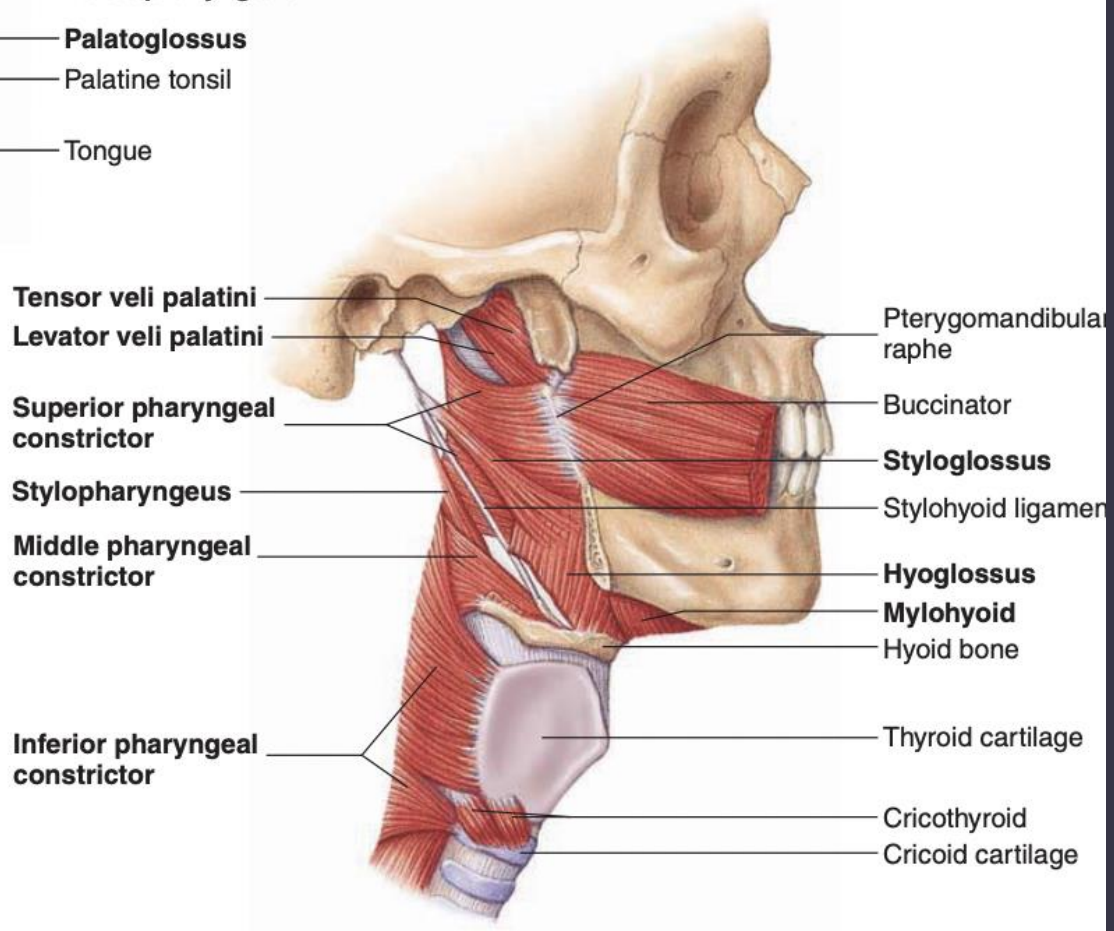
Key features:

- **Cartilages:** thyroid, cricoid, arytenoid, epiglottis
- **Vocal cords (true vocal folds):** vibrate to produce sound
- **Vestibular folds (false vocal cords):** protect airway





(a) **Anterior view**



(b) **Lateral view**

TONGUE MOVEMENT

Roles in Mastication:

- Moves food around in the mouth
- Works with **buccinator** to keep food between the teeth during grinding
- Pushes food upward to the **palate** and backward toward the **pharynx** to initiate swallowing

Roles in Speech:

- Changes shape to modify sounds and articulate words

TABLE 10.6 Tongue Muscles (see figure 10.11)

Muscle	Origin	Insertion	Nerve	Action
Intrinsic Muscles				
Longitudinal, transverse, and vertical (not illustrated)	Within tongue	Within tongue	Hypoglossal	Change tongue shape
Extrinsic Muscles				
Genioglossus (jĕ′nī-ō-glos′ŭs)	Mental protuberance of mandible	Tongue	Hypoglossal	Depresses and protrudes tongue
Hyoglossus (hī′ō-glos′ŭs)	Hyoid	Side of tongue	Hypoglossal	Retracts and depresses side of tongue
Styloglossus (stī′lō-glos′ŭs)	Styloid process of temporal bone	Tongue (lateral and inferior)	Hypoglossal	Retracts tongue
Palatoglossus (pal-ă-tō-glos′ŭs)	Soft palate	Tongue	Pharyngeal plexus	Elevates posterior tongue

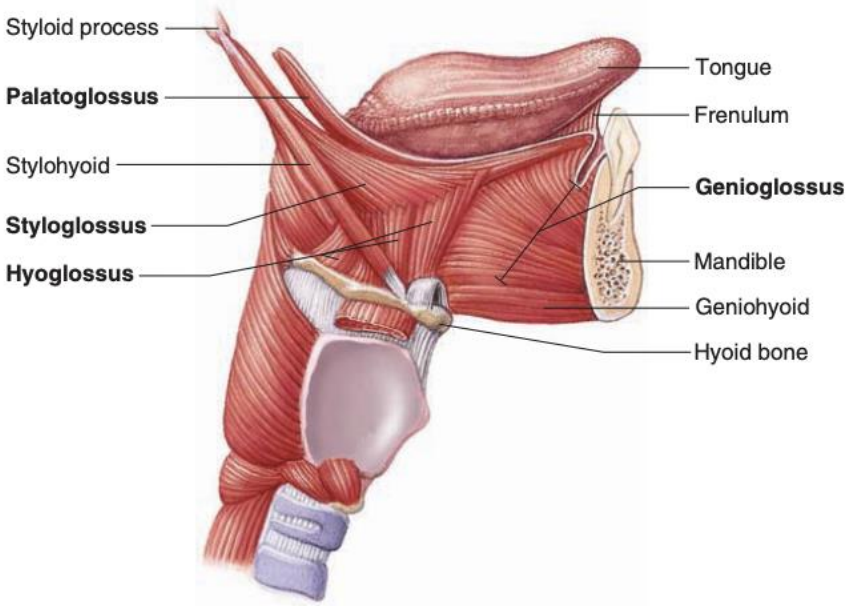


FIGURE 10.11 Tongue Muscles
Right lateral view. (Muscle names in bold are tongue muscles.)

EYE MUSCLES

The eyeball rotates within the orbit to allow vision in a wide range of directions.

Innervation:

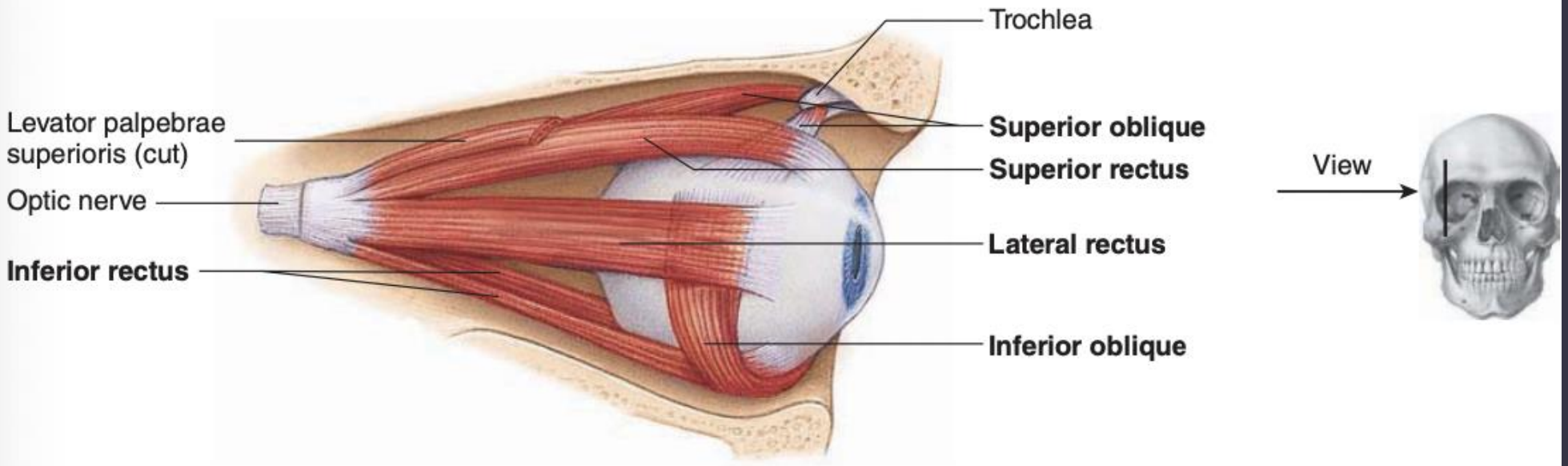
- Oculomotor nerve (CN III) – all except **lateral rectus** (CN VI) and **superior oblique** (CN IV)

TABLE 10.8 **Muscles Moving the Eye (see figure 10.13)**

Muscle	Origin	Insertion	Nerve	Action
Oblique				
Inferior	Orbital plate of maxilla	Sclera of eye	Oculomotor	Elevates and laterally deviates gaze
Superior	Common tendinous ring	Sclera of eye	Trochlear	Depresses and laterally deviates gaze
Rectus				
Inferior	Common tendinous ring	Sclera of eye	Oculomotor	Depresses and medially deviates gaze
Lateral	Common tendinous ring	Sclera of eye	Abducens	Laterally deviates gaze
Medial	Common tendinous ring	Sclera of eye	Oculomotor	Medially deviates gaze
Superior	Common tendinous ring	Sclera of eye	Oculomotor	Elevates and medially deviates gaze

TABLE 10.8 Muscles Moving the Eye (see figure 10.13)

Muscle	Origin	Insertion	Nerve	Action
Oblique				
Inferior	Orbital plate of maxilla	Sclera of eye	Oculomotor	Elevates and laterally deviates gaze
Superior	Common tendinous ring	Sclera of eye	Trochlear	Depresses and laterally deviates gaze
Rectus				
Inferior	Common tendinous ring	Sclera of eye	Oculomotor	Depresses and medially deviates gaze
Lateral	Common tendinous ring	Sclera of eye	Abducens	Laterally deviates gaze
Medial	Common tendinous ring	Sclera of eye	Oculomotor	Medially deviates gaze
Superior	Common tendinous ring	Sclera of eye	Oculomotor	Elevates and medially deviates gaze



(b) **Lateral view**

MUSCLES MOVING THE VERTEBRAL COLUMN

- **Extend, laterally flex, and rotate** the vertebral column
- **Maintain erect posture**

1. SUPERFICIAL GROUP

- connect the vertebrae to the ribs

A. Erector spinae group
B. Iliocostalis
C. Longissimus - most of the muscle mass in the lower back
D. Spinalis

2. DEEP GROUP

- attach between the spinous and transverse processes of individual vertebrae

A. Interspinales
B. Intertransversarii
C. Multifidus
D. Psoas minor
E. Rotatores

TABLE 10.9 Muscles Acting on the Vertebral Column (see figures 10.5, 10.6, 10.14, and 10.15)

Muscle	Origin	Insertion	Nerve	Action
Superficial				
Erector spinae (ē-rek′tŏr, ē-rek′tōr spī′nē; divides into three columns)				
Iliocostalis (il′ē-ō-kos-tā′lis)	Sacrum, ilium, and lumbar spines	Ribs and vertebrae	Dorsal rami of spinal nerves	Extends vertebral column
Cervicis (ser-vī′sis)	Superior six ribs	Transverse processes of middle cervical vertebrae	Dorsal rami of thoracic nerves	Extends, laterally flexes, and rotates vertebral column
Thoracis (thō-ra′sis)	Inferior six ribs	Superior six ribs	Dorsal rami of thoracic nerves	Extends, laterally flexes, and rotates vertebral column
Lumborum (lum-bōr′ŭm)	Sacrum, ilium, and lumbar vertebrae	Inferior six ribs	Dorsal rami of thoracic and lumbar nerves	Extends, laterally flexes, and rotates vertebral column
Longissimus (lon-gis′i-mŭs)				
Capitis (ka′pī-tis)	Upper thoracic and lower cervical vertebrae	Mastoid process	Dorsal rami of cervical nerves	Extends head
Cervicis (ser-vī′sis)	Upper thoracic vertebrae	Transverse processes of upper cervical vertebrae	Dorsal rami of cervical nerves	Extends neck
Thoracis (thō-ra′sis)	Ribs and lower thoracic vertebrae	Transverse processes of upper lumbar vertebrae and ribs	Dorsal rami of thoracic and lumbar nerves	Extends vertebral column
Spinalis (spī-nā′lis)				
Cervicis (ser-vī′sis; not illustrated)	C6–C7	Spinous processes of C2–C3	Dorsal rami of cervical nerves	Extends neck
Thoracis (thō-ra′sis)	T11–L2	Spinous processes of middle and upper thoracic vertebrae	Dorsal rami of thoracic nerves	Extends vertebral column
Semispinalis (sem′ē-spī-nā′lis)				
Cervicis (ser-vī′sis)	Transverse processes of T2–T5	Spinous processes of C2–C5	Dorsal rami of cervical nerves	Extends neck
Thoracis (thō-ra′sis)	Transverse processes of T5–T11	Spinous processes of C5–T4	Dorsal rami of thoracic nerves	Extends vertebral column
Splenius cervicis (splē′nē-ŭs ser-vī′sis)	Spinous processes of C3–C5	Transverse processes of C1–C3	Dorsal rami of cervical nerves	Rotates and extends neck
Longus colli (lon′gŭs kō′lī; not illustrated)	Bodies of C3–T3	Bodies of C1–C6	Ventral rami of cervical nerves	Flexes neck

Muscle

Superficial

Erector spinae
(ē-rek′tŏr, ē-rek′tōr spī′nē; divides into three columns)

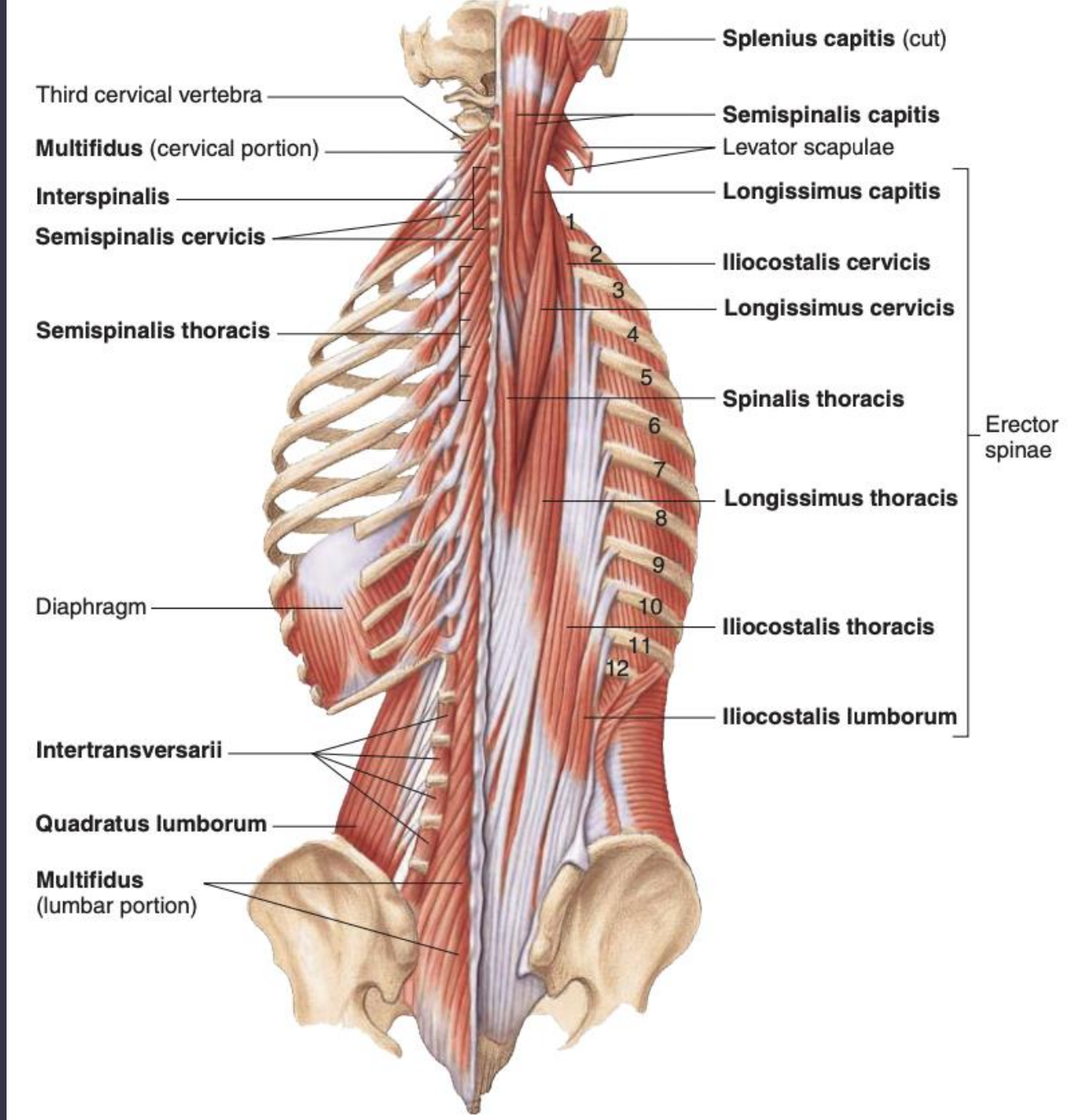
- Iliocostalis (il′ē-ō-kos-tā′lis)
  - Cervicis (ser-vī′sis)
  - Thoracis (thō-ra′sis)
  - Lumborum (lum-bōr′ŭm)
- Longissimus (lon-gis′i-mŭs)
  - Capitis (ka′pĭ-tis)
  - Cervicis (ser-vī′sis)
  - Thoracis (thō-ra′sis)
- Spinalis (spī-nā′lis)
  - Cervicis (ser-vī′sis; not illustrated)
  - Thoracis (thō-ra′sis)

Semispinalis (sem′ē-spī-nā′li

- Cervicis (ser-vī′sis)
- Thoracis (thō-ra′sis)

Splenius cervicis (splē′nē-ŭs ser-vī′sis)

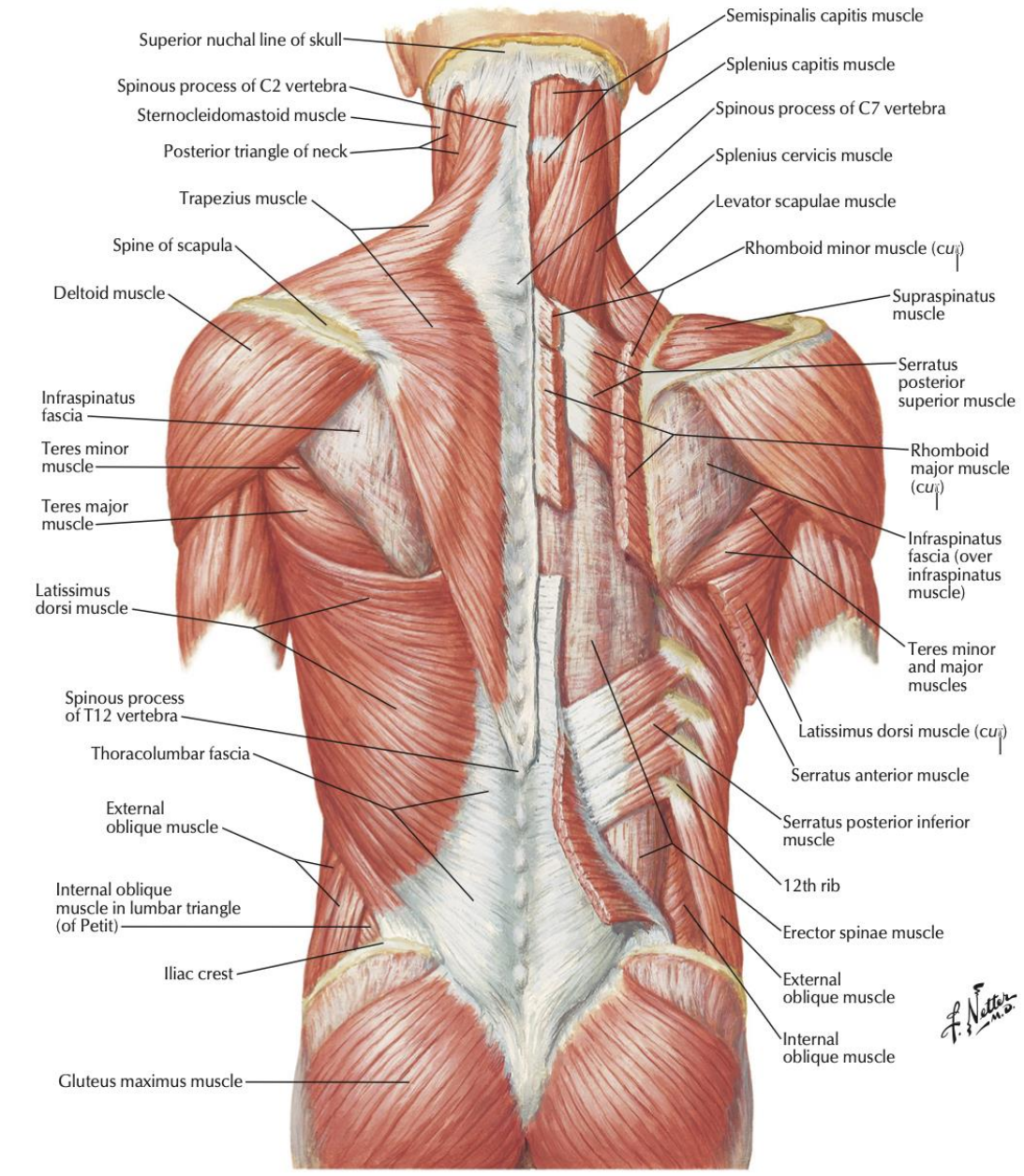
Longus colli (lon′gŭs kō′lī; not illustrated)



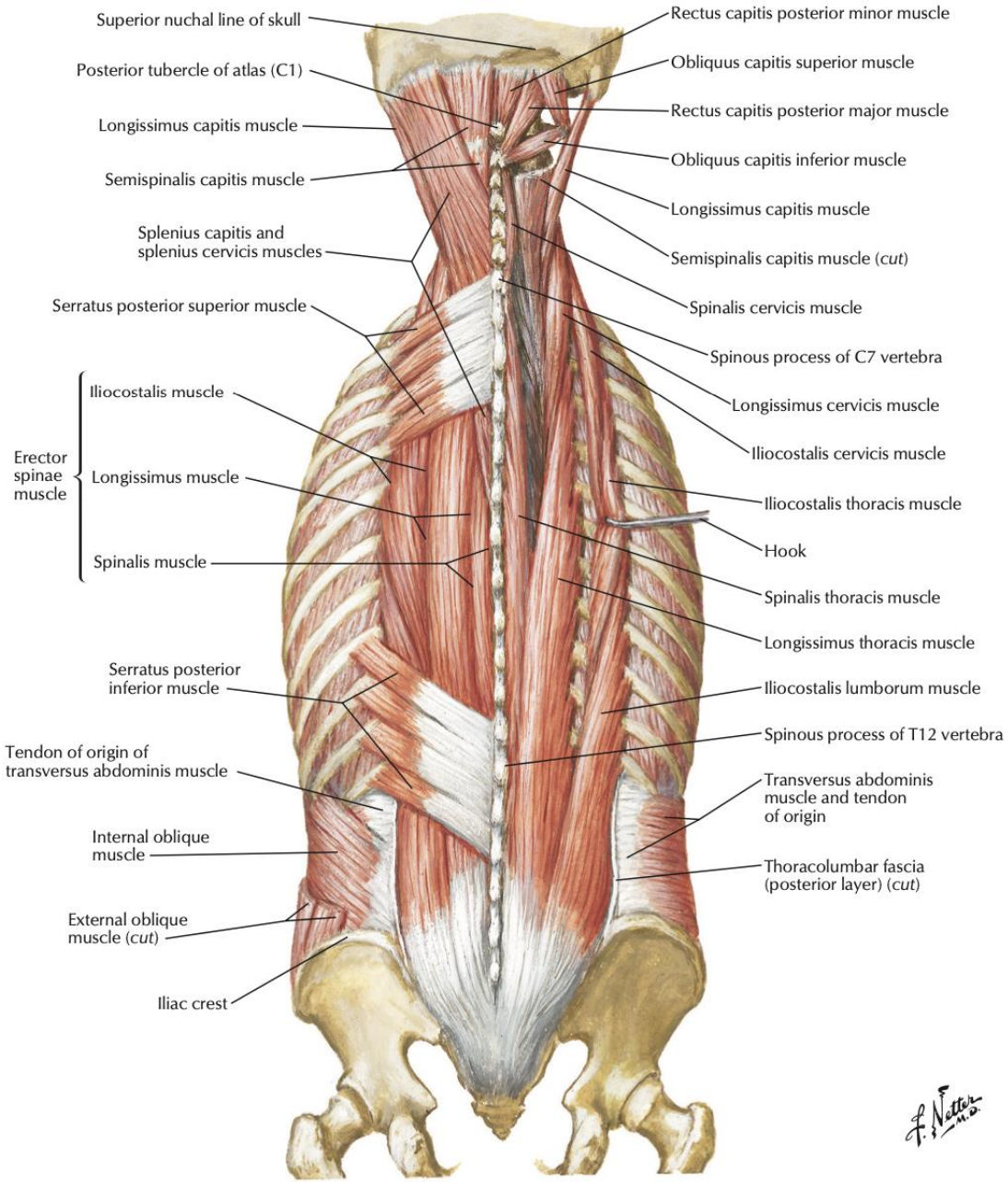
Posterior view

Muscles of Back: Superficial Layers

See also Plates 29, 250



Muscles of Back: Intermediate Layers



Muscles of Back: Deep Layers

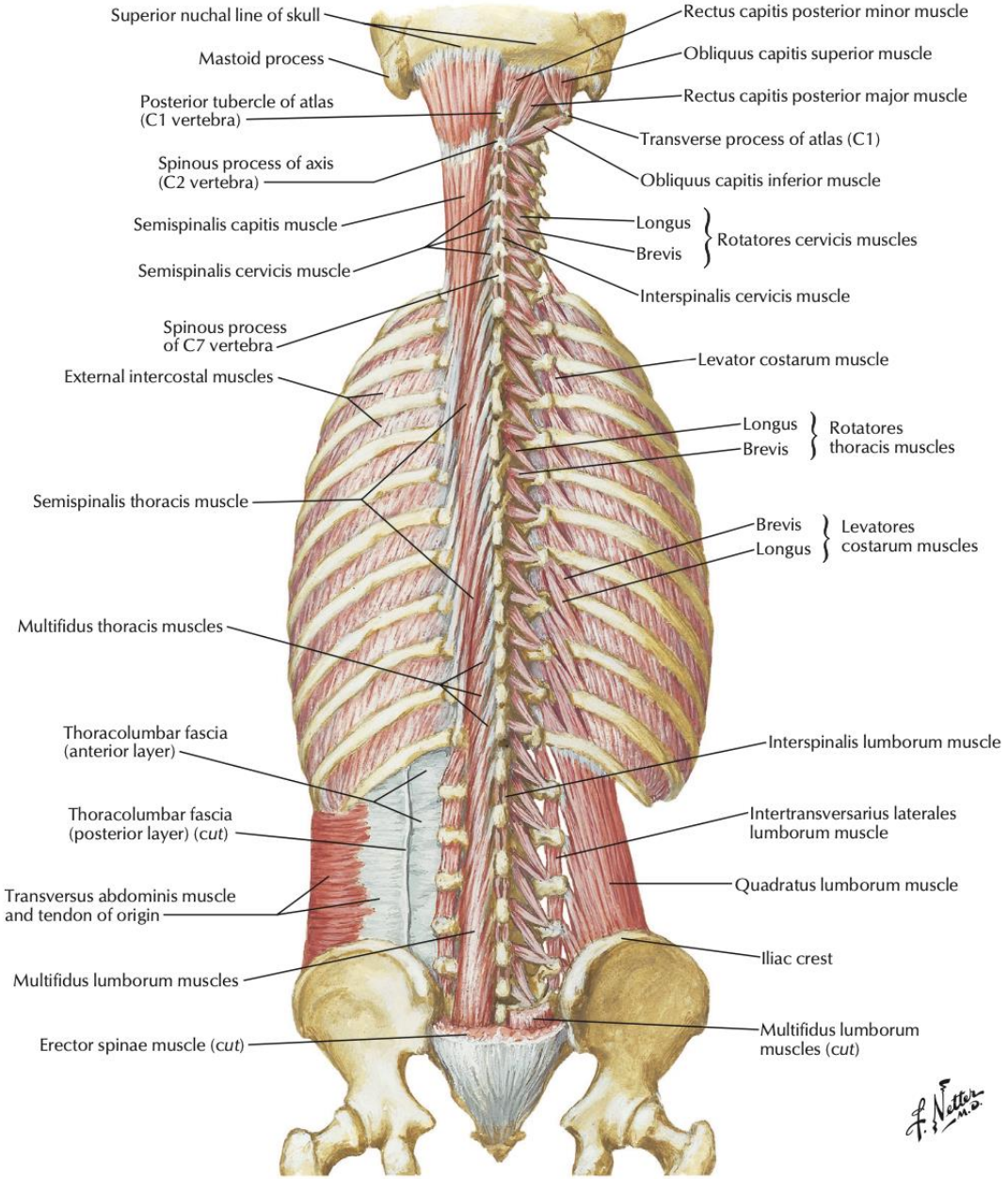


TABLE 10.9 **Muscles Acting on the Vertebral Column (see figures 10.5, 10.6, 10.14, and 10.15)**

Muscle	Origin	Insertion	Nerve	Action
Deep				
Interspinales (in-ter-spī-nā′lēz)	Spinous processes of all vertebrae	Next superior spinous process	Dorsal rami of spinal nerves	Extends back and neck
Intertransversarii (in-ter-trans′ver-săr′ē-ī)	Transverse processes of all vertebrae	Next superior transverse process	Dorsal rami of spinal nerves	Laterally flexes vertebral column
Multifidus (mŭl-tif′i-dŭs)	Transverse processes of vertebrae; posterior surface of sacrum and ilium	Spinous processes of superior vertebrae	Dorsal rami of spinal nerves	Extends and rotates vertebral column
Psoas minor (sō′as mī′ner)	T12–L1	Pectineal line near pubic crest	L1	Flexes vertebral column
Rotatores (rō-tā′tōrz)	Transverse processes of all vertebrae	Base of spinous process of superior vertebrae	Dorsal rami of spinal nerves	Extends and rotates vertebral column

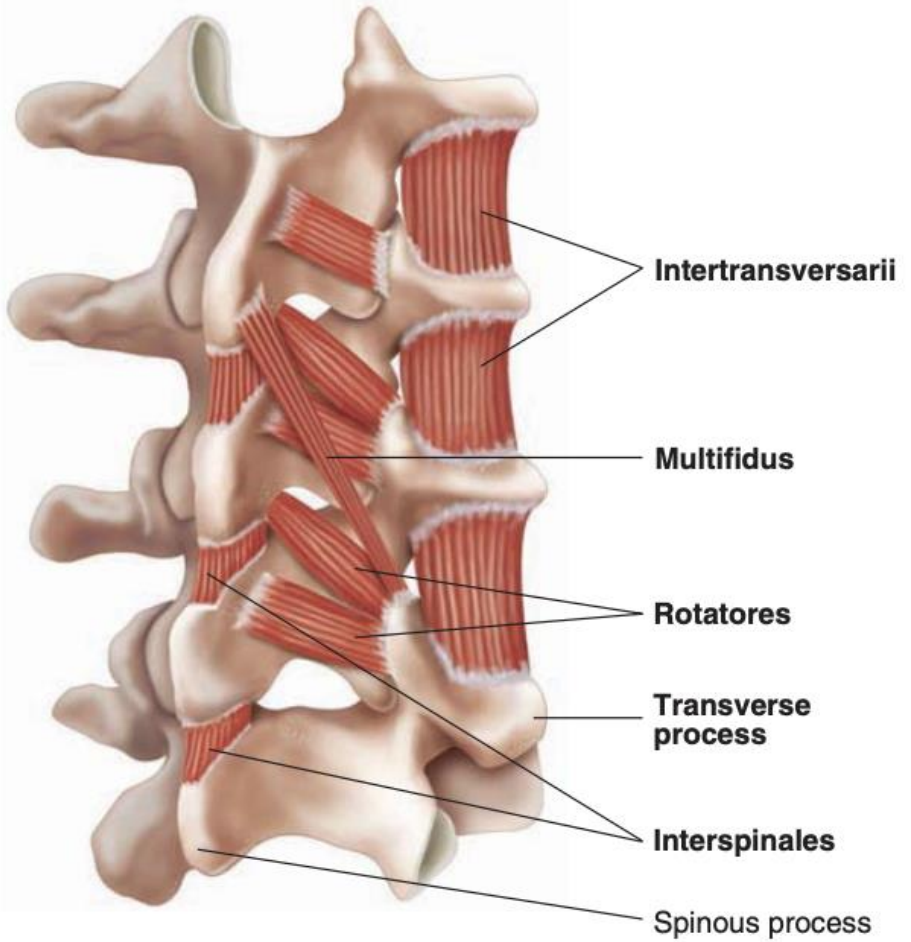


FIGURE 10.15 Deep Muscles Associated with the Vertebrae

Deep

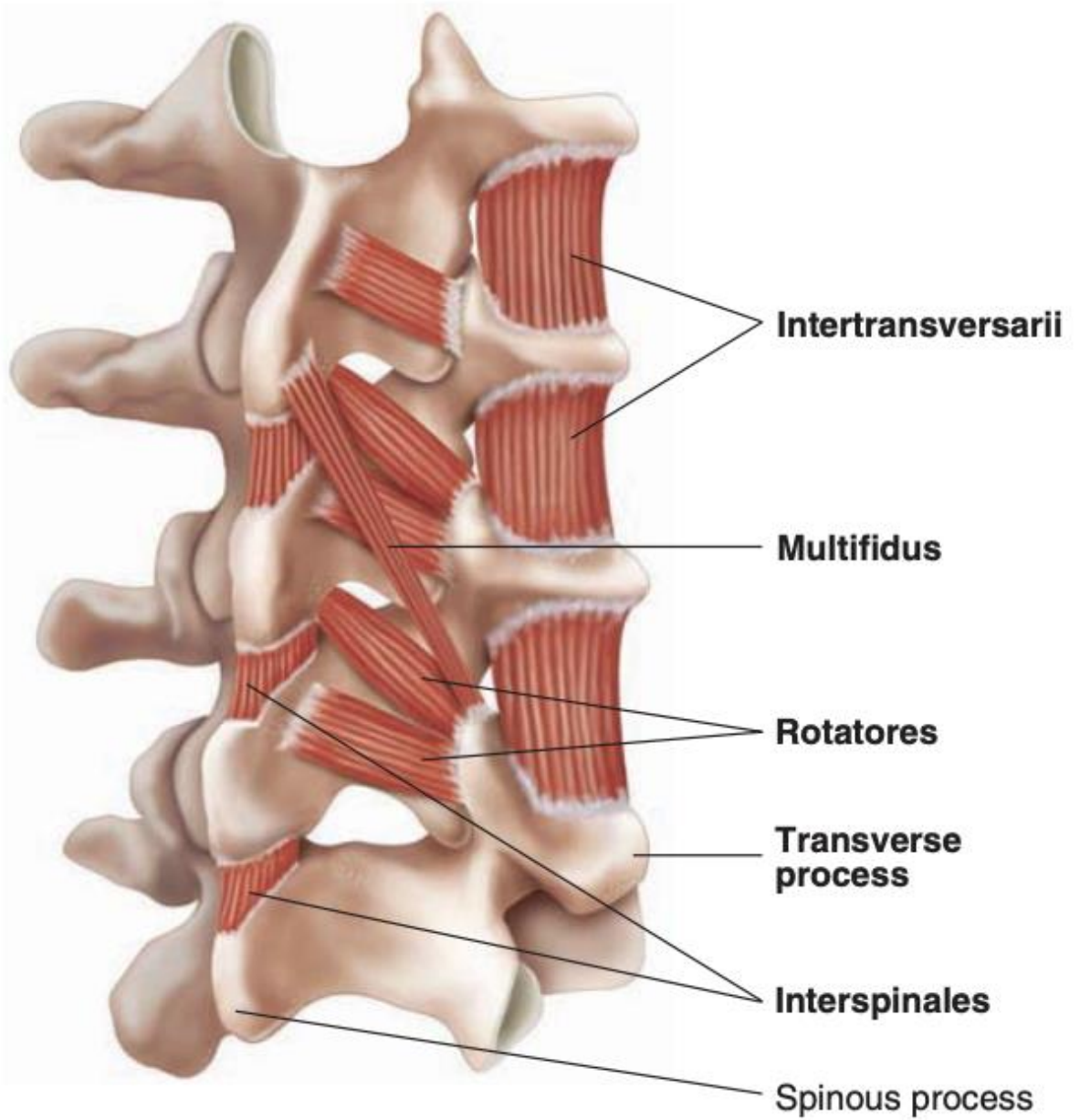
Interspinales
(in-ter-spī-nā′lēz)

Intertransversarii
(in-ter-trans′ver-săr′ē-ī)

Multifidus
(mŭl-tif′i-dŭs)

Psoas minor (sō′as mī′ner)

Rotatores (rō-tā′tōrz)



Posterolateral view

FIGURE 10.15 **Deep Muscles Associated with the Vertebrae**

THORACIC MUSCLES

Thoracic Muscles – Role in Breathing

- **Primary function:** assist in **respiration** (inhalation and exhalation)

1. Diaphragm

- Dome-shaped muscle separating thoracic and abdominal cavities
- **Primary muscle for quiet, normal breathing**
- Controlled by the **phrenic nerve (C3–C5)**

2. Intercostal Muscles

- Located between the ribs
- **External intercostals:** elevate ribs → inspiration
- **Internal intercostals:** depress ribs → forced expiration

3. Scalenus Group

- Elevates the **first two ribs** during **forceful inhalation**

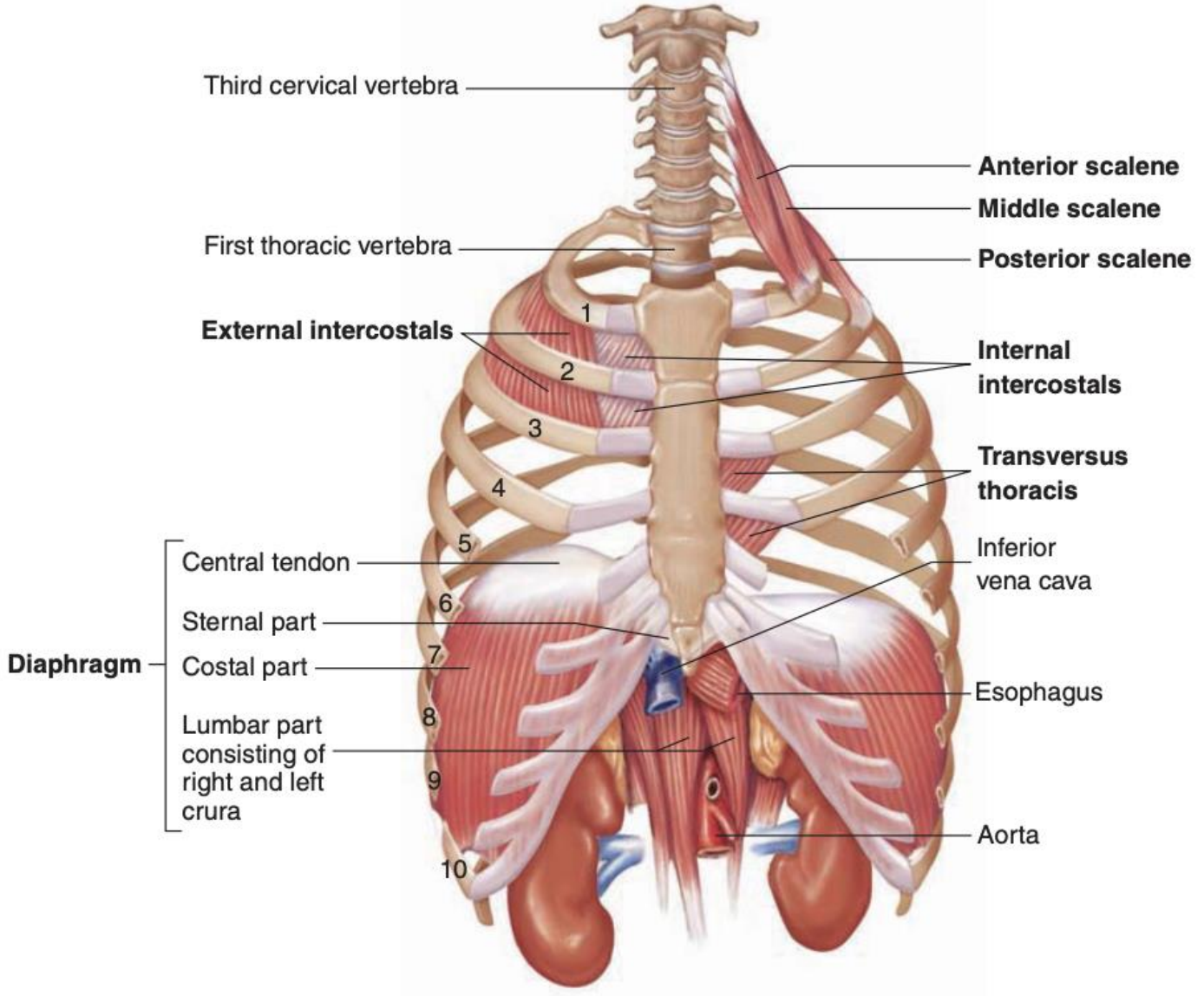
4. Serratus Posterior Group

- **Superior portion:** elevates ribs during inspiration
- **Inferior portion:** depresses ribs during forced expiration

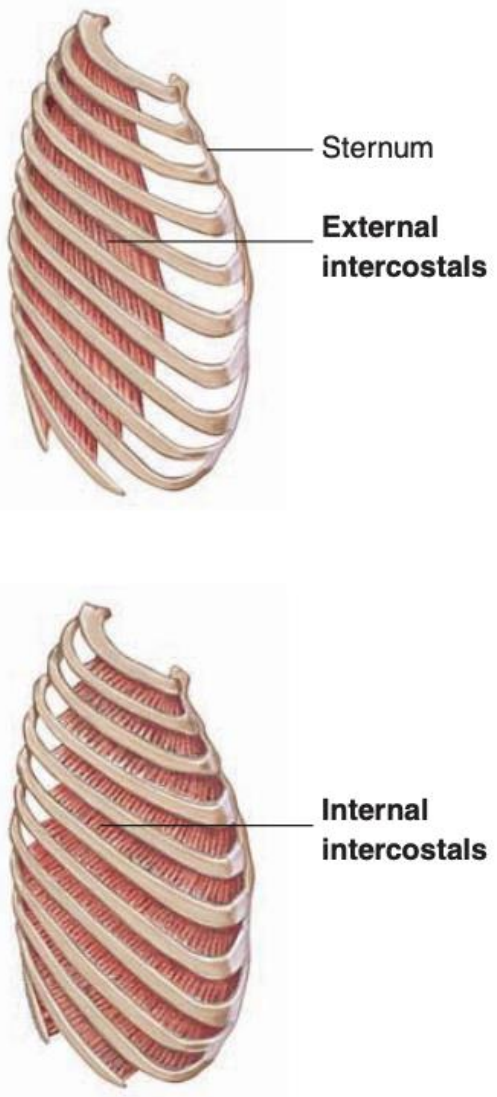
5. Transversus Thoracis

- Weakly depresses ribs, assists in **expiration**

Muscle
Diaphragm
Intercostalis (in′ter-kos-ta′lis)
External
Internal
Scalenus (skā-lē′nŭs)
Anterior
Medial
Posterior
Serratus posterior (sĕr-ā′tŭs)
Inferior (not illustrated)
Superior (not illustrated)
Transversus thoracis (trans-ver′sus thō-ra′sis; not illustrated)



(a) **Anterior view**



(b) **Lateral view**

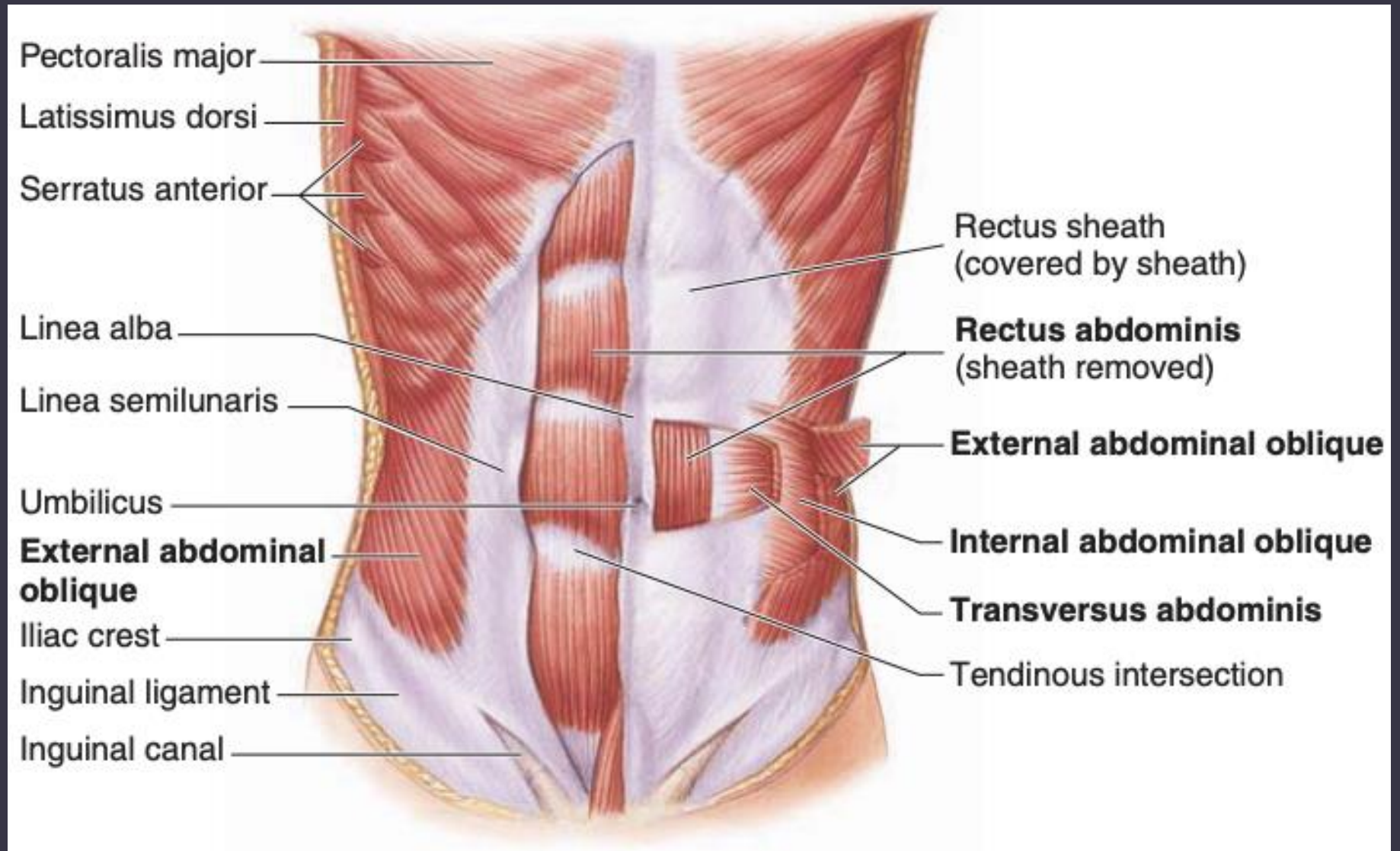
TABLE 10.10 Muscles of the Thorax (see figure 10.16)

Muscle	Origin	Insertion	Nerve	Action
Diaphragm	Interior of ribs, sternum, and lumbar vertebrae	Central tendon of diaphragm	Phrenic	Inhalation depresses floor of thorax
Intercostalis (in′ter-kos-ta′lis)				
External	Inferior margin of each rib	Superior border of next rib below	Intercostal	Quiet inhalation elevates ribs
Internal	Superior margin of each rib	Inferior border of next rib above	Intercostal	Forced exhalation depresses ribs
Scalenus (skā-lē′nŭs)				
Anterior	Transverse processes of C3–C6	First rib	Cervical plexus	Elevates first rib
Medial	Transverse processes of C2–C6	First rib	Cervical plexus	Elevates first rib
Posterior	Transverse processes of C4–C6	Second rib	Cervical and brachial plexuses	Elevates second rib
Serratus posterior (sĕr-ā′tŭs)				
Inferior (not illustrated)	Spinous processes of T11–L2	Inferior four ribs	Ninth to eleventh intercostals and subcostal	Depresses inferior ribs and extends vertebral column
Superior (not illustrated)	Spinous processes of C6–T2	Second to fifth ribs	First to fourth intercostals	Elevates superior ribs
Transversus thoracis (trans-ver′sus thō-ra′sis; not illustrated)	Sternum and xiphoid process	Second to sixth costal cartilages	Intercostal	Decreases diameter of thorax

ABDOMINAL WALL MUSCLES

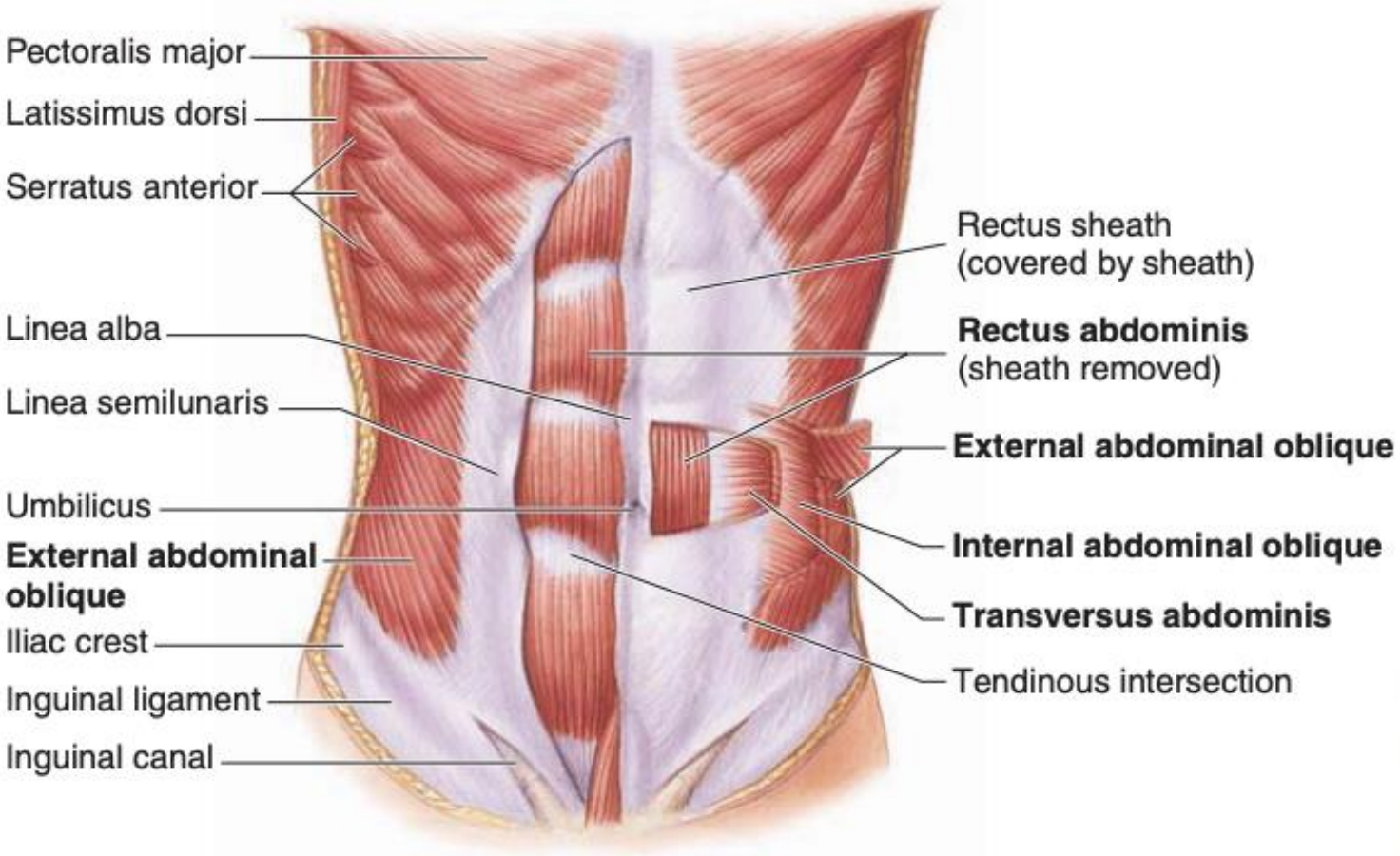
- **Functions:**
  - Flex and rotate the vertebral column
  - When spine is stationary: decrease **abdominal and thoracic cavity volume**
  - Assist in **forced exhalation, vomiting, defecation, coughing, and childbirth**
- **Structure:**
  - **Crisscross layering** of muscles forms a strong anterior wall
  - Provides **protection and support** for abdominal viscera
  - Unlike the back, **anterior wall is not supported by bone**

Muscle
Anterior
Rectus abdominis (rek′tŭs ab-dom′i-nis)
External abdominal oblique
Internal abdominal oblique
Transversus abdominis (trans-ver′sŭs ab-dom′i-nis)
Posterior
Quadratus lumborum (kwah-drā′tŭs lŭm-bōr′ŭm)

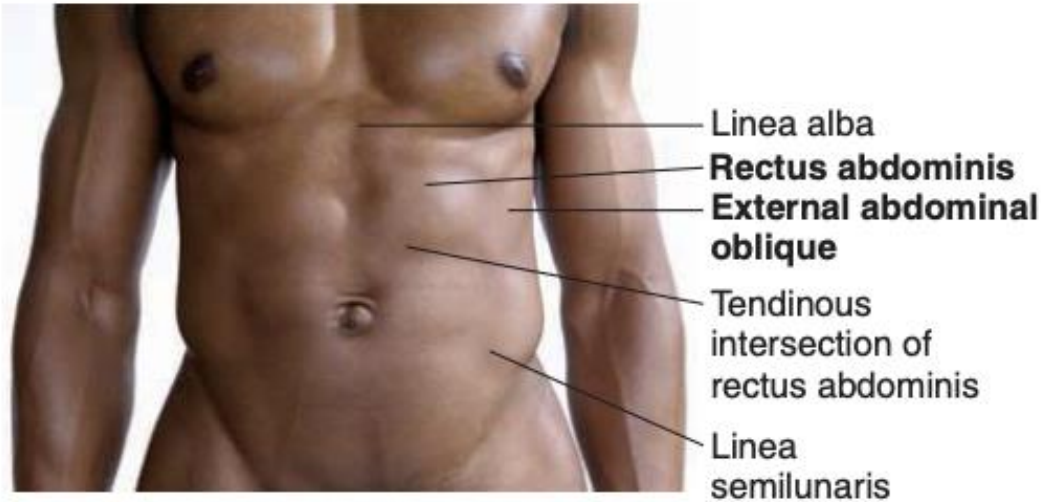


(a) **Anterior view**

FIGURE 10.17 **Anterior Abdominal Wall Muscles**



(a) **Anterior view**



(b) **Anterior view**

FIGURE 10.17 Anterior Abdominal Wall Muscles

(*a*) Windows in the side reveal the various muscle layers. (*b*) Surface anatomy of anterior abdominal muscle. (Muscles of the abdominal wall are in bold.)

ABDOMINAL WALL MUSCLES

Posterior:

Quadratus lumborum

- **Action:**
  - **Lateral flexion** of vertebral column
  - **Depresses 12th rib** during forced exhalation
  - Helps **stabilize the lumbar spine and pelvis**

Muscles of Back: Deep Layers

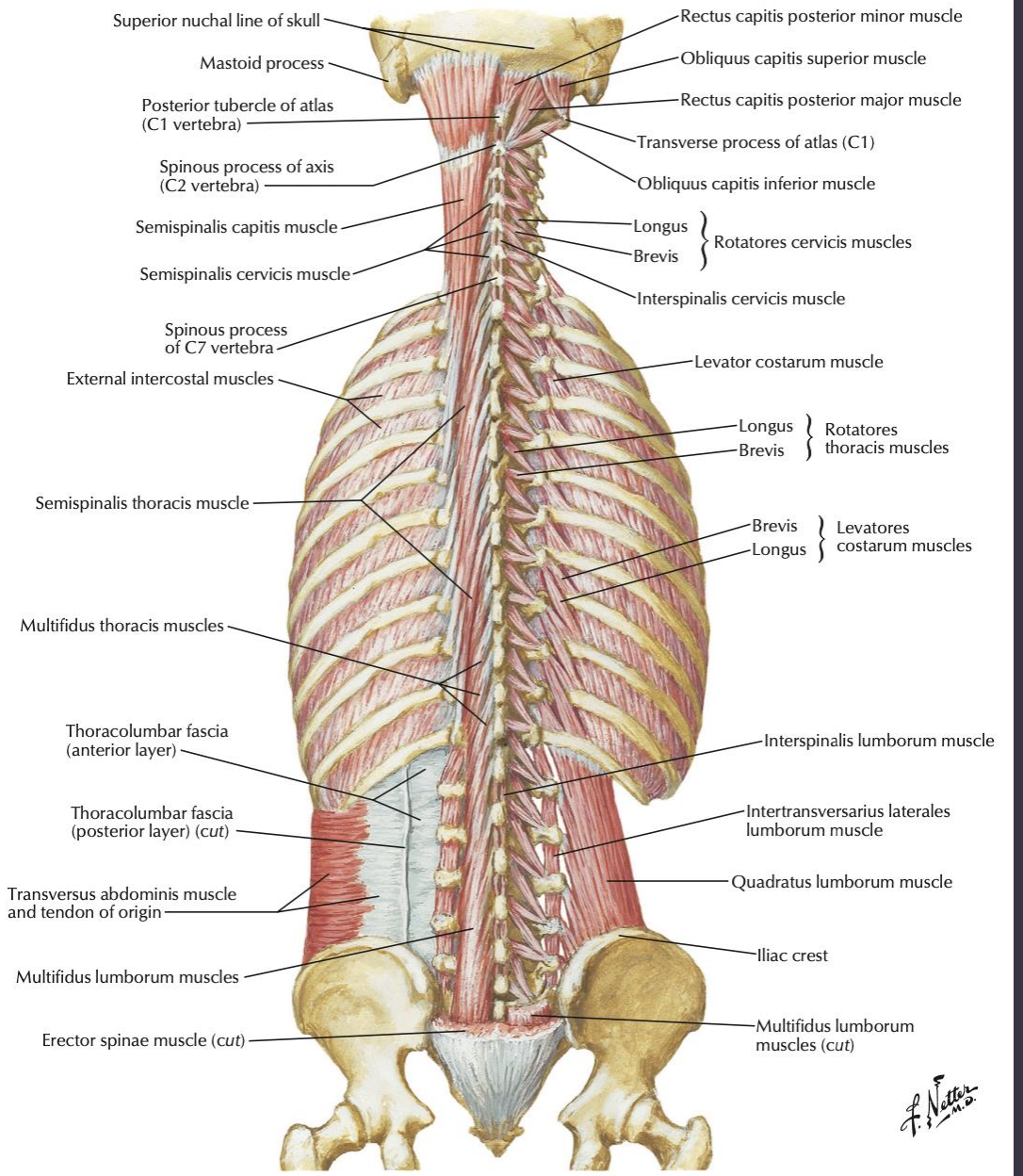
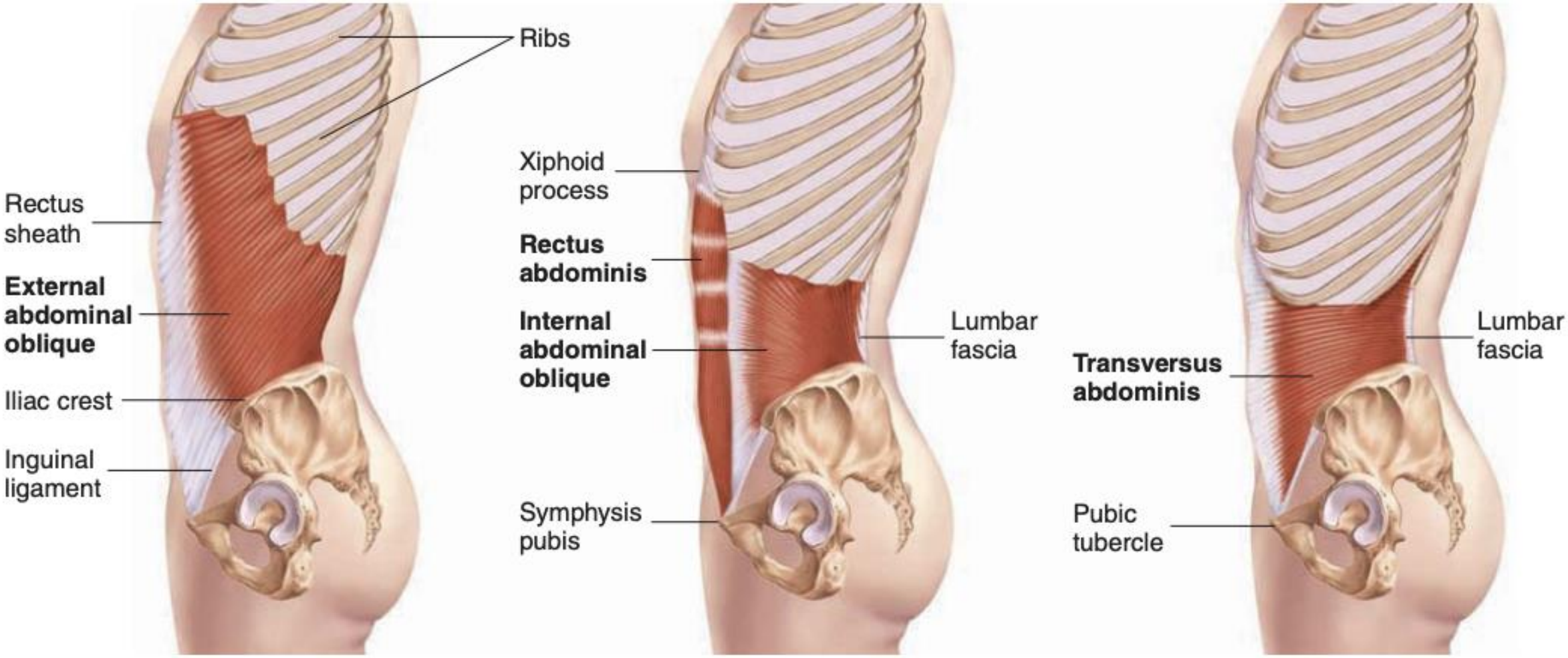


TABLE 10.11 Muscles of the Abdominal Wall (see figures 10.3*a*, 10.17, and 10.18)

Muscle	Origin	Insertion	Nerve	Action
Anterior				
Rectus abdominis (rek′tŭs ab-dom′i-nis)	Pubic crest and symphysis pubis	Xiphoid process and inferior ribs	Branches of lower thoracic	Flexes vertebral column; compresses abdomen
External abdominal oblique	Fifth to twelfth ribs	Iliac crest, inguinal ligament, and rectus sheath	Branches of lower thoracic	Flexes and rotates vertebral column; compresses abdomen; depresses thorax
Internal abdominal oblique	Iliac crest, inguinal ligament, and lumbar fascia	Tenth to twelfth ribs and rectus sheath	Lower thoracic	Flexes and rotates vertebral column; compresses abdomen; depresses thorax
Transversus abdominis (trans-ver′sŭs ab-dom′i-nis)	Seventh to twelfth costal cartilages, lumbar fascia, iliac crest, and inguinal ligament	Xiphoid process, linea alba, and pubic tubercle	Lower thoracic	Compresses abdomen
Posterior				
Quadratus lumborum (kwah-drā′tŭs lŭm-bōr′ŭm)	Iliac crest and lower lumbar vertebrae	Twelfth rib and upper lumbar vertebrae	Upper lumbar	Laterally flexes vertebral column and depresses twelfth rib



(b) **Lateral views**

FIGURE 10.18 Anterior Abdominal Wall Muscles

(*a*) Cross section superior to the umbilicus. (*b*) Abdominal muscles shown individually. (Muscle names are in bold.) AP|R

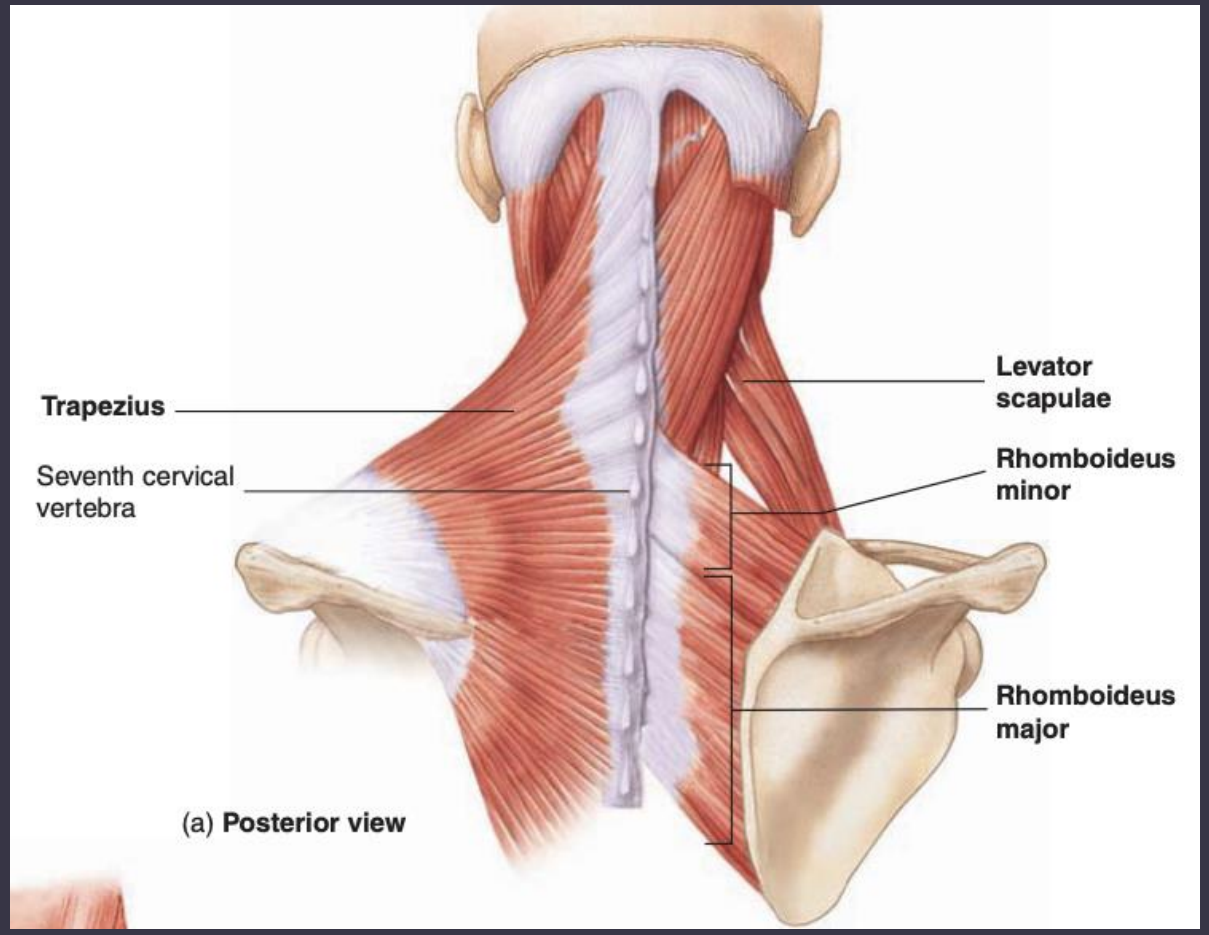
UPPER LIMB MUSCLES

SCAPULAR MOVEMENTS

Function:

- Attach **scapula to thorax**
- Move scapula for **upper limb mobility**
- Act as **fixators** to stabilize scapula during arm movements

1. **trapezius**
2. **levator scapulae**
3. **rhomboideus major**
4. **rhomboideus minor**
5. **serratus anterior,**
6. **pectoralis minor**



UPPER LIMB MUSCLES

SCAPULAR MOVEMENTS

1. trapezius
2. levator scapulae
3. rhomboideus major
4. rhomboideus minor
5. serratus anterior,
6. pectoralis minor

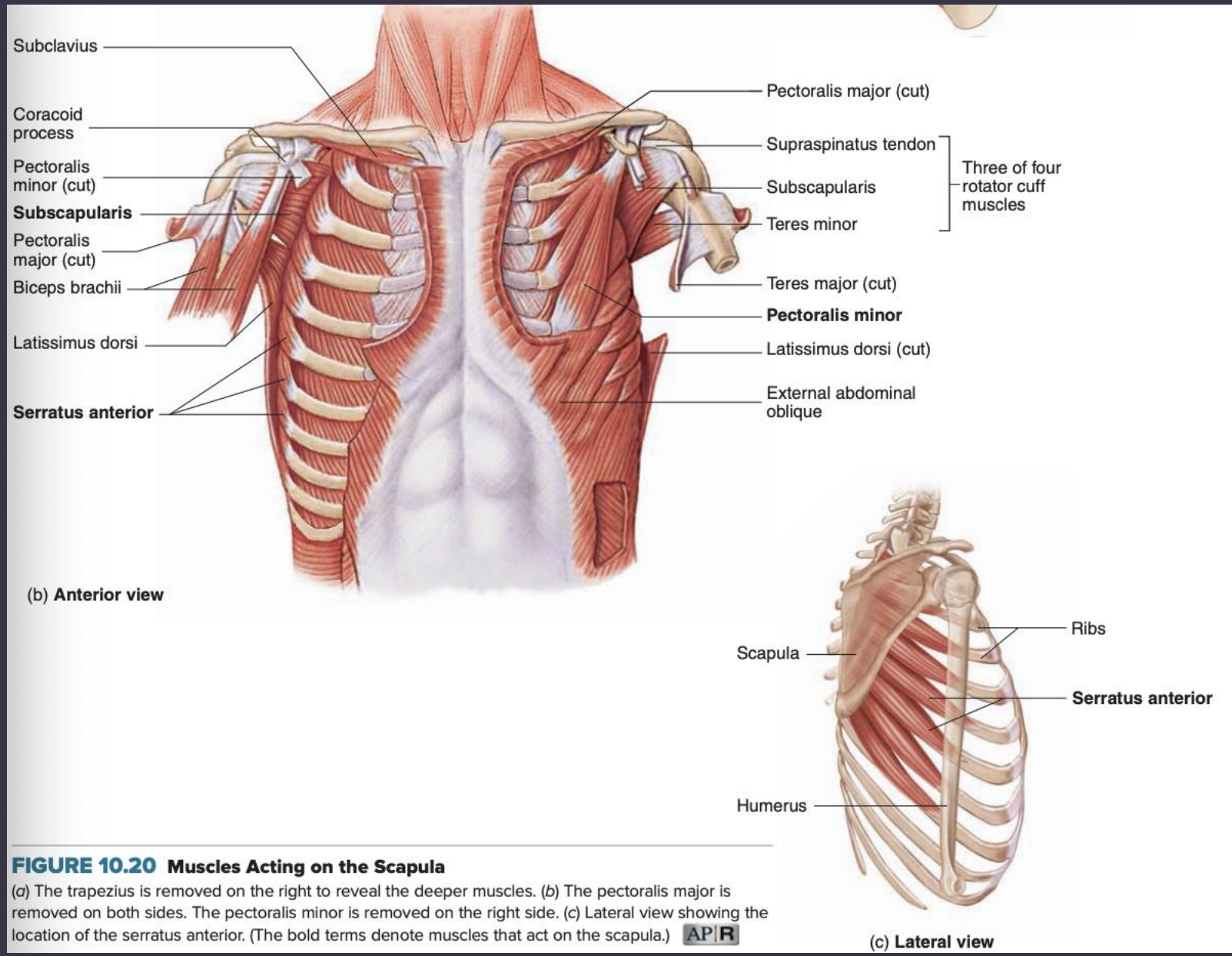


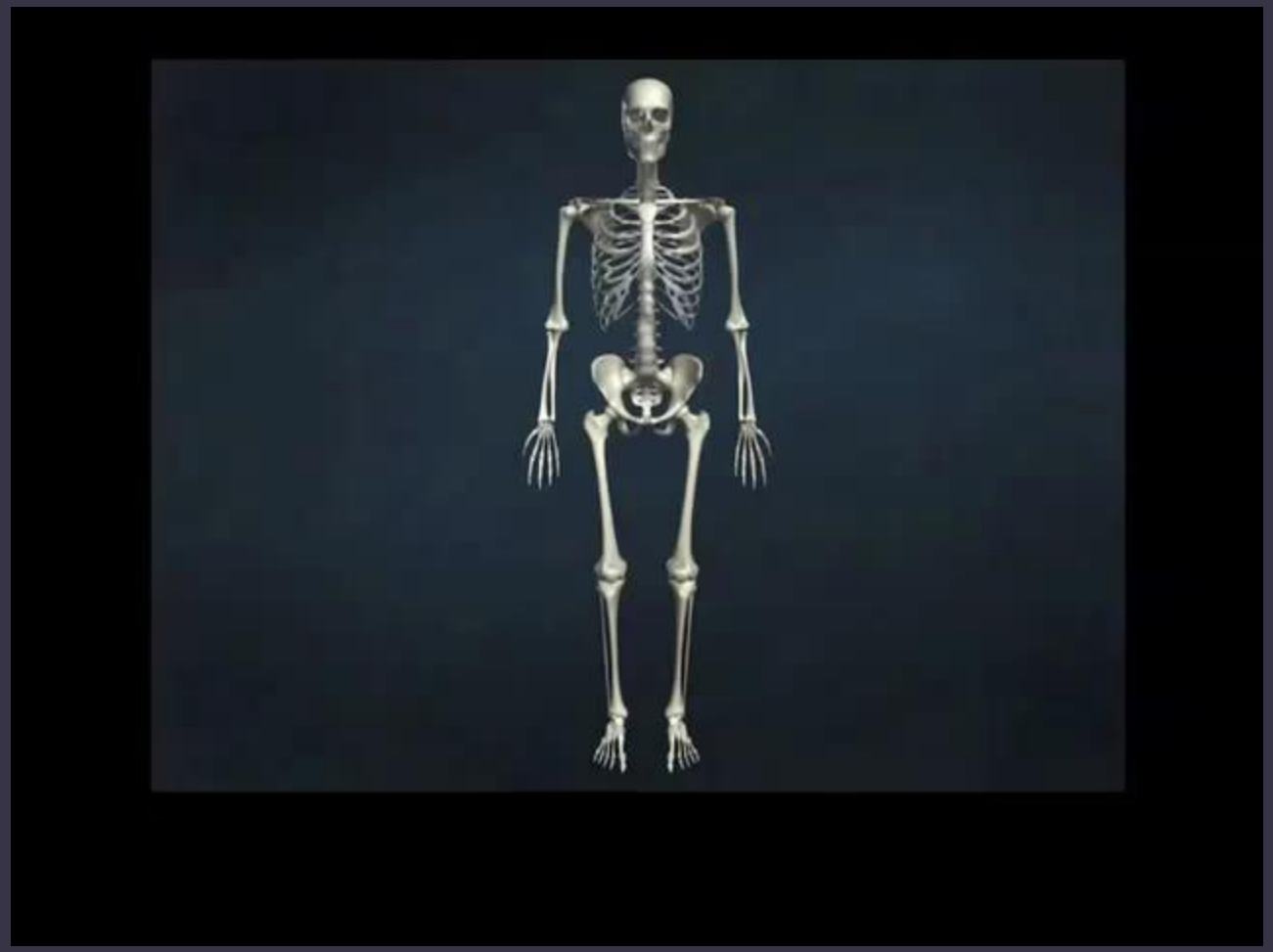
FIGURE 10.20 Muscles Acting on the Scapula
(*a*) The trapezius is removed on the right to reveal the deeper muscles. (*b*) The pectoralis major is removed on both sides. The pectoralis minor is removed on the right side. (*c*) Lateral view showing the location of the serratus anterior. (The bold terms denote muscles that act on the scapula.) AP|R

MUSCLES ACTING ON THE SCAPULA

TABLE 10.13 **Muscles Acting on the Scapula (see figure 10.20)**

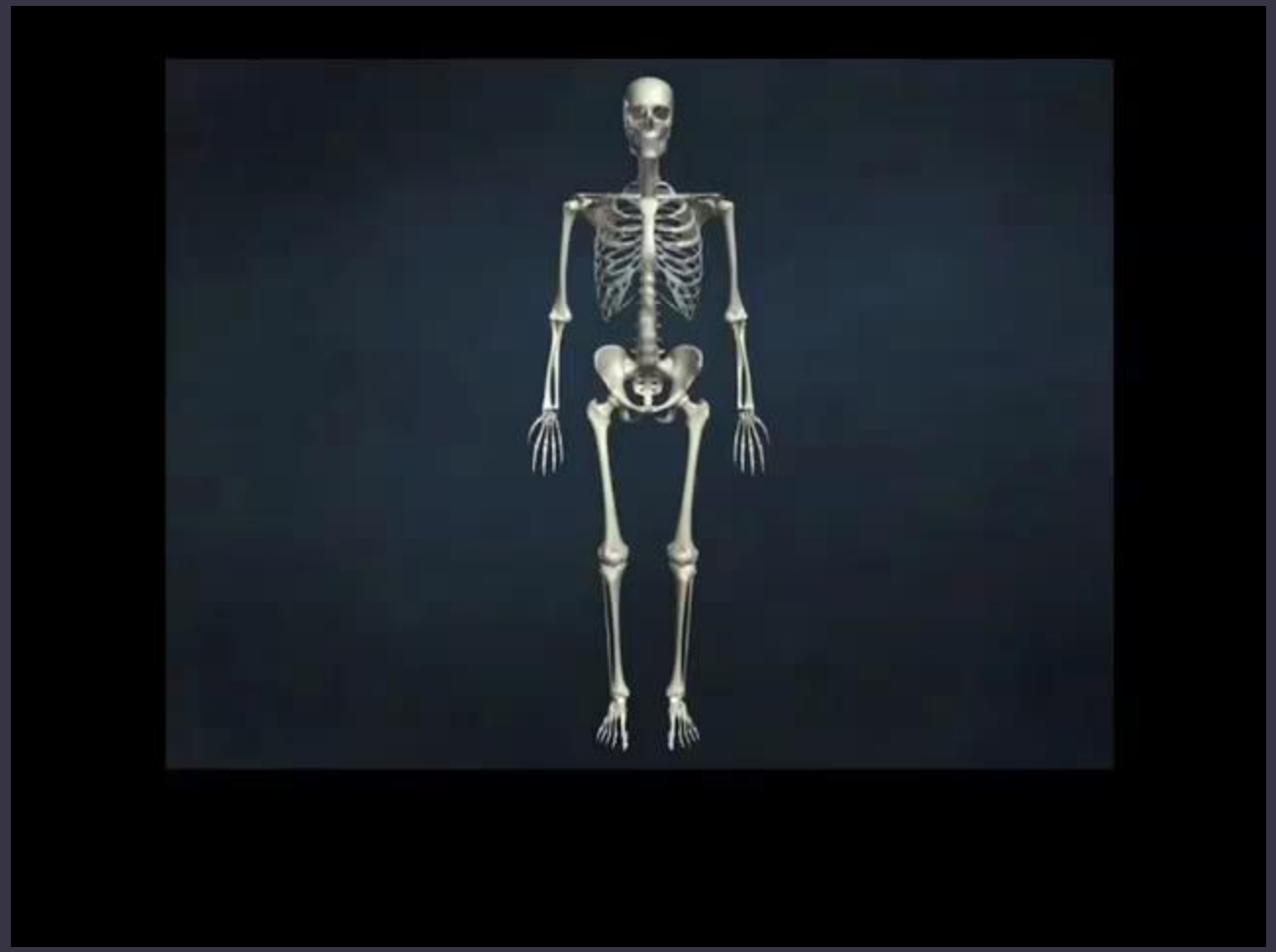
Muscle	Origin	Insertion	Nerve	Action
Levator scapulae (lē-vā′tor, le-vā′ter skap′ū-lē)	Transverse processes of C1–C4	Superior angle of scapula	Dorsal scapular	Elevates, retracts, and rotates scapula; laterally flexes neck
Pectoralis minor (pek′tō-ra′lis)	Third to fifth ribs	Coracoid process of scapula	Medial pectoral	Depresses scapula or elevates ribs
Rhomboideus (rom-bō-id′ē-ŭs)				
Major	Spinous processes of T1–T4	Medial border of scapula	Dorsal scapular	Retracts, rotates, and fixes scapula
Minor	Spinous processes of C6–C7	Medial border of scapula	Dorsal scapular	Retracts, slightly elevates, rotates, and fixes scapula
Serratus anterior (ser-ā′tŭs)	First to eighth or ninth ribs	Medial border of scapula	Long thoracic	Rotates and protracts scapula; elevates ribs
Subclavius (sŭb-klā′vē-ŭs)	First rib	Clavicle	Subclavian	Fixes clavicle or elevates first rib
Trapezius (tra-pē′zē-ŭs)	External occipital protuberance, ligamentum nuchae, and spinous processes of C7–T12	Clavicle, acromion process, and scapular spine	Accessory and cervical plexus	Elevates, depresses, retracts, rotates, and fixes scapula; extends neck

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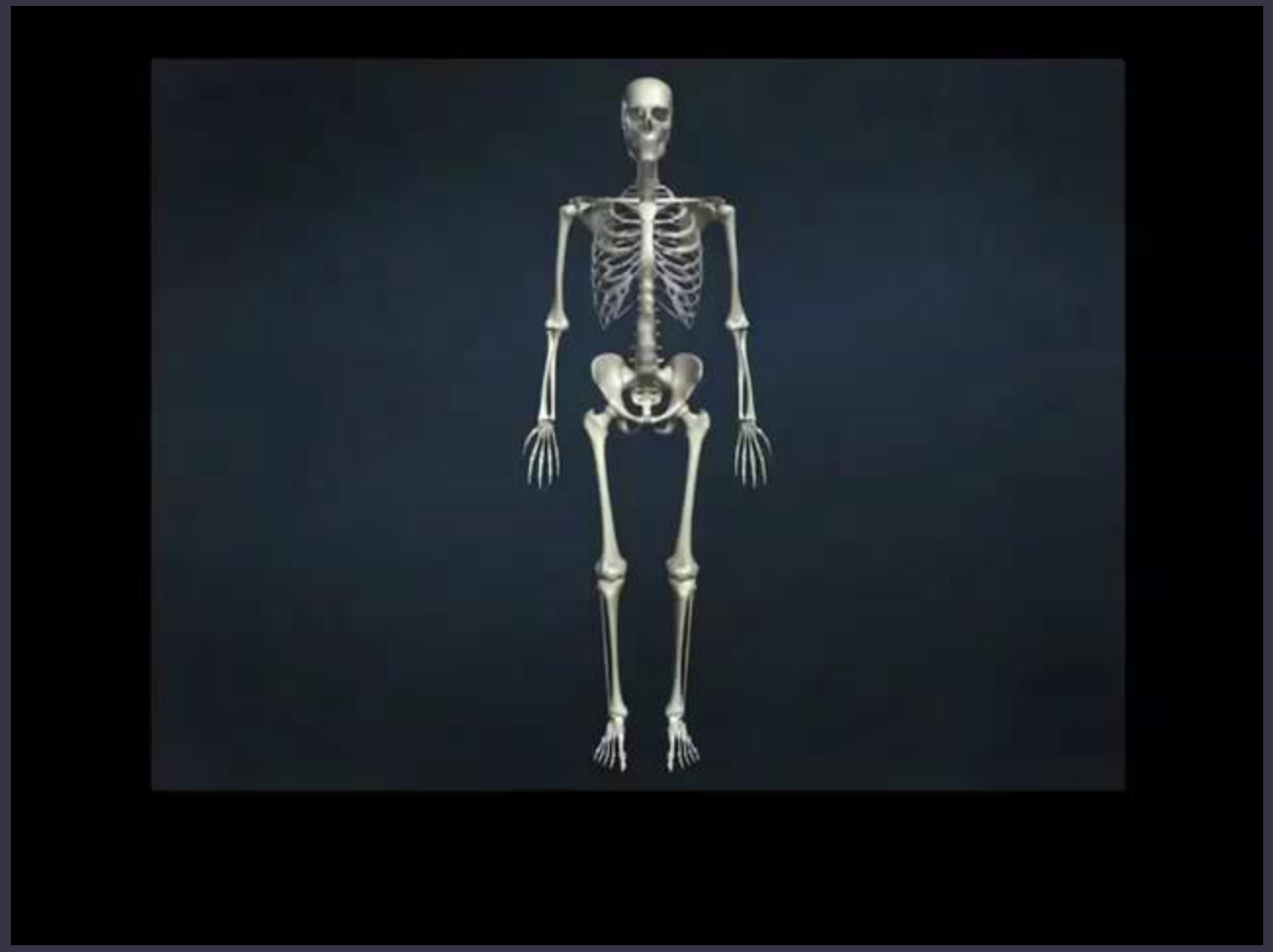
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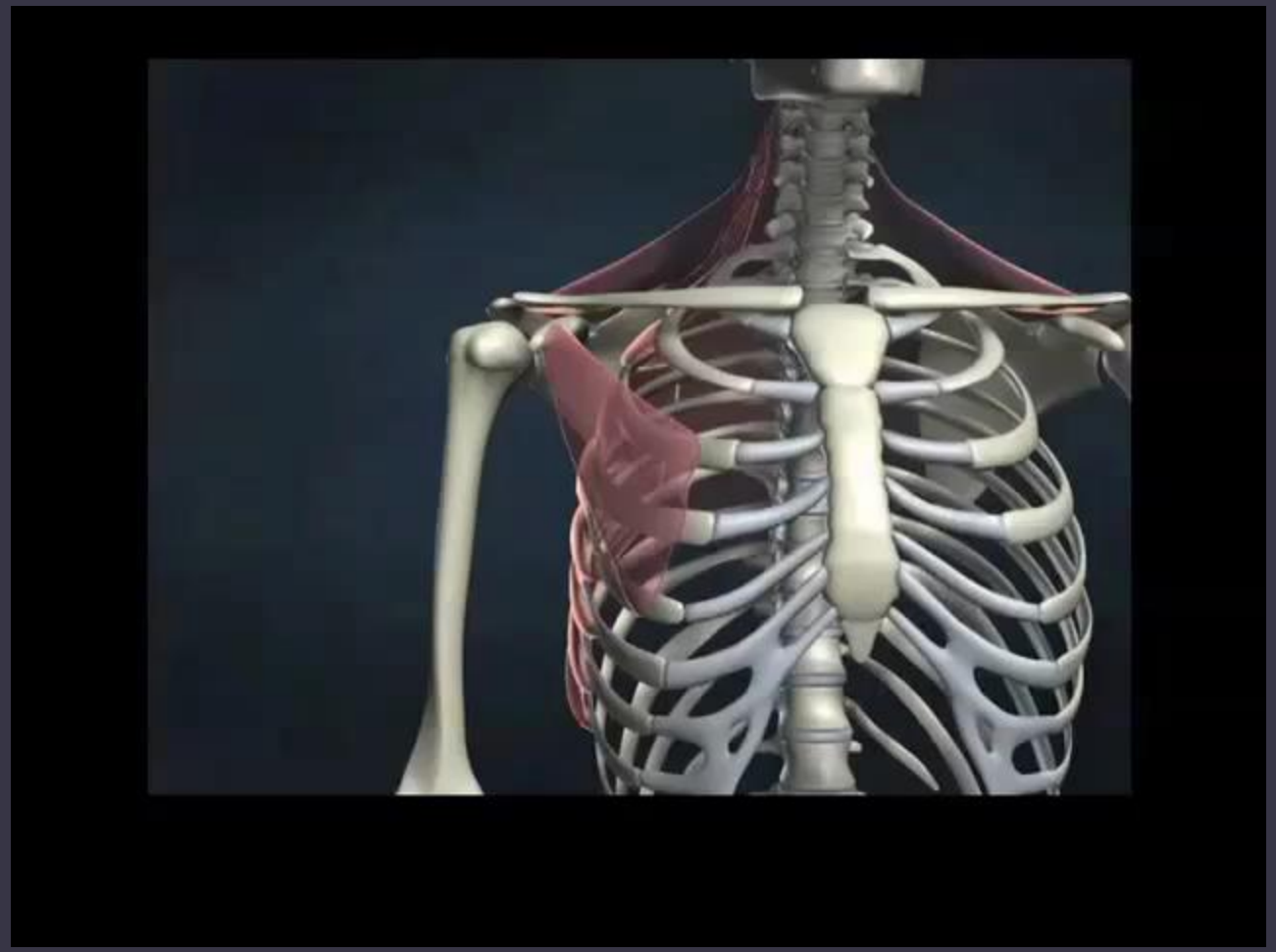
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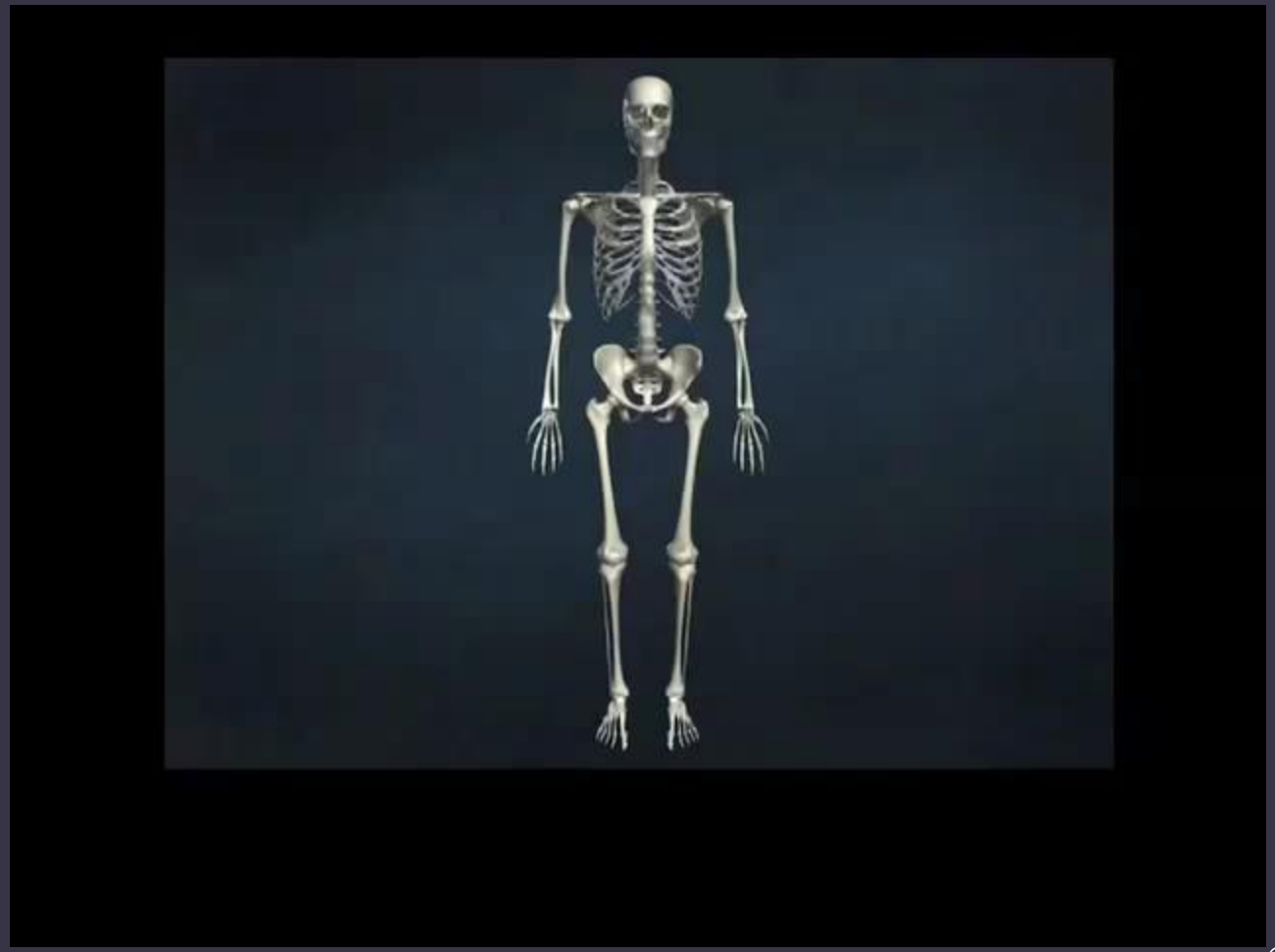
MUSCLES ACTING ON THE ARM

TABLE 10.14 **Muscles Acting on the Arm (see figures 10.20–10.23)**

Muscle	Origin	Insertion	Nerve	Action
Coracobrachialis (kōr′ă-kō-brā-kā-ā′lis)	Coracoid process of scapula	Midshaft of humerus	Musculocutaneous	Adducts arm and flexes shoulder
Deltoid (del′toyd)	Clavicle, acromion process, and scapular spine	Deltoid tuberosity	Axillary	Flexes and extends shoulder; abducts and medially and laterally rotates arm
Latissimus dorsi (lă-tis′i-mŭs dōr′sī)	Spinous processes of T7–L5; sacrum and iliac crest; inferior angle of scapula in some people	Medial crest of intertubercular groove	Thoracodorsal	Adducts and medially rotates arm; extends shoulder
Pectoralis major (pek′tō-rā′lis)	Clavicle, sternum, superior six costal cartilages, and abdominal aponeurosis	Lateral crest of intertubercular groove	Medial and lateral pectoral	Flexes shoulder; adducts and medially rotates arm; extends shoulder from flexed position
Teres major (ter′ēz, tēr′ēz)	Lateral border of scapula	Medial crest of intertubercular groove	Lower subscapular C5 and C6	Extends shoulder; adducts and medially rotates arm
Rotator Cuff				
Infraspinatus (in-fră-spī-nā′tŭs)	Infraspinous fossa of scapula	Greater tubercle of humerus	Suprascapular C5 and C6	Laterally rotates arm; holds head of humerus in place
Subscapularis (sŭb-skap-ū-lā′ris)	Subscapular fossa	Lesser tubercle of humerus	Upper and lower subscapular C5 and C6	Medially rotates arm; holds head of humerus in place
Supraspinatus (soo-pră-spī-nā′tŭs)	Supraspinous fossa	Greater tubercle of humerus	Suprascapular C5 and C6	Abducts arm; holds head of humerus in place
Teres minor (ter′ēz, tēr′ēz)	Lateral border of scapula	Greater tubercle of humerus	Axillary C5 and C6	Laterally rotates and adducts arm; holds head of humerus in place

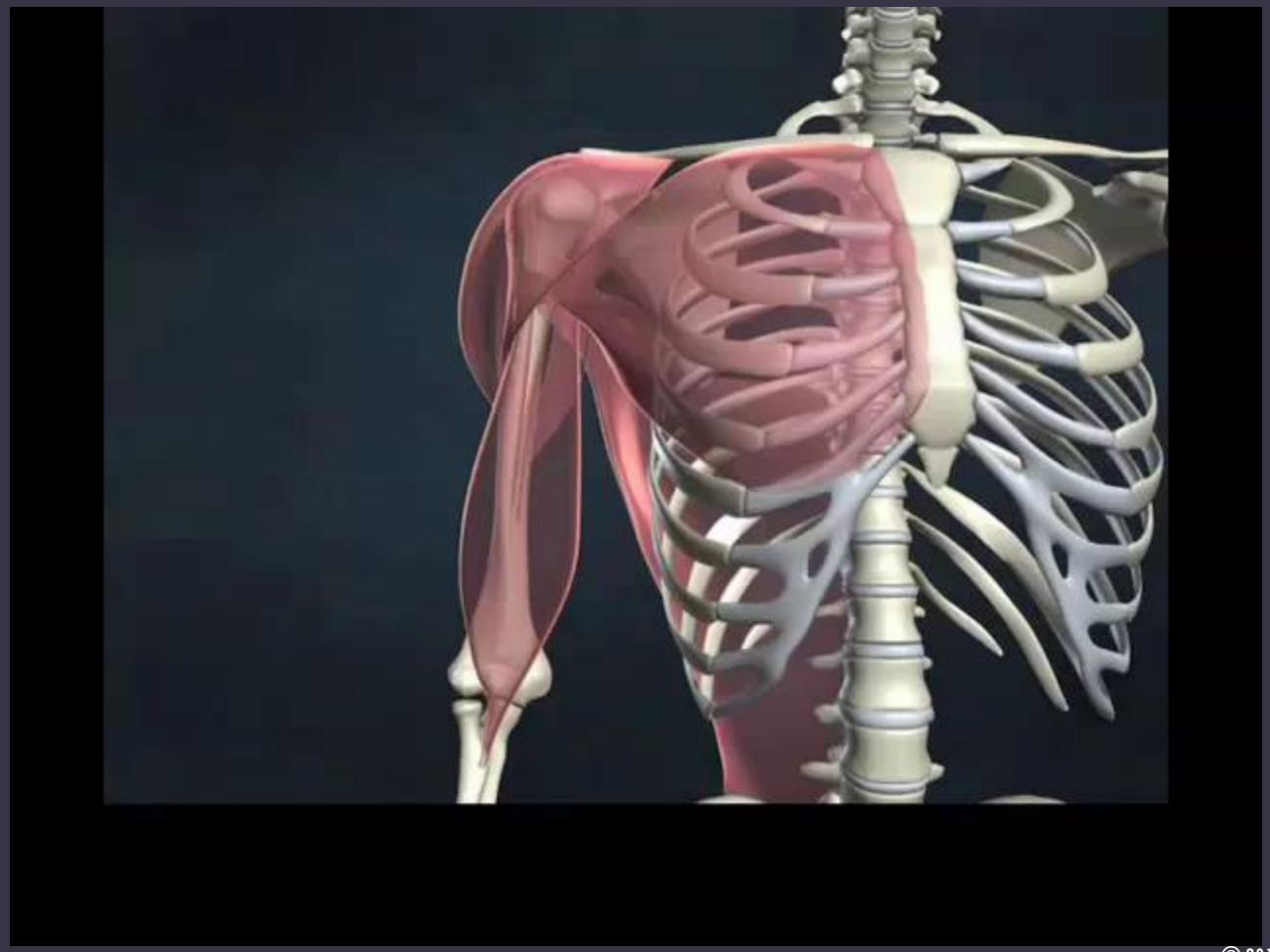
The arm is attached to the thorax by several muscles, including the **pectoralis major** and the **latissimus dorsi**

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MUSCLES ACTING ON THE ARM

Muscle
Coracobrachialis (kōr′ă-kō-brā-kā-ā′lis)
Deltoid (del′toyd)
Latissimus dorsi (lă-tis′i-mŭs dōr′sī)
Pectoralis major (pek′tō-rā′lis)
Teres major (ter′ēz, tēr′ēz)
Rotator Cuff
Infraspinatus (in-fră-spī-nā′tŭs)
Subscapularis (sŭb-skap-ū-lā′ris)
Supraspinatus (soo-pră-spī-nā′tŭs)
Teres minor (ter′ēz, tēr′ēz)

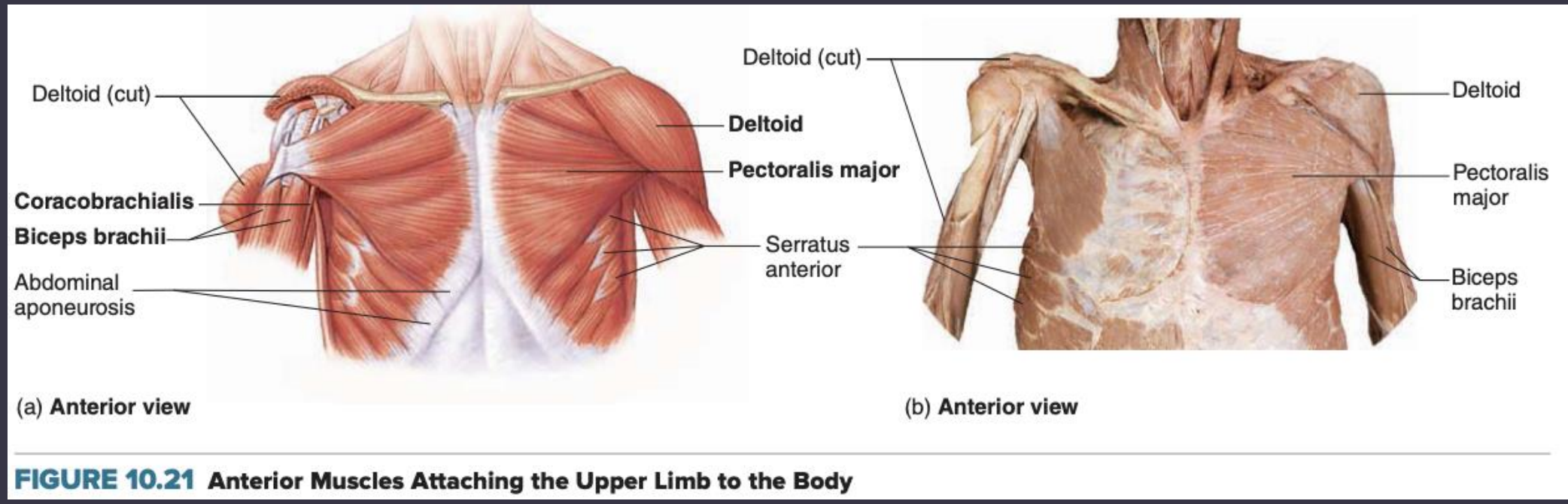


FIGURE 10.21 **Anterior Muscles Attaching the Upper Limb to the Body**

Muscle
Coracobrachialis (kōr′ă-kō-brā-kā-ā′lis)
Deltoid (del′toyd)
Latissimus dorsi (lă-tis′i-mŭs dōr′sī)
Pectoralis major (pek′tō-rā′lis)
Teres major (ter′ēz, tēr′ēz)
Rotator Cuff
Infraspinatus (in-fră-spī-nā′tŭs)
Subscapularis (sŭb-skap-ū-lā′ris)
Supraspinatus (soo-pră-spī-nā′tŭs)
Teres minor (ter′ēz, tēr′ēz)

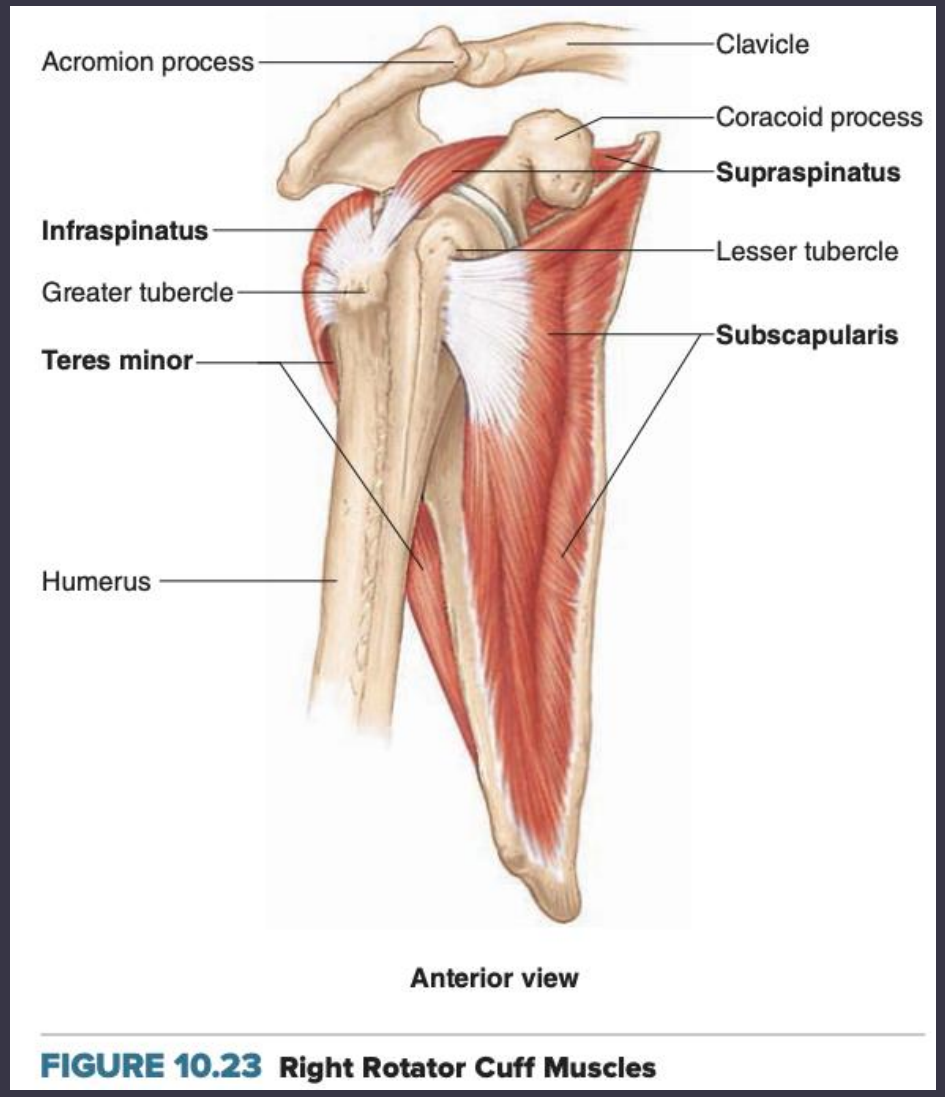


FIGURE 10.23 Right Rotator Cuff Muscles

Function:

- Stabilize the **glenohumeral joint**
- Control **rotation and abduction** of the humerus

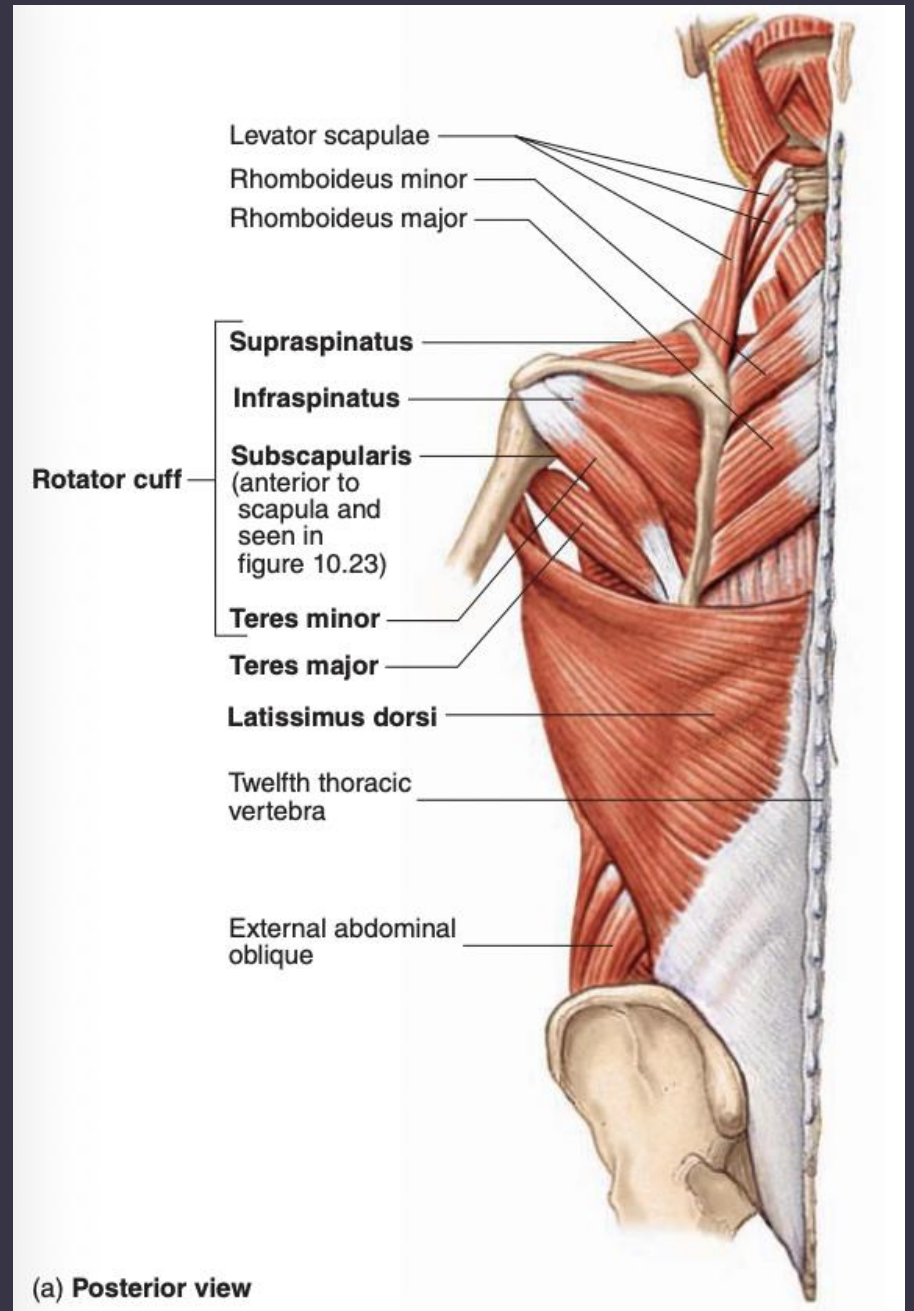


FIGURE 10.22 Posterior Muscles Attaching the Upper Limb to the Body

DELTOID MUSCLE – CLINICAL SIGNIFICANCE

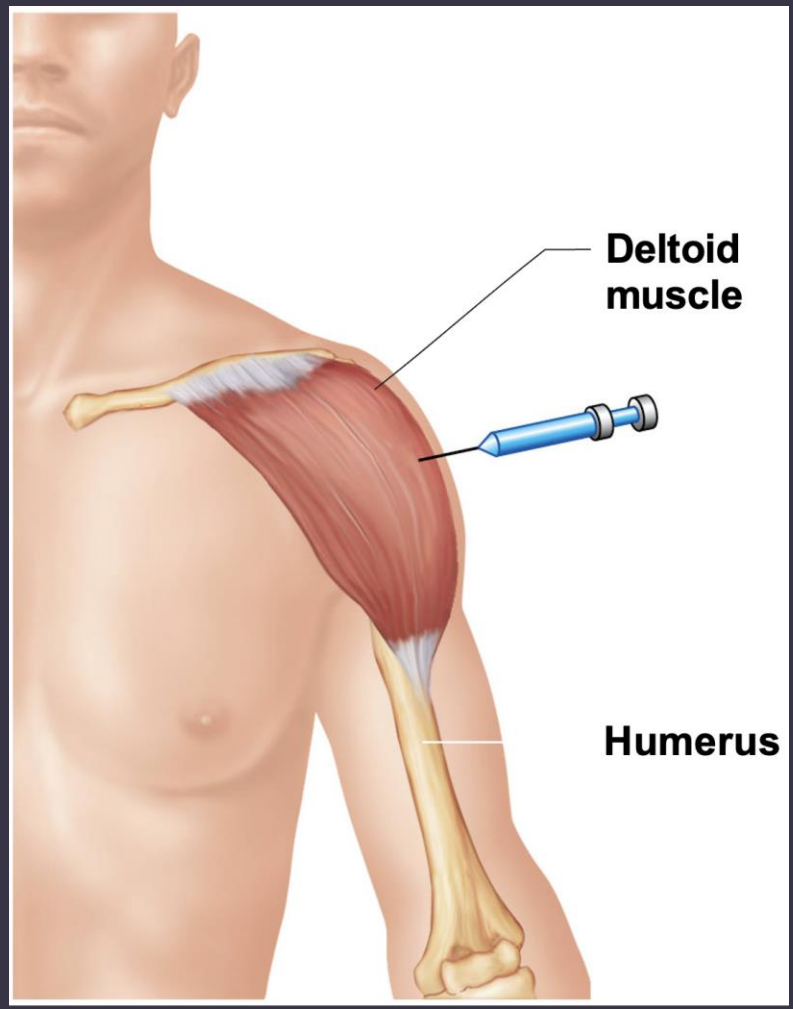
Location: Lateral shoulder, covering the glenohumeral joint

Function:

- Abducts, flexes, and extends the arm at the shoulder

Clinical Use:

- **Common site for intramuscular (IM) injections**
- Accessible, thick, and well-vascularized muscle
- Typically used for vaccines and small-volume injections



MUSCLES ACTING ON THE SHOULDER AND ARM

TABLE 10.15 Summary of Muscle Actions on the Shoulder and Arm

Flexion	Extension	Abduction	Adduction	Medial Rotation	Lateral Rotation
Deltoid	Deltoid	Deltoid	Pectoralis major	Pectoralis major	Deltoid
Pectoralis major	Teres major	Supraspinatus	Latissimus dorsi	Teres major	Infraspinatus
Coracobrachialis	Latissimus dorsi		Teres major	Latissimus dorsi	Teres minor
Biceps brachii	Pectoralis major		Teres minor	Deltoid	
	Triceps brachii		Triceps brachii	Subscapularis	
			Coracobrachialis		

ARM AND FOREARM MOVEMENTS

EXTENSION OF THE ELBOW

1. triceps brachii
2. anconeus

FLEXION OF THE ELBOW

1. brachialis
2. biceps brachii
3. brachioradialis

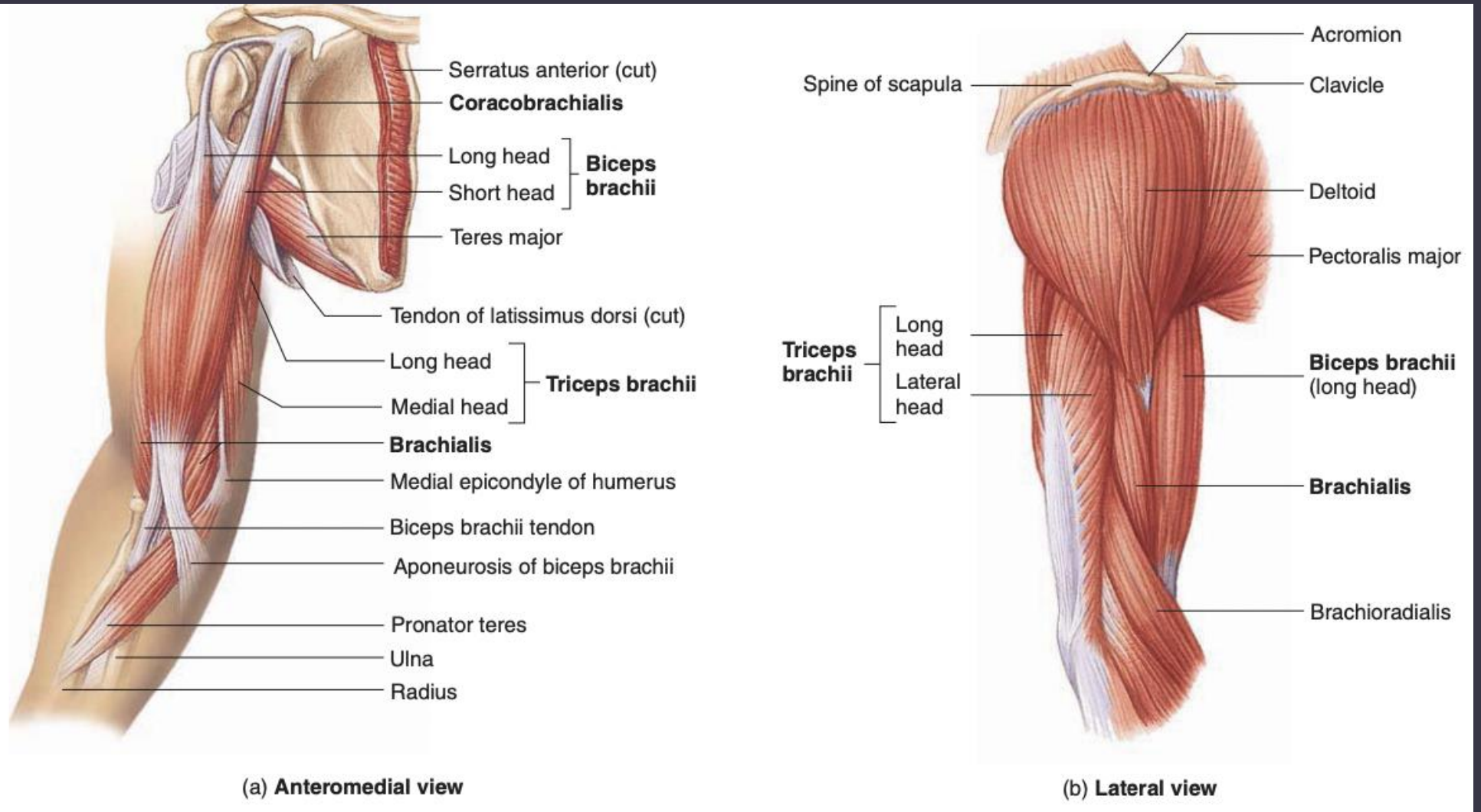
SUPINATORS (TURN PALM UP)

1. supinator
2. biceps brachii

PRONATION (PALM DOWN)

1. pronator quadratus
2. pronator teres

CONTRIBUTORS OF ARM AND FOREARM MOVEMENTS



CONTRIBUTORS OF ARM AND FOREARM MOVEMENTS

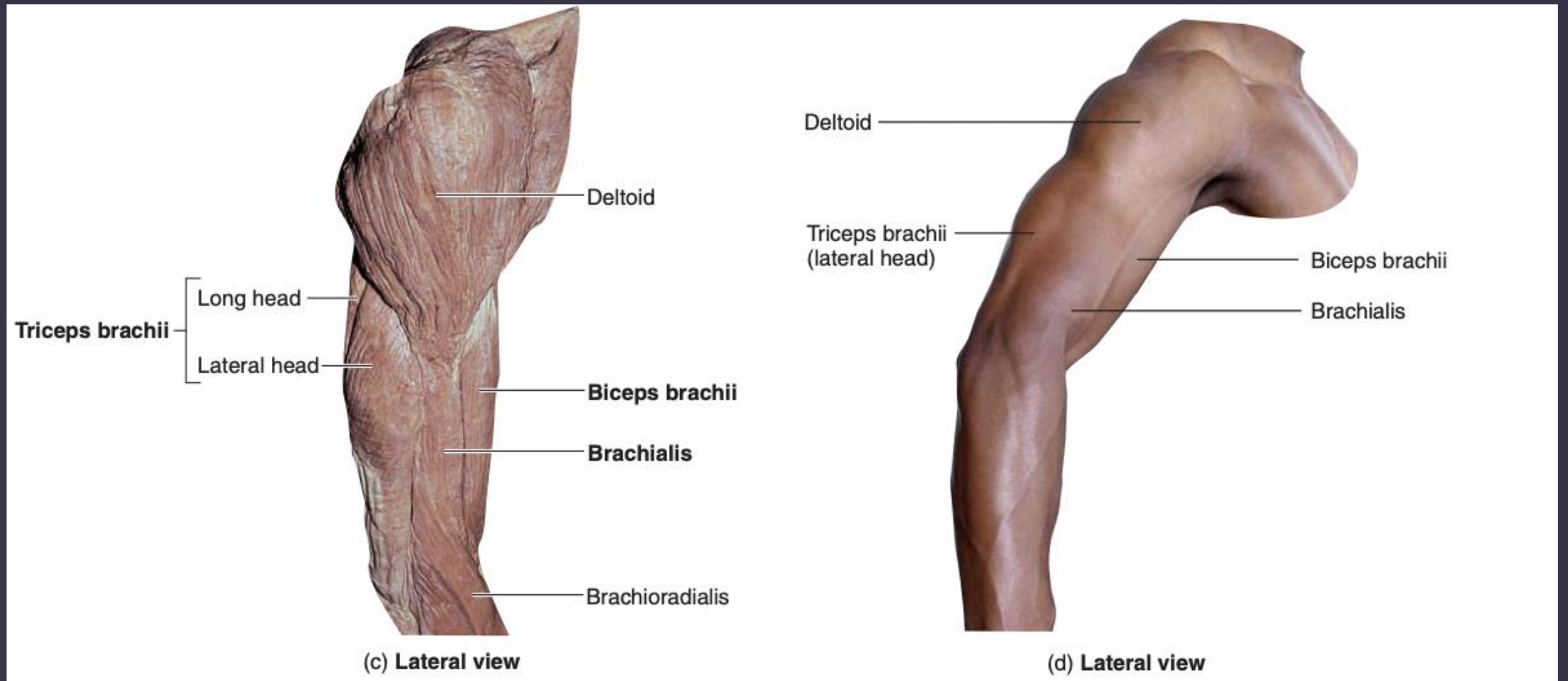


FIGURE 10.24 Lateral Right Arm Muscles

(*a*, *b*) The right shoulder and arm. (*c*) The right shoulder and arm muscles of a cadaver. (*d*) Surface anatomy of the right shoulder and arm. (Names of arm muscles are in bold.) AP|R

FOREARM MOVEMENT

ANTERIOR GROUP

- Most of the anterior forearm muscles are responsible for **flexion of the wrist and fingers** (make a fist)

POSTERIOR GROUP

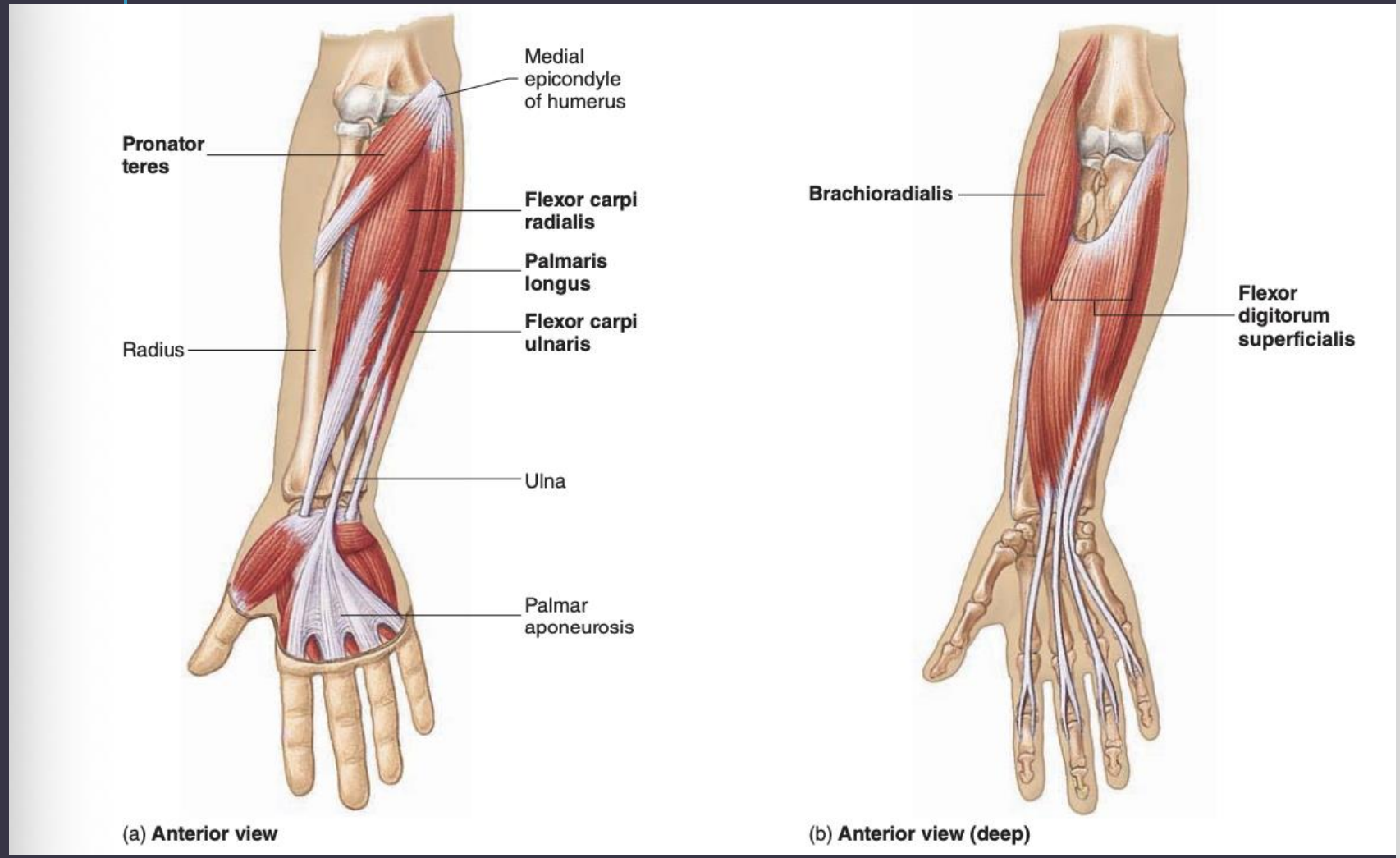
- Most of the posterior forearm muscles cause **extension of the wrist and fingers** (open the fist)

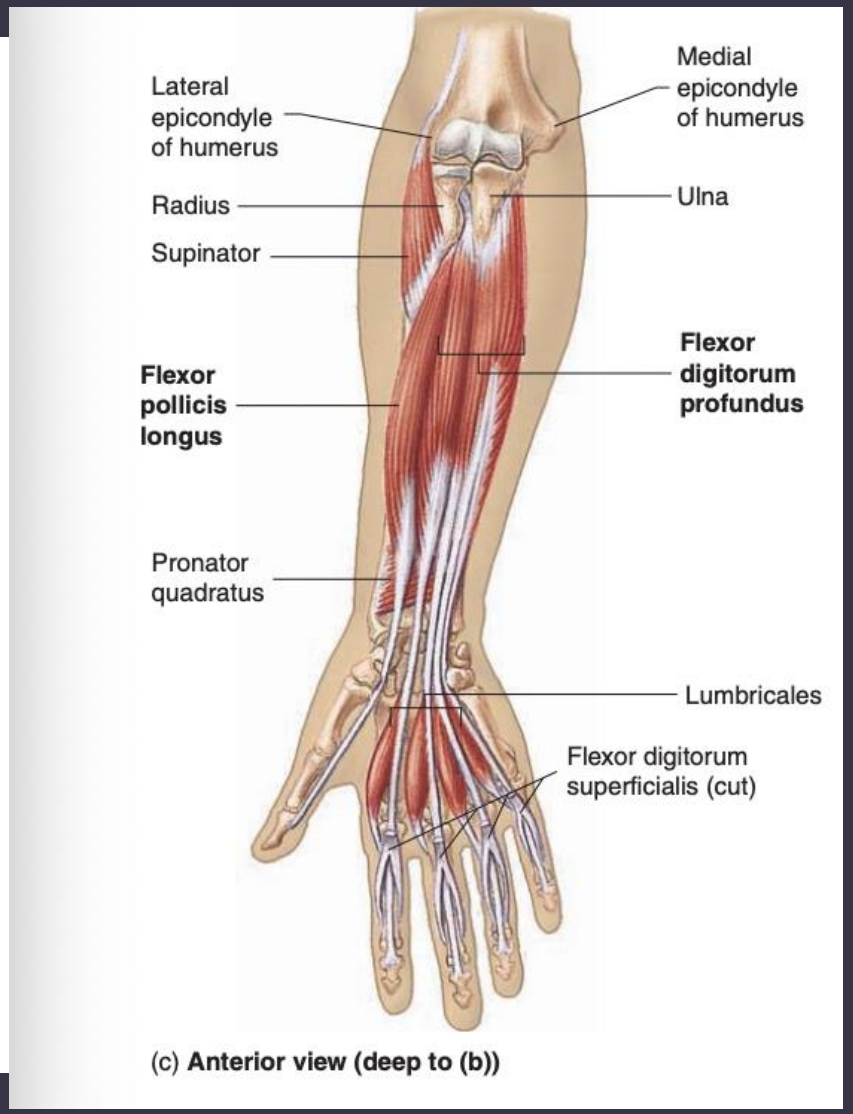
FLEXORS OF THE WRIST

TABLE 10.17 Muscles of the Forearm Acting on the Wrist, Hand, and Fingers (see figures 10.25 and 10.26)

Muscle	Origin	Insertion	Nerve	Action
Anterior Forearm				
Flexor carpi radialis (kar′pī rā-dē-ā′lis)	Medial epicondyle of humerus	Second and third metacarpal bones	Median	Flexes and abducts wrist
Flexor carpi ulnaris (kar′pī ŭl-nā′ris)	Medial epicondyle of humerus and ulna	Pisiform, hamate, and fifth metacarpal bones	Ulnar	Flexes and adducts wrist
Flexor digitorum profundus (dij′i-tōr′ŭm prō-fŭn′dŭs)	Ulna	Distal phalanges of digits 2–5	Ulnar and median	Flexes fingers at metacarpophalangeal joints and interphalangeal joints and wrist
Flexor digitorum superficialis (dij′i-tōr′ŭm soo′per-fish-ē-ā′lis)	Medial epicondyle of humerus, coronoid process, and radius	Middle phalanges of digits 2–5	Median	Flexes fingers at interphalangeal joints and wrist
Flexor pollicis longus (pol′i-sis lon′gŭs)	Radius	Distal phalanx of thumb	Median	Flexes thumb
Palmaris longus (pawl-mā′ris lon′gŭs)	Medial epicondyle of humerus	Palmar fascia	Median	Tenses palmar fascia; flexes wrist

FLEXORS OF THE WRIST

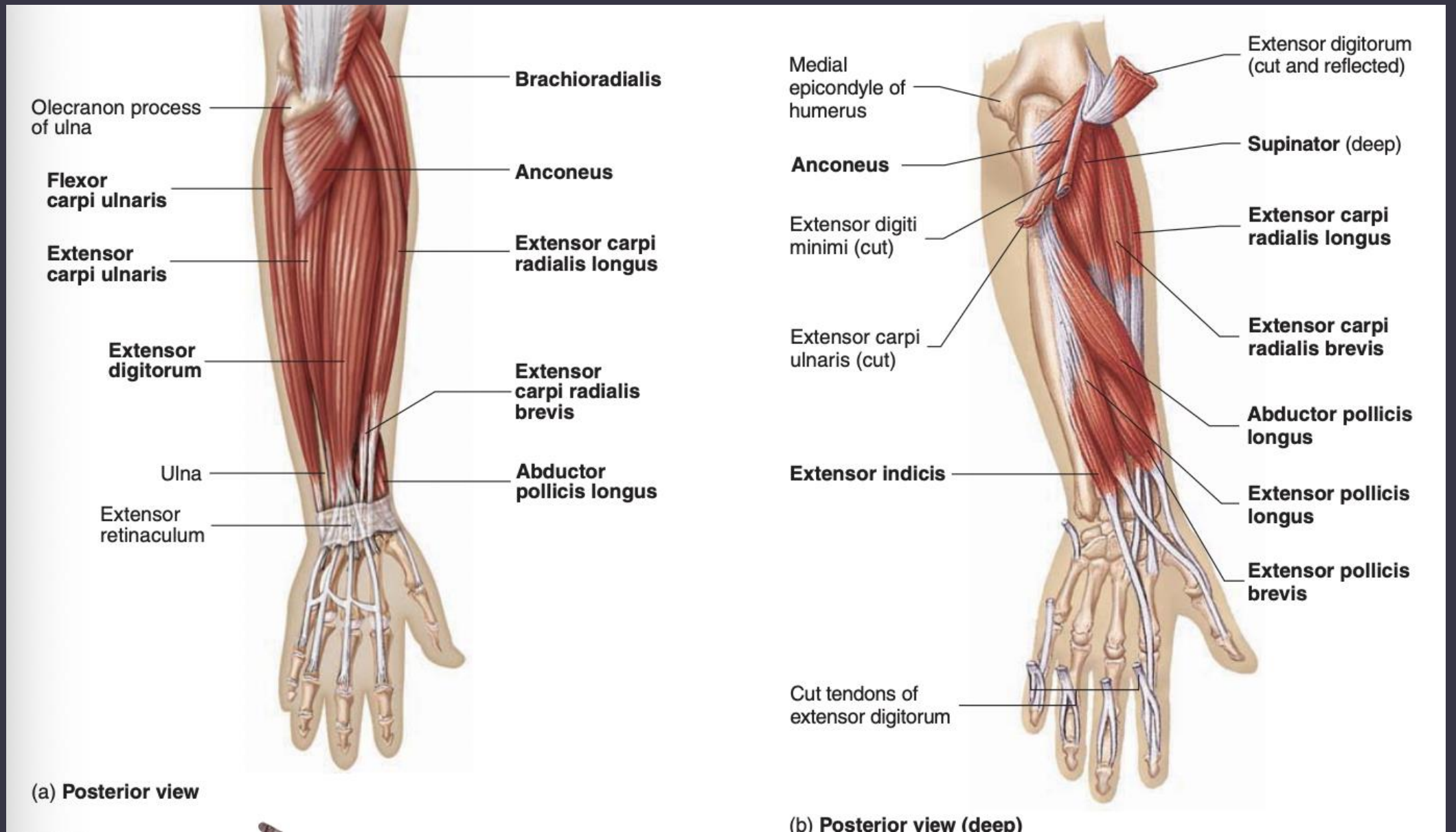




EXTENSORS OF THE WRIST

Posterior Forearm				
Abductor pollicis longus (pol′i-sis lon′gŭs)	Posterior ulna and radius and interosse-ous membrane	Base of first metacarpal bone	Radial	Abducts and extends thumb; abducts wrist
Extensor carpi radialis brevis (kar′pī rā-dē-ā′lis brev′is)	Lateral epicondyle of humerus	Base of third metacarpal bone	Radial	Extends and abducts wrist
Extensor carpi radialis longus (kar′pī rā-dē-ā′lis lon′gus)	Lateral supracondylar ridge of humerus	Base of second metacarpal bone	Radial	Extends and abducts wrist
Extensor carpi ulnaris (kar′pī ŭl-nā′ris)	Lateral epicondyle of humerus and ulna	Base of fifth metacarpal bone	Radial	Extends and adducts wrist
Extensor digiti minimi (dij′i-tī min′i-mī)	Lateral epicondyle of humerus	Phalanges of digit 5	Radial	Extends little finger and wrist
Extensor digitorum (dij′i-tōr′ŭm)	Lateral epicondyle of humerus	Extensor tendon expansion over phalanges of digits 2–5	Radial	Extends fingers and wrist
Extensor indicis (in′di-sis)	Ulna	Extensor tendon expansion over digit 2	Radial	Extends forefinger and wrist
Extensor pollicis brevis (pol′i-sis brev′is)	Radius	Proximal phalanx of thumb	Radial	Extends and abducts thumb; abducts wrist
Extensor pollicis longus (pol′i-sis lon′gŭs)	Ulna	Distal phalanx of thumb	Radial	Extends thumb

EXTENSORS OF THE WRIST



THUMB MOVEMENT

- **Function:** Move the thumb for gripping, pinching, and fine manipulation

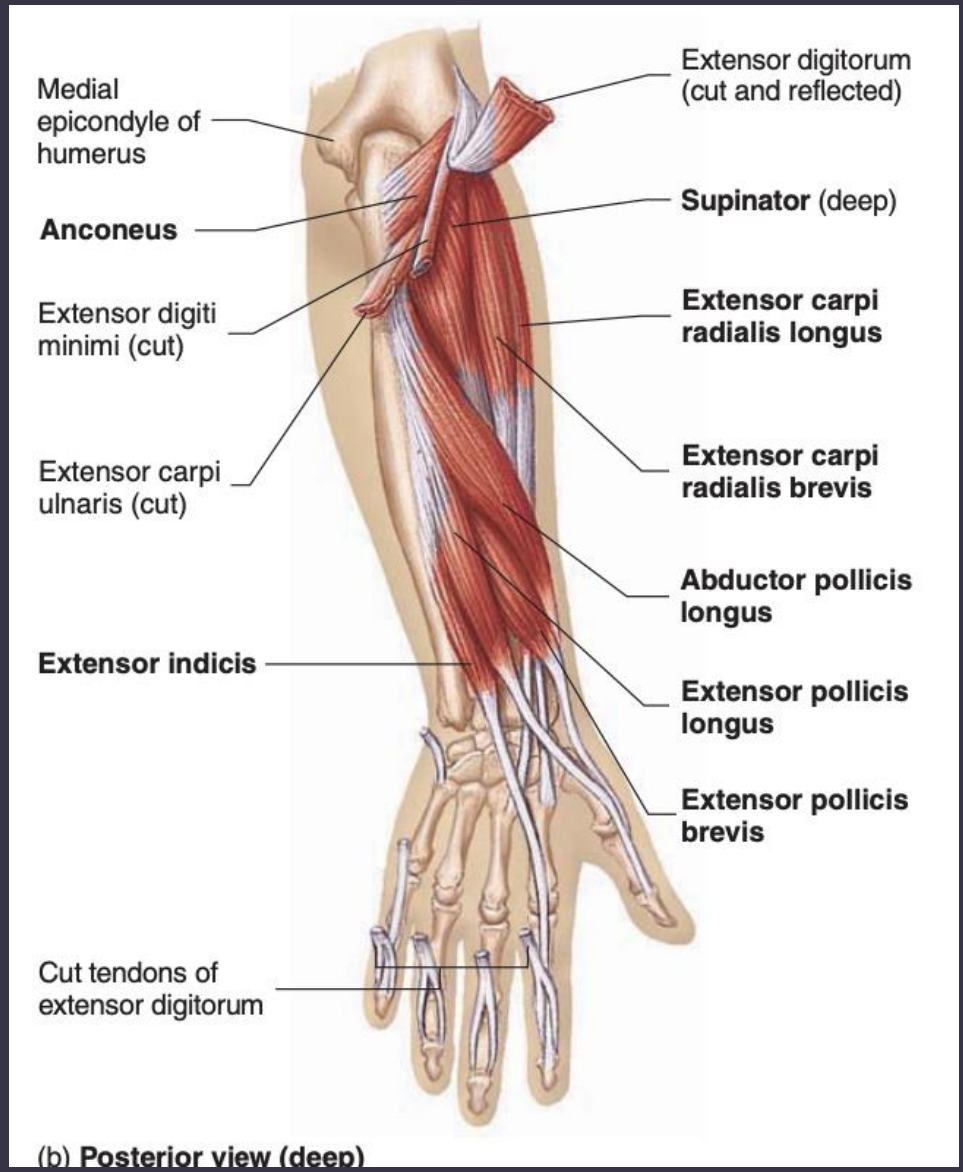
Key Muscles:

1. **Abductor pollicis longus** – abducts the thumb, assists in extension
2. **Extensor pollicis longus** – extends the distal phalanx of the thumb
3. **Extensor pollicis brevis** – extends the proximal phalanx of the thumb

APL

EPL

EPB



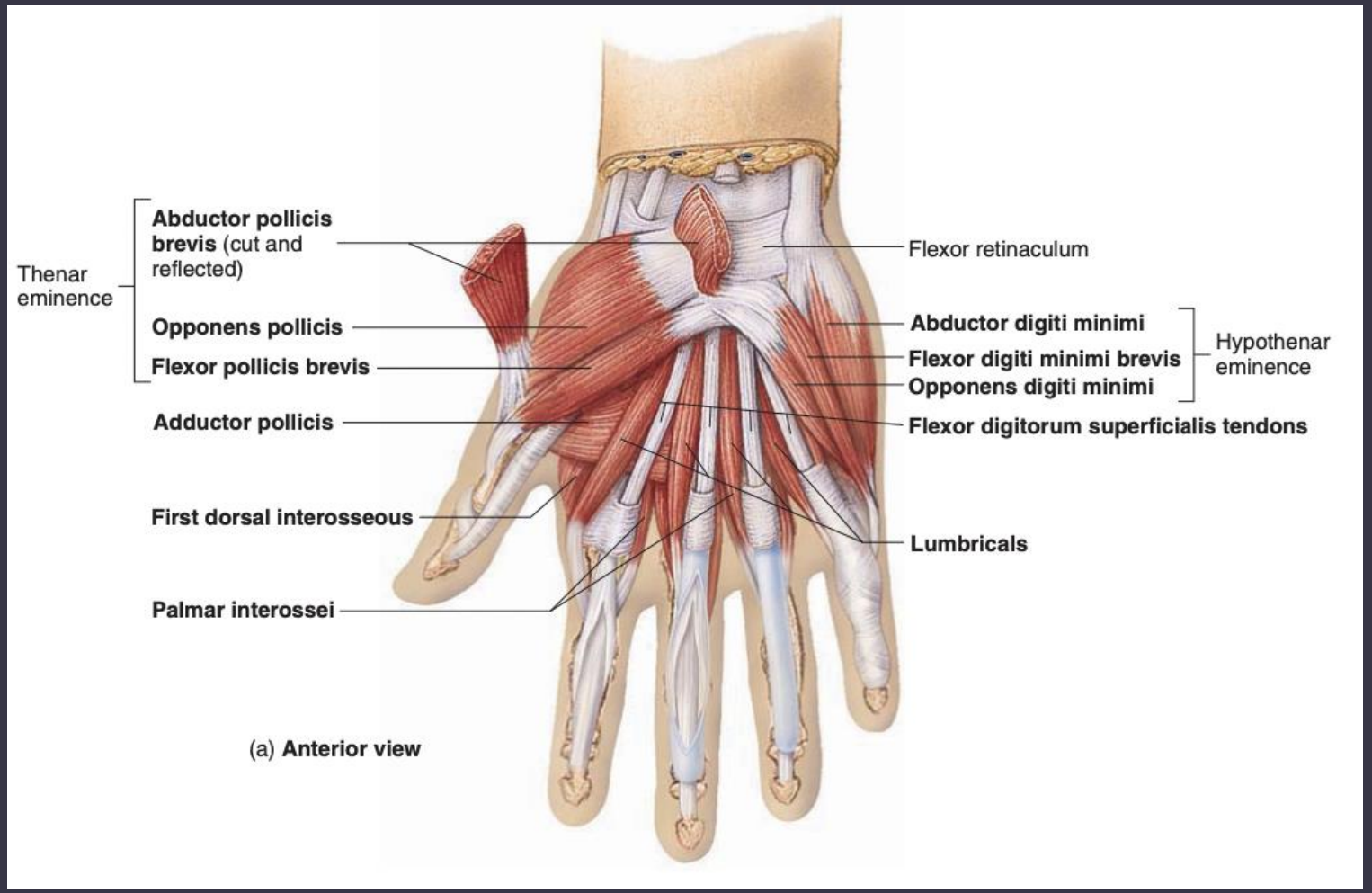
INTRINSIC HAND MUSCLES – FINGER MOVEMENT

Function: Move the fingers and assist with precise hand movements

Location: Entirely within the hand (do not cross the wrist)

Key Groups:

1. **Thenar muscles** – move the **thumb**
2. **Hypothenar muscles** – move the **little finger**
3. **Lumbricals** – flex metacarpophalangeal joints, extend interphalangeal joints
4. **Interossei muscles**



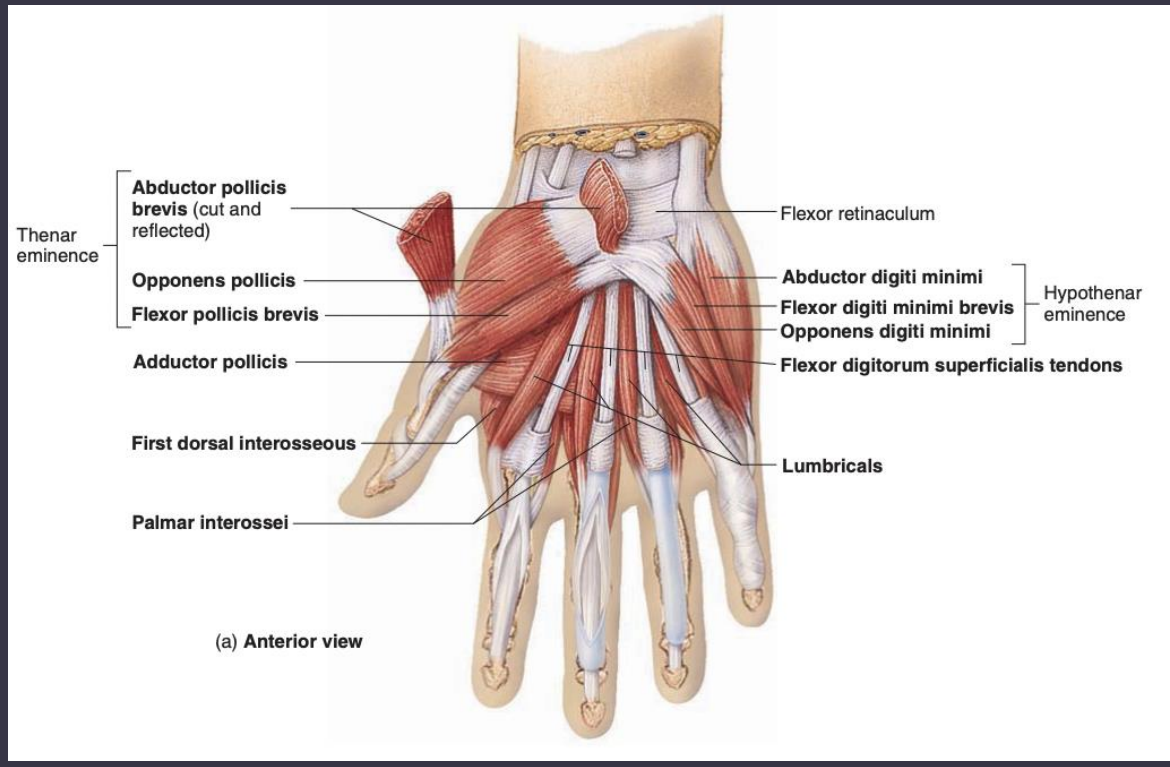
INTRINSIC HAND MOVEMENT
THENAR AND HYPOTHENAR EMINENCES

1. Thenar Eminence (Radial Side – Thumb Muscles)

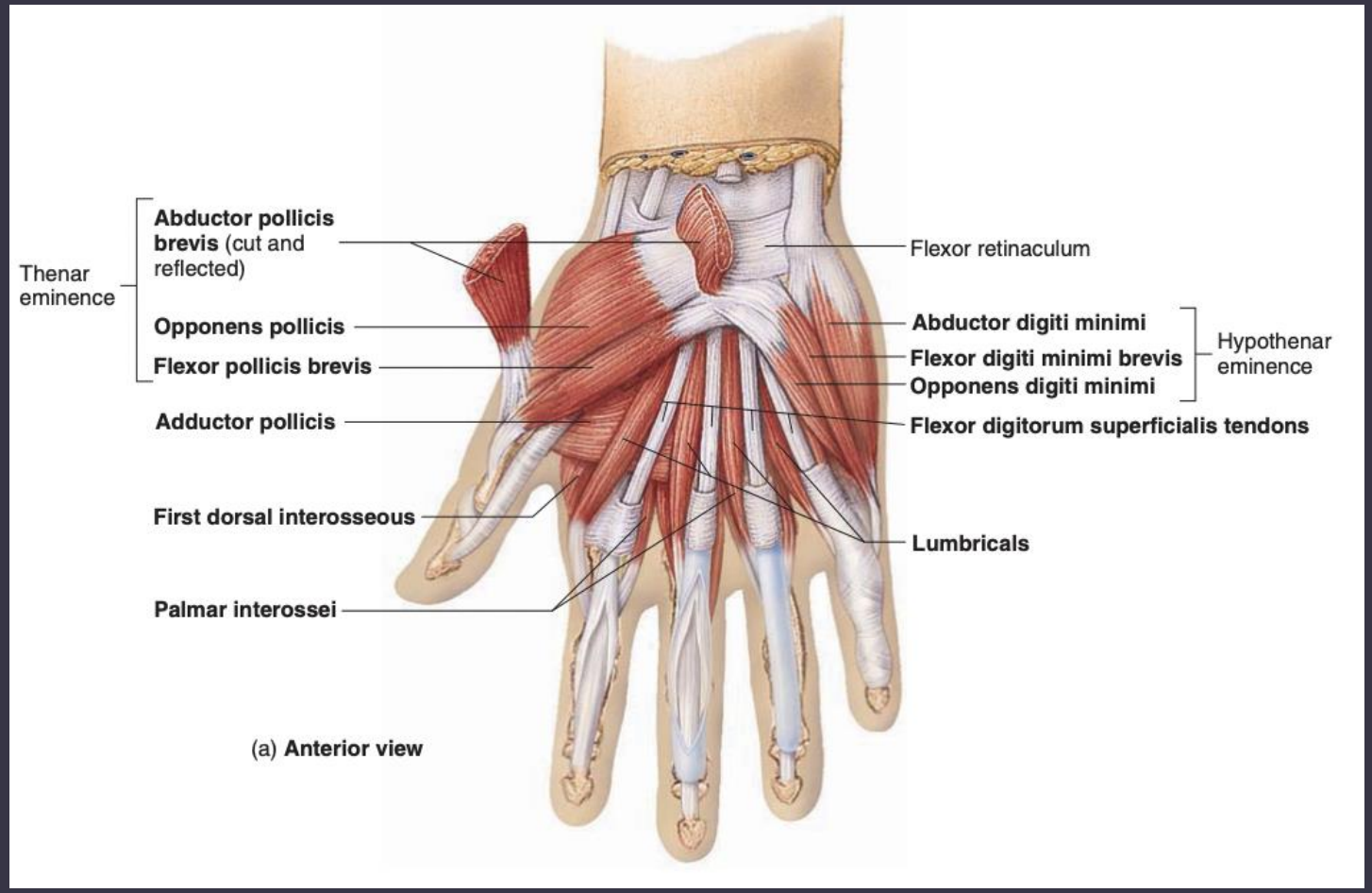
- **Flexor pollicis brevis** – flexes the thumb
- **Abductor pollicis brevis** – abducts the thumb
- **Opponens pollicis** – opposes the thumb to touch other fingers

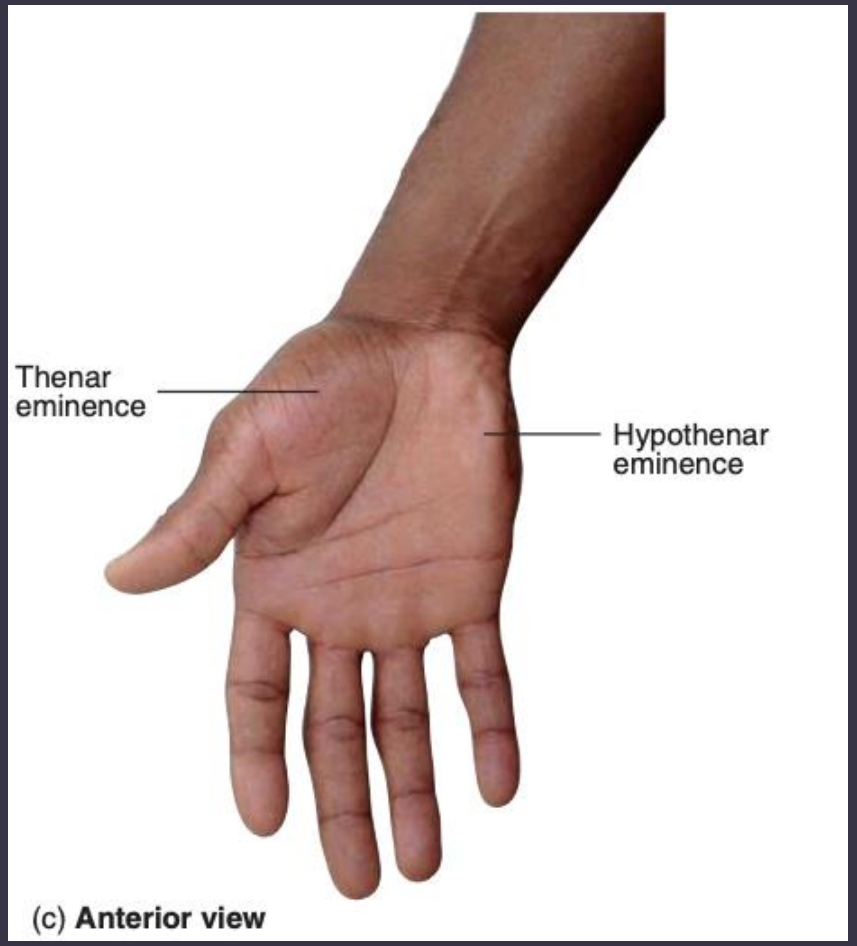
2. Hypothenar Eminence (Ulnar Side – Little Finger Muscles)

- **Abductor digiti minimi** – abducts the little finger
- **Flexor digiti minimi brevis** – flexes the little finger
- **Opponens digiti minimi** – rotates little finger toward thumb (opposition)



INTRINSIC HAND MOVEMENT





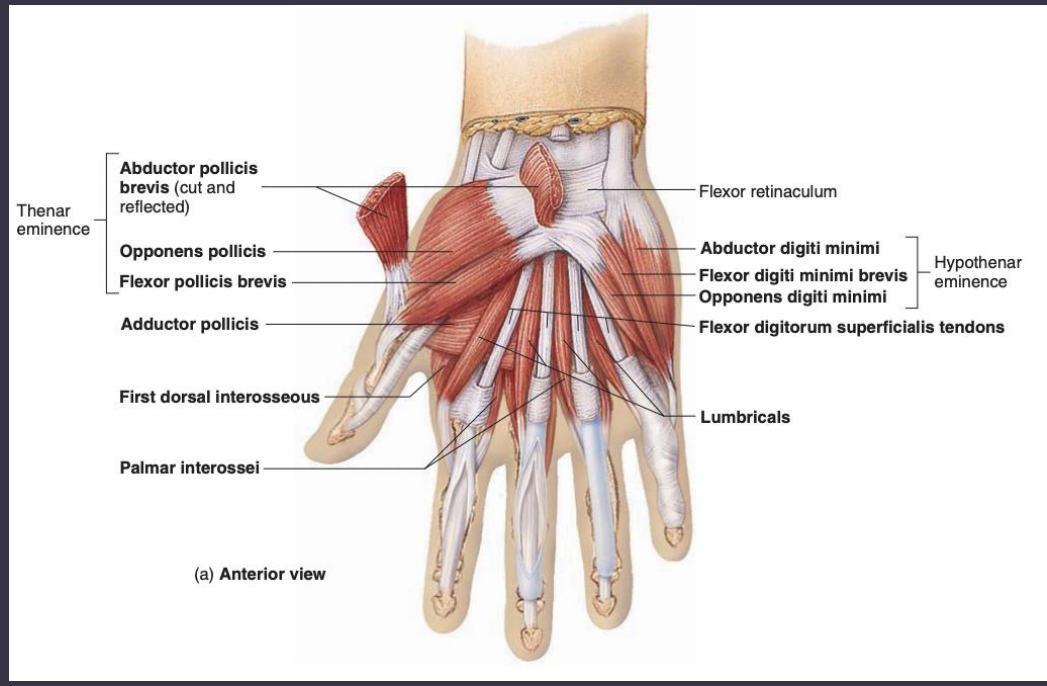
INTRINSIC HAND MOVEMENT
FINGER ABDUCTION AND ADDUCTION

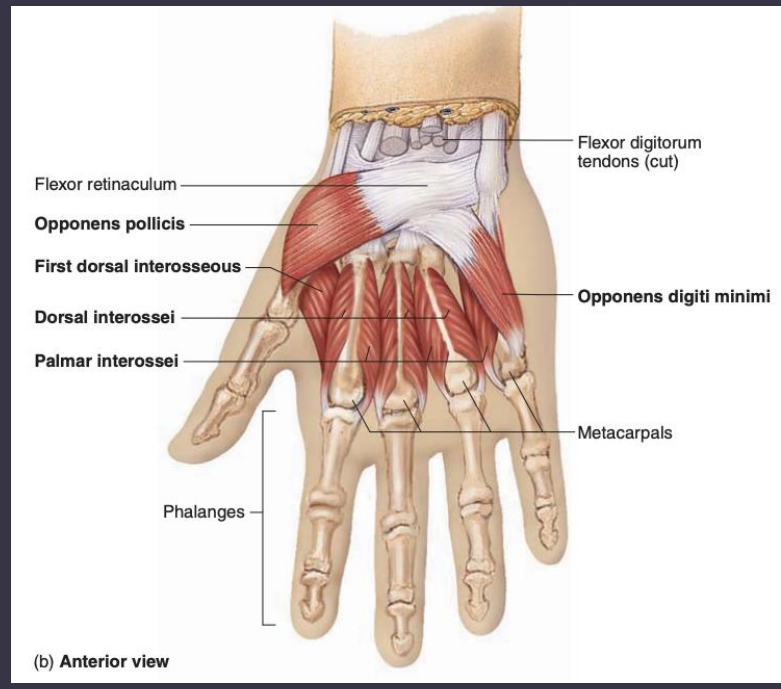
1. Abduction of Fingers (Spreading Fingers Apart)

- **Dorsal interossei** – abduct fingers at metacarpophalangeal (MCP) joints
- **Abductor digiti minimi** – abducts the little finger

2. Adduction of Fingers (Bringing Fingers Together)

- **Palmar interossei** – adduct fingers at MCP joints

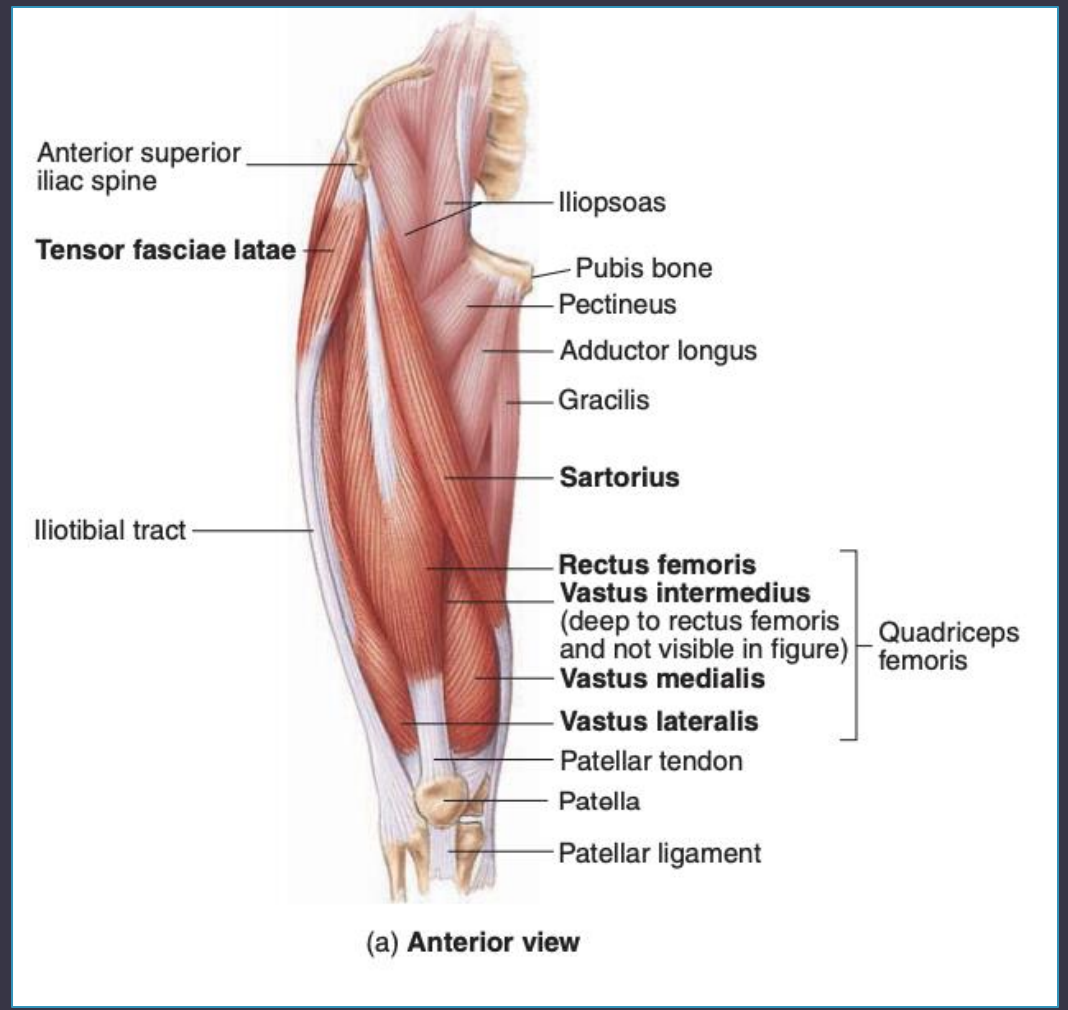




HIP AND THIGH MUSCLES (ANTERIOR, POSTEROLATERAL, DEEP)

Anterior Hip and Thigh Muscles – Hip Flexion

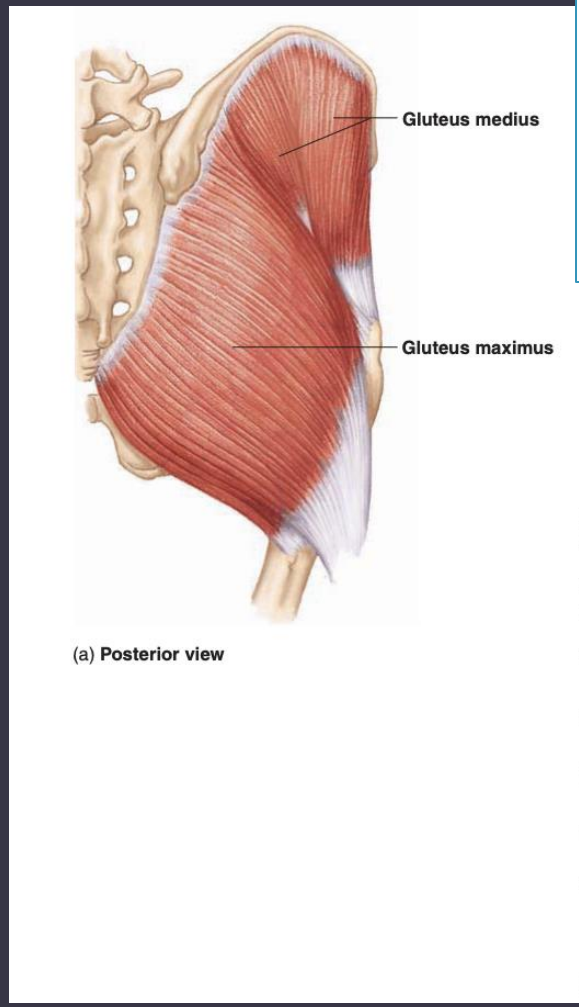
- **Function:** Flex the hip; when thigh is fixed, flex the trunk on the thigh
- **Key Muscles:**
  - **Iliacus** – originates from iliac fossa; flexes and stabilizes hip
  - **Psoas major** – originates from lumbar vertebrae; flexes hip and trunk
- **Collectively: Iliacus + Psoas major = Iliopsoas**, the primary hip flexor

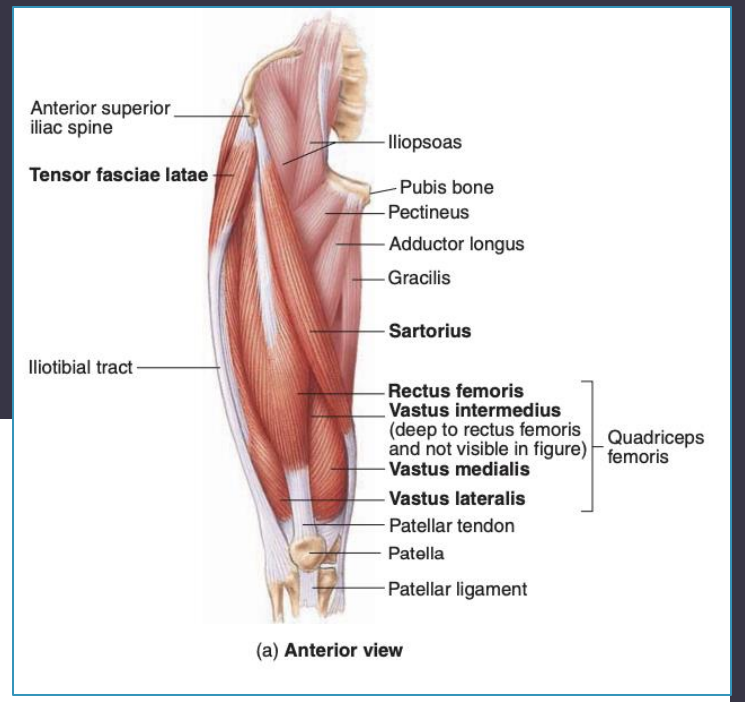


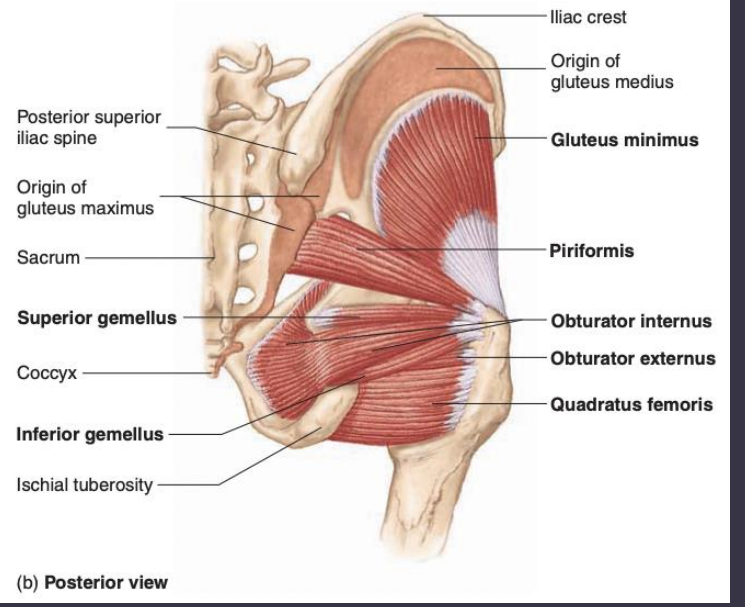
HIP AND THIGH MUSCLES (ANTERIOR, POSTEROLATERAL, DEEP)

Posterolateral Hip and Thigh Muscles

- **Function:** Extension, abduction, and lateral rotation of the hip; stabilization of the pelvis
- **Key Muscles:**
  - Gluteus maximus
  - Gluteus medius
  - Gluteus minimus
  - Tensor fasciae latae





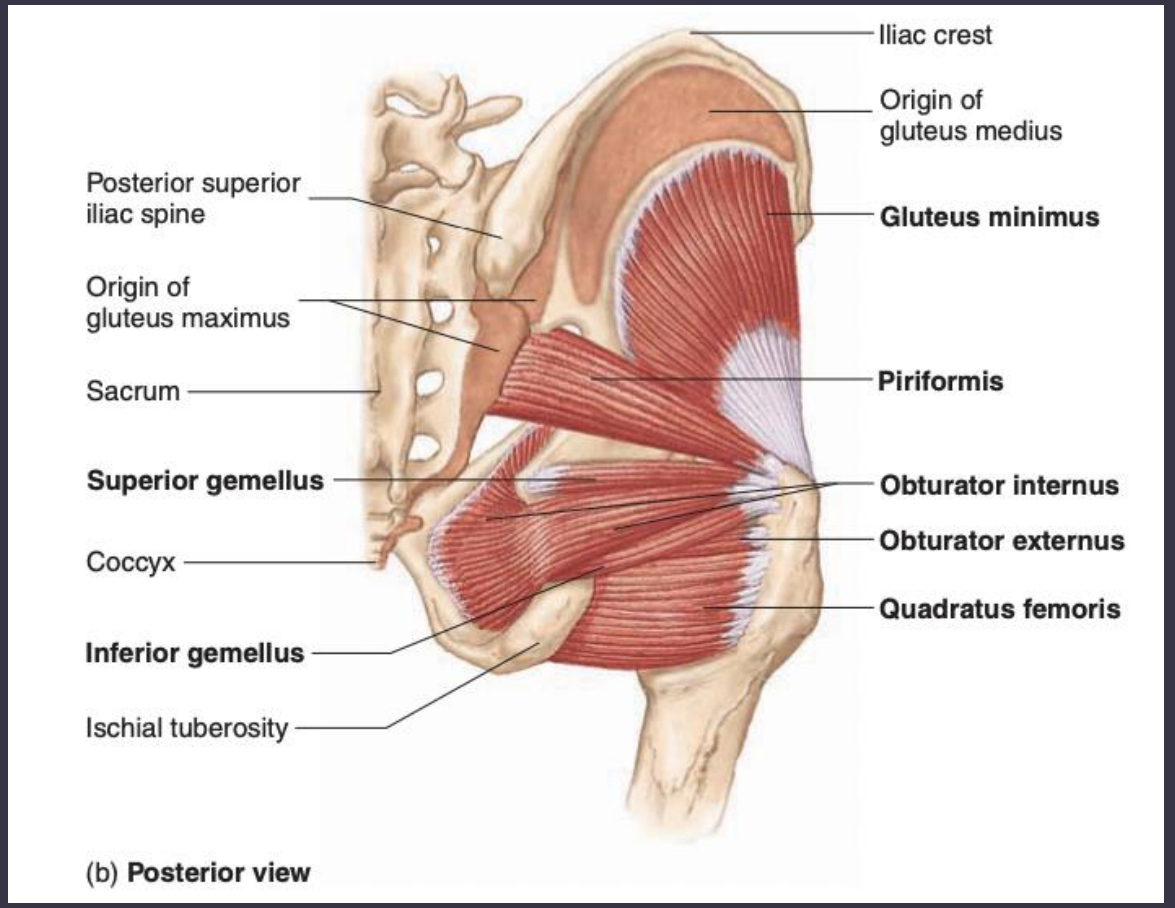


HIP AND THIGH MUSCLES (ANTERIOR, POSTEROLATERAL, DEEP)

Deep Hip Muscles – Lateral Rotators

- **Function:** Laterally rotate the thigh; assist gluteus maximus in hip stabilization

- **Key Muscles:**
  - Gemellus (inferior, superior)
  - Obturator (externus, internus)
  - Piriformis
  - Quadratus femoris



LEG MOVEMENTS (MUSCLES OF THE THIGH)

ANTERIOR COMPARTMENT

- flex the hip
- and/or extend the knee

- **Quadriceps femoris**
- **Sartorius**

MEDIAL COMPARTMENT

- adduct the thigh

- **Adductor brevis**
- **Adductor longus**
- **Adductor magnus**
- **Gracilis**
- **Pectineus**

POSTERIOR COMPARTMENT

- extend the hip and flex the knee

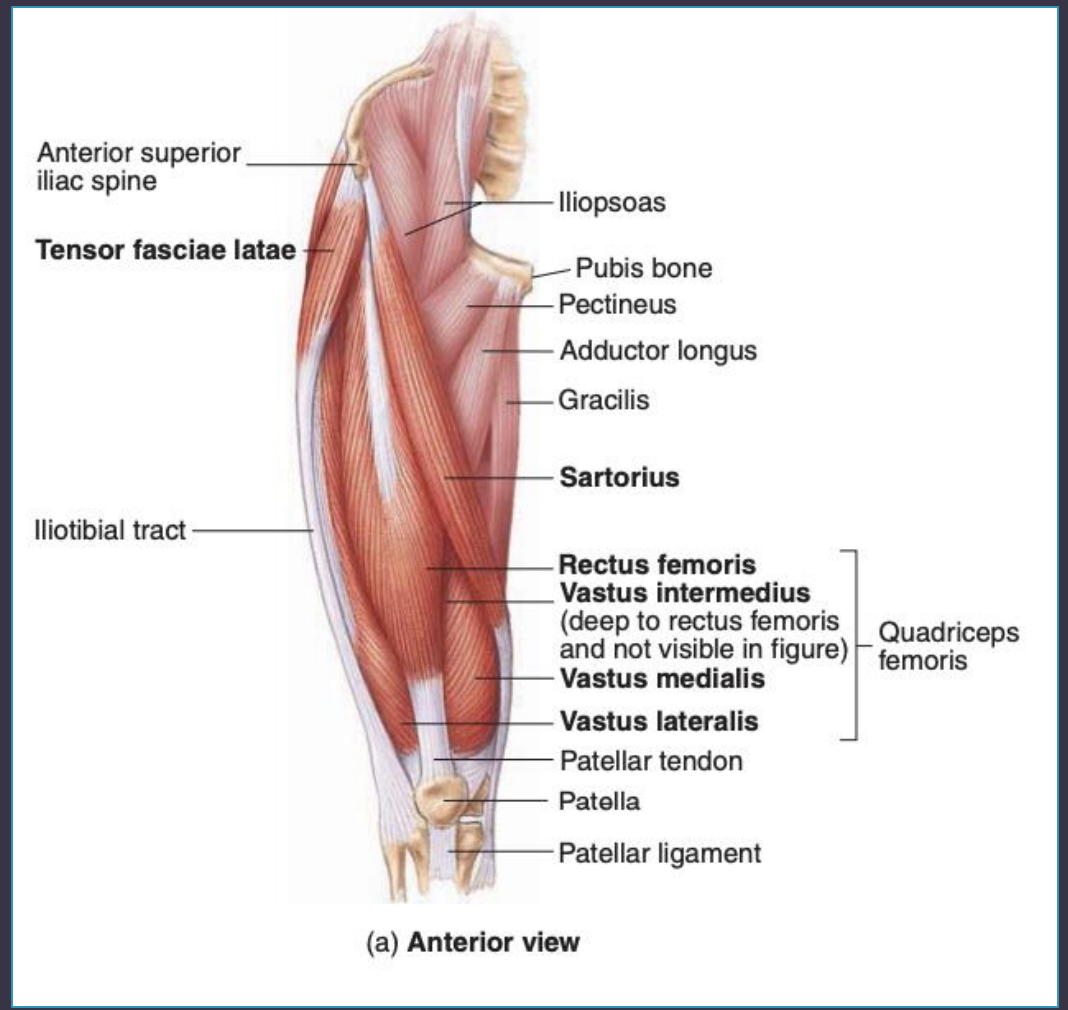
- **Biceps femoris**
- **Semimembranosus**
- **Semitendinosus**

LEG MOVEMENTS (MUSCLES OF THE THIGH)

ANTERIOR COMPARTMENT

- **Function:** Flex the hip and/or extend the knee

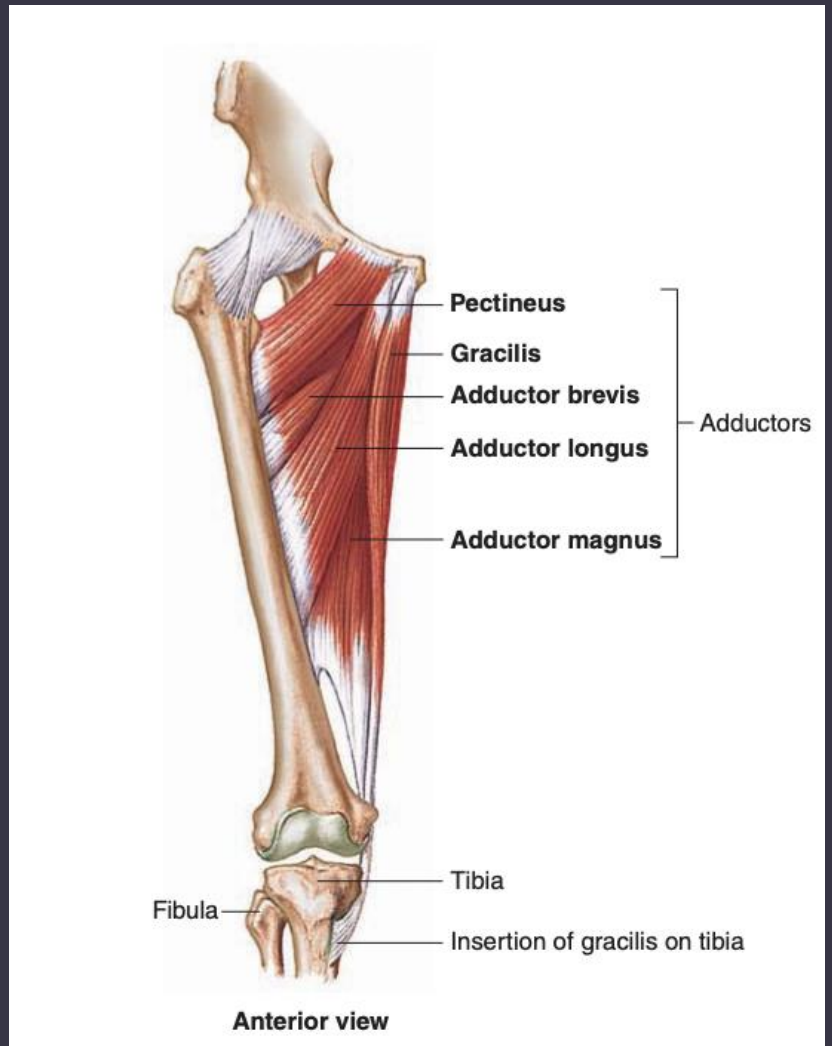
- **Quadriceps femoris**
  - rectus femoris, vastus lateralis, vastus medialis, vastus intermedius
- **Sartorius**



LEG MOVEMENTS (MUSCLES OF THE THIGH)

MEDIAL COMPARTMENT

- **Function:** Adduct the thigh; stabilize pelvis during walking and standing
- **Adductor brevis**
  - **Adductor longus**
  - **Adductor magnus**
  - **Gracilis**
  - **Pectineus**

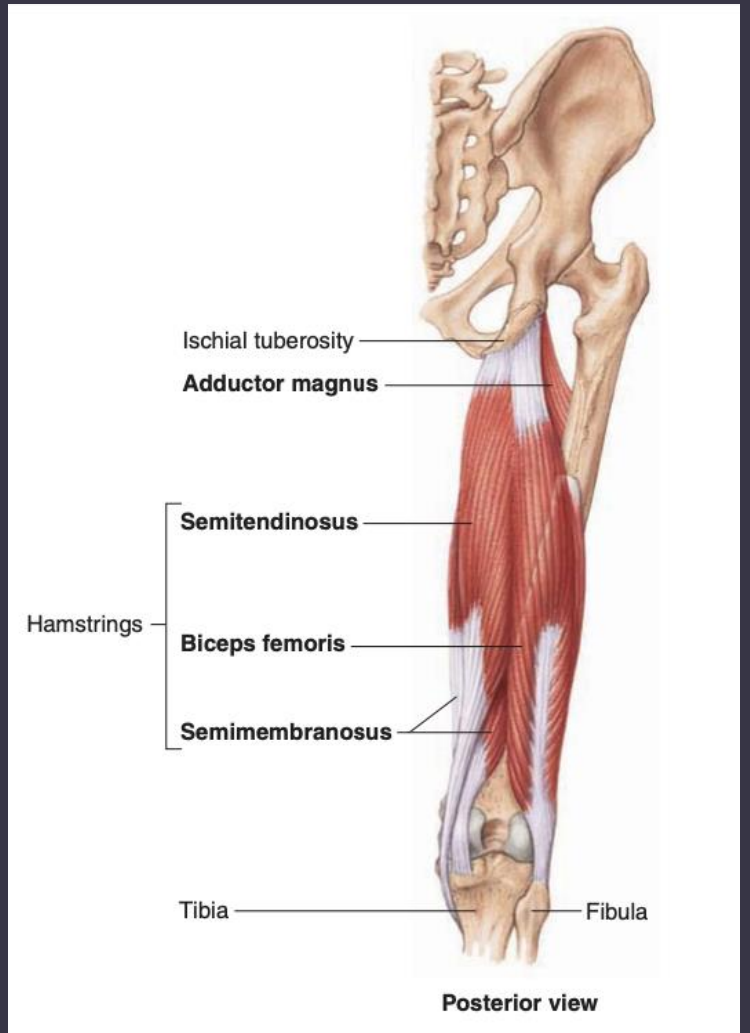


LEG MOVEMENTS (MUSCLES OF THE THIGH)

POSTERIOR COMPARTMENT

Function: Extend the hip and flex the knee- hamstrings

- **Biceps femoris**
- **Semimembranosus**
- **Semitendinosus**



ANKLE, FOOT AND TOE MOVEMENT

EXTRINSIC FOOT MUSCLES

1. Anterior Compartment (Extensors)

- Function: dorsiflexion of the foot, extension of toes, inversion and eversion

2. Posterior Compartment

- **Superficial posterior:** plantar flex the foot
- **Deep posterior:** plantar flex and invert the foot; flex the toes

3. Lateral Compartment

- Function: primarily **eversion of the foot;** assist in plantar flexion

INTRINSIC FOOT MUSCLES

- Function: flex, extend, abduct, and adduct the toes
- Arrangement: similar to **intrinsic muscles of the hand**
- Support arches of the foot and enable fine movements of toes

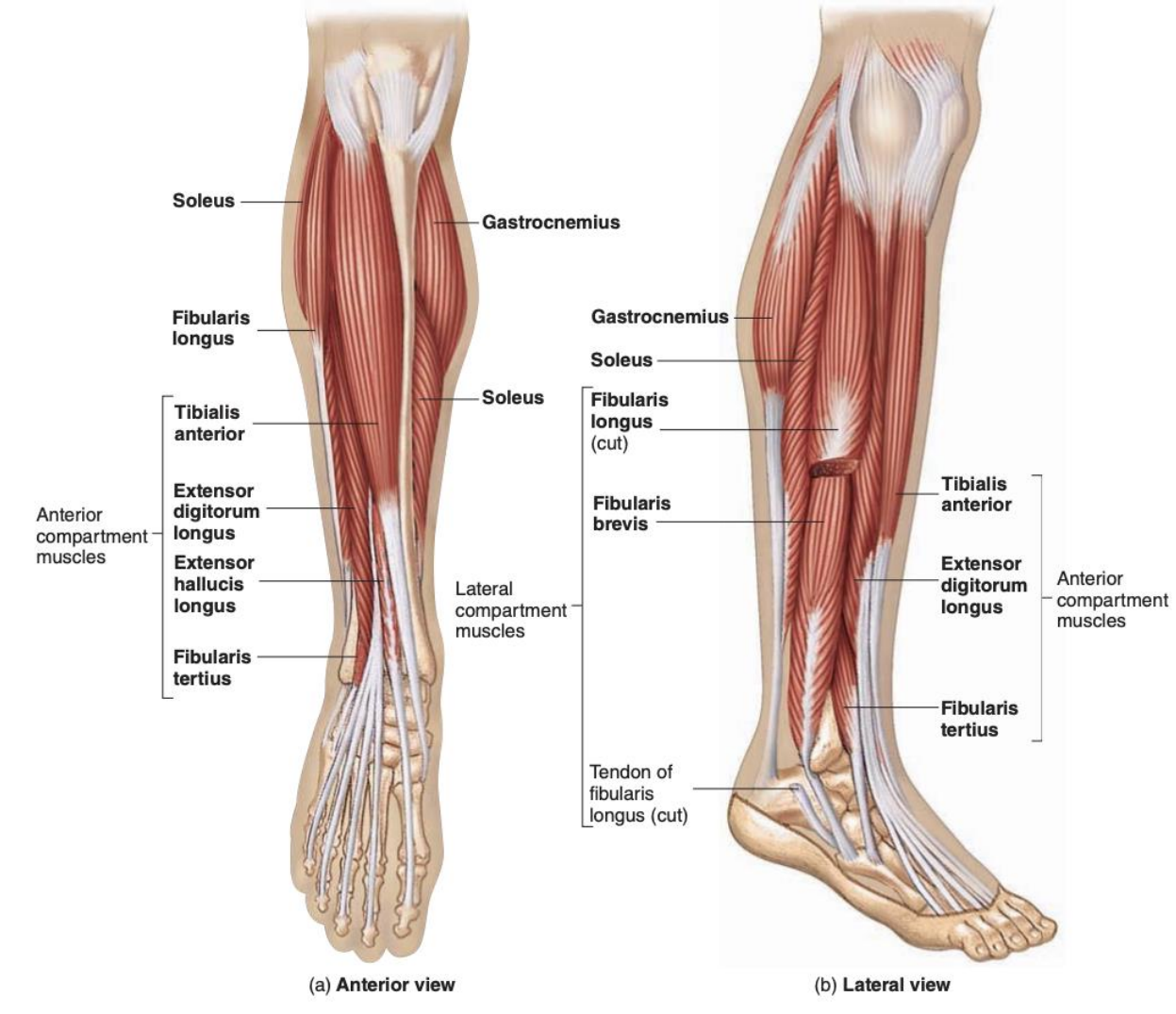
ANKLE, FOOT AND TOE MOVEMENT

EXTRINSIC FOOT MUSCLES

1. Anterior Compartment (Extensors)

- Function: dorsiflexion of the foot, extension of toes, inversion and eversion

Extensor digitorum longus
Extensor hallucis longus
Tibialis anterior
Fibularis tertius



ANKLE, FOOT AND TOE MOVEMENT

EXTRINSIC FOOT MUSCLES

2. Posterior Compartment

- Superficial posterior - plantar flex the foot

Gastrocnemius

Plantaris

Soleus

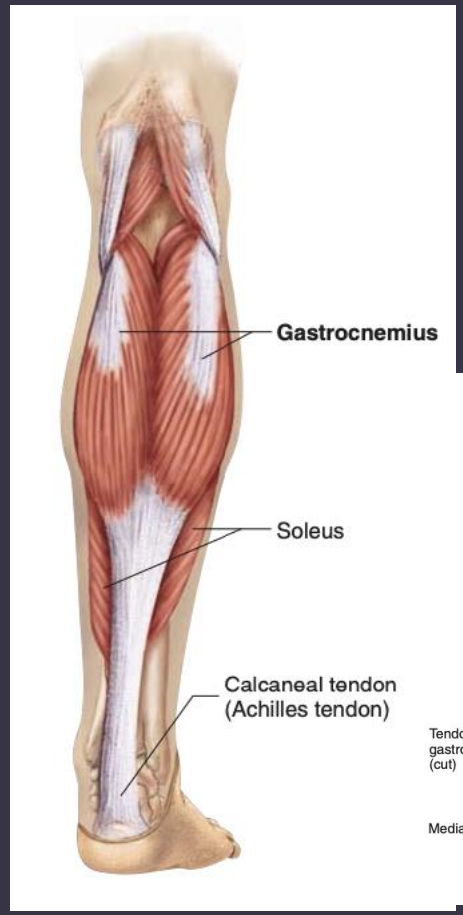
- Deep posterior - plantar flex and invert the foot and flex the toes

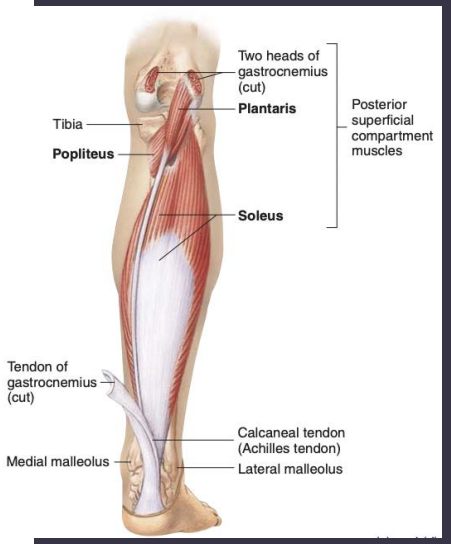
Flexor digitorum longus

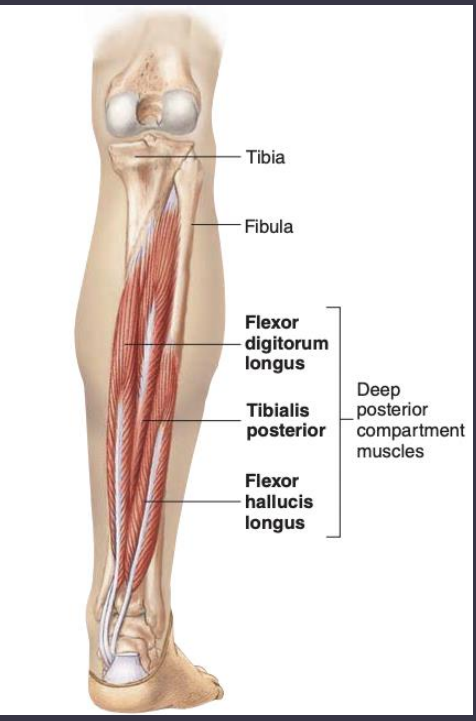
Flexor hallucis longus

Popliteus

Tibialis posterior







ANKLE, FOOT AND TOE MOVEMENT

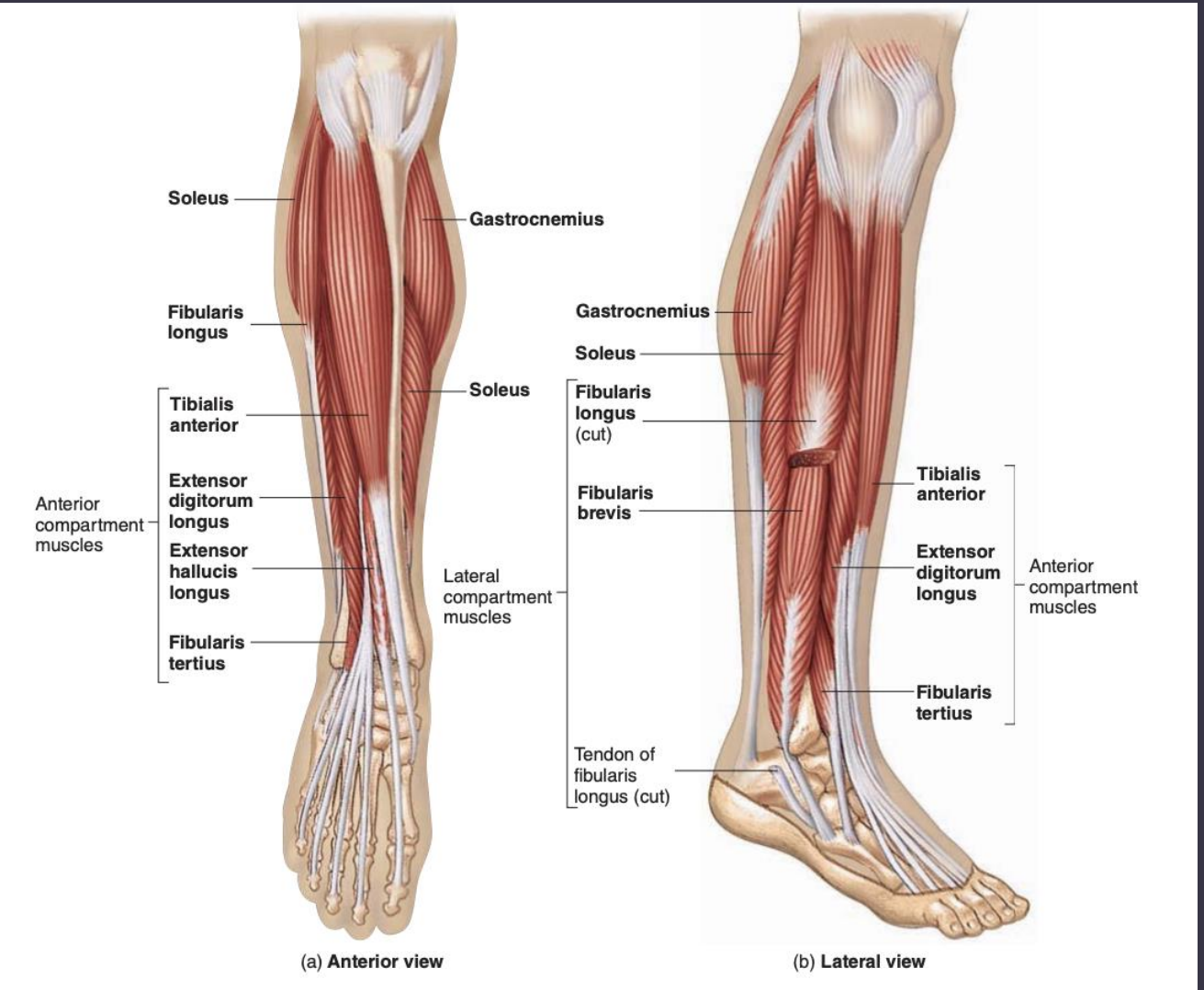
EXTRINSIC FOOT MUSCLES

3. Lateral compartment

- Function: primarily **eversion of the foot**; assist in plantar flexion

Fibularis brevis

Fibularis longus



ANKLE, FOOT AND TOE MOVEMENT

INTRINSIC FOOT MUSCLES

- Located within the foot itself

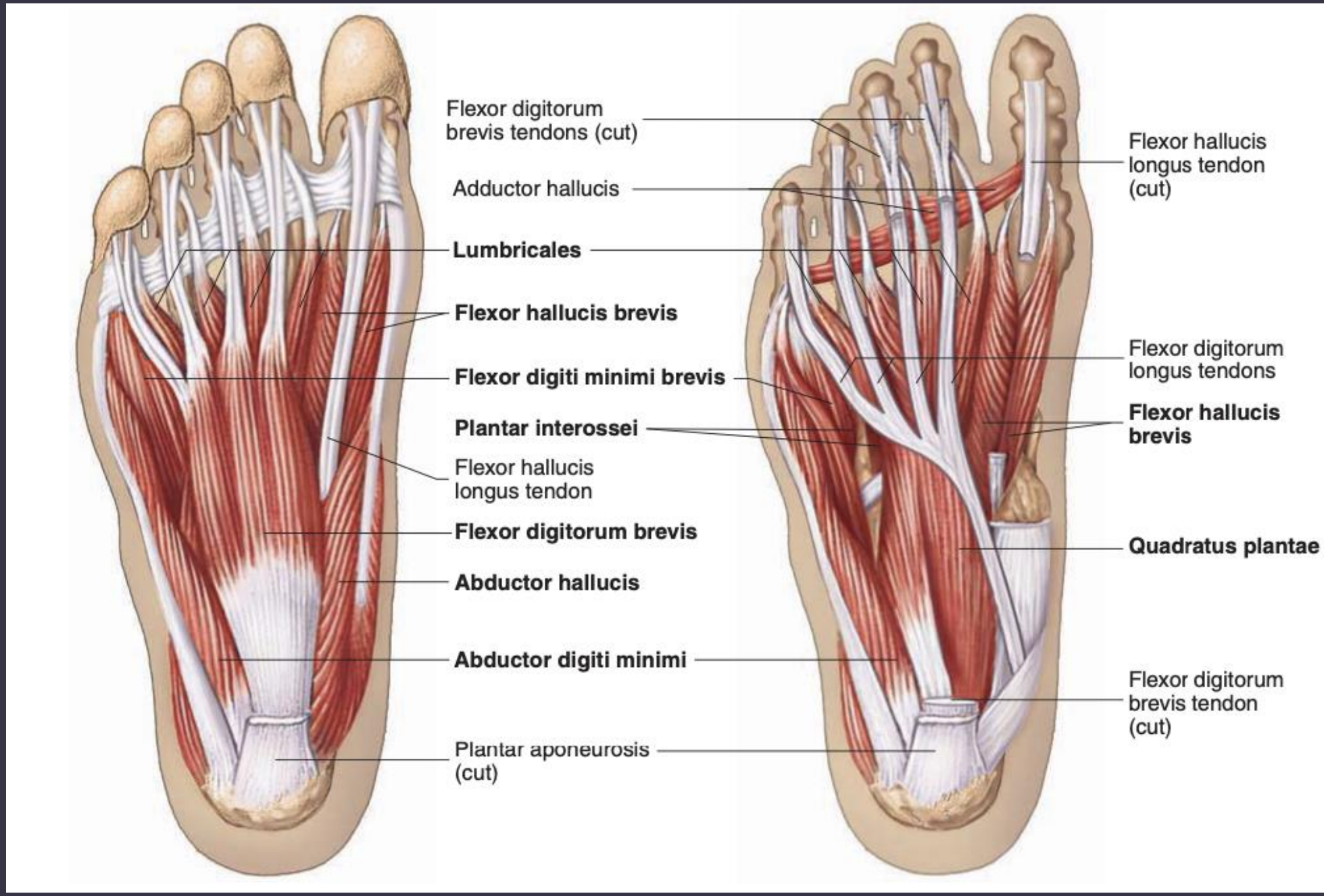
- **F**lex, extend, abduct, and adduct the toes

- **A**rranged in a manner similar to that of the intrinsic muscles of the hand

TABLE 10.23 Intrinsic Muscles of the Foot (see figure 10.36)

Muscle	Origin	Insertion	Nerve	Action
Abductor digiti minimi (ab-dŭk′ter, ab-dŭk′tōr dij′i-tī min′ĭ-mī)	Calcaneus	Proximal phalanx of fifth toe	Lateral plantar	Abducts and flexes little toe
Abductor hallucis (ab-dŭk′ter, ab-dŭk′tōr hal′i-sis)	Calcaneus	Base of proximal phalanx of great toe	Medial plantar	Abducts great toe
Adductor hallucis (a-dŭk′ter, a-dŭk′tōr hal′i-sis; not illustrated)	Lateral four metatarsal bones	Proximal phalanx of great toe	Lateral plantar	Adducts great toe
Extensor digitorum brevis (dij′i-tōr′ŭm brev′is; not illustrated)	Calcaneus	Four tendons fused with tendons of extensor digitorum longus	Deep fibular*	Extends toes
Flexor digiti minimi brevis (dij′i-tī min′ĭ-mī brev′is)	Fifth metatarsal bone	Proximal phalanx of digit 5	Lateral plantar	Flexes little toe (proximal phalanx)
Flexor digitorum brevis (dij′i-tōr′ŭm brev′is)	Calcaneus and plantar fascia	Four tendons to middle phalanges of four lateral toes	Medial plantar	Flexes lateral four toes
Flexor hallucis brevis (hal′i-sis brev′is)	Cuboid; medial and lateral cuneiforms	Two tendons to proximal phalanx of great toe	Medial and lateral plantar	Flexes great toe
Dorsal interossei (in′ter-os′e-ī; not illustrated)	Metatarsal bones	Proximal phalanges of digits 2, 3, and 4	Lateral plantar	Abduct second, third, and fourth toes
Plantar interossei (plan′tăr in′ter-os′e-ī)	Third, fourth, and fifth metatarsal bones	Proximal phalanges of digits 3, 4, and 5	Lateral plantar	Adduct third, fourth, and fifth toes
Lumbricales (lum′bri-kā-lēz)	Tendons of flexor digitorum longus	Extensor expansion of digits 2–5	Lateral and medial plantar	Flex proximal and extend middle and distal phalanges
Quadratus plantae (kwah′drā′tŭs plan′tē)	Calcaneus	Tendons of flexor digitorum longus	Lateral plantar	Assists flexor digitorum longus in flexing lateral four toes

INTRINSIC FOOT MUSCLES



INTRINSIC FOOT MUSCLES

ABDUCTION OF 2ND, 3RD 4TH TOES

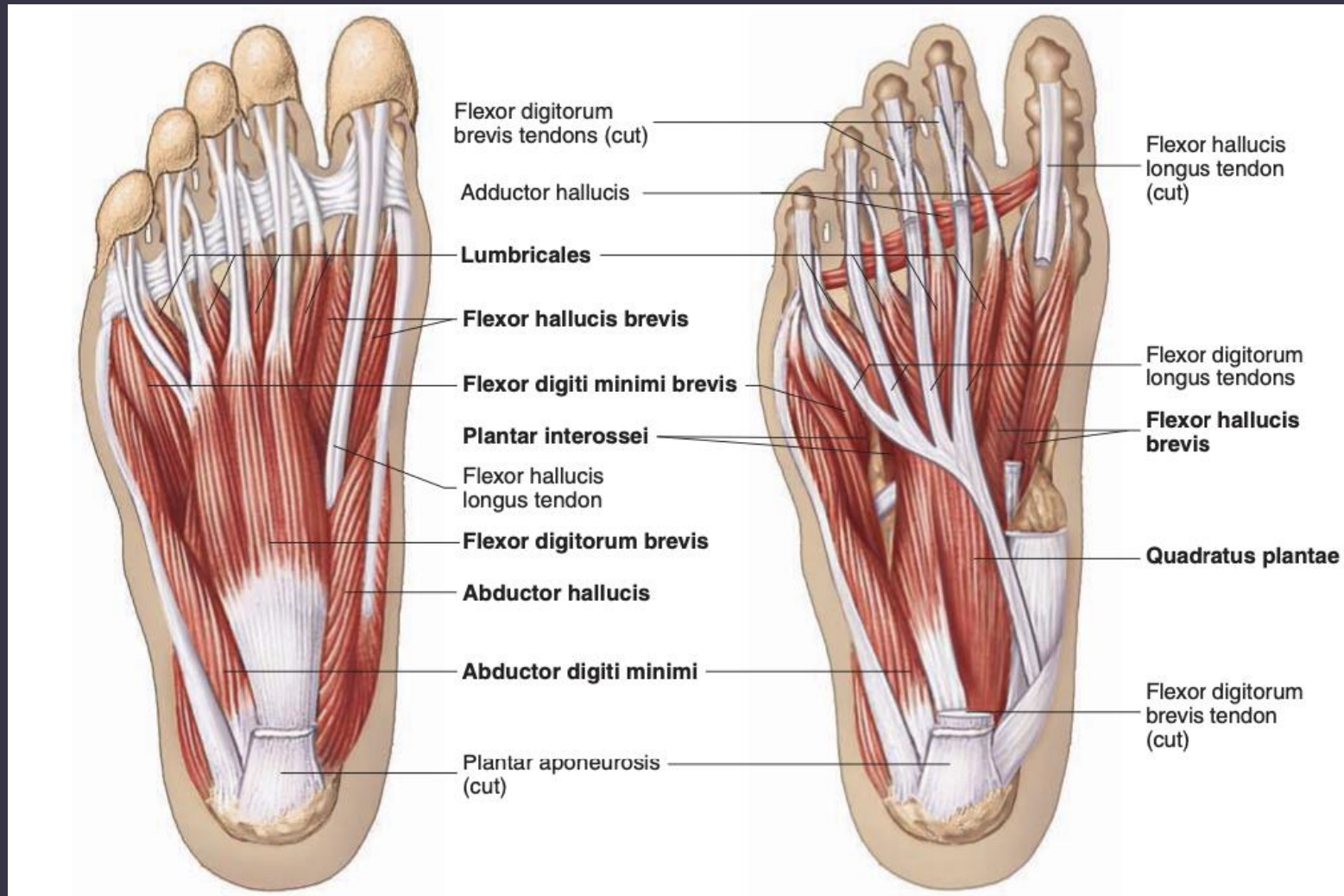
- Dorsal interossei

ABDUCTION OF THE 5TH TOE

- Abductor digiti minimi

ADDUCTION OF 3RD, 4TH, 5TH TOES

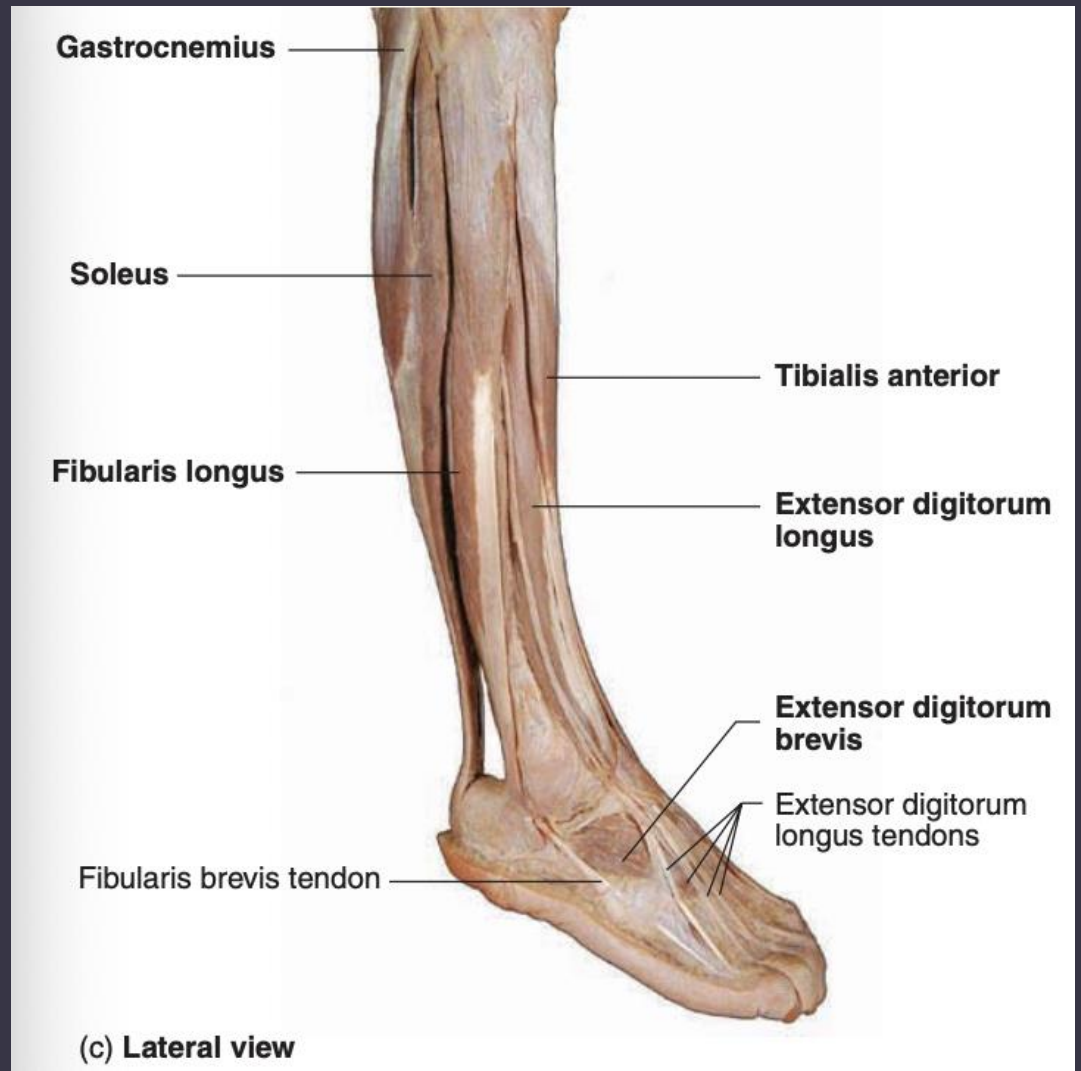
- Plantar interossei



OTHER FOOT MUSCLES

EXTEND THE 4 LATERAL TOES

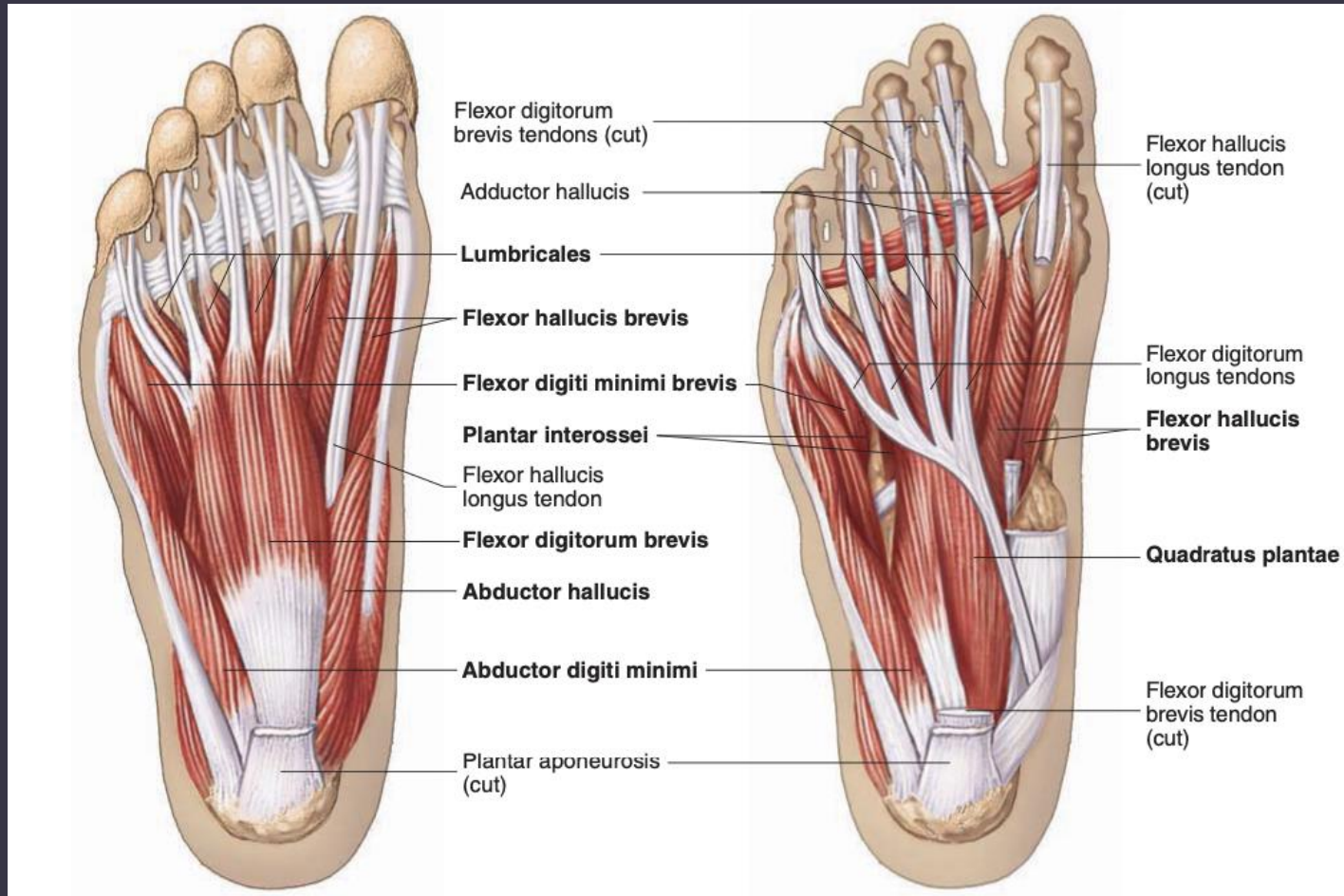
- **Extensor digitorum brevis**
- **Quadratus plantae**
- **Lumbricales** (middle and distal phalanges)
- **Extensor digitorum longus** (anterior leg muscle)



INTRINSIC FOOT MUSCLES

FLEX THE TOES

- **Flexor digiti minimi brevis** (little toe)
- **Flexor digitorum brevis** (lateral 4 toes)
- **Lumbricales** (proximal phalanges)
- **Quadratus plantae** (lateral 4 toes)

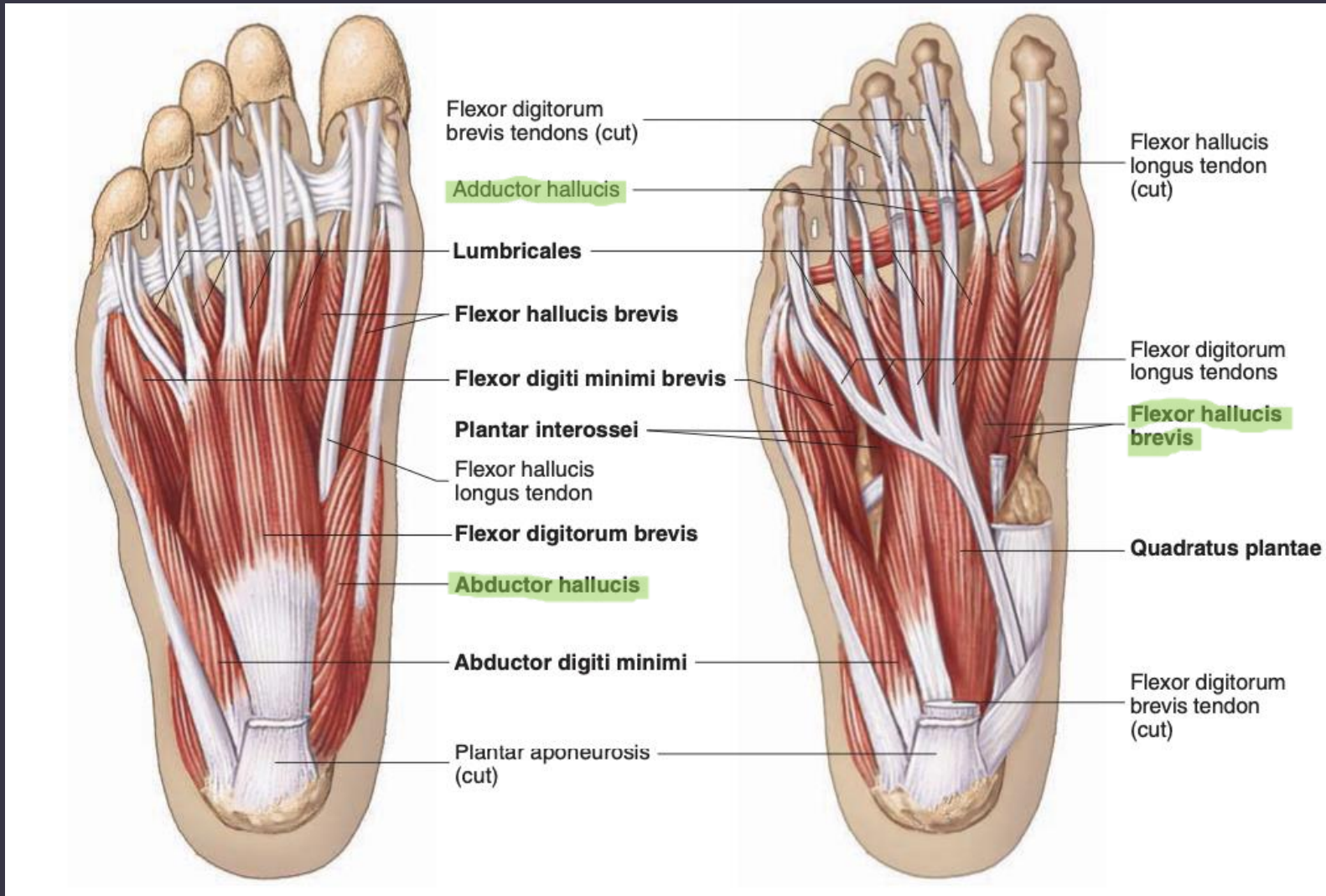


INTRINSIC FOOT MUSCLES
GREAT TOE

ABDUCT THE GREAT TOE
- Abductor hallucis

ADDUCT THE GREAT TOE
- Adductor hallucis

FLEX THE GREAT TOE
- Flexor hallucis brevis



DEVELOPMENTAL ASPECTS OF THE MUSCULAR SYSTEM

- **Nervous system maturation** → increasing muscular control
- Muscle control develops:
  - **Superior** → **Inferior** (head before legs)
  - **Proximal** → **Distal** (shoulder before fingers)
- **Exercise is essential** for muscle health
  - Without use → **atrophy** (loss of muscle mass)
  - With vigorous use → **hypertrophy** (muscle enlargement)
- **Aging effects:**
  - Progressive loss of muscle mass
  - Muscles become more sinewy (stringy)
  - Regular exercise slows decline and maintains strength



THANK YOU FOR LISTENING!